they did. The only immediate goal of the surgery is to help the patient. If Iridium goes bankrupt or the Shuttle fails to drive down launch cost, then we didn't do a good job*. This is the fundamental purpose of mission utility analysis—to evaluate and quantify what it means to meet the needs of the end user.

Essentially all missions go through mission analysis and evaluation stages many times. The literature base describing this process has increased significantly in recent years. Fortescue et al. [2003], Pisacane and Moore [2005], Larson et al. [2009], and Chesley et al. [2008] discuss the process for generic missions; Larson and Pranke [1999] do so for human missions; Przemieniecki [1993, 2000] does this for defense missions; Kemble [2006] does so for interplanetary missions; and NASA [2007] provides an excellent process summary for NASA missions. Kay [1995] discusses the difficulty of doing this within the framework of a political democracy and Sarsfield [1998], Sandau [2006], and Wertz and Larson [1996] provide specific techniques applicable to reducing mission cost, which is also discussed in Chap. 13.

5.1 Introduction to Mission Analysis

Mission analysis is the process of quantifying the system parameters and resulting performance. *Mission utility analysis*, described in Sec. 5.4, is the process of quantifying how well the system meets its overall mission objectives. Recall that the mission objectives themselves are not quantitative. However, our capacity to meet them should be quantified as well as possible in order to allow us to make intelligent decisions about whether and how to proceed. *Mission requirements,* introduced in Chap. 3 and discussed in more detail in Chap. 6, are the numerical expressions of how well the objectives must be met. At their best, they represent a balance between what we want and what is feasible within the constraints on the system and, therefore, deriving them should be a central part of the mission analysis activity. In practice, requirements are often set by a process outside of the space mission and unrelated to what is real or achievable. Therefore, it is often the case that mission analysis is concerned with how and how well externally defined mission requirements can be met. Nonetheless, in principle, mission analysis should be the process by which we define and refine both the system definition and mission requirements in order to meet our broad objectives at minimum cost and risk.

A key component of mission analysis is documentation, which provides the organizational memory of both the results and reasons. It is critical to understand fully the choices made, even those that are neither technical nor optimal. We may choose to apply a particular technology for political or economic reasons, or may not have enough manpower to investigate alternatives. In any case, for successful analysis, we must document the real reasons so others can reevaluate them later when the situation may be different. Technical people often shy away from nontechnical reasons or try to justify decisions by exaggerating their technical content. For example, as a result of our preliminary FireSat II analysis, we may choose a circular orbit at 700 km at an inclination of 60 deg because this is a good mid-range starting point. If so, we should document this as the reason, rather than trying to further justify these parameters or simply leave the reason undefined. If we have documented the reason, then later we, or others, can refine the choice for the best altitude and inclination, rather than having to live by choices for which there is no documented justification.

5.1.1 The Mission Analysis Hierarchy

I like to think of the mission analysis process as a huge spreadsheet model of a space system. Along the top of the spreadsheet matrix are the various parameters and alternatives that one might assess, such as power, orbit, and staffing levels for ground stations. Along the left are the various system parameters that flow from the mission choices, such as the orbit period, or coverage parameters, and along the bottom row are the system's quantitative outputs, indicating its performance, effectiveness, cost, and risk. The matrix itself would capture the functional relationships among the many variables. We would like to wiggle any particular parameter—such as the altitude, or the number of people assigned to the ground station—and determine the effect on all of the other parameters. In this way, we could quantify the system's performance as a function of all possible variables and their combinations.

Fortunately for the continuing employment of mission analysts, the above spreadsheet model does not yet exist.† Instead, we analyze as many reasonable alternatives as possible so we may understand how the system

* What does "we" mean in this context? Obviously most aerospace engineers weren't directly involved in either Iridium or the Space Shuttle. Nonetheless, all of us collectively set the rules and define the "culture" in which we all work. It is this culture that has let it be acceptable in some sense, for the Shuttle to miss its recurring cost projections by a factor of 20 or so and for Iridium to go bankrupt, rather than make a good return on investment for those that put up the money. Our surgeon analogy continues to work here. Surgeons, collectively aren't responsible for the death of a single patient. But they do make the rules under which surgeons operate. If that death causes a reassessment of those rules, it's still not good, but at least the community is responding. The fundamental question for mission utility and space mission engineering is whether our community is responding appropriately and adequately to the fact that programs cost too much and take too long.

† In due course, system engineers may become technologically obsolete. The challenge to future system engineers will be to stay ahead of the computer models in being creative and innovative. One of the elements most often missed by automated processes is how to think about the problem in a totally different way. For example, can we drive down the cost of a human mission to Jupiter by first putting a settlement on the Moon and using those resources to build low-cost interplanetary systems? These are the types of questions that are very difficult to answer by simply adjusting parameters in a model.

behaves as a function of the principal design features, i.e., the system drivers (Chap. 4). This approach does not imply that we are uninterested in secondary detail, but simply recognizes that the mission analysis process, much like the space system we are attempting to analyze, is ultimately limited in both cost and schedule. We must achieve the maximum level of understanding within these limits.

If the resources available for concept exploration are limited, as is nearly always the case in realistic situations, then one of the most critical tasks is to intelligently limit the scope of individual analyses. We want to be able to compute approximate values for many parameters and to determine at what level of detail we should reasonably stop. In practice, this is made difficult by the continuing demand for additional detail and depth. Thus, we would like to be able to determine and make clear to others what elements of that detail will significantly affect the overall system performance and what elements, while important, can reasonably be left to a more detailed design phase.

We use two main methods to limit the depth of analysis in any particular area. The first is to clearly identify each area's system drivers by the methods in Sec. 4.4 and to concentrate most of the mission analysis effort on these drivers. The second is to clearly identify the goal of the system study and to provide a level of detail appropriate to that goal. This approach leads to a *mission analysis hierarchy*, summarized in Table 5-1, in which studies take on increased levels of detail and complexity as the activity progresses. The first three types of studies are meant to be quick with limited detail and are not intended to provide definitive results. The last three are much more complex ways to select an alternative to provide the best system performance. Each of these types of analysis is discussed below.

5.2 Studies with Limited Scope

The first three types of analyses in Table 5-1 provide methods for undertaking a *quick-look assessment*. They provide limited detail, but can frequently be done quickly and at low cost. Consequently, these quick-look assessments are important in any situation that is funding-limited or time-limited, which is very often the case in real mission analysis. We will briefly outline these methods here. However, nearly the entire book is devoted to the process of making initial estimates, which is the basic goal of limited scope studies. We want to be able to understand whether or not a particular project is feasible, and to get some idea of its size, complexity, and cost. Doing this requires that we be able to make numerical estimates and undertake limited studies in order to develop insight into the nature of the problem we are trying to solve.

The biggest difficulty with limited scope studies is the tendency to believe that they are more accurate than they really are. Thus, it is not uncommon to use a feasibility assessment or point design to establish the requirements for a mission in such detail that, in practice, the point design becomes the only alternative which can meet them. As long as we recognize the limited scope of these studies, they have a valuable place in the mission analysis hierarchy and represent one of the most important tools that we can use to understand the behavior of the system we are designing.

5.2.1 Feasibility Assessment

The simplest procedure in the mission analysis hierarchy is the *feasibility assessment*, which we use to establish whether a particular objective is achievable and to place broad bounds on its level of complexity. Frequently, we can do a feasibility assessment simply by comparison with existing systems. Thus, we are reasonably convinced that FireSat II is feasible because most FireSat II tasks could be performed by existing Earth resources satellites. Similarly, it is feasible to land people on the Moon and return them safely to Earth because we have done so in the past.

We can also determine whether a particular goal is feasible by extrapolating our past experience. Is it feasible to send people to Mars and bring them back safely? Here we need to look at the principal differences

Table 5-1. The Mission Analysis Hierarchy. During the preliminary design phase, these help us decide how detailed studies should be.

Analysis Type	Where Discussed	Goal	
Feasibility Assessment	5.2.1	To establish whether an objective is achievable and its approximate degree of complexity	Quick, limited detail
Sizing Estimate	5.2.2	To estimate basic parameters such as size, weight, power, and cost	
Point Design	5.2.3	To demonstrate feasibility and establish a baseline for comparison of alternatives	
Trade Study	5.3.1	To establish the relative advantages of alternative approaches or options	More detailed, complex trades
Performance Assessment	5.3.2	To quantify performance parameters (e.g., resolution, timeliness) for a particular approach	
Utility Assessment	5.4	To quantify how well the system meets overall mission objectives and satisfies the needs of the end user	

between a Mars mission and a Lunar mission. These differences include a longer flight time and higher gravity at Mars than on the Moon, and, therefore, a need for greater radiation protection during the mission and a higher lift-off velocity to leave Mars. These factors make the job more challenging and possibly more expensive than going to the Moon, but there is nothing about the Mars mission that makes it inherently impossible. Getting to Mars is feasible. The problem is being able to do so at modest cost and risk such that we are willing to both pay for the mission and accept the risk.

The third method of providing a feasibility assessment is to provide a very broad design of how such a mission might be accomplished. For example, in the 1970s, Gerard O'Neill of Princeton University proposed building large space colonies at the Lagrange points between the Earth and the Moon. (See Sec. 2.1.9 and O'Neill [1974].) No mission of this scope had ever been undertaken (and still hasn't), and it certainly was not a straightforward extrapolation of any of our normal space experience. O'Neill and his colleagues proceeded to establish the feasibility by developing a variety of alternative designs for such space colonies. While the work done was far in excess of a simple feasibility assessment, it clearly established that such colonies were feasible and gave at least an estimate of the scope of the problem.

5.2.2 Sizing Estimate

The purpose of the *sizing estimate* is to provide an estimate of basic mission parameters such as size, weight, power, or cost. We can do sizing estimates in much the same manner as the feasibility assessment: by analogy with existing systems. Thus, if we are aware of an Earth observation system that has resolution and information characteristics comparable to what we believe are needed for FireSat II, we can use these parameters to give us an initial estimate of the FireSat II parameters.

We can also provide a quantitative estimate of key mission parameters by scaling the parameters from existing missions or payloads in order to obtain estimates of the component sizes for our particular mission. This scaling process is described in Sec. 17.2.6 for space payloads, and in Sec. 14.5 for the spacecraft as a whole. The process of sizing by scaling existing hardware is an extremely powerful approach to estimating what it will take to achieve mission objectives. It is of use not only during the conceptual design process, but throughout the hardware design, definition, and development phases to evaluate the system design as it evolves. If scaling existing systems leads to the suggestion that a particular component should be twice as heavy as the current design suggests, this gives us reason to look very closely at the current design and to try to determine whether or not any factors have been overlooked. We assume that designers of previous systems did a reasonable job of optimizing their system. If the current design is significantly different, either better or worse, then we would like to understand the reasons for these differences. It may be that the technology has advanced to the point where a better design is possible or it may be that the current preliminary design has simply forgotten a key element. This process is a good way to gain confidence in the overall design as we proceed.

As the design develops, more and more accurate sizing estimates come from the scaling process. We proceed by breaking down the system into components and sizing individual components based on scaling estimates with prior systems. Thus, we may initially estimate the system as a whole divided into a spacecraft and payload. As the design becomes more detailed, we will break down the spacecraft into its subsystems and components and estimate the size, weight, and power of each element based upon scaling from prior systems or engineering estimates of the new system to be built. Similarly, we initially size the payload or ground station by comparison with existing systems and eventually by building a list of all the components and undertaking similar sizing estimates for each component. As introduced in Chap. 3, this process of creating a list of components and estimating parameters for each is known as budgeting and is described in more detail in Sec. 14.5.

5.2.3 Point Design

A *point design* is a design, possibly at a top level, for the entire system that is capable of meeting the broad mission objectives. We refer to it as a point design if we have not attempted to optimize the design to either maximize performance or minimize weight, cost, or risk. The point design serves two basic purposes. It demonstrates that the mission is feasible, and it can be used as a baseline for comparison of alternatives. Thus, if we can establish a point design for FireSat II that meets mission objectives with a spacecraft that weighs 500 kg and costs $50 million, then we can use this as a comparison for later systems. If other systems cost more, weigh more, and don't perform as well, then we will abandon those alternatives in favor of the original baseline. If we continue to improve the design so that the cost and risk decrease, then we will let the baseline evolve to take into account the new approaches.

A point design is valuable because we can do it quickly and easily. There is no need to optimize any of the parameters associated with the design unless it is necessary to do so to meet mission objectives. This gives us a sense of whether it will be easy or hard to meet the mission objectives and what are likely to be the most difficult aspects. One of the biggest problems in a point design is taking it too seriously at a later stage. We are always likely to regard work that we have done as representing the best approach, even though we may not have been aware of alternatives. The key issue here is to make use of point designs but at the same time to recognize their limitations and to continue to do trades to reduce overall cost and risk and to look for alternative approaches to meet mission objectives better and at lower cost.

5.3 System Trade Studies and Performance Assessments (Step 8)

Step 8 of our overall Space Mission Engineering process defined in Table 3-1, is to conduct performance assessments and system trades. Deciding whether to proceed with a mission should be based on a strawman system concept or a point design that shows that the mission objectives can be met within the assigned constraints. Of course, the initial point design is probably not the best solution, and we would ordinarily consider a number of alternatives. This process of looking at alternatives is the *system trade process*. Generally there are three types of systems trades that are done, or should be done:

- Single effect trades (easy to do)
- Multiple effects trades (hard to do)
- New and unique approaches (very hard to do)

The system trade process evaluates different broad concepts to establish their viability and impact on performance, cost, risk, and schedule. We then combine the system trade results with the performance assessment (Sec. 5.3.2) and the mission utility analysis (Sec. 5.4) to provide input for concept selection (Sec. 5.7).

5.3.1 System Trade Studies

System trades consist of analyzing and selecting *system drivers*, i.e., key system parameters defined in Sec. 4.4, which largely determine mission performance, cost, risk, and schedule. We use these parameters to define a mission concept and mission architecture that can then be used for performance analysis and utility analysis as described in Secs. 5.3.2 and 5.4. The key system trades are those that define how the system works and determine its size, cost, and risk. Typically, the key system trades will be in one of the major areas defined in Table 5-2. Here the first question should always be: What are the real objectives of the mission? In addition, it would be wise to review the broad objectives of the community involved, as discussed in Chap. 2. What are the long-term goals and objectives of the community and how will this mission support those goals? For example, space telescopes of all sorts are typically out to find things that have never been seen before, which makes defining the system parameters a bit challenging. For military systems, our goal is ordinarily to provide support to the warfighter or gather intelligence about the enemy.

Table 5-2. Principal Areas for Key System Trades. See Table 5-3 for examples.

Area	Principal Issue	Where Discussed
Mission Objectives	What are we really trying to accomplish?	Sec. 4.1
Critical Requirements	What are the requirements that are hard to meet and, therefore, drive cost, risk, and schedule?	Sec. 4.4
Mission Concept	How do we get the information and data and how do we get the results to the end user?	Sec. 4.1
Subject	What do we want to look at or measure to get the information we want?	Sec. 15.2
Type and Complexity of Payloads	What spectral band do we work in and how do we measure it? How many payloads?	Sec. 15.1
Orbit	A specialized orbit or one of a continuum?	Chap. 10
Number of Spacecraft	A single satellite or a constellation?	Chap. 10

Table 5-3. Examples of Representative Key System Trades. Although these system trades are critical, we can't expect numerically precise answers to our system design problem.

Trade Area	FireSat II	SCS	James Webb Space Telescope	Aircraft Detection
Mission Objectives	Save lives and property or reduce fire fighting costs?	Support the warfighter?	Discover stuff we've never seen? If so, how do we build the telescope?	Traffic control or smuggling interdiction?
Critical Requirements	What's a forest fire? What's the needed resolution?	Small omni ground antenna?	How cold a detector? How large an aperture?	How small a plane do we need to detect and track?
Mission Concept	Identify forest fires on board or on the ground?	Circular orbits with fore and aft fixed crosslinks?	How do we deploy and control the mirror?	Is it a cooperative target? How do we cover the areas of interest?
Subject	Heat, fire, or smoke?	Likely a handheld receiver.	Something we've never seen and don't know.	Commercial planes or small private planes?
Payload Type and Complexity	What wavelength? How many spectral bands?	What band? How many?	How many sensors and instruments?	IR, visible, radar, RF?
Orbit	LEO or GEO	LEO, HEO, MEO, GEO	Lagrange point or closer and cheaper?	LEO, HEO, MEO?
Number of Spacecraft	1 or a constellation?	Probably a constellation	Probably only 1	1 or a constellation?

Table 5-3 shows typical examples of key system-trades for several representative missions. For essentially all missions, specification of the critical requirements will be a key system trade. For the FireSat II mission, the subject is probably the heat from the fire itself and the payload is probably an IR sensor. Thus, the principal system trades are probably the mission objectives, mission concept, the resolution and coverage requirements, the orbit, and whether it's a single satellite or a constellation. For a mission such as the James Webb Space Telescope, there will be only 1 telescope, the orbit is of marginal importance, and the subject is moderately well defined, if only very poorly known. Here the principal trades will be the resolution and pointing requirements, the payload, and the mission concept. Communications satellite systems are normally in geosynchronous orbit with a well-defined concept of operations. Here the only real trade is with the required traffic load, the subject, and the size and complexity of the payload.

Truly innovative approaches—those that really change how we think about a problem—typically involve finding a new option among these key system trades. Motorola's Iridium program and subsequent low-Earth orbit communications constellations represented a new way of thinking about using satellites for communications. These have a very different concept of operations and different orbits from traditional systems. Similarly, Chap. 13 presents an innovative approach to thinking about FireSat II that provides a totally different concept of operations and type of payload. Innovative solutions are never easy to come by. To try to find them, a good place to start is with the key system trade areas given in Tables 5-2 and 5-3.

We cannot normally do system trades in a straightforward numerical fashion. Choosing a different concept of operations, for example, will result in changes in most or all of the mission parameters. Consequently, the fact that Option A requires twice the power of Option B may or may not be critical, depending on the orbit and number of satellites for the two options. We need to look at the system as a whole to understand which is better.

The best approach for key system trades is a utility analysis as described in Sec. 5.4. We use the utility analysis to attempt to quantify our ability to meet mission objectives as a function of cost. We then select the option that fulfills our objectives at the lowest cost and risk. As described in Sec. 5.7, this is still not a straightforward numerical comparison, but does have quantitative components.

The simplest option for system trades is a list of the options and the reasons for retaining or eliminating them. This allows us to consider the merits and demerits at a high level without undertaking time-consuming trades. This also allows our list to be challenged at a later date. We should go back to our key system trades on a regular basis and determine whether our assumptions and conclusions are still valid. It is this process of examination and review that allows us to introduce technical innovation and new ideas. It is a process that must occur if we are to drive down the cost of space systems.

Table 5-3◀, Fig. 5-0, Eq. 5-0

Single Effect Trades

For the purpose of trade studies, system drivers generally divide into two categories—those for which more is better (*single effect drivers*) and those with multiple effects. By far the easier to deal with are the "more is better" single effect drivers, for they simply require us to ask: "What is the cost of achieving more of the commodity in question?" For example, in a space-based radar, added power improves performance but costs more money. Thus, the designer will want to understand how much performance is available for how much power. A second example is coverage. For virtually any Earth-oriented system, including our FireSat II example, more coverage means better performance at higher cost. Increasing coverage ordinarily means adding satellites or, perhaps, increasing a single satellite's coverage by increasing its altitude or the range of its sensors. Therefore, we often do a coverage trade considering performance vs. number of satellites, substituting the latter for cost. Assessing performance as a function of power or coverage may take considerable work, but it is relatively easy to present the data for judging by the funding organization, the users, or other decision makers.

Multiple Effect Trades

System drivers and critical requirements that cause multiple effects demand more complex trade studies. The basic problem here is that pushing parameters one way will improve some characteristics and degrade others. In trades of this type, we are looking for a solution that provides the best mix of results. Examples of such trade studies include instrument design, antenna type, and altitude. Each antenna style will have advantages and disadvantages, so we must trade various possible solutions depending upon the end goals and relative importance of different effects.

In trades with multiple effects, selecting the correct independent parameter for each trade is critical. Consider, for example, selecting either a reflector or a phased-array antenna for a space-based radar [Brookner and Mahoney, 1986]. From the radar equation, we know that a principal performance parameter for a radar is the antenna aperture. All other things being equal, larger antennas will provide much better performance. Thus, for our example, we might choose to compare reflector and phased-array antennas of equal aperture. On this basis, we would choose the phased array because its electronic steering makes it more agile than a reflector antenna, which must be mechanically steered. But our choice becomes more complex when we recognize that weight typically limits large space structures more than size does. Generally, we can build a reflector larger than a phased array for a given weight. Based on weight, a reflector may have considerably more power efficiency and, therefore, be a better radar than a phased-array system. Thus, we would have to trade the better performance of a larger reflector vs. the better agility of a smaller phased array. Depending upon the application, the results may be the same as for an aperture-based trade

Table 5-4. System Trade Process for Parameters with Multiple Effects. The example is the altitude trade for the FireSat II mission. See also Fig. 5-1.

Step	FireSat II Example	Where Discussed
1. Select trade parameter (typically a system driver)	Altitude	Sec. 10.2
2. Identify factors which affect the parameter or are affected by it	Coverage Deployment strategy and coverage evolution Orbit period Time in view Eclipse fraction Response time Number of spacecraft needed Launch capability Resolution Radiation environment Communications Lifetime	Sec. 10.2.1 Sec. 10.6.2 Sec. 9.1.5, App. I Sec. 8.4 Sec. 8.2.4 Sec. 10.2.1.4 Sec. 10.6.1 Secs. 10.3.2, 26.1 Sec. 10.2.2.1 Chap. 7 Secs. 16.2, 21.1 Secs. 3.4, 9.4.4, Chap. 30
3. Assess impact of each factor	Can launch up to 1,800 km Best coverage above 600 km Resolution—lower is better Survivability—higher is better, but not much	—
4. Document and summarize results	Launch Coverage Resolution Survivability	Fig. 5-1
5. Select parameter value and possible range	Altitude = 700 km Range = 600 to 800 km	In text

or reverse. The important point is the critical nature of selecting the proper independent variable in system trades. To do so, we must find the quantities that inherently limit the system being considered. These could be weight, power, level of technology, cost, or manufacturability, depending on the technology and circumstances.

Table 5-4 summarizes the system trade process for parameters with multiple effects. Typically the trade parameter is one of our system drivers. We begin by identifying what performance areas or requirements affect or are affected by the trade parameter. For example, the altitude of the spacecraft will have a key effect on coverage, resolution, and survivability and will be limited by launchability, payload weight, communications, and radiation. We next assess the effect in each of these areas and document and summarize the results, generally **without** trying to create a numerical average of different areas. We use the summary to select the parameter value and a possible range. Although the process is complex and may not have a well-defined answer, it is not necessarily iterative unless we find that the results require fundamental changes in other system parameters.

Altitude trades are perhaps the most common example of a trade in which multiple influences push the parameter in different ways. We would normally like to move the satellite higher to achieve better coverage, better survivability, and easier communications. On the other hand, launchability, resolution, and payload weight tend to drive the satellite lower. The radiation environment dictates specific altitudes we would like to avoid, and the eclipse fraction may or may not play a crucial role in the altitude trade. We must assess each of these effects and summarize all of them to complete our trade study. One possible summary is a numerically weighted average of the various outcomes, such as three times the coverage in square nautical miles per second divided by twice the resolution in furlongs. Although this provides a convenient numerical answer, it does not provide the physical insight or conceptual balance needed for intelligent choices. A better solution is to provide the data on all of the relevant parameters and choose based on inspection rather than numerical weighting.

The FireSat II altitude trade provides an example of this process. For FireSat II, neither survivability nor communications is a key issue, but coverage does push the satellite upward. On the other hand, payload weight and good resolution tend to push the satellite lower. Figure 5-1 shows the results of a hypothetical FireSat II altitude trade. Notice that each parameter has various possible outcomes. Altitudes above or below a certain value may be eliminated, or we may simply prefer a general direction, such as lower altitude providing better resolution. Based on these results, we select a nominal altitude of 700 km for FireSat II and a possible range of 600 to 800 km. This selection is neither numerical nor purely qualitative. We have tried to balance the alternatives sensibly, but not in a way that we can numerically justify.

New and Unique Approaches

By far the most challenging system trades, and also those that have the most potential benefit, are trades that involve new and unique approaches to system design. This is where potential major breakthroughs come about that can dramatically alter the way we do business in

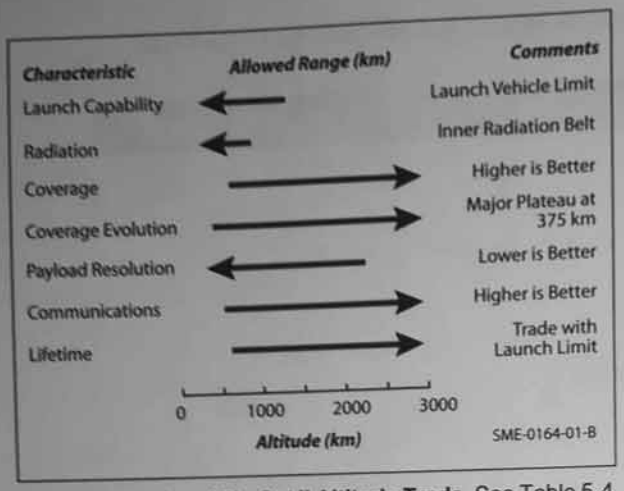


Fig. 5-1. Results of FireSat II Altitude Trade. See Table 5-4 for a list of altitude trade issues. Political constraints and survivability were not of concern for the FireSat II altitude trade.

space. Typically, it's very difficult to make major breakthroughs occur, in part because they may end up being more difficult and less advantageous than was first anticipated (i.e. the reusable Space Shuttle and its goal of dramatically reducing launch cost) and in part because it is difficult in the very conservative space community to move new and innovative approaches beyond the study phase and into real systems (such as the O'Neill space colonies described in Sec. 2.1.9).

An example of this, that also illustrates the potential pitfalls, is the Iridium communications constellation discussed earlier (in the chapter) and in Sec. 2.1.4. Traditionally, communications systems had been thought of as either geosynchronous or Molniya orbits. Iridium took a very different approach and used a LEO constellation instead which, unfortunately, did not work out as intended. On the other hand, both GlobalStar and ORBCOMM are now working successfully in LEO, and Iridium itself has regained sufficient business that an Iridium Next constellation is under development.

A new and unique approach that is working very well is the development of CubeSats, initially intended for educational purposes, but now expanding into highly functional small satellites for a variety of purposes. (See Secs. 2.1.8 and 25.4.)

In the communities that actually fund and build space systems, the acceptance of new approaches moves with all the rapidity of a glacier moving uphill. Nonetheless, we very much need new ideas and new approaches if we are to change the way business is done in space. As we saw in Chap. 1, space missions cost too much and take too long, and we need to find ways to reverse that trend. We will return to this, and discuss explicitly how to do it, in Chap. 13.

5.3.2 Performance Assessments

A *performance assessment* is a numerical analysis of how well the system will perform in specific areas. Computing the area search rate, the available data rate, or the resolution of an optical system are all the result of a performance assessment.

Quantifying performance demands an appropriate level of detail. Too much detail drains resources away from other issues; too little keeps us from determining the important issues or causes us to assess the actual performance incorrectly.

To compute system performance, we use three main techniques:

- System algorithms
- Analogy with existing systems
- Simulation

System algorithms are the basic physical or geometric formulas associated with a particular system or process, such as those for determining resolution in diffraction-limited optics, finding the beam size of an antenna, analyzing a link budget, or assessing geometric coverage. Table 5-5 lists system algorithms typically used for space mission analysis. System algorithms provide the best method for computing performance. They provide clear traceability and establish the relationship between design parameters and performance characteristics. Thus, for FireSat II, we are interested in the resolution of an on-orbit fire detector. Using the formula for diffraction-limited optics in Chap. 17, we can compute the achievable angular resolution from the instrument objective's diameter. We can then apply the geometric formulas in Chap. 8 to translate this angular resolution to resolution on the ground. This result gives us a direct relationship between the altitude of the FireSat II spacecraft, the size of the payload, the angles at which it works, and the resolution with which it can distinguish features on the ground. Similarly, communications systems follow a very-well defined *link budget*, that can determine with

Table 5-5. Common System Algorithms Used for Quantifying Basic Levels of Performance. These analyses use physical or geometrical formulas to determine how system performance varies with key parameters.

Algorithm	Used For	Where Discussed
Link Budget	Communications and data rate analysis	Sec. 16.2
Diffraction-limited Optics	Aperture sizing for optics or antennas; determining resolution	Sec. 17.1
Payload Sensitivity	Payload sizing and performance estimates	Sec. 17.2
Radar Equation	Radar sizing and performance estimates	Cantafio [1989]
Earth Coverage, Area Search Rates	Coverage assessment; system sizing; performance estimates	Chap. 8, App. C
Mapping and Pointing Budget	Geolocation; instrument and antenna pointing; image sensing	Sec. 8.6

good precision, just how fast we will be able to get data back from Jupiter, given the power, wavelength or frequency, and the diameter of the antennas on the ground and on the spacecraft.

System algorithms are powerful in that they show us directly how performance varies with key parameters. However, they are inherently limited because they presume the rest of the system is designed with fundamental physics or geometry as the limiting characteristic. For FireSat II, resolution could also be limited by the optical quality of the lens, by the detector technology, by the spacecraft's pointing stability, or even by the data rates at which the instrument can provide results or that the satellite can transmit to the ground. In using system algorithms, we assume that we have correctly identified what limits system performance. But we must understand that these assumptions may break down as each parameter changes. Finding the limits of these system algorithms helps us analyze the problem and determine its key components. Thus, we may find that a low-cost FireSat II system is limited principally by achieving spacecraft stability at low cost. In this case, our attention would be focused on the attitude control system and on the level of resolution that can be achieved as a function of system cost.

The second method for quantifying performance is by comparing our design with existing systems. In this type of analysis we use the established characteristics of existing sensors, systems, or components and adjust the expected performance according to basic physics or the continuing evolution of technology. The list of payload instruments in Chap. 15 is an excellent starting point for comparing performance with existing systems. We could, for example, use the field of view, resolution, and integration time for an existing sensor and apply them to FireSat II. We then modify the basic sensor parameters such as the aperture, focal length, or pixel size, to satisfy our mission's unique requirements. To do this, we must work with someone who knows the technology, the allowable range of modifications, and their cost. For example, we may be able to improve the resolution by doubling the diameter of the objective, but doing so may cost too much. Thus, to estimate performance based on existing systems, we need information from those who understand the main cost and performance drivers of that technology.

The third way to quantify system performance is simulation, described in more detail in Sec. 5.4.2. Because it is time-consuming, we typically use simulations only for key performance parameters. However, simulations allow much more complex modeling and can incorporate limits on performance from multiple factors (e.g., resolution, stability, and data rate). Because they typically provide much less insight, however, we must review the results carefully to see if they apply to the specific situation we have in mind. Still, in complex circumstances, a simulation may be the only acceptable way to quantify system performance. A much less expensive method of simulation is the use of commercial mission analysis tools or even Excel or MATLAB models.

5.4 Mission Utility and Figures of Merit—Is the Mission Worthwhile? (Step 9)

Is the mission worth doing? Does it meet the fundamental objectives of the end user or the people who paid to create it? Does FireSat II save lives and property and reduce fire fighting costs sufficiently to warrant flying the mission? *Mission utility analysis* is the process of trying to provide a quantitative assessment of the degree to which the system meets the basic objectives it is attempting to fulfill. This is used to (1) provide quantitative information for decision making, and (2) provide feedback on the system design. Ultimately, an individual or group will decide whether to build a space system and which system to build based on an overall assessment of whether the system can meet its mission objectives at an acceptable cost and risk and do so better than alternative approaches. As discussed in Sec. 5.7, this does not mean the decision is, or should be, fundamentally technical in nature. However, even though basic decisions may be political, economic, or sociological, the best possible quantitative information from the mission utility analysis process should be available to support them.

Mission utility analysis also provides feedback for the system design by assessing how well alternative configurations meet the mission objectives. FireSat II shows how this process might work in practice. Mission analysis quantifies how well alternative systems can detect and monitor forest fires, thereby helping us to decide whether to proceed with a more detailed design of several satellites in low-Earth orbit or a single larger satellite in geosynchronous orbit. As we continue these trades, mission utility analysis establishes, for example, the probability of being able to detect a given forest fire within a given time, with and without FireSat II, and with varying numbers of spacecraft. For FireSat II, the decision makers are those responsible for protecting the forests of the United States. We want to provide them with the technical information they need to determine whether they should spend their limited resources on FireSat II or on some alternative. If they select FireSat II, we will provide the technical information needed to allow them to select how many satellites and what level of performance to provide.

5.4.1 Performance Parameters and Measures of Effectiveness

The purpose of mission analysis is to quantify the system's performance and its ability to meet the ultimate mission objectives. Typically this requires two distinct types of quantities—performance parameters and measures of effectiveness. *Performance parameters*, such as those shown in Table 5-6 for FireSat II, quantify how well the system works, without explicitly measuring how well it meets mission objectives. Performance parameters may include coverage statistics, power efficiency, or the resolution of a particular instrument as a function of nadir angle and altitude. In contrast, *measures of effec-*

Table 5-6. Representative Performance Parameters for FireSat II. By using various performance parameters, we get a better overall picture of our FireSat II design.

Performance Parameter	How Determined
Smallest detectable fire	Analysis
Resolution	Analysis
Orbit average area coverage rate	Simulation
Fire location accuracy	Analysis
Temperature of the fire	Analysis
Average number of times per day the fire is observed	Simulation

tiveness (MoEs) or *figures of merit* (FoMs) quantify directly how well the system meets the mission objectives. For FireSat II, the principal MoE will be a numerical estimate of how well the system can detect forest fires or the consequences of doing so. This could, for example, be the probability of detecting a given forest fire within 6 hours, or the estimated dollar value of savings resulting from early fire detection. Table 5-7 shows other examples for FireSat II.

Table 5-7. Representative Measures of Effectiveness (MoEs) for FireSat II. These Measures of Effectiveness help us determine how well various designs meet our mission objectives.

Goal	MoE	How Estimated
Fire Detection	Probability of detection vs. time since start of fire (milestones at 6, 12, and 24 hours)	Simulation
Prompt Knowledge	*Time late*—time from observation to availability to the fire captain	Analysis + Budget
Containment	Probability of containment vs. time	Simulation
Save Property and Reduce Cost	Value of property saved plus savings in fire fighting costs	Simulation + Analysis
Save Lives	Estimate of the number of lives saved as a result of having FireSat II available	Scenario Evaluation

We can usually determine performance parameters unambiguously. For example, resolution is determined by an optical analysis as discussed in Sec. 17.3 and the level of coverage for any point on the Earth's surface is determined by analysis or simulation. The probability of detecting and containing forest fires (i.e., MoEs) better measure our end objective, but are also much more difficult to quantify. It may depend on how we construct scenarios and simulations, what we assume about ground resources, and how we use the FireSat II data in conjunction with other data sources to fight fires.

Clearly, performance parameters and MoEs are unique to each mission and depend on the particular objectives that we are trying to achieve and constraints that we must live with. Nonetheless, in general, we would like to minimize cost and risk, maximize the performance, and obtain the results in a timely manner. Building a satellite to help the warfighter in a specific war is of very little utility if the war is over by the time the satellite is launched. Table 5-8 lists some of the most common performance parameters that are likely to be useful in evaluating mission utility for any mission. Of course, many of these will not apply to a specific mission and we are likely to need some additional mission-specific parameters that aren't on the list, but the table provides a reasonable place to start.

Good measures of effectiveness are critical to successful mission analysis and design. If we cannot quantify the degree to which we have met the mission objectives, there is little hope that we can meet them in a cost-effective fashion. The rest of this section defines and characterizes good measures of effectiveness, and Sec. 5.4.2 shows how we evaluate them.

Good measures of effectiveness must be

- Clearly related to mission objectives
- Understandable by decision makers
- Quantifiable
- Sensitive to system design (if used as a design selection criterion)

MoEs are useless if decision makers cannot understand them. "Acceleration in the marginal rate of forest-fire detection within the latitudinal coverage regime of the end-of-life satellite constellation" will likely need substantial explanation to be effective. On the other hand, clear MoEs which are insensitive to the details of the system design, such as the largest coverage gap over one year, cannot distinguish the quality of one system from another. Ordinarily, no single MoE can be used to quantify how the overall system meets mission objectives. Thus, we typically want to provide several MoEs summarizing the system's capacity to achieve its broad objectives.

Measures of effectiveness generally fall into one of three broad categories associated with (1) discrete events, (2) coverage of a continuous activity, or (3) timeliness of the information or other indicators of quality. Discrete events include forest fires, nuclear explosions, ships crossing a barrier, or cosmic ray events. In this case, the best measures of effectiveness are the rate that can be sustained (identify up to 20 forest fires per hour), or the probability of successful identification (90% probability that a forest fire will be detected within 6 hours after ignition). The probability of detecting discrete events is the most common measure of effectiveness. It is useful both in providing good insight to the user community and in allowing the user to create additional measures of effectiveness, such as the probability of extinguishing a forest fire in a given time.

Some mission objectives are not directly quantifiable in probabilistic terms. For example, we may want continuous coverage of a particular event or activity, such as continuous surveillance of the crab nebula for extraneous X-ray bursts or continuous monitoring of Yosemite for temperature variations. Here the typical measure of

Table 5-8. Performance Parameters Applicable for Most Space Missions. For additional details, see Wertz [2008]. These are primarily technical measures that can be evaluated analytically or with a mission simulation.

1. Cost

Primary Performance Parameters

Non-Recurring Engineering (NRE) = full cost of developing the system

Recurring Cost per Mission = ongoing cost of building and launching subsequent units, including launch, spacecraft bus, payload, and operations (often normalized to 1 year)

Secondary Performance Parameters

Cost per Year—not meaningful if the goal is to achieve a particular objective such as return of samples from a comet

Cost per Unit of Performance—here "performance" should be one of the goal-oriented MoEs. ***Cost per data bit*** is a bad idea in most cases—it's results we want, not data volume.

2. Coverage

Observations per Day—may be per satellite or for the system as a whole, if there are multiple satellites

Number of Spacecraft = number of spacecraft required to achieve a given level of coverage or frequency of observations. May change with time as more spacecraft are added to the system.

3. Data Quality or Quantity

Resolution at Nadir—See Sec. 17.3 for definitions and formulas. An alternative measure often used by image analysts is the ***National Imagery Interpretability Ratings Scale (NIIRS)*** that is based on the ability to interpret the resulting images (See Fiete [1999].)

Bandwidth or ***Number of Simultaneous Users***—as discussed throughout the book, there is an ever-increasing demand for more bandwidth

4. Mission Risk

Probability of Successful Deployment—takes into account both launch failures and failure to work on orbit when first deployed

Probability of a Successful Mission—will be much higher if a back-up is available in case of a launch or on-orbit failure

Ability to Reconstitute Assets—note that the back-up system may well be of lesser capability than the primary one which it is replacing. Most important in military missions or human resupply.

5. Flexibility

Orbit Agility—having a high delta V available provides much higher utility for many missions because of the capacity to adjust the orbit to meet changing mission needs

Observation Time Flexibility —this allows the "driver" to adjust both conditions and arrival time at any given location

Multi-Mission Utility = potential to use one asset for more than one mission

6. Responsiveness

Total Response Time (TRT) = total time from when a new request for data is made until the data is delivered to the end user, including launch and on-orbit check out

Mean Response Time = mean time from when a new request for data is made to an on-orbit asset until the data is delivered to the end user

Time Late = time from when an observation is made until the data is delivered to the end user. Measures how "stale" the delivered data is

Development Time = total time to develop, build, and launch a new satellite

effectiveness is some type of coverage or gap statistics such as the mean observation gap or maximum gap under a particular condition. Unfortunately, Gaussian (normal probability) statistics do not ordinarily apply to satellite coverage; therefore, the usual measure of average values can be very misleading. Additional details and a way to resolve this problem are part of the discussion of coverage measures of effectiveness in Sec. 10.2

A third type of measure of effectiveness assesses the quality of a result rather than whether or when it occurs. It may include, for example, the system's ability to resolve the temperature of forest fires. Another common measure of quality is the timeliness of the data, usually expressed as time late, or, in more positive terms for the user, as the time margin from when the data arrives until it is needed. Timeline MoEs might include the average time from ignition of the forest fire to its initial detection or, viewed from the perspective of a potential application, the average warning time before a fire strikes a population center. This type of information, illustrated in Fig. 5-2 allows the decision maker to assess the value of FireSat II in meeting community needs.

The space mission engineer is unlikely to know from looking at Fig. 5-2 whether this FireSat II data will have a high level of utility to fire fighters. But it is likely that someone who manages fire fighting activities and distri-

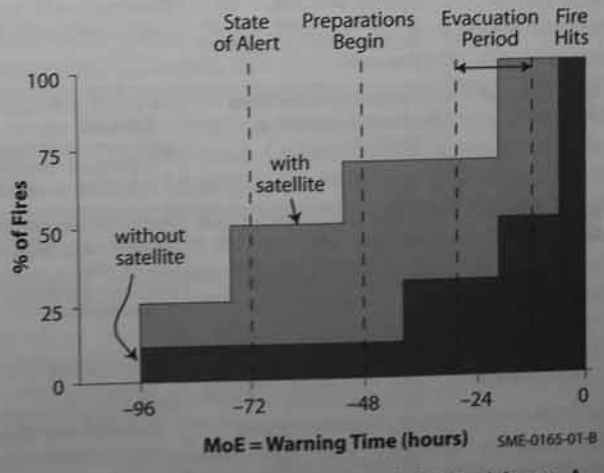


Fig. 5-2. Forest Fire Warning Time for Inhabited Areas. A hypothetical measure of effectiveness for FireSat II.

bution of fire fighting resources, will understand clearly the level of value that additional warning time brings. It may be that the level of advanced warning provided by FireSat II will allow equipment to be put in place and people evacuated in time to make a major difference in the cost and risk of fighting some fires, or it may be that the extra time has very little value if all that is needed is sufficient time for evacuation. Irrespective of the result, the type of information shown in Fig. 5-2 is what is needed to truly judge the utility of FireSat II and, therefore, whether it is worth the cost.

Because MoEs measure the ability of the system to satisfy the mission objectives, they are typically much more specific to each individual mission. In most cases, the objectives of an educational mission or an alt-space mission are very different than those of a military or a commercial mission, as discussed in Sec. 2.1. Most of the mission-specific MoEs fall into 4 broad categories reflecting the impact of the system on the ultimate mission objectives:

- ***Measures of Utility*** = how useful the mission is at meeting the objectives
- ***Measures of Success*** = how successful the mission is at meeting its objectives
- ***Impact on Outcome*** = whether the mission changes the ultimate outcome of events sufficiently to justify its cost
- ***Level of Preparedness*** = whether the mission allows us to be better prepared for future events

Table 5-9 gives some examples of these *goal-oriented MoEs* for several of the space mission communities identified in Chap. 2. As with performance parameters, the selection of specific MoEs depends on the mission objectives. In identifying MoEs, we want to look carefully at the mission objectives and ask what a good measure would be of the level at which the mission achieves these objectives.

5.4.2 Evaluating Mission Utility

In analyzing mission utility, we try to evaluate the measures of effectiveness numerically as a function of cost and risk, but this is hard to do. Instead, we typically use principal system parameters, such as the number of satellites, total on-orbit weight, or payload size, as stand-ins for cost. Thus, we might calculate measures of effectiveness as a function of constellation size, assuming that more satellites cost more money. If we can establish numerical values for meaningful measures of effectiveness as a function of the system drivers and understand the underlying reasons for the results, we will have taken a major step toward quantifying the space mission engineering process.

Recall that mission utility analysis has two distinct but equally important goals—to aid design and provide information for decision making. It helps us improve the mission design by examining the relative benefits of alternatives. For key parameters such as payload type or overall system power, we can show how utility depends on design choices, and therefore, intelligently select among design options.

Mission utility analysis also provides information that is readily usable by decision makers. Generally those who determine funding levels or whether to build a par-

Table 5-9. Goal-Oriented Measures of Effectiveness (GO-MoEs). Columns 2, 3, and 4 are representative space communities from Chap. 2. Column 5 is a specific example of a type of application mission.

Category	Military	Science	Education	Disaster Monitoring
Measures of Utility	How will the mission impact our ability to predict, protect, respond, retaliate, restore, rescue, contain, or limit collateral damage	Amount and importance of scientific data collected Ability to collect data not previously possible	Ability to design, build, test, and launch student experiments Cost of educational space experiments	Accuracy and speed of damage assessment Speed of getting the data to the people that need it
Measures of Success	Reduction in Casualties Reduction in combat duration	Detection of things never before detected Discovery of objects or processes that were previously unknown or not understood	Number of student experiments launched Number launched while the student is still a student	Rapidly locating and assessing damage Finding people in need of help or rescue in time to help or rescue them
Impact on Outcome	Change the outcome of the battle or war Tip the balance against future terrorist attacks	Prove or disprove theories Develop new scientific ideas or concepts Discover previously unknown phenomena	Quality of astronautics and space science education Number of students going into astronautics or space science	Number of lives saved Value of property saved Reduction in amount or duration of human suffering (e.g., getting food quicker to where it is needed)
Level of Preparedness	Ability to respond to changing world events or terrorist attacks Ability to prevent future conflicts by having advanced warning	Ability to respond to unexpected scientific events Ability to gain knowledge from transient events (e.g., nearby nova explosion)	Ability to educate the next generation of scientists and engineers Ability to demonstrate to children and the public the value of space science and astronautics	Ability to respond rapidly to the next major natural or man-made disaster Ability to prevent disasters (e.g., earthquake prediction; hurricane forecasting)

Table 5-9, Fig. 5-2◀, Eq. 5-0

ticular space system do not have either the time or inclination to assess detailed technical studies. For large space programs, decisions ultimately depend on a relatively small amount of information being assessed by individuals at a high level in industry or government. A strong utility analysis allows these high-level judgments to be more informed and based more on sound technical assessments. By providing summary performance data in a form the decision-making audience can understand, the mission utility analysis can make a major contribution to the decision-making process.

5.4.2.1 Mission Utility Simulation

Typically, the only effective way to numerically evaluate mission utility is to use a mission utility simulation designed specifically for this purpose. This is **not** the same as a *payload simulator*, which evaluates performance parameters for various payloads. For FireSat II, a payload simulator might compute the level of observable temperature changes or the number of acres that can be searched per orbit pass. In contrast, the *mission simulator* assumes a level of performance for the payload and assesses its ability to meet mission objectives. The FireSat II mission simulator would determine how soon forest fires can be detected or the amount of acreage that can be saved per year.

In principle, mission simulators are straightforward. In practice, they are expensive and time consuming to create and are rarely as successful as we would like. Attempts to achieve excessive fidelity tend to dramatically increase the cost and reduce the effectiveness of most mission simulators. The goal of mission simulation is to estimate measures of effectiveness as a function of key system parameters. We must restrict the simulator as much as possible to achieving this goal. Overly detailed simulations require more time and money to create and are much less useful, because long run times keep us from running them enough for effective trade studies. The simulator must be simple enough to allow making large number of runs, so we can collect statistical data and explore various scenarios and design options.

The mission simulation should include parameters that directly affect utility, such as the orbit geometry, motion or changes in the targets or background, system scheduling, and other key issues, as shown in Fig. 5-3. The problem of excessive detail is best solved by providing numerical models obtained from more detailed simulations of the payload or other system components. For example, we may compute FireSat II's capacity to detect a forest fire by modeling the detector sensitivity, atmospheric characteristics, range to the fire, and the background conditions in the observed area. A detailed payload simulation should include these parameters. After running the payload simulator many times, we can, for example, tabulate the probability of detecting a fire based on observation geometry, time of day, and size of the fire. The mission simulator uses this table to assess various scenarios and scheduling algorithms. Thus, the mission simulator might compute the mission geometry,

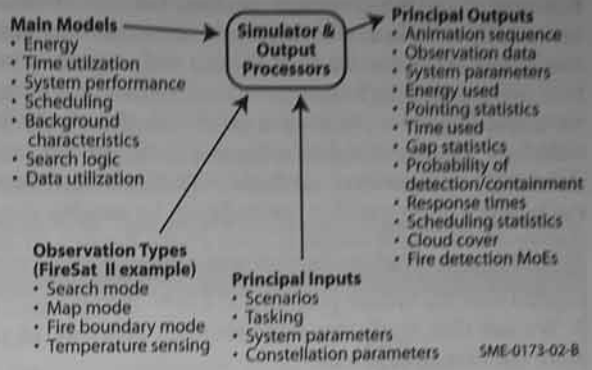


Fig. 5-3. Typical Mission Utility Simulation Inputs, Models, and Outputs for FireSat II.

time of day, and fire size, and then use the lookup table to determine the payload effectiveness.

With this method, we can dramatically reduce repetitive computations in each mission simulator run, do more simulations, and explore more mission options than with a more detailed simulation. The mission simulator should be a collection of the results of more detailed simulations along with unique mission parameters such as the relative geometry between the satellites in a constellation, variations in ground targets or background, and the system scheduling or downlink communications. Creating sub-models also makes it easier to generate utility simulations. We start with simple models for the individual components and develop more realistic tables as we create and run more detailed payload or component simulations.

In a constellation of satellites, scheduling is often a key issue in mission utility. The constellation's utility depends largely on the system's capacity to schedule resource use appropriately among the satellites. At the end of a single simulation run, the system should collect and compute the statistics for that scenario, generate appropriate output plots or data, and compute individual measures of effectiveness, such as the percent of forest fires detected in that particular run.

The next step is to run more simulations using new start times or otherwise varying the conditions for the scenarios. Changing the start times alters the relative timing and geometry between the satellites and the events they are observing, thus, averaging results caused by these characteristics. Collecting statistics on multiple runs is called a *Monte Carlo simulation*. For example, we might average the percentage of forest fires detected over different runs with different timing, but using the same scenario, to estimate the overall probability of detecting forest fires—our ultimate measure of effectiveness. The mission simulator should accumulate output statistics and prepare output plots over the Monte Carlo runs.

Frequently, in running mission simulations, we must choose between realistic and analytical scenarios. Realistic scenarios usually are too complex to help us understand how the system works but are still necessary to satisfy the end users. On the other hand, simple scenarios

illuminate how the system is working but do not show how it will work in a real situation. The best answer is to use simple scenarios for analysis and realistic scenarios to assess mission performance. In the FireSat II example, we might begin by studying a single satellite to determine how it behaves and then expand to a more complex simulation with several satellites. We might also start evaluating a multi-satellite constellation by looking at its response to a simple situation, such as one fire or a small group of uniformly distributed fires. This trial run will suggest how the system performs and how changes affect it. We can then apply this understanding as we develop more realistic simulations.

A related problem concerns using a baseline scenario to compare options and designs. Repeating a single scenario allows us to understand the scenario and the system's response to it. We can also establish quantitative differences by showing how different designs respond to the same scenario. But this approach tends to mask characteristics that might arise solely from a particular scenario. Thus, we must understand what happens as the baseline changes and watch for chance results developing from our choice of a particular baseline scenario.

Finally, mission simulations must generate usable and understandable information for decision makers—information that provides physical insight. Two examples are strip charts of various system characteristics and animated output. A *strip chart* plot is similar to the output of a seismograph or any multi-pin plotter, in which various characteristics are plotted as a function of time. These characteristics might include, for example, whether a particular satellite is in eclipse, how much time it spends in active observation, and the spacecraft attitude during a particular time step. Plots of this type give a good feel for the flow of events as the simulation proceeds.

A valuable alternative for understanding the flow of events is looking at an animation of the output, such as a picture of the Earth showing various changes in the target, background, and observation geometry as the satellites fly overhead. Thus, as Fig. 5-4 illustrates, an animated simulation of FireSat II output could be a map of a fire-sensitive region with areas changing color as fires begin, lines showing satellite coverage, and indications as to when fires are first detected or when mapping of a fire occurs. Animation is not as numerical as statistical data, but it shows more clearly how the satellite system is working and how well it will meet broad objectives. Thus, mission analysts and end users can assess the system's performance, strengths, and short-comings, and the changes needed to make it work better.

5.4.2.2 *Scenario Evaluation—The Hawaii Disaster of 2025*

Many of the MoEs in Table 5-9 are very difficult to evaluate numerically in large part because they aren't intended to be engineering oriented, but are meant to measure how well the mission objectives are being met. Realistically, there probably is no way to estimate via

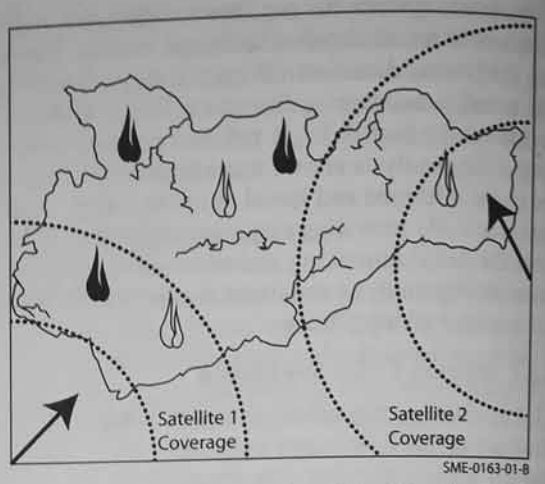


Fig. 5-4. Hypothetical Animation Output for FireSat II Mission Utility Simulator. Color displays are very valuable for animation sequences because we need to convey multiple parameters in each frame.

simulation the number of lives saved, the impact on the outcome of the war, or the importance of the resulting science. One of the best approaches to dealing with this is *scenario evaluation* in which we look at a real or hypothetical scenario and try to evaluate how that scenario would have been different had the system being evaluated been in place. Typically we would look to people who have been through the scenario if it's a real historical event, or similar ones if it's a hypothetical scenario, to estimate what the consequences would have been had the space system been in place. The advantage of using a real historical event or sequence of events is that at least for the baseline case, i.e., what happened without the system in place, the results are reasonably well understood. In addition, this type of evaluation is meaningful to decision makers because it is easy to discuss what actually happened and contrast that with what could have happened.

The scenario evaluation can be done either in conjunction with a simulation or, less precisely, by simply estimating the results. As an example, consider the great Hawaiian disaster of October 10, 2025 [Wertz, 2008]. On the evening of October 9, an earthquake and resulting tsunami struck in the ocean east of Japan with damage to all of the major airports and most homes, businesses, and infrastructure. Communications were badly disrupted and it was difficult to assess the damage or to find people that needed help, needed to be moved, had been washed out to sea, or were in imminent danger due to potential aftershocks. The tsunami also headed for Hawaii where evacuations of low ground began immediately. While major population centers were largely evacuated, it wasn't possible to get word to all of the residents and it was feared that many in rural areas may have been washed out to sea when the tsunami hit Hawaii.

By order of the President, HawaiiSat was launched from KSC 10 hours after the earthquake hit to provide both visible and IR images. The satellite carried an 0.5 meter diameter imager. Following a 4 deg inclination

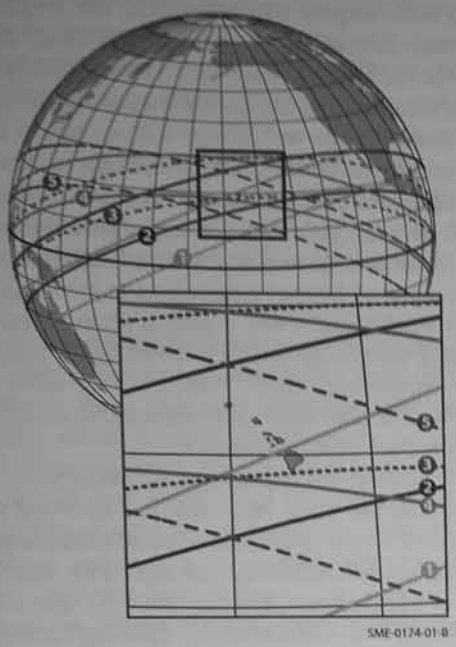


Fig. 5-5. Swath of the First 5 Orbits for the Hypothetical HawaiiSat Mission in October, 2025. Hawaii will be visible on 5 or 6 orbits/day with the observation times moving approximately 35 minutes earlier per day.

change at the equator, the satellite was placed in a 250 km altitude circular orbit at an inclination of 24.5 deg. The ground swath of HawaiiSat's first 5 orbits are shown in Fig. 5-5. The first images of the damage at a low elevation angle were returned on the first orbit about 80 minutes after launch. The second pass, 90 minutes later, flew almost directly over the islands and gave very good images of the damage and allowed the first estimate of the drift rate and direction of debris floating at sea by comparison with images from the first orbit. The next 2 orbits provided images at shallow angles for the big island and more nearly overhead for the rest of the islands. The fifth orbit again passed nearly overhead, 6 hours after the first orbit. 5 or 6 orbits of imaging occurred on subsequent days, moving about 35 minutes earlier per day.

The spacecraft parameter and cost estimates for the above system are based on the Frequent Revisit Surveillance mission discussed by Wertz et al. [2006] and Cox, et al. [2005, 2006]. Based on these parameters, the HawaiiSat performance parameters and MoEs are shown in Table 5-10.

While we don't yet have all of the numbers we need, it seems clear that a responsive disaster monitoring satellite could provide a significant capability to respond, restore, rescue, contain, and limit collateral damage. With this capability the government will be prepared to save lives and property.

In this type of scenario evaluation, some people will argue that existing on-orbit assets could do the job and that HawaiiSat wouldn't be needed. Others will argue that historically that doesn't happen and that existing assets are not in fact made available until well after the immediate need is past. Irrespective of which side of that argument prevails, the scenario evaluation approach, particularly when combined with real historical events and some mission simulations, can provide a great deal of insight into the utility of both existing and proposed systems and their strengths and weaknesses. This then leads to a more informed decision as discussed in Sec. 5.7.

Table 5-10. HawaiiSat Performance Parameters and MoEs.

Parameter	Value
Cost	
NRE	$40 M
Recurring Cost/Mission	$17.3 M
Coverage	
Observations/Day	5 to 6
Number of Spacecraft	1
Data Quality	
Nadir Resolution	0.3 m
Risk	
Deployment—Prob. of Success	90%
Mission—Prob. of Success	99%
Flexibility	TBD
Responsiveness	
Total Response Time	5 min
Mean Revisit Time	85 min for 6 to 7.5 hrs
Time Late	5 min
GO-MoEs	
Need a mission simulation and discussion with emergency response personnel to evaluate these critical measures	

5.5 Defining the Baseline Mission Concept, Revising Requirements, and Evaluating Alternatives (Steps 10–12)

Steps 10 to 12 in the Space Mission Engineering process from Table 3-1 are to define the baseline mission concept and architecture, revise the quantitative requirements and constraints, and evaluate alternatives. A *baseline mission*, like the point design defined in Sec. 5.2.3, is simply a mission definition that, in a broad sense, fulfills the overall mission objectives. By *mission definition* we mean what is being used for each of the 8 elements of the mission architecture plus the basic concept of operations. For FireSat II we want to know broadly how we're going to detect and monitor forest fires and how we're going to get that information to the end user. Ultimately, we are concerned with the firefighter in the truck and how he or she gets the data they need to do their job better and safer. Although we will work a great many technical details as we go through the definition of the mission, we should return often to the end user. Are we meeting their needs in a timely fashion at minimum cost and risk? If we aren't convinced that this is the case, then we should go back and work on the problem until we are. Space mission engineering is a highly iterative process and it is rarely the case that the

first approach is the best, and our first ideas may not even work that well.

Table 5-11 summarizes the process for selecting the baseline, trading on requirements, and evaluating alternative mission concepts. We begin by going back to the broad mission objectives and constraints that define what the mission is all about (Sec. 3.3). What is it that we're trying to achieve? Are there major constraints on how we are allowed to do that, such as the use of a particular technology or building on prior experimental missions? Who are the people that will ultimately pay for the mission, who are the people that will use the data that it generates, and what are their goals and objectives? This leads us to Step 2 where we identify the space community or communities that are being served. Irrespective of the details of a particular mission, a mission being done for the education community will almost certainly have a different set of objectives and constraints than will a military mission or a science mission meant to explore the depths of the Universe.

Table 5-11. Process for Selecting A Baseline Mission and Evaluating Alternatives. See text for discussion. See Sec. 5.6 for specific examples for the FireSat II and SCS mission.

Step	Where Discussed
1. List mission objectives and constraints	Sec. 3.3
2. Identify the space community being served and the end users and their broad objectives and approaches	Sec. 2.1
3. Determine which is the most important criteria for the mission—performance, cost, risk, or schedule	Sec. 1.1, Chap. 3
4. For each element of the mission architecture plus the mission concept, select an approach consistent with Step 3. This is a ***Point Design*** and the ***Initial Baseline***.	Sec. 4.1 to 4.3, 5.2.3
5. List major alternative approaches	Chap. 4
6. Look for the critical requirements that drive performance, cost, risk, or schedule (may be different for alternative approaches)	Sec. 4.4
7. For the baseline and each major alternative, begin the process of the Trading on Requirements	Sec. 5.5
8. Select a ***Revised Baseline*** and Major Alternatives for further analysis	—
9. Document and Iterate. This is a highly iterative process and documenting the reasons for choices is key.	—

The third step is to determine the most important criteria for the mission—performance, cost, risk, or schedule. Is the group that is funding the mission dramatically risk averse, such that we want to use largely processes and hardware that have worked in the past, or are they open to new ideas and approaches that might be able to dramatically drive down cost or provide a new level of performance? Throughout much of the history of the space program, the emphasis on most missions was avoiding risk, largely irrespective of the consequences on cost and schedule. (A common quote in the space industry is "If it works, all else is forgiven.") As discussed in Chap. 1, this has brought us to a space program that, in many cases, we can no longer afford and that doesn't generate results quickly enough to have sufficient utility to many end users. An asteroid warning system is of very little use if it is cancelled because of excessive cost or delayed until after an asteroid hits New York. Because of budget constraints in nearly all space programs worldwide, there is now a strong tendency to ask whether we can do things differently, or adopt hardware or processes from one of the other space communities or non-space applications, in order to significantly reduce cost or schedule.

Having created the framework for what we're trying to achieve and the basic rules within which we need to work, the next step is to define an initial baseline mission that can do the job, although perhaps with less performance or at a higher cost, greater risk, or longer schedule than we would like. We do this by defining each of the 8 elements of the mission architecture:

- Subject
- Payload
- Spacecraft Bus
- Ground Segment
- Mission Operations
- Orbit
- Command, Control, and Communications Architecture
- Launch Segment

Each of these is defined in more detail in Table 4-1. In addition to the elements of the mission architecture, we need to define the broad mission concept—i.e., how all of the mission elements work together to meet the needs of the end user. As discussed in Secs. 4.2 and 5.6, the mission concept for FireSat II revolves primarily around how (and where) we identify a forest fire and distinguish it from a really big barbecue and how we get that information to the firefighter in the truck. The method of doing these assessments are defined in Chap. 4 and in the discussion of the quick, limited detail studies (feasibility assessment, sizing estimate, and point design) in Sec. 5.2.

In addition to the initial baseline, we also want to list the major alternatives. These will nearly always involve one of elements of the mission architecture or the mission concept. For FireSat II one of the major alternatives will be the orbit. Do we use one or more small satellites in LEO or a single large satellite in GEO? We will get at the process of numerically evaluating these alternatives shortly, but in this step we simply want to identify them. This is a part of the creative process of space mission engineering. What are the alternative approaches that we might use to increase performance or drive down cost,

risk, or schedule? Are there processes or hardware from other space communities or from the non-space arena that might work well for this mission? Do the advances in computing power and image recognition and analysis software allow us to identify forest fires on-board the spacecraft and go directly from the spacecraft to the end user or the fire management station? We select as our initial baseline the approach that appears to be most consistent with what is important to this user—performance, cost, risk, or schedule.

In order to evaluate alternative approaches, we first identify the *critical requirements* that serve to drive the performance, cost, risk, and schedule. The process of identifying these critical requirements is defined in Sec. 4.4. Note that both the critical requirements and the system drivers may be different for different approaches. Two key issues here are to define the **reasons** for each requirement and to look for requirements that drive the cost, but don't improve performance, risk, or schedule. **Our goal is to adjust both the system parameters and the mission requirements to achieve our mission objectives at the minimum cost, risk, and schedule, or at least within a cost, risk, and schedule that we can afford, given that cost and schedule are likely to increase as the mission definition proceeds.**

The process of mission definition is much like an individual buying a new car. What we want to buy is a combination of what we would like and what we can reasonably afford and is likely to be strongly influenced by what is available in the marketplace. Is there new, lightweight mirror technology that can help us build a larger telescope at much lower mass and cost than was possible the last time we built one? Will new computer technology let us do more data processing on board such that we can simply send down the results and not all of the data needed to get there? What we want is to help the customer become an intelligent consumer.

The numerical processes for evaluating the major alternatives are the system trade studies, performance assessments, and mission utility analyses defined in detail in Secs. 5.3 and 5.4. The performance parameter trades are typically straightforward analyses that are relatively easy to do. These can help greatly when we are looking at a simple trade like orbit altitude or the amount of power or the resolution that is required to do the mission. Mission utility assessments are likely to be much harder, but also more useful. For FireSat II, when we compare the single geosynchronous satellite with multiple small satellites in LEO, the two alternatives will have very little in common, except the ability to detect forest fires. The system drivers and critical requirements will be different for both and they will have very different operational approaches and hardware. The only realistic way to make a comparison between them is to look at the mission utility that each can achieve as a function of cost, risk, and schedule.

5.6 Examples: FireSat II and SCS

As an example, we would like to go through the definition of a baseline system and the identification of major alternatives and system trades for both FireSat II and the SCS constellation defined in Sec. 3.2. As discussed in Sec. 3.3, the primary objective of FireSat II is **to detect, identify, and monitor forest fires throughout the United States, including Alaska and Hawaii in near-real time.** and for the Supplemental Communications System, SCS, it is **to provide low-cost, low-data-rate, real-time supplemental communications to existing military systems and aide in natural disaster relief with continuous connectivity to CONUS.** For both missions there are several possible secondary objectives, but these may change with the definition of the system and, therefore, are more important later when we begin the evaluation of alternatives. The principal constraints on both systems are those that are implied by the primary mission objectives—to be low cost and to be in place at the time that they are needed. In addition, we expect that military systems will want secure communications, while this is not likely to be of concern for the natural disaster relief mission.

For FireSat II, the principal community being served is the Earth applications community and the end user is the forest service and fire fighting community. Other than being regular users of GPS, they have no connection with space activity and no real interest in space per se. Thus, they will be interested in FireSat II only to the extent that it solves a problem that they can't otherwise solve (such as early detection of fires in remote areas) or solves problems more economically or substantially better than the traditional non-space solutions. If monitoring forest fires from space can get them data sufficiently quicker or better than other approaches such that they can more quickly contain fires, reduce fire fighting costs, and save property and lives (including reducing the risk to the firefighters themselves), then there will be some interest. But it is also likely that they will be wary of a system that costs 10's of millions of dollars or more (possibly much more) and that is well outside their area of expertise. There is likely to be a level of (possibly unspoken) concern on their part that FireSat II is more of a gimmick to support the space industry than a real and useful tool for identifying and fighting fires. This, in turn, means that clearly demonstrating mission utility to a group that fundamentally doesn't know or care about space will be critical.

In contrast, SCS is intended to serve two communities, both of which already use space assets and recognize a very real need for better communications in remote locations. The military community uses space assets for a great many functions and, along with the intelligence community, is one of the largest users, builders, and developers of space assets. The disaster relief community (part of the Earth applications community) works on a much lower budget, but has made good use of the Surrey-developed Disaster Monitoring Constellation,

Table 5-11◀, Fig. 5-5◀, Eq. 5-0

which provided the first images published in *Aviation Week* of the aftermath of Hurricane Katrina. In both cases, the end user is likely an individual in a remote area with limited communications equipment (much like the firefighter in the truck) who needs communications with the outside world as reliably and continuously as possible to save lives and accomplish their basic mission.

For all 3 groups involved (fire fighting, disaster relief, and military), low-cost is the most important criteria for both FireSat II and SCS. The fire fighting and disaster relief communities have very little budget to devote to "external" systems, such as space missions. The military community has substantial resources for communications, but already has large, expensive systems in place. The only reason for SCS is to provide additional communications at low cost. Because the systems will be low-cost and most likely SmallSats, they do not need to be as risk averse as traditional, large, multi-billion dollar satellites. Nonetheless, risk is important because the budget is limited and replacing the systems would be painful.

Defining an initial baseline for each system will require looking at each of the 8 mission elements and the mission concept. Assuming we have done this initial analysis using each of the chapters defined in Sec. 4.1, the preliminary results for FireSat II and SCS are shown in Table 5-12. In some cases, the likely lowest cost approach has been selected as the baseline (the payload and mission concept for FireSat II and the orbit for SCS) and in some cases a more traditional approach has been selected as the baseline (the ground station approach for both). So long as we continue the system trade and system evaluation process, the choice of the initial baseline isn't critical. The baseline will evolve as we continue to make trades and evaluate alternatives. The risk here is that we may try to fix the baseline in concrete and treat it as somehow better than the alternative approaches and this is what we very much need to avoid. After a few rounds through the system trade process we will learn more about which approaches are better and what the real goals and objectives are, assuming that the customer or end user are a part of the trade process, as they should be.

Our next step is to identify the critical requirements that will tend to drive performance, cost, risk, and schedule. These will tend to be different for the different alternatives. For the baseline FireSat II mission, the critical requirements for performance, cost, risk, and schedule are most likely the revisit interval since that drives the number of satellites, and the mission concept, since that determines the complexity of both ground and on-board processing. If we look at the GEO option for FireSat II, revisit interval won't really matter since we can see CONUS continuously. The requirements that will matter for GEO are the resolution (which could lead to needing a bigger telescope) and the need to cover

Table 5-12. Definition of the Initial Baseline and Major Alternatives for FireSat II and SCS. While alternatives exist for essentially all of the elements, we are only concerned here with the most important alternatives that will have a major impact on performance, cost, risk, schedule, or the mission concept.

Element	FireSat II Baseline	FireSat II Major Alternatives	SCS Baseline	SCS Major Alternatives
Subject	Heat from fire	—	Handheld receiver	Cell phone with wire antenna to boost signal
Payload	IR sensor	Possibly multiple spectral bands	Software defined radio; bands TBD	—
Spacecraft Bus	SmallSat bus, body-mounted arrays, integral propulsion, Earth pointing	May want steerable payload to look specifically at forests	200 kg SmallSat bus with deployable arrays and fixed crosslinks fore and aft	Mass of the spacecraft can be a continuous variable depending on the equipment and launch
Ground Segment	Dedicated ground station in CONUS	Commercial, pay-per-pass ground system	2 dedicated ground stations for redundancy	Commercial, pay-per-pass ground system
Mission Operations	Single facility doing both fire detection and commanding	Single facility doing commanding and monitoring of the on-board fire detection	Single facility doing both payload and constellation control	—
Command, Control, and Communications Architecture	Direct link from ground segment to FireSat II	—	Continuous link from ground station in CONUS to all of the satellites	—
Orbit	2 satellites in LEO orbit at inclination = 60 deg	1 satellite in GEO stationed over CONUS	3 satellites in circular, equatorial orbits at 20,000 km; supplemented with 3 more equatorial or 3 polar	TBD satellites in elliptical orbits—not as good, but more traditional
Launch Segment	Pegasus or Minotaur I	Secondary launch on a larger vehicle	All 3 satellites launched on a single EELV	Single launches on Pegasus or Minotaur 4
Mission Concept	Fire detection by trained analysts at mission ops facility; fire data relayed back through FireSat II and down to end user	On-board fire detection and monitoring sent directly to end user with data going from time-to-time to monitoring activity	Fixed crosslinks provide continuous connectivity to CONUS with only 1 hop from anywhere within 50 deg of the equator	—

northern Alaska, which may be too far north to be easily visible from GEO. If GEO becomes a viable option as the analysis proceeds, we may choose to argue that forest fires are less of a problem in northern Alaska and, therefore, the latitude that we need to reach can be limited.

For the SCS baseline constellation the critical requirements will be the capability of the hand held receiver, the spacecraft bus mass (which determines the size and power of the payload), and the need to overcome jamming for the military mission since this could dramatically reduce the available bandwidth. In the SCS alternatives we have chosen to list the possibility of doing something to boost the gain of an ordinary cell phone so that either military personnel or relief workers could use their cell phone which they are much more likely to have with them than some extraneous piece of equipment, particularly in the case of emergency relief operations. It would be an enormous asset if we could do this, but we don't know if it can be done. Therefore, we will need to devote analysis time, and most likely quite a bit of test time, to see what is possible and reasonable to do under adverse conditions in remote areas.

For both FireSat II and SCS we have listed commercial, pay-per-pass ground stations as a potential low-cost option to significantly drive down the ground system and operations cost. In this model, data can be brought down and commands sent back up via any of a series of commercial ground stations around the world, typically for a few hundred dollars per pass. Data is then put on a secure internet link and, if needed, encrypted on board the spacecraft before being transmitted. The satellite system then becomes simply another node on the secure internet. This has the potential of being both lower cost and more reliable than dedicated ground stations, but it tends to be a non-traditional approach (although service provided ground stations have been in operation for decades) and, therefore, may be **perceived** as too high a risk.

As part of evaluating alternatives we will make use of all of the various analytic techniques defined in Secs. 5.2, 5.3, and 5.4 including performance assessments, systems trades, and utility analysis. For FireSat II we can do system trade studies for many of the alternative approaches:

- Use of single vs. multiple spectral bands
- Earth pointing vs. steerable payload
- Choice of ground station approach
- Launch vehicle options (we want to keep more than 1 option available in order to allow negotiation for the lowest price)
- Basic mission concept of ground vs. on-board identification and monitoring of forest fires

The key need of the mission utility analysis will be twofold—to evaluate the results of multiple satellites in LEO vs. a single satellite in GEO and to convince ourselves, the customer, and the end user that FireSat II will really have substantial mission utility and be worth the estimated cost and risk. This will, of course, depend very much on what the cost turns out to be.

For SCS we can probably do a simple performance assessment for the issue of spacecraft mass and system trades for the

- Orbit selection
- Choice of ground station approach
- Launch vehicle options (as with FireSat II, we want to keep more than 1 option available in order to allow negotiation for the lowest price)

The cell phone option will probably require substantial analysis and testing to see whether it is workable and may go forward or may be discarded as impractical. Here the main purpose of a utility analysis is to clearly show the customer (and possibly Congress) that the end result is well worth the probable cost.

Having done our systems trades and utility assessment we then go back and either revise or confirm our initial baseline approach. Here we need to do two things. We need to document the choices we have made and the studies and analysis that have led to them so that they can be revisited from time to time as needed. Second, and perhaps equally important, is to not believe our results. There will be many more trades and analyses as we go forward. We haven't even started yet to take into account potential secondary objectives for either mission and these may take on a more important role, particularly if one or more of the secondary missions becomes involved in funding the program. For example, GPS did not have sufficient political support to be funded when it first came about. For this reason the Nuclear Detection System (NDS) was added and provided the margin needed to generate sufficient support in Congress for the system to come about. This means that we must continue to remain open to trades and alternatives as the system becomes better defined. And we must always go back to the needs of the end user—the firefighter in the truck, the soldier in an isolated area, or the relief worker in a remote disaster area somewhere in the world. Are we truly satisfying their needs at minimum cost and risk?

5.7 Deciding Whether a Mission Should Proceed

This section is concerned not with the detailed engineering decisions for a space mission, but with the broad trades involved in defining the overall mission—whether to proceed with it and what concept to use. Decisions for space missions fall into three broad categories: (1) go/no-go decision on proceeding with the mission; (2) selection of the mission concept; and (3) detailed engineering decisions, which are generally described throughout this book.

In principle, the go/no-go decision depends on only a few factors, the most important of which are:

- Does the proposed system meet the overall mission objectives?

- Is it technically feasible?
- Is the level of risk acceptable?
- Are the schedule and budget within the established constraints?
- Do preliminary results show this option to be better than nonspace solutions?

In addition to the above technical issues, a number of nontechnical criteria are ordinarily equally or more important in the decision-making process:

- Does the mission meet its political objectives?
- Are the organizational responsibilities acceptable to all of the organizations involved in the decision?
- Does the mission support the infrastructure in place or contemplated?

For example, a mission may be approved to keep an organization in business, or it may be delayed or suspended if it requires creating an infrastructure perceived as not needed in the long term. The mission analysis activity must include nontechnical factors associated with space missions and see that they are appropriately addressed.

The top-level trades in concept selection are usually not fully quantitative, and we should not force them to be. The purpose of the trade studies and utility analysis is to make the decisions as informed as possible. We wish to add quantitative information to the decisions, not quantify the decision making. In other words, we should not undervalue the decision-maker's judgment by attempting to replace it with a simplistic formula or rule.

Table 5-13 shows how we might try to quantify a decision. Assume that a system costs $500 million, but an improvement could save up to $300 million. To save this money, we could use option A, B, or C. Option A would cost $50 million, but the probability of success is only 70%; B would cost $120 million with 99% probability of success; C would cost $200 million with a 99.9% probability of success.

Which option should we select? The table gives the cost if successful, the cost if the improvement fails, and the expected values of both the cost and net savings. By numbers alone, we would select option B with an expected savings of $177 million. However, reasonable and valid cases can be made for both A and C. In option A, we risk only $50 million, and, therefore, are minimizing the total cost if the improvement succeeds or if it fails. In fact, the $620 million cost of failure for option B may be too much for the system to bear, no matter the expected savings. Option C provides a net savings of "only" $100 million, but its success is virtually certain. Although savings for this option are less dramatic, it does provide major savings while minimizing risk. In this case, we may assume the cost to be a fixed $400 million, with failure being so unlikely that we can discount it. Option B, of course, balances cost and risk to maximize the expected savings.

Table 5-13. Mathematical Model of a Hypothetical Decision Process for a New Space System (costs in $M). Numerically, we would choose B or A' if it were available. As discussed in the text, any of the choices may be best depending on the decision criteria.

Current Cost	$500M
Potential Savings if Improvement is Successful	$300M

Option	Cost of Improvement	Probability of Success	Total Cost if Successful	Total Cost if Failed	Expected Total Cost	Expected Savings
A	$50M	70%	$250M	$550M	$340M	$160M
B	$120M	99%	$320M	$620M	$323M	$177M
C	$200M	99.90%	$400M	$700M	$400.3M	$99.7M
A'	$50M	80%	$250M	$550M	$310M	$190M

Suppose, however, that option A had an 80% probability of success as shown in A', rather than the original 70% probability. In this case, the expected savings of A' would increase to $190 million, and would make it the preferred approach in purely mathematical terms. However, most individuals or groups faced with decisions of this sort are unlikely to change from option B to A' based solely on the increase in estimated probability to 80%. Their decisions are more likely to depend on perceived risk or on minimizing losses. Using non-mathematical criteria does not make the decisions incorrect or invalid, nor does it make the numerical values unimportant. We need quantitative information to choose between options but we do not have to base our decisions exclusively on this information.

As a second example, we can apply the results of utility analysis to concept selection for FireSat II. In particular, the number of satellites strongly drives the cost of a constellation. If we select the low-Earth orbit approach for FireSat II, how many satellites should the operational constellation contain? More satellites means better coverage and, therefore, reduces the time from when a fire starts until it is first detected. Consequently, one of our key parameters is the *time late*, that is, the time from when a fire starts until the system detects its presence and transmits the information to the ground. Figure 5-6 plots the hypothetical time late vs. the number of satellites for FireSat II. The details of such plots will depend on the latitude under consideration, swath coverage, altitude, and various other parameters. However, the characteristic of increasing coverage with more satellites eventually reaches a point of diminishing returns. This will normally be true irrespective of the coverage assumptions.

If we assume an initial goal for time late of no more than 5 hours, we see from the plot that a system of 6 satellites can meet this goal. Alternatively, a 4-satellite system can achieve a time late of 6 hours. Is the smaller time late worth the increased number of satellites and the money to

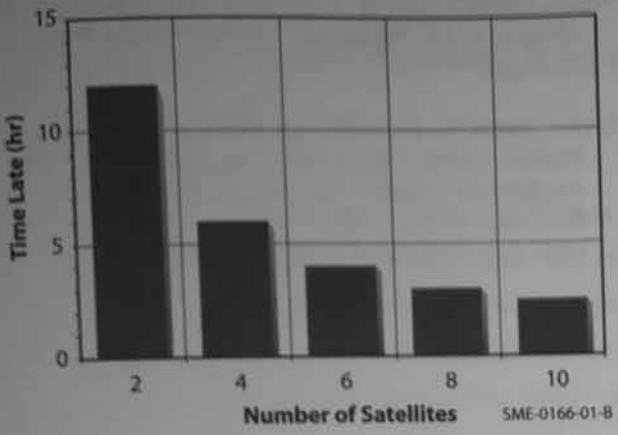


Fig. 5-6. Hypothetical Coverage Data for FireSat II. See text for definitions and discussion. As discussed in Sec. 10.10, growth in coverage typically comes in increments or plateaus. These are assumed to be 2-satellite increments for FireSat II.

build them? Only the ultimate users of the system can judge. The additional warning may be critical to fire containment and, therefore, a key to mission success. However, it is also possible that the original goal was somewhat arbitrary, and a time of **approximately** 5 hours is what is really needed. In this case, fire-fighting resources could probably be used better by flying a 4-satellite system with 6 hours time late and applying the savings to other purposes. Again, mission utility analysis simply provides quantitative data for intelligent decision making.

Of course, we must remember that the number of FireSat II satellites will depend not only on the utility analysis but also on politics, schedules, and resources. The public must see FireSat II as an appropriate response to the problem, as well as an efficient use of scarce economic resources compared to, for example, more fire fighters. In addition, a satellite system may serve several missions, with multiple mission criteria and needs. Just as we cannot apply only one criterion to some system drivers, we may not be able to balance numerically the several criteria for mission selection. Instead, the developers, operators, and users must balance them using the insight gained from the system trades and mission utility analysis.

Having undertaken a round of system trades, evaluated the mission utility, and selected one or more baseline approaches, we are ready to return to the issue of system requirements and their flow-down to various components. Chapter 6 treats this area, which is simply the next step in the iterative process of exploring concepts and defining requirements.

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6 Formal Requirements Definition

Jeffery J. Puschell, *Raytheon*

6.1 The Requirements Definition Process

6.2 Budgeting, Allocation, and Flow-Down

6.3 Introduction to Error Analysis

6.4 Specifications and Requirements Documentation

6.5 System Engineering Tools—*Lloyd Purves*

6.6 The Role of Standards in Space Systems Development—*Frederick A. Slane*

6.7 Are Requirements Needed?—Capability-Based vs. Requirements-Based Systems

The intent of the formal requirements definition process is to describe a system that can actually do what it is intended to do. Space missions frequently begin with a broad set of qualitative, sometimes vaguely defined mission requirements—to find evidence of water on Mars, provide high speed Internet access for ships at sea, or improve navigation for military forces. Ultimately these mission needs flow by a creative and sometimes magical process called *requirements definition* into a large set of very specific quantitative requirements on all elements of a space system—e.g., the spacecraft must have 2 star trackers, each weighing less than 0.5 kg, using less than 1.3 W of average power, with an 8×13 deg field of view, capable of detecting 7th magnitude stars with an angular accuracy of 0.004 deg (1σ) using on-board pattern recognition processing over a 15 year lifetime in geosynchronous orbit. To be effective, these requirements must be tied to the highest level system performance and mission requirements and to lower level subsystem requirements through performance allocations and error budgets. In addition, space system performance requirements must be verifiable through ground-based hardware and software testing and analysis.

This requirements definition process often ends up defining the cost and complexity of the system, too. Consequently, the requirements definition process may be the single most important part of space mission engineering for establishing performance, cost, and risk of the system. To produce lower cost, more effective systems, we should put as much or more thought and care into defining the requirements as we do into designing the system to fulfill these requirements.

If most of the system cost is effectively determined at the end of detailed requirements definition, then the requirements definition process itself must become the principal focus in the process of reducing cost, especially near the beginning of system development. In early phases of space mission engineering, system optimization, with an eye toward cost reduction, comes about primarily by trading performance requirements versus cost and mission effectiveness. This process defines and revises requirements in an iterative process to minimize cost and risk of the system as a whole, while simultaneously meeting mission needs. Our broad mission objective may be to find evidence of water on Mars, but, to design a system to meet that need, that objective is translated into a set of performance requirements to map 95% of the surface of Mars at infrared wavelengths in one year at a resolution of 50 m. This translation of mission needs into requirements determines space system performance, cost, and risk. Will the scientific return of our mission be significantly degraded by a resolution of 75 m or 200 m mapped over three years? What are the cost implications of each of these changes? In the end, system requirements definition must be completed by design, manufacturing, supply chain and test system engineers working in close coordination with mission stakeholders. In some cases, it may be necessary to select detailed requirements that embody a sensible compromise between mission objectives and mission affordability. Only when we truly understand the nature of this compromise, i.e., what it costs to achieve a given level of mission performance, can we begin to make rational decisions on defining the system requirements.

Too often, detailed system performance requirements definition is left entirely to the end user of the system or to mission stakeholders. The government or organization that is financing the mission establishes detailed requirements and then asks candidate suppliers to bid on a system to meet those requirements at the lowest cost. However, defining technical performance requirements in advance of thoroughly trading performance requirements with cost means that the requirements will have been largely fixed before the impact on cost is completely understood. Furthermore, if requirements trade studies are made without benefit of validation that can be achieved only by building and testing candidate hardware and software in mission relevant environments, system level cost and risk versus performance cannot be

thoroughly understood and the probability of significant cost and schedule growth increases within context of a fixed set of requirements. If a better process is to allow requirements to evolve as the system definition changes in response to better understanding of the detailed and validated relationship between mission value and system performance, cost and risk, then we should always be looking for ways to adjust the requirements to reduce cost and optimize mission value versus cost. This chapter introduces that process.

Requirements definition consists of two distinct but interrelated functions:

- A mission and customer-driven process to define mission objectives and relate those objectives to top level and detailed system and subsystem requirements and specifications
- An analytical process that breaks down requirements into constituent components with creation of error budgets, performance budgets, and cost budgets to track progress and ensure that the system objectives are met.

The following section, deals with the mission and customer-driven requirements definition process. Sections 6.2 and 6.3 focus on the process of allocating top-level system requirements to system components and mathematically analyzing associated allocation and error budgets. Specifications and requirements documents are discussed in Sec. 6.4. System engineering tools for capturing and managing relationships between requirements at all levels are discussed in Sec. 6.5. Section 6.6 discusses the role of standards in space system development. Finally, Sec. 6.7 examines whether requirements are really needed at all.

6.1 The Requirements Definition Process

System requirements are the quantitative expression of an approach for meeting mission objectives. These requirements are recorded in a set of system requirements documents or system specifications. Frequently, space systems customers create system requirements by first developing a straw man or reference mission design, that is, a detailed design description that the defining organization believes is fundamentally capable of meeting mission objectives. Thus, if the broad goal is to look for evidence of water on Mars, a hypothetical system of satellites to provide mapping over some defined period of time may be described. If the objective is to create high speed global communications, a constellation consisting of a specific number of low-Earth orbit satellites each with a number of communications channels capable of meeting broad mission objectives may be described. The characteristics of this straw man design are then used to create system specifications, i.e., so many satellites are required at a specific altitude over the surface of Mars with instruments capable of making certain measurements with specified angular resolution. The fundamental problem with this approach is that although

Table 6-1. Critical Issues in Requirements Development Note that the reason is rarely included in the specification, but must be documented if requirements trades are to be performed. See text for discussion.

Critical Features of System or Subsystem Requirements:
- Should be based on the fundamental mission objectives
  1. Mission or payload derived
  2. Flow down from basic requirements
- Should be part of the system trade process
- Should state what is to be done (e.g., pointing, mapping, and timing), rather than how to do it (orbit, attitude, and on-board clock)
- Are the quantitative expressions of how well the objectives are met, recorded in the system specification

Types of Requirements:
- *Functional requirements* = how well it must perform
- *Operational requirements* = how it is to be used
- *Constraints* = what limitations are imposed on the system

Elements which Should be Documented for Each Requirement in the System or Subsystem Specification:
- *Function* = what is to be done
- *Performance requirement* = how well it has to be done
- *Verification* = how the performance is to be verified
  1. Inspection 2. Test 3. Analysis

Key Element which Should be Documented (though typically omitted from the system specification):
- *Reason = why* it is required

the original system designers may have been very clever and knowledgeable, they do not necessarily have all of the necessary or appropriate knowledge to determine how best to meet mission needs at the lowest possible cost. Consequently, if system requirements are expressed in terms of specific parameters of the reference design, i.e., so many satellites or star trackers with a specific field of view or limiting magnitude, then any further trades may be effectively eliminated. The defining organization has essentially created the system design and fixed the system cost and made it more difficult to reduce that cost.

On the other hand, if system requirements at all levels specify what needs to be achieved rather than how to achieve it, then the system trade process can continue and the customer may be able to take advantage of new knowledge, new developments or clever techniques from throughout the space system development community. For example, resolution on the ground may be specified rather than satellite altitude and instantaneous field of view of an instrument. This leaves open the trade for using a large number of satellites at a lower altitude, or a smaller number at higher altitudes, or perhaps even a smaller number at lower altitudes and taking longer to collect the required data. Similarly, for the star tracker, required field of view or resolution should not be specified, but simply the desired end result; a required attitude accuracy. Even better, given spacecraft attitude is of little interest to a typical end user, a better parameter to specify is geolocation knowledge or pointing knowledge. Requirements should be specified in mission relevant

terms such as pointing and mapping rather than in system design terms of how to achieve mission requirements such as orbit and attitude. By specifying what we want, and not how to achieve it, we can leave open other options that have the potential of reducing cost and risk or perhaps improving performance at minimal increase in cost or risk, in order to obtain the best system at the lowest cost. These and other critical issues in the requirements development process are summarized in Table 6-1.

There are three basic types of requirements:

- *Functional requirements* that define what a system is supposed to do and how well it must do it
- *Operational requirements* that define how the system is to be used
- *Constraints* that define limitations imposed on the system such as operating environment and requirements imposed by other system elements to minimize electromagnetic interference

The requirements and constraints for any specific mission will be driven by fundamental mission objectives and by how the mission is implemented. Nevertheless, whether the mission is an international scientific research program, a commercial venture or an emergency program responding to urgent need, its requirements will likely be developed within the broad categories shown in Table 6-2 for the FireSat II mission that is discussed throughout this book.

Requirements are based primarily on fundamental mission objectives. What must the system do to satisfy mission objectives? Mission objectives drive requirements for the payload more than any other part of the system. Nevertheless, mission objectives impact other system elements, too. For example, spacecraft orbit, attitude stability and timing requirements flow from needs that can vary widely according to mission ranging from measuring the Earth's infrared radiation across the entire globe every day at 500 m resolution to providing high speed Internet access

Table 6-2. Example of Top Level Mission Requirements. We typically subdivide these top-level requirements into more specific requirements applicable to specific space missions.

Requirement	Factors which Typically Impact the Requirement	FireSat II Example
	FUNCTIONAL	
Performance	Primary objective, payload size, orbit, pointing	0.12 K sensitivity at 300 K 500 m resolution 500 m location accuracy
Coverage	Orbit, swath width, number of satellites, scheduling	Daily coverage of 750 million acres within continental US
Responsiveness	Communications architecture, processing delays, operations	Send registered mission data within 30 min to up to 50 users
Secondary Mission	As above	Land and sea surface temperature, high resolution water vapor imagery and crude winds over the continental US
	OPERATIONAL	
Duration	Experiment or operations, level of redundancy, altitude	Mission operational at least 10 yrs
Availability	Level of redundancy	98% excluding weather, 3-day maximum outage
Survivability	Orbit, hardening, electronics	Natural environment only
Data Distribution	Communications architecture	Up to 500 fire-monitoring offices + 2,000 rangers worldwide (max. of 100 simultaneous users)
Data Content, Form, and Format	User needs, level and place of processing, payload	Location and extent of fire on any of 12 map bases, average temperature for each 30 m^2 grid
	CONSTRAINTS	
Cost	Manned flight, number of spacecraft, size and complexity, orbit	< $20M/yr + R&D
Schedule	Technical readiness, program size	Initial operating capability within 5 yrs, final operating capability within 6 yrs
Regulations	Law and policy	NASA mission
Political	Sponsor, whether international program	Responsive to public demand for action
Environment	Orbit, lifetime	Natural
Interfaces	Level of user and operator infrastructure	Comm. relay and interoperable through NOAA ground stations
Development Constraints	Sponsoring organization	No unique operations people at data distribution nodes

across Africa to 0.3 m diameter receivers to providing 1 m navigation accuracy worldwide to commercial users. A commercial end user of navigation data does not care about the attitude stability of the satellites providing the basic information, only that her location is known to within the needed accuracy and timeliness.

Specific mission implementation approaches drive requirements definition, also. For instance, a specific mission implementation may require the ability to point the spacecraft solar arrays at the Sun, communicate with the ground station, or get within 2 km of a distant asteroid. These requirements are typically, though not necessarily, less stringent than the mission or payload derived requirements. However, many implementation driven requirements apply to all mission phases, whereas mission driven requirements frequently apply only to the operational phase. Thus implementation driven requirements, though often less stringent, may drive the range over which orbit and attitude systems, for example, must operate or the various conditions under which data must be provided. A typical implementation driven function would be to regain control of the spacecraft if it tumbles. This implies the need to have low accuracy attitude in a variety of orientations such that it is possible to recover the system when it is in a non-nominal mode. This might imply use of very coarse Sun sensors covering a large fraction of the sky, or an Earth sensor that is capable of picking up the Earth as the spacecraft tumbles as derived requirements related to specific implementations.

Constraints of various types drive requirements definition, too. For example, an infrared radiometer operating on board a dedicated remote sensing satellite in geosynchronous orbit will have radiation and electromagnetic interference constraints that are quite different from a radiometer operating on board a multifunctional satellite with high power communications transmitters in medium Earth orbit. A very different type of constraint is an international mission may be constrained to use a launch vehicle provided by the nation funding the mission, which could limit payload size and mass. Communications missions are constrained by their ability to secure the radio frequency licenses required to operate in customer nations, which can impact not only system design but also mission operating areas.

For each requirement, regardless of its origin or type, we want to select it to meet mission needs in a way that minimizes cost, schedule, and risk for the overall system development and operation. Once again, this argues in favor of making the requirements definition process itself part of the system trade process.

There are two basic types of trades:

- Trades on the overall level of performance
- Trades on how the performance is achieved

Most often the level of performance is defined as a specific number, such as radiometric sensitivity at a specific wavelength, spacecraft attitude accuracy or location knowledge accuracy. Frequently, these values are fixed by the nature of the system procurement and associated requirements definition process. That is, in order to be fair to all of the candidate suppliers, the defining agency picks a number which they believe to be appropriate and then asks suppliers to achieve this value at minimum cost. Neither the customer nor the supplier wants to change the number. The customer has gone to some lengths to specify this number, and would like to have it achieved, although of course, at very low cost. The supplier doesn't want to appear non-responsive to the customer's needs, and therefore will go to great lengths to achieve the specified performance, even if it drives up the cost or risk of the end system. A more effective approach for optimizing end system performance with respect to cost and schedule is to maintain an ongoing conversation in the early stages of system procurement on the implications of specific requirements on system performance, cost, and risk. To be most effective, this conversation must occur within context of validated, well understood relationships between performance and cost derived from hardware already flown or tested in mission relevant environments. It is this well-informed dialog and trading back and forth which too often are omitted in the requirements definition process.

The trades on how the performance is achieved are also critical. Ideally, these should allow the assessment and use of well understood alternative design options to reduce cost or to meet other elements of the system performance. Thus a system requiring high-accuracy pointing may specify the use of star trackers to achieve the required attitude accuracy. On the other hand, it may be possible to achieve an acceptable level of accuracy using a combination of Earth sensors and gyros to average the noise from the Earth's horizon. This is an example of a trade process that may be able to achieve the fundamental objectives of the mission at lower cost and risk.

Most people in the space business recognize that in order to minimize cost, requirements definition should be part of the overall system trade process, and yet this is rarely done for several reasons. As discussed above, suppliers do not wish to be perceived as non-responsive or less competitive and therefore frequently do not want to disagree with the customer over performance requirements or even suggest that certain requirements may not be achievable. But minimizing the cost of the system implies that we need to have this basic debate between desired performance and cost.

A second reason for not trading on requirements is that it frequently involves moving requirements, and therefore budget, between various elements of the system, which may be owned by different customer organizations. For example to achieve a required level of geolocation accuracy, we may want to consider changing from tight attitude requirements and loose orbit requirements to the other way around. That would end up putting additional requirements on the orbit, and therefore the ground system, in order to save money on the spacecraft. However, these distinct parts of the system

are typically done by different organizations operating in very different ways. Therefore, to save money we may need to take money from one organization and give it to another. This can prove to be an unpopular process.

If it is politically difficult to trade on requirements, how do we go about doing it? The best approach is to make it a very explicit and open process. Almost every mission is divided into a series of phases as the definition of the system becomes more mature and complete. Each of these phases usually begins with a briefing or technical meeting defining the status of the system at that time. Thus, to define status of the program, we might have a Concept Design briefing or a kick-off for the Design and Development phase. This can occur at the system level, subsystem level or component level. The key issue for requirements trades is to have the first portion of each such briefing devoted explicitly to requirements trades. Ask the system engineer either for the prime contractor or the component manufacturer to explicitly define the source of the requirements that they are working to and indicate which of those requirements are principally responsible for impacting performance, cost, risk, and schedule. This allows the customer to examine trades on key requirements with the objective of reducing overall system cost with respect to achieving some set of mission objectives. The key point is that this process makes trading on requirements an open and defined engineering process subject to review and detailed examination in much the same fashion as the system design itself is examined and reviewed.

To complete effective system trades we need to identify the critical key performance requirements, i.e., those that drive mission performance, cost, risk and schedule. One approach for identifying key performance parameters is to examine top-level system requirements and develop different system concepts for meeting those requirements. Critical requirements will frequently depend on the specific implementation being considered and often concept exploration will help isolate the requirements that drive performance, cost and schedule. We then study the impact of changing each of the fundamental requirements. Regardless of how we define the key performance parameters exactly, the objective is to let the requirements evolve in order to achieve the mission objectives at minimum cost and risk. To do this we need to concentrate our system trades around the critical requirements, so that we are continuously examining and re-examining those requirements that have the highest impact on performance, cost and risk.

Specifications

Specifications document system requirements. Every documented requirement must include three things:

- *Function* = what is to be done
- *Performance requirements* = how well it is to be done
- *Verification* = how the performance is to be verified

We can verify performance of the system as a whole by allocating top-level requirements to subsystems and components and then verifying performance of each of the components to show that the final system will work as designed. In many cases, system level performance verification for some requirements occurs by testing the system as it is intended to fly, to the greatest extent possible.

A simple example of system level performance verification is to examine a top-level requirement to use a specific launch vehicle capable of launching a certain total spacecraft mass to the mission required orbit. This total mass is allocated to each of the various subsystems and, in turn, to the various components within those subsystems. This will be reflected in the specification documents for the components, and all of the masses will be added up in a single mass budget. The masses of the components will first be verified by analysis and ultimately by weighing all of them as the spacecraft is built. In doing this, we will determine that the final spacecraft will in fact meet its mass requirement.

One major element that is almost always missing from the typical specification document is the rationale for the requirement and its priority with respect to other requirements. Specification documents tend to be a demand from the customer to the supplier and therefore generally do not reflect the reasons why specific requirements are imposed. Nonetheless, if we are to continue the system trade process and make it a part of the overall process of system design, then we must know why specific requirements are imposed. Otherwise there is no way to go back and determine if a less stringent or more stringent requirement is appropriate. Consequently, even though it does not fit well within the contractual hierarchy, providing the reasons for the various requirements given in the specification is extremely important to the process of trying to drive down system cost.

In order to be useful in terms of whether the system can meet its broad requirements, each individual requirement must be capable of being verified. This verification is typically done in one of three ways: inspection, test or analysis. Thus, we may determine that a particular system has redundant components simply by inspection. Alternatively, we may test the star tracker to determine the resolution or we may verify adequate performance of a structural element by analysis. Whatever method we choose, we need to specify how all requirements are to be verified.

The general classes of system specifications are shown in Fig. 6-1. In smaller programs, many of these specification documents are not created in order to reduce the very substantial cost associated simply with documentation. On the other hand, failure to document those system characteristics that are critical to determining performance can ultimately drive up cost dramatically. Particularly important is the *Interface Requirements Document*, because it formalizes an interface between people or organizations that may not be working closely together. For example, it may be the job of the navigation computer to compute the position and

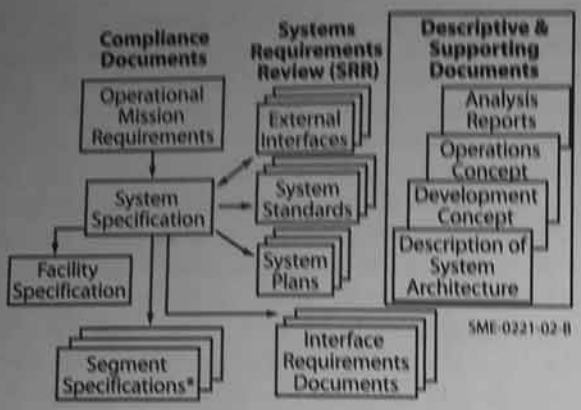


Fig. 6-1. Typical Specification Documents. While not all of these documents exist for all programs, the information contained in each should be recorded so that it is available for review and is agreed to by everyone involved in the project.

velocity of the spacecraft. How that computation is done is typically of little consequence to the end user. Nonetheless, it is important to specify how that information will be communicated to those systems which need it. The interface requirements document could specify, for example, the data rates, units, and word characteristics associated with the information generated by the navigation system.

As stressed above, a key issue in requirements definition, and therefore in creating specification documents, is to specify what is to be done and not how it is to be done. No mission requires star trackers. Missions require accurate geolocation and pointing knowledge that can be derived in part from accurate attitude data. How this is achieved across the system design should be left to organizations or individuals assigned the responsibility of achieving it.

6.2 Budgeting, Allocation, and Flow-Down

Having established the top-level system requirements, the next step in the requirements definition process is to budget, allocate, or flow-down these requirements to the components which make up the overall system. This section summarizes the broad process of doing that. The following section and App. G present a detailed mathematical description of how error budgets are analyzed. Section 8.6 provides a detailed numerical example for mapping and pointing budgets which are a principal area of interest for space mission engineering.

The budget is the numerical list of the components into which the top-level resource or requirement is divided along with the rules on how the components are to be combined to meet the system requirement. The margin is the planned difference between the system requirement and the sum of the components. Margin should be tracked and controlled at the system level to be available to meet implementation needs. Table 6-3 provides a list of items frequently budgeted in space missions. Generally almost anything that should be budgeted represents either a cost-driving resource or an element that is necessary to make the mission successful. For example, mass, power, and propellant are cost-driving commodities that are budgeted on virtually all missions. Communications systems would have a pointing budget to ensure that the communication antennas are correctly pointed and a link budget to ensure that there is adequate margin in the data transmission. An Earth observation mission would add a radiometric sensitivity budget to establish the accuracy with which spectral radiance emitted by the Earth can be measured. If the time limit for obtaining data is critical, as might be the case for military surveillance or emergency data processing, then a time budget would be established to monitor and control this.

Table 6-3. Items Frequently Budgeted in Space Missions. Overall budgets are typically established at the systems level and flow-down or allocated to the subsystems. See Wertz and Larson [1999] for a discussion of each of the budgets listed here.

Primary	Secondary
Weight	Subsystem Weight Power Propellant
Geolocation or System Pointing Errors	Pointing and Alignment Mapping, Attitude Control, Attitude Determination Position Determination
Timing	Coverage Communications Operations Processing
Availability	Reliability Operational Availability
Cost	Development Cost Deployment Cost Operations and Maintenance Cost

There are two basic types of space system budgets. *Commodity budgets* allocate any of the finite resources available on the spacecraft, such as mass, power, propellant or data rate. These budgets are straightforward; there exists an available amount and that quantity is allocated in some appropriate fashion among the subsystems that need it. Thus, the spacecraft mass will be allocated to all of the subsystems, including, for example, the attitude control system. The attitude control mass budget will then be allocated to the various components, which will in turn be allocated to the constituent parts of the components. Verification of the commodity budgets is also straightforward. We weigh each component and ultimately the spacecraft. Commodity budgets are often broken down further. For example, the mass budget may be broken down into different masses during different mission phases, and the power budget is frequently broken down into orbit average and peak power requirements.

The second and more complex type of budget is the *error budget* which breaks down the required accuracy on any measurement or control requirement, such as the mapping errors, pointing errors, or absolute timing errors. These are discussed in more detail in Secs. 6.3 and 8.6. The principal cause of complexity for these budgets,

Table 6-4. Simplified Pointing Error Budget for a Space Telescope. (All values in degrees.) In reality, there would be substantially more elements in the pointing budget as shown, for example, in Fig. 6.2. Note that if the errors had been summed literally, the total of 0.0155 deg would exceed the requirement and a major design enhancement would be required.

Random Errors	
Star Sensor Noise	0.002
Star Centroid Calculation	0.003
Pixel-to-pixel Variations	0.002
Thermal Variations	0.0015
RSS of Random Errors	0.0044
Bias Errors	
Star Sensor Mounting Errors	0.002
Payload Mounting Error	0.002
Payload Detector Mounting Error	0.003
RSS of Bias Errors	0.0041
Sum of Random and Bias Errors	0.0085
Requirement	0.010
Margin	0.0015

and a potential major cost driver for the system as a whole, is the process of how error budgets are to be added. If the elements that make up the error budget are random and uncorrelated then a root sum square (RSS) is the correct approach. If the error components are fully correlated, or if they are components that will physically add, then they should be added linearly. For attitude systems, for example, the most common result is to RSS the short-term noise, RSS the bias sources and then add the two results. However, in a worst-case situation the bias terms would add linearly. The correct approach to doing this is a mathematical question, discussed in more detail in Sec. 6.3.

Table 6-4 shows a simplified pointing error budget for a space telescope. In a real mission there would be substantially more elements in the budget, as shown in Fig. 6-2 on geopositioning error. However, this simple example illustrates one of the fundamental problems in error budgets. The user wishes to point the telescope to within 0.01 deg. If we RSS the random terms and bias terms separately, and then add the results, we're able to meet this requirement with some margin. On the other hand, if all of the terms were simply added linearly, the requirement would not be met, and we would have to either change the requirement or change the system design, presumably at significantly higher cost. In the end, how we treat error budgets depends on what the mission is trying to achieve. Is it a scientific mission in which we are trying to establish confidence levels? A military mission, where we are looking for the worst case that might be taken advantage of by an opponent? Or a commercial mission, in which we are trying to provide a customer with services of a given quality? In each case, we need to work with the user or the customer in order to determine their mission intent and create the most appropriate error budget and budgeting process.

Figure 6-2 shows an example of the allocation of a geopositioning error from a top-level mission requirement down to detailed component designs. As can be seen from the figure, this allocation is both complex and involves both the spacecraft and the mission control or data processing segments. Each step of the operation includes at least a small source of additional error. This may be round-off errors in the computer, errors in the uncertainty of coordinate transformation, or errors in physical hardware or their calibration. The importance of the budgeting process itself is to reach agreement on the list of error sources such that we can then find numerical values appropriate to each source, to ensure ourselves that the ultimate mission requirement will be met.

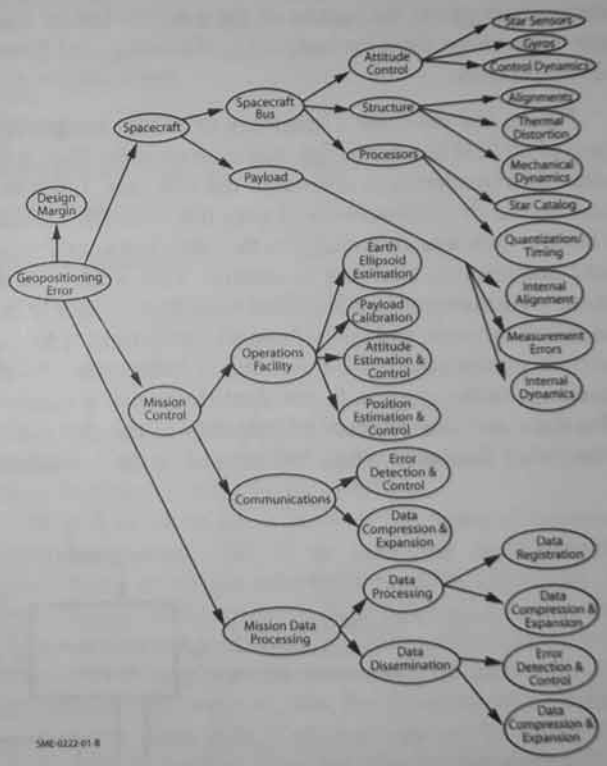


Fig. 6-2. Allocation of Geopositioning Error with Margin from Mission Requirements to Component Design Level.

Budgeting is both a mathematical process, as discussed in Sec. 6.3, and a political process in which various individuals and organizations agree on how much of any of the budgeted elements is to be incorporated within their particular subsystem. Each subsystem would of course like to have as much of the mass, power and precision budgets as possible in order to make their job less expensive and less complex. In principle, we should be willing to change the allocation among the various budgets in order to make the entire system perform as well as possible for a minimum cost. However, this may require shifting resources between competing organizations or making changes to contractual require-

ments. For example, it may be possible to do additional calibration on an Earth sensor in order to improve its accuracy to make up for a lack of accuracy in the GPS receiver. However, since these two components will typically be built by different manufacturers, changing this allocation could represent a contractual change for both manufacturers which could make it expensive for the prime contractor to achieve the end goal. Consequently, most system and subsystem engineers will work hard to both achieve an appropriate balance in the original budget and to provide sufficient margin to accommodate additional error sources that arise. This political process of balancing error budgets is summarized in Fig. 6-3. This chart was originally created by Mike Williams [1992], who coined the phrase "negotiation of terrible injustices" to give a sense of the political negotiations that go on during the budgeting and allocation process. It certainly suggests the nature of the political battles that can often ensue from the budgeting, allocating, and flow-down process.

Figure 6-4 provides a summary of a time budget for providing data in a forest fire detection mission. Here it is clear that timeliness is critical to the end user. Information on the fire's location is of very little interest a week after the data was collected. On the other hand, millisecond accuracy is also not necessary. This implies that there is a time-span in which data must be provided to be useful to the user, and that we can allocate this time to the various components of the system, including the orbital coverage of the spacecraft, the identification of the forest fire data, and the delivery of this data to the end user. Note also that there may be more than one timeline involved. For example, we may have one timeline from the time that a detectable fire starts until it is first seen by the system, and a second, probably much shorter, timeline from when a fire is detected by the system until that data is reported to a firefighter on the ground. Even here, there can be substantial contractual questions such as whether "detection" of a forest fire represents appropriate data being collected on board the spacecraft, or the process of identifying a specific forest fire which may occur somewhat later. The critical issue for system design is to create this budget such that issues can be defined and resolved as early in the process as possible. Consequently, the detailed budgeting process is an important one for the organizations involved, as well as for the system engineering process of ensuring that the mission objectives can be met.

6.3 Introduction to Error Analysis*

In this section we discuss how error budgets are to be handled numerically—i.e., how component errors are to be "summed" to yield the total error or system error. In a sense, the requirements allocation problem discussed in Sec. 6.2 is the inverse of the summation problem. Requirements allocation assigns to each component an allowable error. Here, we want to know how to combine the errors to check that the resulting system error meets requirements.

* Adapted from Geoffrey N. Smit, "Introduction to Error Analysis" Sec. 5.4 in Wertz [2001].

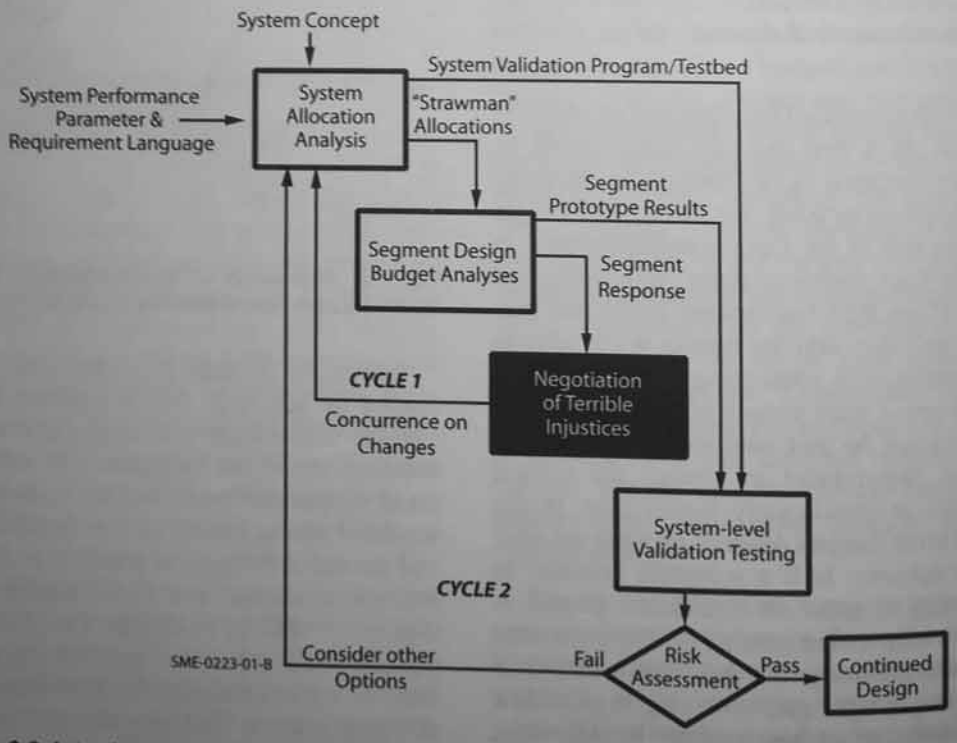


Fig. 6-3. Introduction to Error Analysis.

Table 6-4◀, Fig. 6-3, Eq. 6-0

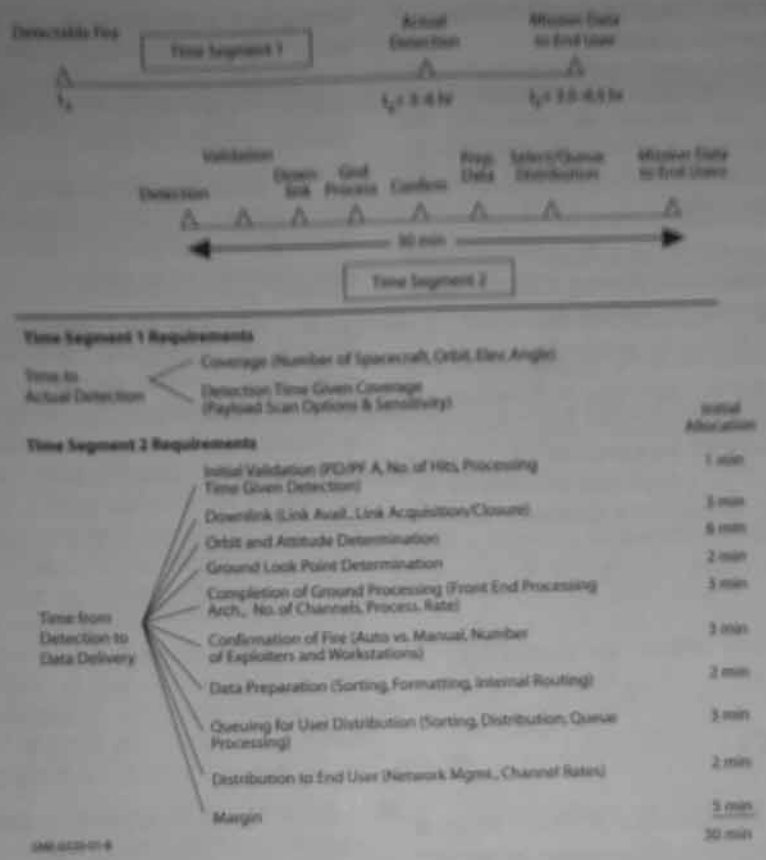


Fig. 6-4. Allocation of Data Timeline.

System error analysis involves the following steps:

1. **Modeling the system** and identifying the error sources i.e., the places where errors are introduced.
2. **Determining the sensitivity** of the system error to each of the error sources.
3. **Characterizing the distribution** for each of the error sources.
4. **Combining the sources** to determine the system error.

Accomplishing **Step 1** adequately is crucial and is system specific. In realistic systems, the models for error analyses can be dismayingly complex. Our treatment here will be as simple as possible, to bring out the main features.

Step 2 is really an integral part of Step 1, but we have separated it out for emphasis. Step 1 normally gives us a deterministic relationship between error sources and the system error. For want of a better term, we shall call this relationship the *error equation*. Typically, the system error will not be dominated by the effects from a single source. Usually, there are numerous error sources, and an efficient design will tend to balance their effects. In some situations, a system may have singularities leading to high sensitivity to a particular input. In the normal situation (no singularities, reasonably balanced response to the various errors), the error equation can be represented as a sum in which the output error is the algebraic sum of the inputs. (See, for this example, Taylor [1997].)

Step 3 involves knowledge of the types of components being used, and of the operating environment. Since many errors are inherently random, some sort of probabilistic characterization must be given. Usually, the component errors are specified in terms of standard deviations. For example, a star tracker may be accurate to 30 arcseconds (3σ), implying that the distribution is Gaussian and that about 99% of all readings will be within 30 arcseconds of the true direction. Ideally, the probability distribution of the errors could be given, but this level of information is usually not available.

From the error equation, we can easily bound the system error. If $e = e_1 + e_2 + \ldots + e_n$, where e is the system error and $e_1, e_2, \ldots, e_n$ are the component errors, then $|e| \leq |e_1| + |e_2| + \ldots + |e_n|$. This bound can form the basis for an error budget. Normally, however, it will be too conservative, i.e., it will demand that we keep the e_i considerably smaller than is necessary to achieve a given bound on e. We must exploit the statistical nature of the errors to make the design affordable while still meeting the error budget.

Step 4 addresses the mathematical problem of estimating the statistical properties of the system error, given those of the components and the error equation connecting them. The summation calculation can be approached in various ways. The actual analysis undertaken and the

degree of detail included will depend on the system and on the amount of information available:

- In some cases (e.g., when the probability distribution of the e_i are normal or uniform), the distribution of the sum can be evaluated analytically. (See Feller [1971], Meyer [1972], Papoulis [2002], and Parzen [1992].)
- When this is not possible (or efficient) the output distribution can be generated using a Monte Carlo simulation as described in Table 6-6.
- In many cases, generation of the system probability distribution may not be necessary or feasible, e.g., when the component distributions are unknown. It is possible to estimate the system standard deviation directly in terms of the component standard deviations. This not only makes fewer demands for detail in the input data, it is also far simpler, computationally. For most error budgeting, this is the preferred approach.

It is shown in App. G (where the terminology is defined) that if $e = e_1 + e_2 + \ldots + e_n$, then

$$\sigma^2 = \sum_i \sigma_i^2 + 2\sum_{i>j}\left[\mathrm{Cov}\left(e_i \cdot e_j\right)\right] \qquad (6\text{-}1)$$

where σ is the standard deviation of e, σ_i is the standard deviation of e_i and $\mathrm{Cov}(e_i, e_j)$ is the covariance between e_i and e_j. If the covariances are zero (i.e., if the errors are independent), this reduces to

$$\sigma = \left(\sigma_1^2 + \ldots + \sigma_n^2\right)^{1/2} \qquad (6\text{-}2)$$

This is called the root sum square or RSS. If the correlations are unity, so that the covariances equal the products of the corresponding standard deviations, then σ reduces to the sum of the σ_i. If the correlations are partial, we get a combination rule between the RSS and the sum. This is a standard result for the combination of standard deviations, and it forms the basis for most error analyses. (See Feller [1971], Meyer [1972], Papoulis [2002], and Parzen [1992].)

The procedure for undertaking an error analysis for the purpose of error budgeting is summarized in Table 6-5. A related procedure for a Monte Carlo error analysis is given in Table 6-6.

We have been describing the error addition process as though the system and the errors were static. In most cases this is not true, and the dynamics of the system and the time structure of the component errors should be accounted for to obtain an accurate error estimate. At the simplest level, this involves looking at the frequency spectra of the various errors, and noting that in some cases this allows us to reduce the effective system error variance. (See the boxed example.) On a more sophisticated level, the system dynamical equations can be manipulated to provide an equation for the propagation of the system error covariance matrix which is then optimized. The error budget is then formulated in terms of this optimal error. The degree of sophistication adopted will depend on how far along in the program we are and on the program's financial budget.

Note that there are two distinct "flows" in the error analysis and budgeting process:

- The desire for less conservatism, and hence more accuracy, leads us to increasing levels of sophistication in our modeling of the system.
- The need to keep the calculations manageable leads us to simpler ways of characterizing the errors. This aspect of the flow is often driven by a lack of data on the error performance of the components.

Frequency Distribution of Errors

Any system-level measurement, whether it involves calculating direction to an image pixel or reading spectral radiance from a radiometer, can be thought of as selecting a single sample from within a statistical ensemble of all possible measurements. This ensemble includes noise effects from all system elements. Each noise component may have its own temporal or frequency characteristics. The sampling process that leads to estimates of most probable measurement value, error and overall probability distribution function occurs as a time series of measurements. Therefore, sampling rate, collection duration and other details of how these measurements are collected over time may affect measurement value and estimated error.

For example, measurement samples taken closer together in time may be more similar in value than those taken farther apart in time. In those cases, estimated probability distribution of the measurement depends on the time scale used in collecting the samples. At any given time, only a limited part of the total statistical ensemble may actually be accessible in the sampling process. That region of accessibility can move around within the overall ensemble in a random manner that in itself represents a source of noise. Over relatively short time periods, a smaller measurement variance than the entire ensemble range will be found along with a slowly varying bias that can be tracked and eliminated from the error budget. In some cases, the slowly varying bias can be deterministic such as when it is due to thermal or lighting effects that are periodic with spacecraft orbit.

Consider the sources of error associated with calculating direction to an image pixel. These error sources include star sensor measurement error, star sensor mounting error, star catalog error, attitude computation error, payload sensor measurement error, pixel centroid error, payload sensor mounting error and coordinate transformation error. (See Table 5-19 of Wertz [2009].) We would expect errors like star sensor and payload mounting errors to vary much more slowly than others like star sensor measurement error resulting from electronic noise in the detector array. On the shorter time scales of the higher frequency errors, we are able to estimate these biases and correspondingly reduce the system error. See App. G on the book website for a more detailed and thorough discussion of frequency dependent noise.

Get More

Table 6-4◀, Fig. 6-4◀, Eq. 6-2

Table 6-5. Creating Error Budgets. RSS = "Root Sum Square" = Square root of the sum of the squares of the individual components. (See Wertz [2001], Secs. 5.4, 5.5 and App. C for more detail.)

Step	Comments
1. Define the end-to-end measurement process.	Start with basic orbit, attitude, and payload measurements and compute the final output.
2. Identify sources of error in each step.	Break down each step into its lowest constituents.
3. Identify the statistical distribution of each error source.	Typically Gaussian or uniform if the measurements are quantized.
4. Assign a preliminary error value to each error source	If Gaussian, use 3σ error; if quantized use full range.
5. Estimate the maximum probable error by adding all error terms in worst case end-to-end geometry.	If this meets the requirement, quit and go to Step 12. If singularities occur, calculation and implementation may need refinement.
6. Estimate the minimum probable error by RSSing all error terms.	If this doesn't meet the requirement, then need to find a way to reduce error sources or find another solution.
7. If the requirement lies between (5) and (6), proceed to more accurate summation techniques.	Steps 8–11.
8. Determine the frequency associated with each of the error sources.	Over what time span is one measurement set not correlated with the previous one?
9. Divide the error sources into frequency regimes.	Typically, Gaussian noise at the sampling frequency, most error sources at orbit frequency, and some at other frequencies (for example, weather sampling or magnetic field errors).
10. Determine the error within each frequency regime as follows:	Within each frequency regime, there is the potential for errors to be correlated.
10a. Add the correlated errors.*	Look explicitly for where singularities can occur.
10b. RSS the uncorrelated errors and the sums of the correlated errors	Assumes these are statistically independent.
11. RSS the errors for different frequency regimes; this gives an analytic estimate of the total error.	Errors at different frequencies are orthogonal (cf Fourier analysis) and can be RSS'd.
12. Review the steps looking for missing sources of error, such as uncertainty in coordinate transformations. Also conduct an independent search for singularities.	Every coordinate transformation includes an error associated with the definition of the new coordinate frame in terms of the old one.
13. Document and Iterate.	—

* For partial correlations, summing is conservative, RSS is optimistic. For more accurate formulas, see App. G.

A key measurement problem is that correlated error sources can lead to singularities in the measurement process for which the error in the result can become unbounded (i.e., very large). For a geometrical discussion of the source and treatment of measurement correlation, see Sec. 8.6.

Table 6-6. Monte Carlo Error Analysis. This is a numerical approach requiring a large number of runs. Singularities become evident due to a dramatic increase in the spread of the solutions when noise or biases are added to the data. (See Sec. 8.6, Wertz [2001], and App. C for more detail.)

Step	Comments
1. Do steps 1–4 of the analytic approach	See Table 6-5.
5. Assign a representative value for each of the variables involved	Pick both representative values and extremes where singularities can occur.
6. Execute the computation, from initial measurements to final answer; adding error to each computation according to the assigned probability distribution	Near singularities, computations may fail due to an illegal operation (e.g., asin of a number greater than 1).
*7. Evaluate the statistical results of multiple runs**	The statistical distribution of the result is the best estimate of the final error.
8. Check carefully for singularities, correlated measurements, and missing error sources†	Must cover the full range of measurements. If this is done, singularities become evident in the data due to very high noise levels.
9. Iterate and document	

* For ways to enhance the efficiency of this process. For example, see Robert et al. [2010].

† As in the case of an analytic assessment, measurement singularities can dramatically impact the final error and cause divergence in numerical solutions. (See Sec. 8.6)

Error sources can be categorized in various ways: random vs. systematic (or deterministic), additive vs. multiplicative, modeled vs. unmodeled. Some of the classifications commonly used, such as noise vs. bias, are often subjective. The basic idea is to distinguish between effects which can be removed by calibration or filtering and those which cannot and which must therefore be added to the error budget. This distinction will depend on several factors. Some error sources are inherently random (such as thermal noise in electronic systems and optical detectors), while others, such as misalignments, are systematic. In principle, the systematic part can be calibrated out. In practice there are always residuals due to limitations in the calibration technique or due to the myriad of details in the system behavior. The ability to compensate for error sources will depend on how accurately they have been modeled. In very high precision systems, we may be prepared to take into account a large number of details which would simply be ignored and lumped into "noise" in other systems. For example, a star tracker may distort under temperature changes. To some extent we can measure this effect and use an open loop calibration curve to take it out based on a temperature measurement. However, there are also hysteresis effects, so the distortion is not exactly repeatable. This would contribute to the residuals. A more complex model accounting for the thermal time history might be used to further reduce the residuals. There will always be a bound to the degree of detail that can be justified, and the unmodeled effects will usually be treated as "random."

The time scale over which measurements are being taken will also influence how we view different sources. What may appear as a slowly varying bias on one time scale, may look like noise on another. With this in view, the distinction between noise and bias is really based more on how they are treated in a given system. Normally, bias is estimated, either via a separate calibration procedure or some sort of filter, and subtracted out. Noise is usually filtered via a low pass filter. In general, error sources that are outside the frequency band being used for measurements can be removed by various kinds of filters. Error sources within the measurement band can sometimes be removed, provided we understand the system well enough. The errors which would be summed, as described in the rest of this section, would be those which cannot be removed or which are not economical to remove.

Finally, combinations of sensors, and the corresponding combinations of their errors, do not always result in a system error larger than that of the components. In fact, many configurations are set up to allow sensors to complement each other. A simple example would be to put several identical sensors in parallel and take their average. The variance of the output would be $(1/N)$ times the variance of an individual sensor reading. A more realistic example would be combining a gyro with a star sensor.

Table 6-6◀, Fig. 6-4◀, Eq. 6-2◀

6.4 Specifications and Requirements Documentation

Specifications and requirements documents are developed using organizational best practices that are generally consistent with commercial and international standards documented by the Electronics Industries Association (EIA), IEEE, International Organization for Standardization (ISO), and International Electrotechnical Commission (IEC) along with the draft US Department of Defense (DoD) Standard MIL-STD-499C on "System Engineering." The commercial and international standards include EIA-632 on "Processes for Engineering a System," IEEE 1220 on "Standard for Application and Management of the Systems Engineering," EIA-649 on "Configuration Management" and ISO/IEC 15288 on "Systems and Software Engineering—System Life Cycle Processes." The removal of most military standards, including the system engineering standard, as contractual requirements in US DoD developments has led to a different paradigm for verifying excellence in system engineering practice that includes organizational certifications for implementing standards in ISO 9000 and the use of Carnegie Mellon Software Engineering Institute's Capability Maturity Model® Integration (CMMI®) levels to encourage and document continuous process improvements in system development and delivery. The comprehensive *NASA Systems Engineering Handbook* (NASA/SP-2007-6105 Rev 1) provides an excellent guide to fundamental systems engineering concepts and techniques, including requirements definition and documentation. All of these best practices and documents emphasize the importance of mission and requirements development and management and place it at the beginning of sound system design processes.

Effective requirements documents must be consistent within the set of program documents, complete relative to the maturity of the system in development and traceable to the highest level mission and system requirements and specifications and to derived lower level requirements and specifications. A consistent performance requirement should be written only once, kept in a readily accessible place and positioned appropriately in the hierarchy of established requirements. System requirements documents and specifications become more complete as the system design and implementation matures. Nevertheless, at each stage of completeness, every level of requirement must satisfy the next higher level. As an example of completeness, the geolocation requirement of the FireSat II mission should address all error sources both in segment and interface documents. Requirements must be rigorously traceable as they are developed, allocated and decomposed or derived, to ensure they are unambiguous and verifiable. The method used for tracing requirements should be specified in program requirements documents. Section 6.5 discusses tools for documenting requirements and managing requirements traceability.

Requirements documents or specifications notionally fall into eight classes (Fig. 6-1) that are supported with descriptive documents and analyses that need to be current with the requirements baseline. The *system requirements document* (sometimes called "system requirements specification") should cover every aspect of what the system should do (functions) and how well it should do it (performance requirements). Normally, the system requirements document is developed before subsystem and component requirements documents, because it should establish the requirements basis for these lower level system elements. However, for any given system implementation approach, trade-offs may be required at the system level in response to cost, interface issues, performance limitations, or schedules related to subsystem design approaches.

Requirements reviews are necessary predecessors to system design reviews and important elements in program design reviews. Requirements-related issues must have the same weight as design issues in readiness decisions by a program to proceed to its next step of development. Requirements assessments may identify the need to reassess project drivers, including:

- Relax design constraints
- Reallocate requirements
- Request a change to funding profiles.

System test plans are derived from requirements documents and capture approaches for requirements validation and verification in qualification and acceptance processes. Test plans typically reflect test specifications and identify objectives, environments and levels of assembly at which tests are to be performed.

Requirements specifications are potentially subject to change. Therefore, they should be designated "preliminary" prior to reviews at each stage of design. During formal design phases, while requirements may have to be traded, the specifications must, like design documents, be subject to rigorous configuration management and change control.

In addition, when requirements specifications at a top level govern more than one system element, tailoring a spec to accommodate the specific character of a subsystem may be appropriate, especially if the requirements not directly associated with system performance.

Interface Management

Interface Requirements Documents (IRDs) define all physical, functional and operational interface requirements between different systems and between different elements or components in the same system to ensure hardware and software compatibility. An IRD consists of physical and functional requirements and constraints on hardware and/or software items. The IRD controls the interfaces between interrelated components of a system under development, as well as between the system under development and any external systems (either existing or under development) that comprise a complete architecture. Interfaces between systems must be carefully considered with the complete end-to-end system architecture, even in the early stages of system development. The system level IRD may be referred to or included in the system specification.

Interface Control Documents (ICDs) describe design solutions to specific interface requirements. ICDs are the key to integrating and maintaining relationships between different systems and different segments within the same system. An ICD defines the physical interface between two system elements, including the number and types of connectors, electrical parameters, mechanical properties and environmental constraints and the data interface including all input and output data sent between elements. ICDs covering critical interfaces, such as the spacecraft-to-ground segment for FireSat II, can become highly complex. Two guidelines are important in developing and refining interface documents. Each ICD normally covers only two segments, although multiple elements within segments may require consideration of relationships with other segments. In general, we should avoid designs necessitating such added complexity.

Interface agreements must be documented at every level of design, usually in ICDs. At the system level, project managers or system engineers control the ICDs, whereas individual integrated product team leads control the ICDs at lower system levels. Although the exact content and format of interface documents may vary significantly across different systems and development organizations, ICDs always address physical, data and electrical interfaces and interactions. In addition, other system characteristics such as gyro drift and star tracker performance, for example, including nonlinearities of the transfer function, output axis coupling or star tracker noise, require the same definition level in the ICD so that the mission ground station can correctly calibrate these system elements and use them effectively in processing mission data.

6.5 System Engineering Tools

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A challenge to describing engineering software tools is that their current rate of change is very high. On a roughly annual timescale, individual software programs of this class can become obsolete and even cease to exist. Programs that continue to be useful see their user interfaces, and the platforms on which they run, change at equivalent rates.

As examples of ongoing change, the platforms for engineering software have evolved from mainframe computers, through minicomputers and desktop personal computers, and now largely consist of laptop computers. Moreover, elements of this software are currently beginning a migration to handheld smart phones. Over this time period, the user interface has progressed from alphanumeric punch cards and printouts, to mono-

Table 6-6◀, Fig. 6-4◀, Eq. 6-2◀

chromatic text terminals, to color graphics terminals using pointing devices like mice that interact with clickable, drop-down menus. Voice input and touch screens are now beginning to be used. The actual 3D objects produced by 3D printers (sometimes called stereo-lithography or rapid prototyping) can be considered a new kind of output from engineering design programs.

Although somewhat invisible to the user, there is also an ongoing evolution in the means by which the software for these programs is executed. While laptops typically still store and execute the software of these engineering programs, often the user needs to be connected to a key-server that insures that the user has a valid, unexpired license to use the software. Sometimes the software is not resident on the platform hosting the user interface. Instead, the user interface is often a web browser showing a page that is the portal to a website where the software is either:

- Temporarily downloaded to run on the platform supporting the user interface,
- Run on a computer associated with the website,
- Propagated to a remote, massively-parallel supercomputer,
- In extremely compute-intensive cases (such as searching for SETI signals), propagated to any internet-connected personal computer in the world whose owner in willing to allow his or her computer to be so utilized

Compare that rate of change with the current Soyuz-FG launch vehicle, which is still used as a primary means for supplying the International Space Station with crew and supplies. The Soyuz-FG is the latest of many upgrades of the R-7, a modified ICBM developed by the USSR and used in 1957 to launch the first space mission, Sputnik. The Soyuz still looks much like the original R-7 and utilizes the same basic propulsion system. There is every indication that the Soyuz, or another upgrade of it, will still be in use past 2017, meaning a more than 60-year lifetime.

However, despite the seeming rapid change of engineering software, it is important to understand that its underlying theory and mathematics evolve even more slowly than long-lived launch vehicles. CAD programs still use elements of Euclid's geometry, and trajectory simulators are built on the theories of Kepler, Newton, and Einstein. Thus, the fundamental capabilities of these programs evolve far more slowly than their hosting platforms, computing power, user interfaces and convenience features.

The software tools that are applicable to space mission analysis and design can be usefully divided into two basic groups.

1. A simple set of tools that can be used by a single systems engineer or mission designer, or a small group of generalists, to develop a rough, but frequently adequate, mission concept. The size of these efforts could range from a single week (for a single individual to develop a variant of a mission that he or she is already familiar with) to more than a labor year of effort for three or four generalists to develop a relatively unique and complex mission.
2. More complex and sophisticated software tools that are typically used by specialists, such as the use of NASTRAN by a structural analyst. On larger mission design efforts, typically involving 5 to 10 people working for 6 to 12 months, these more capable tools are frequently used to come up with much more accurate and complete mission concepts. If the mission concept is actually developed into a real mission, these tools will continue to be used throughout the development process, first to support more detailed design and finally to help analyze how closely the performance of hardware in testing comes to its design objectives.

Obviously, it is possible, as frequently happens, that some mixture of the above approaches will be employed. Possibilities include:

- Augmenting the work of generalists with a subset of specialists
- Bringing in needed specialists for short periods to solve specific problems
- Having some generalists with good enough discipline backgrounds that they also function as effective specialists in certain areas.

Simple Tool Sets

In his book, *Space Vehicle Design*, Michael Griffin says that one of the capabilities desirable in a systems engineer is the knowledge and ability to carry out a first order design of any spacecraft subsystem. In a somewhat larger context, it is equally possible for a systems engineer to develop a complete first-order mission design for relatively straightforward missions. Such end-to-end skills are equally useful when space mission design is undertaken by a small group, for the result can be much improved when each person has the knowledge to contribute to and check the design and analysis of any part of the mission.

The essential tools for such an end-to-end capability can be summarized as follows:

- The first and most essential ingredient is a handbook, essentially one such as SME, which covers all basic areas of space flight design in a consistent and quantitative way that is understandable to a person having an education and background equivalent to that of an engineer working on space missions. The values and formulas presented in such a handbook need to be presented in a form that can be conveniently programmed in software.

There are also more specialized handbooks, such as for propulsion and thermal design, and occasionally a systems engineer may want to go to them for additional detail.

- The most important software tool then is a software program that makes it simple to enter and use these formulas and data. Generally, a spreadsheet program, such as Excel, is adequate. Spreadsheets are also useful as simple databases, and documenting the traceability of requirements. The ability for one worksheet to make use of data on another further increases their versatility.
- Occasionally, a simple procedural language, such as Basic, can usefully augment the capabilities of a spreadsheet. Some procedural languages can be integrated with spreadsheets.
- Knowledge of the top-level information about related missions accomplished in the past is generally a requirement for developing any reasonable mission concept. Handbooks such as SME typically only have room to briefly describe a very small number of example missions. Many more missions are described in adequate detail in papers presented at space conferences such as those by the AIAA. Today, relevant papers from such conferences are typically searched for and accessed via Internet browsers such as Explorer, Safari, and Firefox. Downloading a desired paper can require a credit card and fee, but often, companies and educational and research institutions have blanket agreements to allow their users free access to such papers. Search engines such as Google and Bing can often find the abstracts to useful papers, as well as websites and other on-line articles that are free.
- Finally, there are a variety of software tools that are almost critical for effectively documenting and communicating mission designs. Examples are word processing programs, such as Word, and business graphics programs such as Power Point. Frequently, the output of such programs will be converted into a common format, such as PDF, which can then be read on basically any computer with a free, downloadable PDF reader. As a more powerful visualization tool, it would be reasonable to get a simple, inexpensive 3D CAD program that would be used to document the design with 3D solids. Individual snapshots from the CAD program can be converted to 2D PDF images, and some CAD programs can output a 3D geometry in a format that can be viewed in 3D by free, downloadable viewers. In general, a simple website building tool is also needed to document almost any study. This is because often the only practical way to organize and communicate all of the intermediate and final data from a mission design is via a website that can be hierarchically organized, password protected, and provide different access privileges, such as read and write, based on user identification.

Complex Tool Sets



For a discussion of more complex tools see the book website.

6.6 The Role of Standards in Space Systems Development

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Technical Standards Play an Increasingly Important Role

Space Standards are benchmark requirements for space systems, services, operations or other specific views documented by experts in the field, often, but not always, in response to lessons learned. From our example standard below we have the generalized purpose of a standard:

> "This…Standard establishes a common framework and provides a common basis for the interchange of…data. It allows implementing organizations…to proceed coherently with the development of compatible derived standards for the flight and ground systems that are within their cognizance. Derived…standards may implement only a subset of the optional features allowed by the…Standard and may incorporate features not addressed by this…Standard."



Standards are used throughout all life cycle phases and must be accounted for in the space development process. Note that the word "must" here is not regulatory. Simply put, standards effectively include current best practices. While one does not need a standard to do space mission engineering work, failure to consider openly available, current standards does leave the engineer liable in the event of failure.

Some standards are relatively obvious. For example, engineering data standards that are used for transfer among engineering tools. ISO 10303 includes the STEP standards like AP203 for transfer of mechanical drawing information and the relatively new AP233 for systems engineering data. STEP files are commonly used to transfer mechanical drawings between organizations and different CAD tools. Consider that many tools use proprietary data formats for the file structure, and these formats change with time. The result is that it is difficult to read files from even 10 years ago. Using standard formats to store this data will enable us to provide the long-term storage necessary for design information of flight and ground systems that last over multiple decades.

Table 6-6◀, Fig. 6-4◀, Eq. 6-2◀

Other data standards include PDF, RTF, and JPEG. With JPEG or TIFF, we have the ability to store images and view them many years from now—try to view the original image data from Voyager or Lunar Orbiter! The value of this type of engineering standard is well understood. Another is the use of TCP/IP to support space systems. CCSDS uses TCP/IP as one base for their standards work.

Due to the evolving nature of all engineering and scientific knowledge the benchmark contained in standards does change over time (TCP/IP began migration to v.6 approximately in the mid-1990s). It is the responsibility of all of us to update the knowledge in open standards as new lessons are learned, with special emphasis on safety. It is also the responsibility of each of us to use standards as we should use any tool; with judgment on when and how to vary from the benchmark. While a student may blindly apply standards as the use of the tool is learned, the master is expected to know his or her tools and expertly apply them. To extend teaching from the martial arts, a master is not limited by his or her weapons, but uses what is at hand to extend his or her capabilities.

Unfortunately, many engineers and scientists are not familiar with engineering standards; until recently no engineering schools have included engineering standards as part of their curriculum [Bickart, 2007], and scientific curricula will never include something as coarse as engineering topics. Therefore, the scientists and engineers who are often leaders in space systems development or space organizations do not understand open technical standards. A practical and extended education on the topic of standards is needed for the student and the master. Inclusion of the topic of standards in this text is a needed first step.

A small tempest, perhaps a "tempest in a teapot," rages in the space industry today regarding the role of technical standards, their development and application. From the first successful space launch in 1957 each organization and each project within those organizations has created their own engineering basis. It is understandable that in the early stages of spaceflight this was so; there were few standard practices and many unknowns to deal with. Today, our knowledge base is much deeper, with personal, institutional, national and even regional bodies of space expertise. Our industry-wide technical knowledge base should be strong, and one indicator of that is an industry standards base, as is common in other industries. However, there still are those who maintain that open technical standards for space systems are more trouble than they are worth.

Interestingly, the global market value of the space industry in 2009 [The Space Report, 2010] was about $262B (US). This is 3.4 times what Home Depot reported in the same year ($76.9B US). The point of this comparison is that the space industry has barely reached a level of production and service delivery where standardization makes sense. So the question is not *if* we need standards for the space industry. Some capabilities we discuss in this text *require* high volume or significant interoperability. *Where* we need standardization is a much more relevant point of discussion than *if* we need standardization and standards. The question of where we need standards and standardization is answered not by technical arguments, but by economic arguments. The counter question, "Where do we *not* want standards or standardization?" is answered by policy or political arguments. Happily, we will not get into those arguments here, but you need to understand that as the space industry grows, as the components of space infrastructure become ubiquitous to society at large, as spacecraft become smaller, and as modularity in space systems increases, the economic argument for increased standardization and standards *at some level* also grows. Since we are a small industry, we enjoy high visibility and prestige in society, and our political masters find us often to be troublesome, we need the strength in self-governance standards provide.

The small tempest will subside, because we cannot go back to the "good old days" of space projects being run as personal fiefdoms. And the teapot will expand as our industry continues its steady growth. Establishing an intelligent standards base now is a challenge worth our best efforts.

What are the Right Standards to Develop and Use?

Standards cover many topics, and there is no set rule on content. Understanding the answers to the questions of Who? What? Why? Where? When? and How? is very important in space systems engineering and standards are useful in finding quick and reliable answers. It is also true that good standards are useful to a common interest group. The larger the group that actually uses the standard, the more useful it is. Generally standards can cover operations, systems and services areas of any mission. Special areas such as space data and information transfer have been very effective topic areas. Within or across these areas standards effectively address:

- Interfaces of all types
- Form, fit or function of types of components which are used regularly
- Processes or procedures with many contributors/users
- All aspects of verification, mostly dealing with testing
- Other areas

Standards are tools. They are often very technical tools, and they have a formal construction and language. A standards document is a listing, in concise language, of required items on a specific topic. **The points listed in a standard can be required ("normative") and are indicated by use of the word "shall." Some points in a standard are not required but are strongly sup-**

Table 6-6◀, Fig. 6-4◀, Eq. 6-2◀

ported; these are indicated by words such as "should." Often a standard will list possible variations; each variation will include "shall" and "should" types of statements. To see how this works, it is helpful to look at an example. One is provided at the end of this section.

It is also important to understand that there are some things that standards are not. Standards are not

- Rigid lists of requirements invented by people with no expertise in the topic area
- Meant to be unduly burdensome
- Required (unless specified by contract)
- Replacements for responsibility for engineers or scientists to fully address technical issues in their projects
- Removers of options on how technical issues on projects are addressed
- Constraints on technology

Properly applied, standards are not cost drivers. (However, using standards can make it more difficult to recover program schedule by eliminating portions of the test program. Unfortunately the practice of eliminating "unnecessary" testing at the end of the development cycle is uncommon in the space industry.)

The Merits of Standards (how do they help make spacecraft better or cheaper)

There are many reasons for voluntary compliance technical standards. They are primarily a means of self-governance by a technical community or industry. For people within the community they are a means of clearly communicating essential information. For people outside the community they are a means of establishing expectations. Standards are enforced by being referenced in contracts as compliance documents.

Standards are applied to the products and services in an industry, and also to the expectations of organizations in how these products and services aggregate to satisfy customer needs (in other words, organizational processes). A fundamental need addressed by a standard is to identify, quantify and document a basic requirement, so a producer may deliver products and services that meet (or exceed) the "bar" of customer expectations or accepted by the industry. For example, an engineering standard for a spacecraft requires design and verification of the spacecraft to meet the vibration expected through the launch profile (Sec. 26.3). An operational example is the requirement to design space to ground communications consistent with the basic physics in the laws of astrodynamics (Chap. 9 or Sec. 10.5). Open standards are very new to the space industry, so the benefit of common language, definitions, interfaces, verification methods and so forth are not yet realized.

A challenge for people who create space standards is the proper decomposition of products and services in a manner acceptable to yield efficiency, cost effectiveness, acceptable risk, reliability, flexibility and so on. For this reason standards must continually evolve: we continue to learn better ways to build and we continue to learn from mistakes we have made.

Standards promulgate from the most general levels of any industry down to ever increasing levels of detail over time. As the industry matures, it is possible—it is a competitive requirement—to internally control efficiency. It also becomes necessary as an industry grows in size for external controls, such as acceptable levels of risk for manned spaceflight, be negotiated for consistency with risk from other industries. For example, manned spaceflight today assumes risk consistent with experimental aviation. Tomorrow our expectation of risk might be on par with commercial aviation.

Multiple space communities (Chap. 2) need many similar things and it makes sense that re-use between communities keeps cost down. Standards document acceptable, critical re-use characteristics for space products, space services, and space operations. Within the space community at large there are still some communities that are closed (vs. open) in their practices. The manned spaceflight community is, once again, our example, but we will look at the most exclusive manned spaceflight community—NASA: In the area of reliability it has taken monumental shifts in the global manned spaceflight market to force NASA to discuss systems, services or operational technical metrics, rather than the human rating certification process and the related manned system acquisition process.

The Demerits of Standards (ways they can be harmful)

On the demerit side, standards poorly applied tend to potentially drive cost by demanding that things be done which are unnecessary. They may slow breakthroughs by the forced application of standards that aren't applicable to new technology. (For example, on a project building all-composite, cryogenic tanks that have never been built before, people have tried to apply standards for composite overwrap pressure vessels. Due to technological differences these didn't really make sense.) The important point is that standards are developed by professionals for professionals. So in standards application, use professional judgment: if and when necessary tailor the requirements so things make sense. If needed seek professional help.

Tailoring Standards to Meet Specific Mission Needs

Tailoring is something that we do a great deal of to control cost. Since the requirement listed in a standard are created to address a topic in a notional or general sense, it is clear that almost no standard is a perfect fit for a given project. True, the requirements listed are meant to be a minimal set. A practicing engineer must be smart in applying any initial approach to solving program design, fabrication, test, operations, or disposal issues, and that is no less true for applying a standard baseline than it is for applying text book solutions.

Table 6-6◀, Fig. 6-4◀, Eq. 6-2◀

Where Standards Come From

There are many standards (and more are coming). As the example standard demonstrated, the authority for creating a standardized set of requirements rests with some duly formed body. In the example it is the CCSDS, but there are many, many Standards Development Organizations (SDOs). For the space industry a few specific SDOs are listed below, and their function is discussed briefly. While the technical work in SDOs is done by focused expert groups, each standard is reviewed and approved by staff and an executive board.

Which standards are important depends on what you are trying to do. The primary international SDOs for space are:

- The Consultative Committee for Space Data Systems (CCSDS)
  - Originally formed as a collaboration of the space agencies of the world, CCSDS has recently expended its charter to include defense or commercial organizations. Participation is free, but registration is required. See their website for current communications architecture and standards. (CCSDS calls their standards "recommendations")
  - CCSDS maintains a close relationship with ISO and is associated with ISO as ISO TC20/SC13, the Subcommittee for Space Data Systems. Interestingly, CCSDS documents are downloaded for free, while the same document, purchased through ISO with an ISO coversheet, will cost in the range of $70 to $130.
- The International Organization for Standardization (ISO), Technical Committee 20 (Aircraft and Space), Subcommittees 13 (above) and 14, Space Systems and Operations
  - Chartered by the United Nations and comprised of national member bodies (e.g., the American National Standards Institute (ANSI)). Each participating country in a Technical Committee, Subcommittee or Working Group registers to participate. Participation is free. See their website.

The primary US professional SDO for space is:

- The American Institute for Aeronautics and Astronautics (AIAA)
  - This professional organization works through both Technical Committees (to participate in a TC AIAA membership is required) and Committees on Standards (to participate in a CoS AIAA membership is not required). See their website for AIAA standards.

Each of these SDOs develop what are called voluntary compliance standards (VCS). Compliance is only required as specified by the program/project contract. In creating a VCS an SDO assures documents are valid by requiring consensus of the participating experts developing the document, and a public review at the final stage. Consensus requires that all points of view presented be addressed. This means when an expert position is not included, a clear reason is given. In the public review phase, each comment is addressed and the disposition documented. The SDO staff is responsible for bringing discrepancies in the consensus process to the attention of the project lead and the SDO standards executive board.

The primary European regional SDO for space, which operates in a manner similar to NASA in standards development, is the European Cooperation for Space Standardization (ECSS). See their website for ECSS standards.

Of the many available standards, the Consultative Committee on Space Standards (CCSDS) provides several good examples. They are free to download and can be referenced in this text without special permission. The standard on Attitude Data Messages is a CCSDS Recommended Standard (a Blue Book), issued in May 2008. It is a good example to use while referencing Chap. 8. This example defines how the important data on mission geometry is communicated to and from the spacecraft and the ground control center. Because this standard exists, it is no longer necessary for each project to define these terms.

Take a moment to review the excerpt in the boxed example from CCSDS 504.0-B-1. Notice that the document has a publication date. This is important in answering when the standard should be used. Standards must be current to be relevant. A refresh rate of every 5 years is recommended, but as long as even 10 years acceptable. Notice also that the document uses an "authority" statement to identify who the user is meant to be. While this doesn't limit use to only those organizations, it does identify which body is trying to establish some form of self-governance and where the participating experts are from. The procedure for developing this standard follows an established process. This standard has more strength, more authority, within the CCSDS member country space agencies than in a purely commercial activity.

The Statement of Intent and the Forward help the user to understand how and why the standard was developed, when to use it and how to properly apply the requirements contained within. The process for developing open standards is by consensus, which is not a simple democratic process. The requirement for endorsement and use is voluntary by each organization. This type of standard development is termed "voluntary consensus." Resulting standards are "voluntary consensus standards."

It is important to note that standards do not generally define how to fulfill requirements. They simply state the requirement. That is not to say lower level requirements will not be part of this or another standards. Standards may also set conditions for successful use or application.

ATTITUDE DATA MESSAGES
RECOMMENDED STANDARD (excerpts)
CCSDS 504.0-B-1
May 2008

...

STATEMENT OF INTENT

The Consultative Committee for Space Data Systems (CCSDS) is an organization officially established by the management of its members. *The Committee meets periodically to address data systems problems that are common to all participants, and to formulate sound technical solutions to these problems* why the standard was developed. Inasmuch as participation in the CCSDS is completely voluntary, the results of Committee actions are termed **Recommended Standards** and are not considered binding on any Agency. This **Recommended Standard** is issued by, and represents the *consensus* not a simple democratic process of, the CCSDS members. Endorsement of this **Recommendation** is *entirely voluntary* use is voluntary. Endorsement, however, indicates the following understandings:

Whenever a member establishes a CCSDS-related **standard**, this **standard** will be in accord with the relevant **Recommended Standard**. Establishing such a **standard** does not preclude other provisions which a member may develop.

- Whenever a member establishes a CCSDS-related **standard**, that member will provide other CCSDS members with the following information:
  - The **standard** itself.
  - The anticipated date of initial operational capability.
  - The anticipated duration of operational service.
- Specific service arrangements shall be made via memoranda of agreement. Neither this **Recommended Standard** nor any ensuing **standard** is a substitute for a memorandum of agreement

No later than five years from its date of issuance, this **Recommended Standard** will be reviewed by the CCSDS to determine whether it should: (1) remain in effect without change; (2) be changed to reflect the impact of new technologies, new requirements, or new directions; or (3) be retired or canceled. In those instances when a new version of a **Recommended Standard** is issued, existing CCSDS-related member standards and implementations are not negated or deemed to be non-CCSDS compatible. It is the responsibility of each member to determine when such standards or implementations are to be modified. Each member is, however, strongly encouraged to direct planning for its new standards and implementations towards the later version of the Recommended Standard.

See website for complete standard.

3 ATTITUDE PARAMETER MESSAGE (APM)

3.1 OVERVIEW

3.1.1 Attitude information may be exchanged between two participants by sending the attitude state (see reference [E4]) for a specified epoch using an Attitude Parameter Message (APM). The message recipient must have an attitude propagator available that is able to propagate the APM state to compute the estimated attitude at other desired epochs. For this propagation, additional ancillary information (spacecraft properties such as inertia matrix, torque vectors, and maneuver planning data, if applicable) shall be included with the message.

...

3.2.2 APM HEADER

3.2.2.1 Table 3-1 specifies for each header item:

a) The keyword to be used;
b) A short description of the item;
c) Examples of allowed values; and
d) Whether the item is obligatory or optional.

3.2.2.2 Only those keywords shown in Table 3-1 shall be used in an APM header.

Keyword	Description	Examples	Obligatory
CCSDS_APM_VERS	Format version in the form of 'x.y', where 'y' is incremented for corrections and minor changes, and 'x' is incremented for major changes.	1	Yes
COMMENT	Comments (allowed at the beginning of the APM Header after the APM version number). Each comment line shall	This is a comment	No

...

3.3 APM EXAMPLES

Figures 3-1 through 3-8 are examples of Attitude Parameter Messages [Note: only one is included here].

CCSDS_APM_VERS	= 1.0
CREATION_DATE	= 2003-09-30T19:23:57
ORIGINATOR	= GSFC
COMMENT	GEOCENTRIC, CARTESIAN, EARTH FIXED
COMMENT	OBJECT_ID: 1997-068A
COMMENT $ITIM	= 1999 DEC 18 $ original launch time
OBJECT_NAME	= TERRA
OBJECT_ID	= 1997-068A
CENTER_NAME	= EARTH

...

A Final Note: If It's Broke—Fix It!

All open standards organizations are, in fact, open for professional participants. Since poorly written standards or standards requirements affect the entire industry it is important to fix what needs to be fixed. Additionally, standards developers generally want to share information in their specialty area and they make great collaborators on concepts for future (standards) work.

6.7 Are Requirements Needed?— Capability-Based vs. Requirements-Based Systems

A well known and long-standing problem in US space system acquisitions is that final program costs tend to be much higher than initial cost estimates. Testimony by the US Government Accountability Office (GAO) before the Senate in March 2010 described how esti-

mated total costs for some recent space system acquisitions were two to three times greater than initial estimates (Fig. 6-5). Most of these very expensive systems have been or will be delivered late (Fig. 6-6) and some of them will be delivered with diminished capability. While there can be a variety of reasons for large cost growth, late deliveries and reduced technical performance, recent GAO assessments, National Research Council studies and reports by US Congressional committees have all noted that problems with how system requirements are actually defined and managed are among the root causes of poor program execution. Recommended improvements to the requirements definition process included implementing "requirements discipline…to prioritize stakeholder needs and consider the impact of programmatic changes on cost and schedule…understand impact of any change to program requirements…deny requests to change program requirements if their request would unacceptably impact cost, schedule, or system performance." Other recommendations included "…research and define requirements before programs are started and limit changes after they are started…use systems engineering to close gaps between resources and requirements before launching the development process." In 2008, one Congressional committee, the House Permanent Select Committee on Intelligence (HPSCI) recommended migrating away from a requirements-based acquisition approach towards a capabilities-based strategy to improve the US national space strategy. Are requirements the problem? If so, are requirements even needed? What is a capabilities-based system?

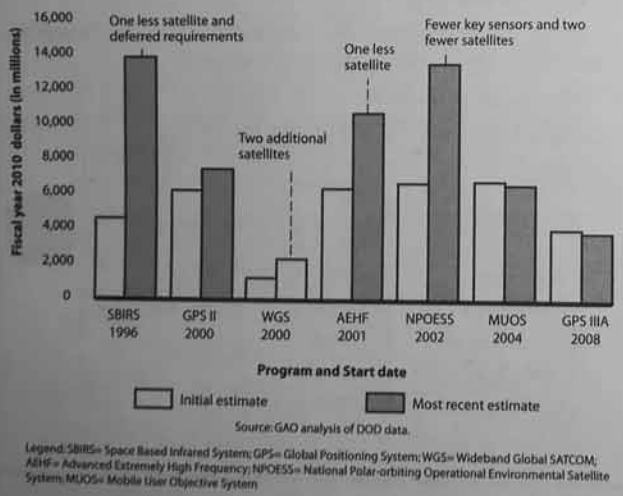


Fig. 6-5. Differences in Total Program Costs from Program Start and Most Recent Estimates (Fiscal Year 2009). (US Government Accounting Office GAO-10-447T)

The US became the world's preeminent space power by designing and building systems based on mission requirements driven by perceived importance and value of that mission much more than cost. For the most part, startling cost growth and program delays didn't matter to

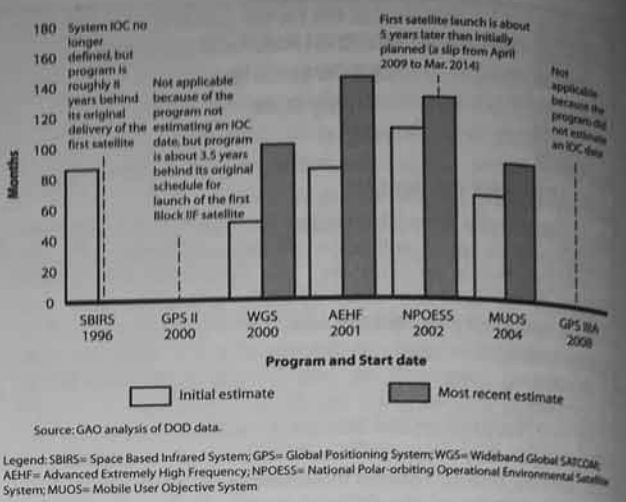


Fig. 6-6. Differences in Total Number of Months to IOC from Program Start and Most Recent Estimates. (US Government Accounting Office GAO-10-447T)

a wealthy nation focused on mission success in a highly competitive and threatening Cold War environment. The mission had overarching importance and the US dazzled the World with the success of Apollo, Corona, Viking and ultimately the Hubble Space Telescope. The NRO's Corona reconnaissance satellite program (Fig. 6-7) required 14 launches to achieve its first mission success back in the late 1950s and 1960s. Would a more cost-constrained and risk-adverse nation be so patient and focused on mission success?

Over time, in an era of American hegemony, a tendency developed to underestimate complexity of mission hardware and software and make optimistic assumptions about how simple it would be to develop and build new mission systems. In many cases, the relationship between mission requirements and capability was not completely understood—especially in early program phases—and the requirements definition process described here, which is based on detailed understanding of relationships between requirements, capability and cost was not followed. In a competitive environment, through a Darwinian process, bidders were motivated and encouraged to argue that mission goals could be achieved as cheaply as possible to maximize return for a given assumed budget. As a result, the relationship between mission objectives and cost in proposals tended to be poorly understood by proposer and reviewer alike, and initial cost estimates were overly optimistic and unrealistic to the point that it became impossible to implement missions within initial budgets.

In a sense, a capabilities-based process responds to this increasingly unacceptable circumstance of unbounded cost based on poorly understood mission-cost relationships by standing the requirements-based process on its head and asking the simple question of what mission requirements can we accomplish with hardware and software systems that we've got already? In this approach, we don't derive system performance

Table 6-6◀, Fig. 6-6, Eq. 6-2◀

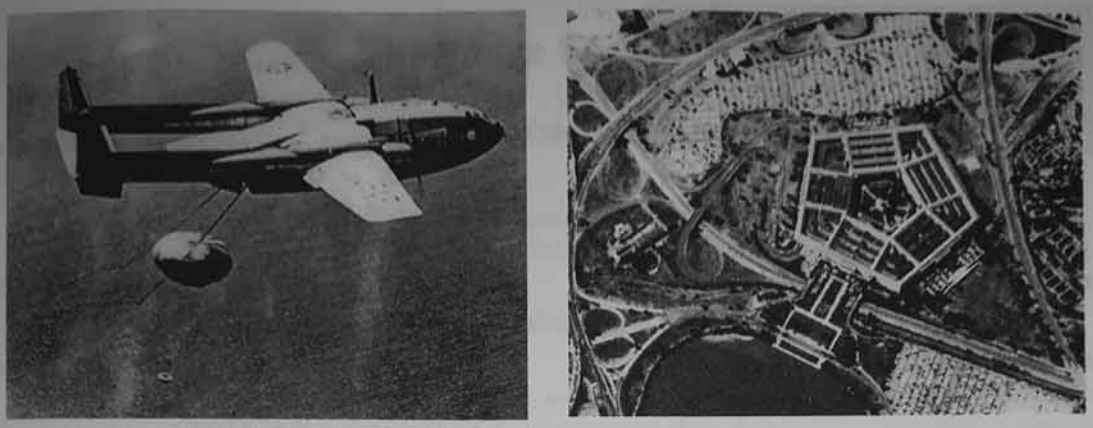


Fig. 6-7. The NRO's Corona Program. This program, declassified in 1995, involved satellites that returned film canisters to Earth in capsules, called "buckets," which were recovered in mid-air by specially equipped aircraft during their parachute descent. After 13 consecutive failures, the program enjoyed a relatively high success rate with 102 out of 144 Corona satellites returning usable imagery. (Courtesy of National Reconnaissance Office)

requirements from mission requirements. Instead, we take existing systems and either use them in new ways or piece them together into new systems and see what we can get for worthwhile missions. In that sense, capabilities-based systems take advantage of existing infrastructure to find new mission value, while delivering systems faster and at lower cost. We still need specifications to describe the system, but the mission requirements are derived from existing capability.

An example of a capabilities-based mission is the measurement of ionospheric and atmospheric physical characteristics like temperature and water vapor distributions made by looking at GPS satellites through the Earth's atmosphere with receivers onboard LEO satellites. This so-called GPS occultation technique has been implemented in a number of international systems like COSMIC (Constellation Observing System for Meteorology, Ionosphere and Climate), illustrated in Fig. 6-8, a joint US-Taiwan mission.

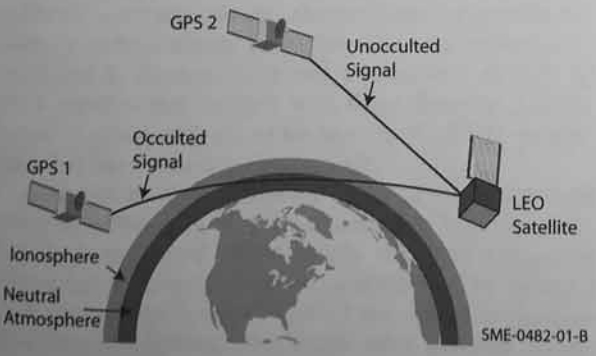


Fig. 6-8. COSMIC Measures Physical Properties of the Atmosphere and Ionosphere through precise measurement of phase delay and amplitude of GPS signals viewed through the atmosphere.

An obvious flaw in the capabilities-driven process is where do the existing capabilities come from if someone does not take bold steps to reach for and develop new technology that can be built into these lower cost systems? Who would build a GPS constellation to enable a COSMIC? Existing capabilities can come from at least two sources. The first is major system developments that are in fact driven by urgent mission needs. In some cases, hardware developed as part of these programs can be built into other systems and in other cases the systems themselves can be used for other purposes, as with GPS occultation. Another approach, possibly with more applicability, is to develop specific hardware and software in dedicated research and development programs that are embedded into broad national strategy roadmaps. Following "acquisition reform" in the 1990s, technology development was folded into operational system development to "save money." Unfortunately, like so many well-intended government initiatives to save money, this move had the exact opposite effect with technology being developed during operational system programs, even after Critical Design Review (CDR), with standing armies of engineers, accountants and program managers waiting for critical technology to be finished, which increased cost and schedule and decreased customer confidence.

Clearly, at least part of the right answer is to return a model that served us well for decades, which involves developing technology in agile research programs that are executed as part of a broader national strategy. In that case, proven, newer technology gets implemented at low risk and with much more predicable cost and schedule into operational systems. Despite the relative comfort and appeal of this approach, in some cases, it's just not good enough. The nation will always have urgent needs that demand technical solutions that go beyond demonstrated hardware and software, but not beyond laws of physics, engineering and production and so we will always have an engine for change that will lead to new capabilities. Even in those cases, though, key enabling technologies need to be developed and proven first before final authority to proceed to develop expensive operational systems makes sense.

Table 6-6◀, Fig. 6-8, Eq. 6-2◀

Applicable Standards:
AIAA S-102-2010: Performance-Based Reliability Modeling Requirements
See website for reference.

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7 The Space Environment

W. Kent Tobiska, *Space Environment Technologies**

7.1 **The Space Environment and Space Weather**
Solar Output; The Upper Atmosphere
7.2 **The Earth's Magnetic Field**
7.3 **Radiation Belts**—***W. Kent Tobiska, Nicola Sarzi-Amade***
7.4 **Microgravity**—***James R. Wertz***
7.5 **Orbital Debris—A Space Hazard**—***David B. Spencer, Ronald A. Madler***
Orbital Debris Environment; Implications of Orbital Debris for Orbit and Spacecraft Design

* With contributions from D.J. Gorney, J.B. Blake, H.C. Koons, M. Schulz, *The Aerospace Corporation*
A.L. Vampola, R.L. Walterscheid, *The Aerospace Corporation*

7.1 The Space Environment and Space Weather

The *space environment* is ambiguously defined as the region beyond the Earth's atmosphere (usually the troposphere) where non-terrestrial energy sources begin to dominate and spacecraft operate. The near-Earth space environment includes the Earth's upper atmosphere, ionosphere, radiation belts, and magnetosphere. The deep space environment extends beyond to include interplanetary space (the *heliosphere*), other planetary and small body regions, and even galactic environments that affect our Solar System.

Space weather is the shorter-term variable impact of the Sun's photons, solar wind particles, and interplanetary magnetic field upon the Earth's environment that can adversely affect space systems. It includes, for example, the effects of solar coronal mass ejections, solar flares, solar and galactic energetic particles, as well as the solar wind. All of these affect Earth's magnetospheric particles and fields, geomagnetic and electrodynamical conditions, radiation belts, aurorae, ionosphere, and the neutral (i.e., non-ionized) thermosphere and mesosphere as well as space systems that interact with these domains. Disturbances to these regions cause unwanted effects on systems that have sensitive components.

Active or passive mitigation of space weather effects can most effectively be done when space weather information is produced automatically and then integrated into space systems and space operations. The systems most susceptible to space weather are related to communications, navigation, low Earth orbit satellite operations, and geosynchronous satellite operations.

Operational space weather is, in essence, a supply chain. It is a system of organizations, people, technology, activities, information and resources that move space weather products or services from suppliers to customers. The raw material of space environment data is processed by models into information and then refined into specialized products for delivery to end customers. Space weather's greatest value is when it is an integrated information layer within broader systems [Tobiska and Bouwer, 2010; Tobiska, 2009].

The near-Earth space and atmospheric environments strongly influence the performance and lifetime of operational space systems by affecting their size, weight, complexity, and cost. Some environmental interactions also limit the technical potential of these systems. They can lead to costly malfunctions or even the loss of components or subsystems [Tribble, 1995; Hastings and Garrett, 1996; DeWitt et al., 1993].

Operating in space can pose significant problems for many spacecraft systems. For example, in vacuum, most organic materials will *outgas*—i.e., generate spurious molecules that may act as contaminants to other surfaces. Even before reaching orbit, particles from the atmosphere may fall onto optical surfaces and degrade the performance of electro-optical instrumentation. Because there is no practical way to clean spacecraft surfaces once the vehicle reaches orbit, maintaining effective contamination control during design and development is a significant issue for most spacecraft [Tribble et al., 1996]. Ultraviolet (UV) degradation of materials, surface charging, deep dielectric charging (see Sec. 7.1.1), communication interference, and unexpected perturbations to orbits and attitudes are other types of effects that can be attributed to space weather.

Above 100 km, a spacecraft or rocket enters the very tenuous upper atmosphere called the thermosphere. In LEO (low-Earth orbit) orbits below 500 km, a spacecraft encounters the constituent species of the atmosphere, i.e., molecular nitrogen and oxygen, and atomic oxygen, hydrogen, and helium, at orbital velocities on the order of 8 km/s. Interactions between the satellite and the neutral atmosphere primarily affects the spacecraft's orbit by causing drag on the surfaces that has a cumulative

effect on the orbital energy and orbital lifetime (See Sec. 9.4.4.). Additional effects include erosion of satellite surfaces due to adsorption of atomic oxygen resulting in thermal, electrical, and possibly structural surface property effects.

7.1.1 Solar Output

Energy is transferred to the Earth from the Sun via three main forms: electromagnetic radiation (photons), charged particles, and magnetic, electric, and gravitational fields. The source of this energy comes from the solar interior where nuclear fusion produces photons that eventually find their way from the core through the *Radiative zone* over the course of a million years. At the Radiative zone–Convection zone boundary (the *Tachocline*), the plasma cools enough to condense into highly ionized atoms and here the solar magnetic field is born.

Magnetic flux, charged particles (ions and electrons), and photons continue their way through the Convection zone to the surface, which is the coolest part of the Sun at ~5770K. Magnetic flux features erupt through the surface in the form of *sunspots* and sunspot groups. Above the surface, the *solar atmosphere* consisting of the *chromosphere*, *transition region*, and *corona* feel the effects of emergent magnetic flux episodes in the form of active regions, plage, flares, and coronal mass ejections (CMEs)†. In these features, solar atmospheric ions and electrons are accelerated by magnetic shearing and eruptions; in these processes, solar X-ray, extreme ultraviolet (EUV), and UV photons [ISO, 2007] are dynamically created by the plasma's ionized states and the ions' relaxation to lower energy states. The solar magnetic field reverses polarity every ~11 years as part of a 22 year solar magnetic cycle, observed since 1610 as the sunspot cycle. The intensities of the X-ray, EUV, and UV irradiances that are the most Earth-effective energies are associated with the sunspot cycle. The solar 10.7 cm radio flux, called the F10.7 *proxy*,‡ has variability that occurs in sync with energy processes in the solar atmosphere and has been used as an indicator for the solar EUV irradiances. Figure 7-1 shows the solar cycle 23–24 variation of the F10.7 proxy. Cycle 24 started in January, 2008.

The integration of spatial radiances from all points on the visible solar disk become the spectral solar irradiances that make their way to Earth in 8 minutes, causing thermospheric density increases, ionospheric enhancements, and spacecraft surface charging (photoelectric effect). The fast solar wind (500–800 km/s) originates in solar coronal holes* and overtakes the slow solar wind (200–500 km/s) that emanates from the active corona. The electrons and protons in the solar wind are bound on trajectories defined by the spiral-like structure of the Sun's magnetic field that extends throughout the heliosphere. They arrive at Earth within 2–3 days. Occasionally, extremely fast solar wind particles from CMEs (800–2,000 km/s), including Solar Energetic Protons (SEPs), blast through the solar wind and can arrive at Earth as early as ½ hour to a few hours after the eruptive event.

At Earth the fast solar wind from the coronal holes, also called *Co-rotating Interactive Regions* (CIRs), can cause minor ionospheric changes. The extremely energetic particles from CMEs can cause major and minor disturbances to the Earth's magnetic field, called *geomagnetic storms* and *substorms* (See Sec. 7.2.), by coupling with, and distorting, the external Earth magnetic field called the *magnetosphere*. On spacecraft, the

† **Plage:** Plage are bright regions in the Sun's middle atmosphere (chromosphere) usually near and above sunspots. These brightenings extend away from the bright chromosphere active regions above the sunspots and are the result of magnetic flux diffusing away from the active region into the quiet regions. They account for part of the EUV emission from the Sun seen at the Earth that causes thermospheric density increases and, thus, increases to satellite drag.
Flare: A solar flare is a large explosion in the Sun's atmosphere that affects all layers of the solar atmosphere (photosphere, chromosphere, and corona). It produces radiation across the electromagnetic spectrum at all wavelengths, from radio waves to gamma rays, although most of the energy goes into X-rays and EUV. Flares occur in active regions above sunspots and brighten along the strong magnetic field lines coming from the Sun's interior and extending into the corona. Flares occur on timescales of minutes to a few hours and their X-ray and EUV radiation affects Earth's ionosphere and thermosphere.
CME: A coronal mass ejection (CME) is a massive explosion in the solar atmosphere associated with a large flare where material (ions and electrons) from the Sun's atmosphere escape into space. CMEs inject a large quantity of energetic ions and electrons into the solar wind. This material, if it follows interplanetary field lines that connect the Sun to the Earth, can cause major disruptions to the Earth's magnetosphere, resulting in geomagnetic storms that cause intense heating of the Earth's polar atmosphere and affect satellite drag.

‡ The output of the Sun at a wavelength of 10.7 cm has often been called the F10.7 *index* or, more correctly, the F10.7 *proxy*, and is an indicator of solar activity. There is nothing magic about 10.7 cm, except that the solar output at this wavelength has been measured continuously since WWII and, therefore, is known to correlate to the Sun spot cycle and solar radio activity over an extended period. The reason that the F10.7 proxy is used for atmospheric drag calculations is that its variation generally replicates the actual heating process for the upper atmosphere, i.e., the absorption of solar EUV photons by oxygen and nitrogen. The more photons absorbed, the more active are the molecules and lower layers expand to higher altitudes, creating the effect of greater densities at higher altitudes. F10.7 has no direct relationship to this process but varies day to day in a manner similar to the solar photons that do affect the atmosphere densities and satellite drag [ISO, 2007].

* Coronal holes are regions where the corona is dark. These features were discovered when x-ray telescopes were first flown above the Earth's atmosphere to reveal the structure of the corona across the solar disc. Coronal holes are associated with open magnetic field lines and are often found at the Sun's poles.

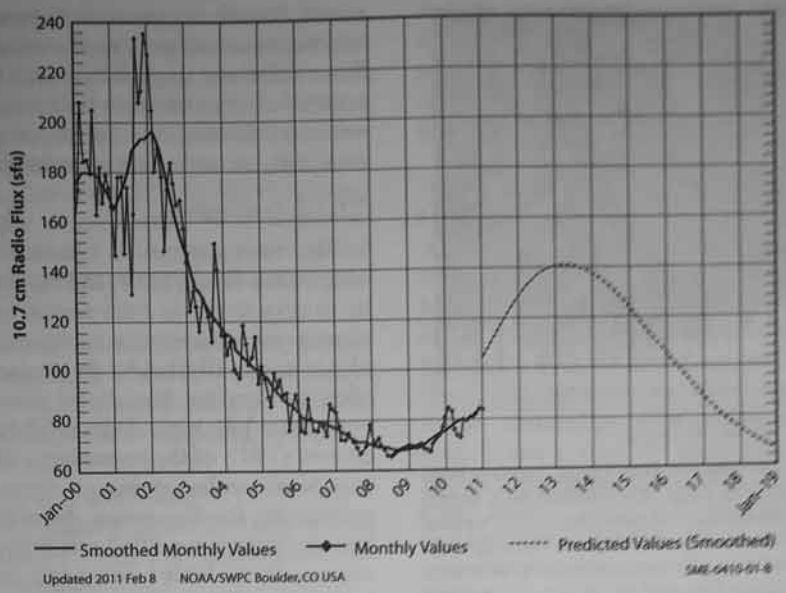


Fig. 7-1. F10.7 Solar Cycle Number 24 Progression from ISES and NOAA SWPC. Shown is the mean F10.7 daily flux in units of sfu = 10^{-22} W m^{-2} Hz^{-1}. (See also Fig. 9-16 in Sec. 9.4.4.)

charged particles in the storms and substorms can cause severe surface charging at geosynchronous orbit as well as deep dielectric charging/discharging. Deep dielectric discharging occurs when electrons of 2–10 MeV penetrate deep into satellite surfaces. Excess charge spreads out evenly on conducting surfaces but results in uneven potential distribution on dielectric surfaces. Potential differences can reach a threshold at which point discharge occurs. Deep dielectric charging occurs inside the surfaces of satellites and discharges can occur to circuits inside the spacecraft. In the outer Van Allen Radiation Belt these events inject increased populations of particles. The Earth's thermosphere experiences density enhancements that heat the upper atmosphere. The ionosphere sustains increased densities from electron precipitation into the polar region and ion drag through the neutral atmosphere (*Joule heating*), thereby affecting radio communications, radar signals, and GPS navigation.

From outside the Solar System, galactic cosmic rays (GCRs) (see Sec. 7.3) from supernovae and black holes reach the Earth, penetrate to the lower atmosphere, and impact nitrogen or oxygen, creating a spray of particles, including neutrons, that then scatter outside the atmosphere and decay to protons and electrons in the inner Van Allen Radiation Belt. The free neutron is in fact radioactive—within about 10 minutes it breaks up into a proton, which captures most of the energy, an electron, and a massless neutrino.

7.1.2 The Upper Atmosphere (Thermosphere and Ionosphere)

7.1.2.1 Thermosphere

The thermosphere (85–500 km altitude range) is the neutral atmosphere in Near-Earth space. EUV irradiance penetrates to different levels of the thermosphere depending upon the energy of photons of a given wavelength, the absorption cross sections of atoms and molecules, and the attenuation of the flux by the thermospheric density. Heating, and subsequent density changes, occurs when an EUV photon is absorbed by a particle. This is called *photoabsorption*. The route to thermal energy deposition proceeds through two major paths: 1) *photoionization* of molecules or atoms, where an electron is stripped from the particle and carries away most of the excess energy as kinetic energy, followed by ion-neutral and photoelectron reactions which convert stored chemical energy into kinetic, or thermal energy, or 2) *photodissociation* of a molecule, where the chemical bond between atoms is broken, with the initial photon energy being transferred into particle kinetic energy. The upper thermosphere is a heat source and is nearly isothermal. By either of these paths, heat produced at high thermospheric altitudes, i.e., above 200 km, is transported down to cooler atmosphere regions. Because the solar X-ray, EUV, and UV irradiances vary with flare (hours), solar rotation (27 days), active region evolution (3–6 months) and solar cycle (10–12 year) changes, this affects the densities of the thermosphere and ionosphere on the same time scales. Figure 7-2 shows the altitude-structured daily neutral density variability in the thermosphere with the decline of solar cycle 23, the episodic solar active region evolution, and the 27-day solar rotation.

In addition to solar X-ray, EUV, and far UV (FUV) heating, the Earth's neutral upper atmosphere absorbs energy in the polar regions as a result of electron precipitation along magnetic field lines and ion drag (*Joule heating*). Terrestrial dynamical processes (winds, waves, and tides) also play a role in transferring energy from region

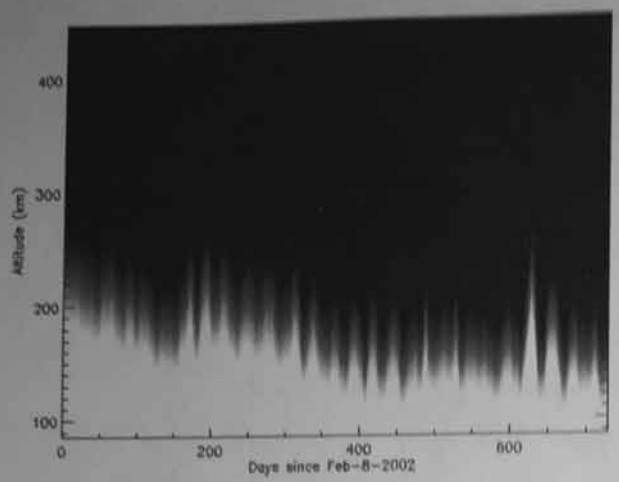


Fig. 7-2. Thermosphere Density for Two Years Starting February 8, 2002 is Modeled with daily SOLAR2000 EUV Irradiances in the 1DTD (1-dimensional, time dependent) Physics Model from Space Environment Technologies. The grayscale gradient indicates exponentially decreasing sub-solar daily densities (light to dark) with altitude from 100 to 450 km. The secular decreasing level of density at every altitude is from the decline of solar cycle 23, the sharp spikes are the density enhancements due to 27-day solar rotation, and the broader periodic density enhancements (several months) comes from solar active region evolution and decay.

to region, resulting in density enhancements or depletions. The main effect of thermospheric density changes is increased or decreased drag on a spacecraft (See Sec. 9.4.4.), attitude perturbations, and atomic oxygen erosion.

Atomic oxygen. Atomic oxygen is the predominant atmospheric constituent from 200 km to 600 km and has important effects upon space systems. Atomic oxygen can react with thin organic films, advanced composites, and metallized surfaces [Visentine, 1988], resulting in degraded sensor performance. For example, Kapton, a material commonly used for insulation and seals, erodes at a rate of approximately 2.8 μm for every 10^{24} atoms m^{-2} of atomic oxygen fluence [Leger et al. 1984]. *Fluence* is the total number of particles that has crossed a unit area. It is related to the *number density* (i.e., the number of particles per unit volume) of atomic oxygen and the satellite velocity.

Atomic oxygen forms when solar far ultraviolet radiation dissociates molecular oxygen. Above 110 km, atmospheric constituents reach a density distribution based on their influence by gravity; each constituent's density varies with altitude according to its *scale height*. The scale height of a constituent is the height change over which the density drops to e^{-1} of its value. In diffusive equilibrium, the scale height is inversely proportional to its molecular weight. Thus, the density of light constituents decreases less rapidly with altitude than the density of heavy constituents, and eventually the light constituents dominate the mixture of gases. Atomic oxygen is lighter than the molecular nitrogen and oxygen; therefore, near 200 km altitude, atomic oxygen becomes the most abundant constituent. The large solar-cycle variation in atomic oxygen means spacecraft materials can be selected based on phasing the mission to the solar cycle. Since there are large differences between solar cycles, material choices made on the basis of average solar maximum conditions may be inappropriate because of the possibility of more extreme solar maximum conditions.

7.1.2.2 Ionosphere

The thermosphere is coupled with the ionosphere where solar X-ray, EUV, and FUV photons have the ability to strip electrons from neutral species. The result is a plasma (*ionosphere*) that is created on the dayside of the planet, and is affected by the magnetic-electric field coupling to form the *Equatorial Anomaly*, and that disappears during the night. Figure 7-3 shows the total electron content (TEC) of the ionosphere. The two bands of electron density enhancement on either side of the magnetic equator are the *Equatorial Anomaly*. Just as the neutral thermosphere is modified primarily by the solar photons, secondarily by Joule heating and charged particle precipitation, and thirdly by winds, waves, and tides, so the same relative importance affects ionospheric variability. Ionosphere and neutral atmosphere coexist together but the densities of ionosphere species are about 6 orders of magnitude less dense than the neutral species. Ionosphere particles (constituents) are formed from the neutral when solar EUV and X-ray photons hit them and knock off electrons—those electrons then become the population of particles in the ionosphere but, when the Sun goes behind the planet, the electrons reattach to the ions, creating neutrals, and collapse the ionosphere.

The main features of the ionosphere include Chapman profiles for each of the ion species that are dominant in different regions. A Chapman profile (or Chapman layer) is the distribution of ionization as a function of height produced solely by absorption of solar radiation. It is named after Sidney Chapman, one of the principal scientists who made early investigations of the ionosphere. In 1931 he derived the simple case of a single gas with an

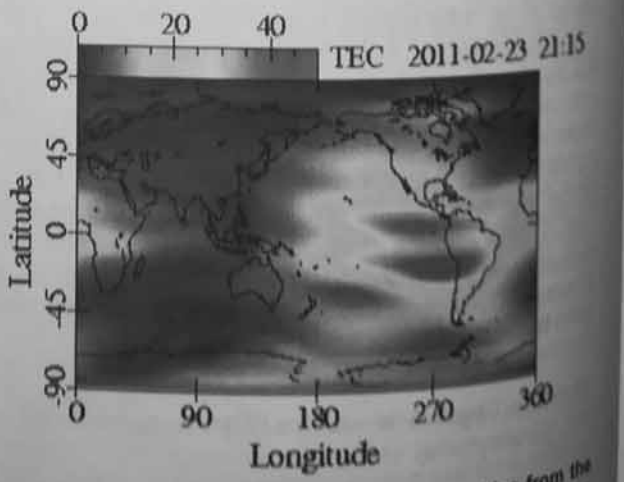


Fig. 7-3. The Ionosphere Total Electron Densities from the USU Space Weather Center GAIM System. Greatest electron densities occur during the dayside and the two bands on either side of the magnetic equator are called the Equatorial Anomaly.

absorption coefficient that is constant for all wavelengths. The Chapman layer theory assumes a single species gas with exponential distribution (therefore with constant scale height) and monochromatic radiation from the Sun for ion production rate, and that there is an equilibrium between the creation of free electrons and their loss by recombination [Goodman, 2005]. These assumptions represent a major simplification with respect to reality; however the Chapman theory is a fundamental baseline for ionospheric profile modeling. The regions of the ionosphere are:

- D-region: 50–85 km, dominated by H_2O^+, formed by solar wavelengths, λ, where $\lambda < 1$ nm (X-ray) or Lyman-α ($\lambda = 121.6$ nm); the main neutral constituent (or species) in the overlapping mesophere is N_2 (78%) followed by O_2 (21%)
- E-region: 85–140 km, dominated by O_2^+, $1 < \lambda < 10$ nm (XUV) or Lyman-β ($\lambda = 102.6$ nm) or CIII ($\lambda = 97.7$ nm); the E-F1 valley is the cross-over between the two regions
- F1-region: 140–250 km, is dominated by NO^+, $10 < \lambda < 91$ nm (EUV), F1 maximum electron density at ~200 km
- F2-region: 250–500 km, dominated by O^+, $10 < \lambda < 91$ nm (EUV), F2 maximum electron density at ~300 km
- Topside ionosphere: 500–1,500 km, dominated by the escape of H^+ and, occasionally, O^+ during periods of intense thermospheric heating due to major flares and geomagnetic storms. The principal neutral species in this exosphere altitude range is H and sometimes O, again depending on thermospheric heating due to flares and CMEs
- The plasmasphere: extending to approximately geosynchronous orbit (~36,000 km) with a falloff of cold (few eV) protons and electrons to interplanetary levels

These charged particles, called the plasma environment, can cause differential charging of satellite components both on the surface and interior of the vehicle. If severe, this charging can exceed breakdown electric fields and the resulting electrostatic discharges may be large enough to disrupt electronic components. More energetic space radiation, such as electrons with energies from about 200 keV to 1.5 MeV, can become embedded in dielectric components and produce electrostatic discharges in cable insulation and circuit boards. This *bulk charging* may disrupt a subsystem's signals or the operation of its devices. Even if mild, the charging may alter the electrical potential of the spacecraft relative to space and affect the operation of scientific instrumentation.

7.2 The Earth's Magnetic Field

The Earth's magnetic field is comprised of the main field created by the iron core dynamo within the planet and the magnetosphere components. The latter include the *ring current*, the *cross-tail current*, and the *magnetopause surface current*. Together, main field and magnetosphere comprise the magnetic system that can affect ground- and space-based systems.

The Earth's magnetic field is roughly dipolar and the combination of Maxwell's Equations, Ohm's Law, the Equation of continuity, and the hydrodynamic equation are used to describe its general character. The main magnetic field can be modeled by the World Magnetic Model (WMM) or the International Geomagnetic Reference Field (IGRF). Both models represent the field from the Earth's core out into space, and both are revised every 5 years. In particular, the IGRF is revised by the International Association of Geomagnetism and Aeronomy (IAGA) and is in its 10th generation (1900.0–2010.0). The IGRF, which is part of in ISO Earth Main Field Standard (ISO 16695) along with the WMM, has a regularly updated website [IGRF, 2011]. An example of the Earth main field is shown in Fig. 7-4. Here, the WMM total intensity at the 2000.0 epoch is shown with contour intervals in 2000 nT and a Mercator projection. The variation in field strength ranges from 24,000–66,000 nT (0.24–0.66 Gauss).



As the dipole field extends beyond the Earth's surface, the magnetic field lines are subjected to perturbations by the plasma of the solar wind. The interaction between the solar wind and the Earth's magnetic field causes the magnetic field on the night side of the Earth to stretch into a very elongated structure known as the magnetotail. Figure 7-5 shows an example of the Earth's magnetosphere system. A thin plasma sheet bifurcates the *magnetotail*, which is the portion of the magnetosphere opposite to the Sun behind the Earth and which extends over 1,000 Earth radii parallel to the velocity flow of the solar wind.

Through the interaction between the solar wind and the Earth's magnetic field, some of the solar wind's kinetic energy is converted to magnetic energy stored in the magnetotail. *Geomagnetic storms* are large disturbances in the near-Earth environment (especially in the geomagnetic field), which are caused by solar wind and interplanetary field structures that originate from solar disturbances such as coronal mass ejections. In these storms there is a strong increase of energetic ions (tens to hundreds of keV) in the ring current region just inside geosynchronous orbit. The *Disturbance Storm Time (Dst) index* is an indicator of the strength of the magnetospheric ring current as it is modulated by the solar wind, and as it perturbs the Earth's main magnetic field. There are also occasionally intense fluxes of relativistic (MeV) electrons that are injected into the outer Van Allen radiation belt. This perturbation on the Earth main field is registered as a decrease in the main field strength and Dst is reported in nT and with the range of +100 to –400. A geomagnetic storm is occurring when Dst reaches –75 and major storms can reach values as low as –300 nT or lower. Because this ground-based Dst measurement from 4 mid-latitude magnetic observatories is an excellent

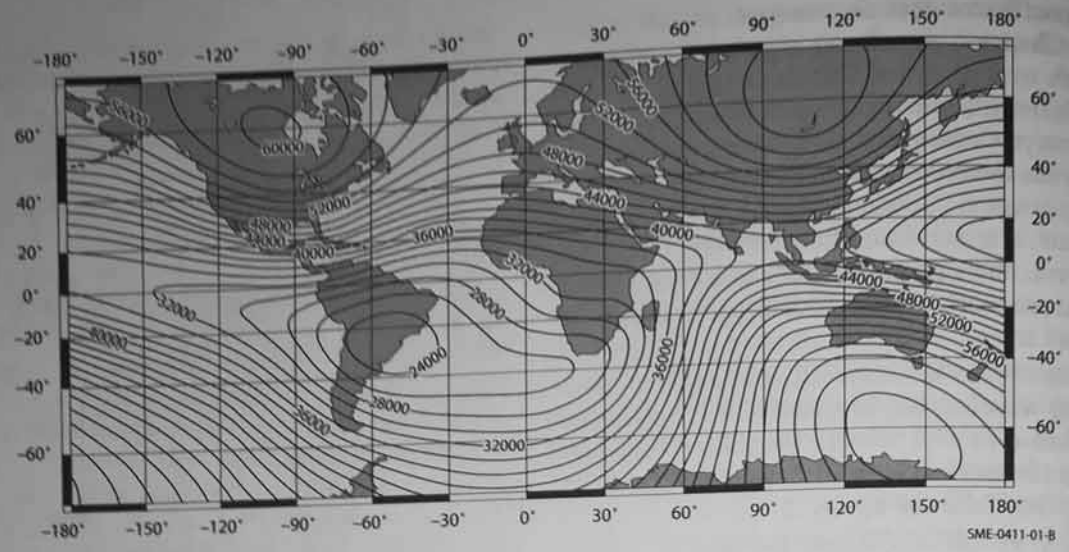


Fig. 7-4. The WMM2000 Earth Main Field Model is Shown with 2000 nT Contour Intervals.

indicator of solar wind influence on the magnetosphere, and hence of the energy from Poynting (electromagnetic) flux and (kinetic) particle precipitation in the polar regions, it provides a very good, time-resolved surrogate for the high latitude heating in the upper atmosphere that affects satellite drag.

Satellite drag calculation is moving toward the ISO and COSPAR methods (NRLMSIS for constituents and JB2008 for mass densities and drag calculations). Dst is used in JB2008 for increasing the atmospheric density (and drag) during geomagnetic storms. The older F10.7 is still useful but less accurate. F10.7 will continue to be used and as the new models become more available engineers can use them. Historically, F10.7 has been used as an indicator, a proxy, of the variable drag on a LEO satellite. Figure 9-17 in Chap. 9 shows the density vs. altitude for various F10.7 values. If those variable densities are translated into LEO satellite lifetimes, Fig. 9-18 shows the altitude as a function of dates for 9 hypothetical satellites launched over a 6-year period. F10.7 and Dst are decoupled; engineers need both and have to use a computation application that both incorporates atmospheric density models and contains the proper methods.

The charged particle motions are affected by the magnetic field's intensities and particle bounce, drift, and *magnetic mirroring* can occur. Magnetic mirroring is the bouncing of a charged particle along the field lines back and forward between the northern and southern geomagnetic hemispheres. The interaction of these particles with the magnetosphere causes magnetospheric variability, which then manifests as geomagnetic variations measured by magnetic observatories at the Earth's surface. The planetary geomagnetic indices of Kp and Ap (where

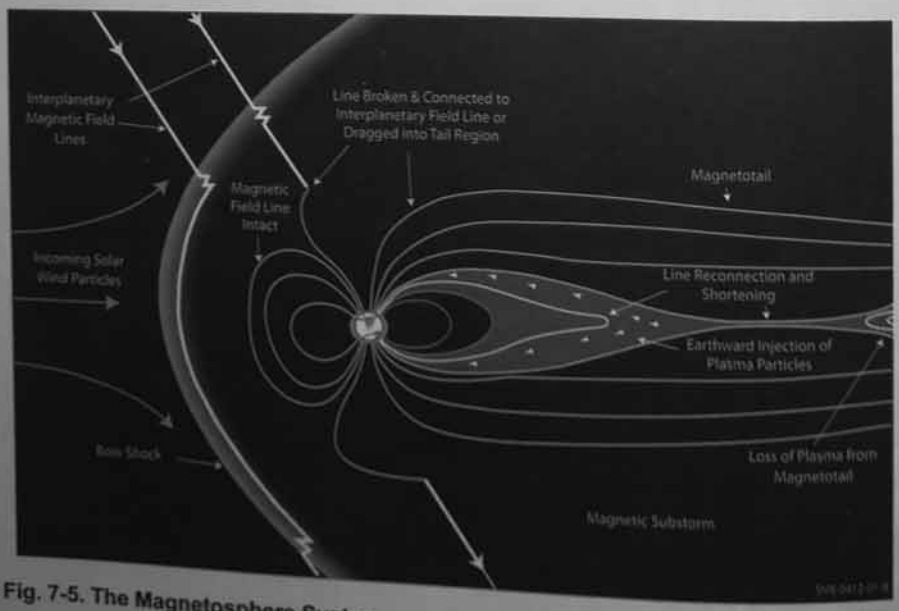


Fig. 7-5. The Magnetosphere System.

Table 7-0, Fig. 7-5, Eq. 7-0

Ap is the daily average of 8 a-indices and Ap is the 3 hour index of geomagnetic activity, equivalent to Kp but on a scale of 0–400 instead of 0–9), and the disturbance storm time (Dst) index for the ring current all provide information about how the solar wind perturbs the magnetosphere, and hence the Earth's main field. They are also proxies for the influx of charged particles along magnetic field lines into the polar regions, i.e., particle precipitation, that combine with high latitude current systems to cause heating and auroral phenomena in the upper atmosphere. Figure 7-6 shows the occurrence of periods of geomagnetic storms during several solar cycles, where Ap* indicates values of Ap ≥ 40, while Fig. 7-7 shows the USGS and Kyoto Dst indices in 2009 with minor storm conditions.

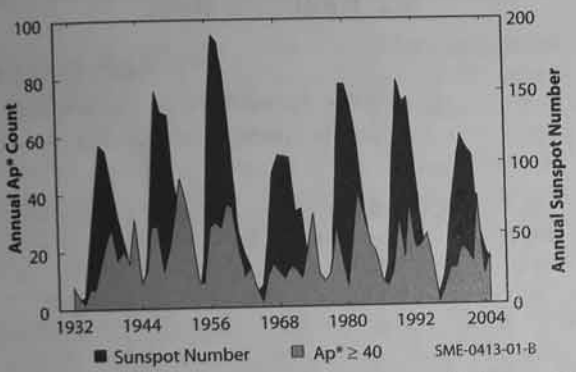


Fig. 7-6. Sunspot (black) vs Magnetic Storm (light gray) for Solar Cycles 17–23.

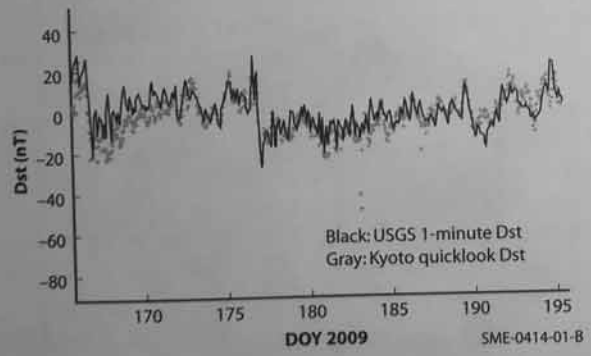


Fig. 7-7. USGS (black) and Kyoto (gray) Dst for Days 165–195 in 2009.

Because this energy cannot build up indefinitely, magnetic substorms dissipate it from time to time. *Magnetospheric substorms* are caused by the dynamic response of the magnetosphere to varying solar conditions. The energy input from the solar wind is governed by the orientation of the interplanetary field and, as long as the magnetospheric region remains stable, energy is stored as magnetic energy. At some critical point the magnetotail becomes unstable and the magnetic energy is released via the *substorm expansion phase*, which involves injection of energetic particles (electrons and ions—tens to hundreds of keV) into the vicinity of the geostationary orbit. These substorms are associated with strong electric currents in the auroral region, rapid fluctuations, and configuration changes of the magnetospheric magnetic field. These substorms produce an energetic plasma (5 to 50 keV) that is injected toward the Earth. This hot plasma can extend into geosynchronous orbits and has the ability to charge the surface of any spacecraft within it to high negative voltages.

The electrical potential of a spacecraft or one of its components is measured with respect to the plasma in which it is immersed where the net current flow is zero. That is, the following currents must cancel each other: (1) the electron current from the plasma to the vehicle, (2) the ion current from the plasma to the vehicle, (3) the secondary electron current, (4) the backscattered electron current, and (5) the photoelectron current. The voltage at the component's surface also depends on the material's capacitance with respect to the surrounding materials, especially the vehicle ground. Because materials have varying secondary emission coefficients and photoelectron currents, their equilibrium potentials also differ. An electrostatic arc occurs if the electric field exceeds the breakdown field along the surface of the material, through the material, or between adjacent materials.

Disturbed geomagnetic conditions near geosynchronous orbit can also lead to charge collection in dielectric (non-conducting) materials. This can result in either spacecraft surface or deep dielectric charging, both of which can also happen in LEO and MEO (medium-Earth orbit) environments. With charge buildup, *electrostatic discharge* (ESD) can occur as *Paschen discharge* and *arcing*† and the result can be a loss of system components or even catastrophic failure of a mission.

Electromagnetic interference (EMI) from such arcs can cause spacecraft to operate erratically. Surface charging detrimental to vehicle operation occurs mainly in orbits where electrons with energies of 10 to 20 keV dominate the electron current from the plasma to the vehicle. At low altitudes this charging occurs only at high latitudes where auroral electrons collide with the vehicle, which is passing through an otherwise cold, low-density plasma. For other low-altitude locations, low energy electrons usually develop enough current to keep electric fields below breakdown levels. In higher orbits, such as geosynchronous, surface charging occurs during magnetospheric substorms between the longitudes corresponding to midnight and dawn. Figure 7-8 shows an example of charging potential and deep dielectric discharge probability at GEO (geosynchronous-Earth orbit) using models built with the statistics of the SCATHA spacecraft. It approximates the

† ESD initiates when the potential between two parts of a spacecraft increases and a sparking potential is reached. Paschen discharge is the sparking potential between two electrodes (or oppositely charged surfaces) in a gas. The sparking potential is a product of the gas pressure and the distance between the electrodes. Arcing is ESD that occurs on a short time scale.

electron-flux distribution during a substorm by summing a cold and a hot Maxwellian distribution. The cold component can have a density of 0.2 cm^{-3} and a temperature of 0.4 keV; the hot component can have a density of about 2.3 cm^{-3} and a temperature of 25 keV. It is the hot component from the magnetosphere plasma sheet that can place additional charge on spacecraft surfaces in GEO, which can then result in surface charging events.



Fig. 7-8. The GEO Alert and Prediction System, GAPS [2011]. Surface Charging Potential and Deep Dielectric Discharge Probability for the GOES-10 Spacecraft on Dec. 1, 2008. The 3-hour Ap is correlated with the SCATHA measurements to provide estimates of the surface charging. The GOES 0.6 MeV electron densities are correlated with deep dielectric anomalies and provide an estimate of deep dielectric discharge probability.

The Fig. 7-8 plot shows the current epoch, real-time electron flux at the GOES satellite position, which is used as an indicator of the general level of electron flux for all GEO spacecraft. This north pole view of the GEO environment shows the local time along the GEO orbit. The inner ring is the –48 hour history of the potential, the solid white ring is the current epoch, and the outermost ring is the +72 hour forecast of the potential for spacecraft surfaces in GEO. Electron fluxes producing a potential of greater than 20 keV can result in increased surface charging events. Design considerations for spacecraft desire to keep the differential charging caused by this plasma well below breakdown potentials, or use materials that can tolerate the electrostatic discharges. GEO spacecraft design considerations for surface charging are detailed in NASA TP 2361 [Purvis et al., 1984].

Design guidelines are available to help reduce differential potentials on vehicle surfaces [Purvis et al., 1984]. For example, we can select candidate materials and conductive coatings, apply numerical or analytical models using their quantifiable characteristics, and determine their differential potentials in space. If one cannot prevent discharges by selecting alternative materials, it is useful to consider alternatives such as special filtering, cabling, or grounding. One can employ coupling models for electromagnetic interference simulation, and test the vehicle for electrostatic discharges in its flight configuration. It is important to note that while differential charging as discussed here is not seen in lower equatorial orbits, the spacecraft potential may be as much as 90% of the solar array voltage more negative than that of the surrounding plasma, depending on the configuration of the spacecraft electrical power supply. This may be a concern on scientific missions, where nonbiased measurements of the plasma environment are desired, and it may also give rise to arcing or other undesirable effects if high voltage power supplies are used [Tribble, 1995].

7.3 Radiation Belts

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The *Van Allen radiation belts* are a permanent hazard to spacecraft. They consist of electrons and ions (mostly protons) having energies greater than 30 keV and distributed non-uniformly within the magnetosphere. As shown in Fig. 7-9, the energetic electrons preferentially populate a pair of toroidal regions centered on the magnetic shells $L \sim 1.3$ (inner zone) and $L \sim 5$ (outer zone).

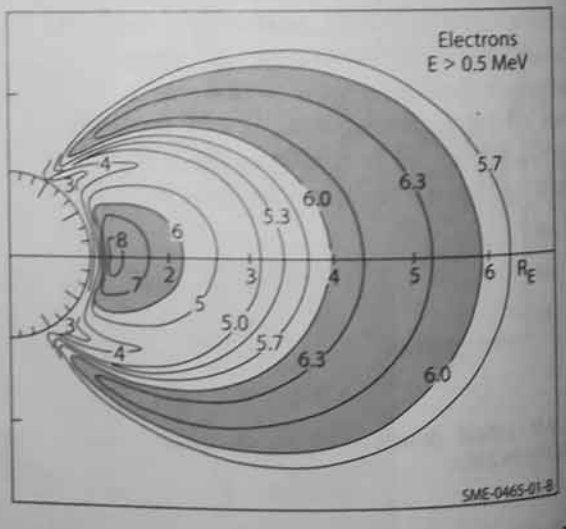


Fig. 7-9. Electron Belts of the Inner and Outer Zones. The numbers on the contours represent the $\log_{10}$ of the integral omnidirectional flux in units of particles $cm^{-2}s^{-1}$. The horizontal axis is the magnetic equator marked in units of Earth radii. Only electrons with energies above 0.5 MeV are included. Adapted from Vette et al., [1966] by Schultz and Lanserotti [1974].

A *magnetic L-shell* is the surface generated by rotating a magnetic field line around the dipole axis. It approximately satisfies the equation

$$R = L \cos^2 \lambda \qquad (7\text{-}1)$$

Table 7-0, Fig. 7-9, Eq. 7-1

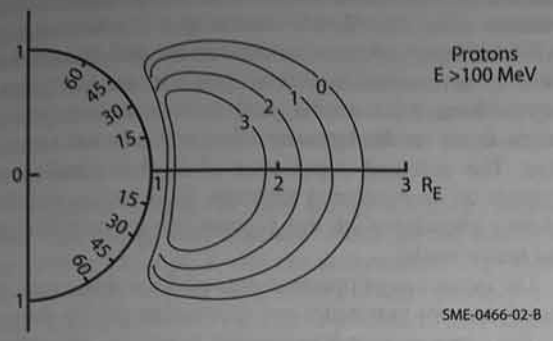


Fig. 7-10. High-Energy Protons in the Inner Zone. Axes and units are the same as on Fig. 7-9. (Note a somewhat different scale.) Data is the omnidirectional flux from the AP8MIN model. Only protons with energies above 100 MeV are considered.

where R is the distance in Earth radii from the idealized point dipole near the Earth's center, and λ is the magnetic latitude. Thus, the L value of a dipolar magnetic shell is its equatorial radius measured in Earth radii. A generalized concept of L, introduced by McIlwain [1961], takes account of the way higher harmonics of the main geomagnetic field perturb the motion of charged particles from their dipolar trajectories. Normally, one uses this concept instead of the dipole L value for mapping the trapped radiation environment. Standard models of the Van Allen belts are available from the National Space Science Data Center. The legacy model, which provides inner zone (L = 1.2–2.4) electron data is AE8, which has an average energy range of 40 keV to 5 MeV [Bilitza et al., 1988]. AE8 has two forms, AE8MIN and AE8MAX, which represent the time-averaged environments during solar minimum and maximum.

Outer-zone electron fluxes vary much more over time than inner-zone fluxes. Indeed, during a major magnetic storm, the equatorial intensity at a given energy and L value may grow by several orders of magnitude in less than a day. Between storms, such flux enhancements usually decay exponentially with an energy-dependent lifetime $\tau \sim 10$ E days at $L > 3$, where E is energy (MeV).

Available proton models are AP8MIN and AP8MAX, which also represent the solar minimum and maximum periods. At solar maximum, the increased atmospheric densities decrease the proton fluxes because the trapped protons collide with the atmosphere at low altitudes. The AP8 models cover the energy range from 100 keV to > 400 MeV. Figure 7-10 shows some of AP8MIN's typical contours for proton fluxes in (R, λ) space. Since the MIN model predicts slightly more flux than the MAX model, it can be used during solar maximum or as a conservative model for long-term missions.

New models for electrons (AE9) and protons (AP9) are in beta-testing with an anticipated release in 2011. Using the energetic proton models (e.g., AP8) is difficult because they are organized in terms of B (local magnetic field intensity) and L. Secular variations in the Earth's magnetic field drive the normally stable energetic protons toward lower altitudes and the models do not take this into account. Thus, if we project the magnetic field more than 10–15 years into the future, our calculations will be invalid for low-altitude orbits [Konradi and Hardy, 1987]. The present and future configurations of the inner-zone proton belt probably will not differ much, provided both are described in terms of L and λ.

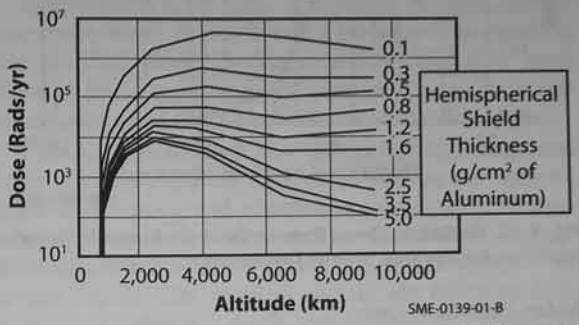


Fig. 7-11. Radiation Dose Rates as a Function of Altitude for Low-Altitude Polar Orbits. Dose rates are shown for several shielding depths.

One can determine the appropriate amount of shielding for future spacecraft by computing the dose rate for the desired orbit as a function of shield thickness. To do so, we must apply a radiation transport computation to the time-averaged space environment. Figure 7-11 shows how various shielding depths affect radiation doses in low-altitude polar orbits. Figure 7-12 shows results for geosynchronous satellites. A *rad* is that amount of radiation which deposits 100 ergs (= 6.25×10^7 MeV) per gram of target material (100 mils of aluminum is equivalent to 0.686 g cm^{-2}). The *total radiation dose* consists of three components: proton dose, electron dose and *bremsstrahlung X-ray dose* produced by the interaction of electrons with the shielding material. In low-Earth orbits, energetic protons in the inner radiation belt contribute most to the total radiation dose. Radiation dose strongly depends on altitude; below 1,000 km the dose increases at approximately the 5th power of the altitude. At geosynchronous altitude the greater than 5 MeV proton dose is negligible and the bremsstrahlung dose dominates the electron dose for shield thicknesses greater than 1 cm.

Figure 7-12 illustrates that protons trapped near a geosynchronous orbit do not have enough energy to penetrate 10 mils of aluminum. Nevertheless, many trapped protons and heavier ions in this region of space have energies around 10–200 keV. These lower-energy ions can harm space systems differently than penetrating radiation. By depositing their energy in the spacecraft skin, the lower-energy ions can cause a temperature rise sufficient to significantly enhance the infrared background. Heat loads of up to 0.5 Wm^{-2} are possible. These same low-energy ions can degrade the effectiveness of paints and protective glass by breaking chemical bonds in their surface layers. Unfortunately, we cannot shield against these effects.

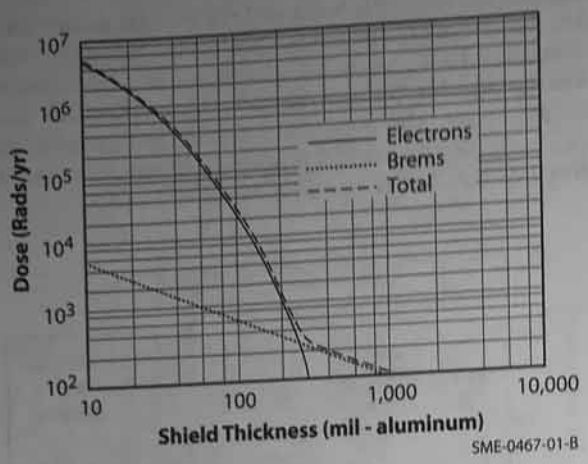


Fig. 7-12. Radiation Dose Rate in Geosynchronous Equatorial Orbit for Various Shield Thicknesses.

Solar Particle Events

Solar particle events (SPEs) occur in association with solar flares. SPEs are rapid increases in the flux of energetic particles (~1 MeV to above 1 GeV) lasting from several hours to several days. On the average, only a few SPEs occur per year; however, they have important consequences for space systems and for humans in space. For example, they degrade solar array elements, increase background noise in many types of electro-optical sensors, and cause illnesses in astronauts.

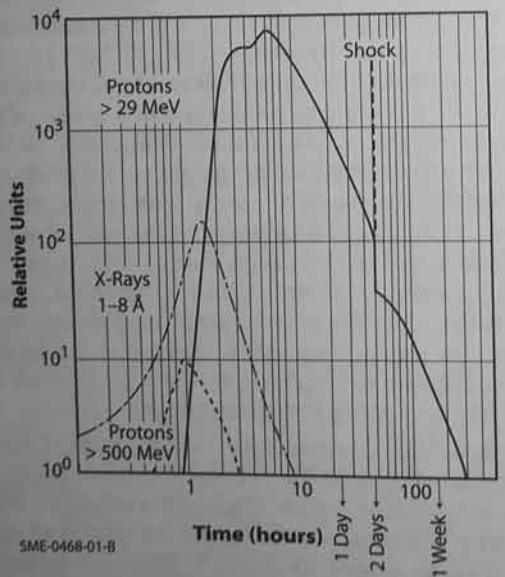


Fig. 7-13. Typical Time Evolution of a Solar Particle Event Observed on Earth.

Figure 7-13 shows the typical evolution over time of a solar particle event observed near Earth. The profile of a solar particle event depends on the time evolution of the originating solar flare, how long the energetic particles take to diffuse within the solar corona and how the particles propagate within the interplanetary medium. Protons of relativistic energies arrive at Earth within minutes after the flare's occurrence. Lower energy (~10's of MeV) protons are slowed by diffusion within the solar corona and by interactions with the interplanetary medium. After a solar particle event occurs, proton fluxes decay to background noise values over several days. The practical importance of an individual event depends on its maximum intensity, its duration, and the relative abundance of the highest-energy components and heavy nuclei.

The frequency of proton events peaks within a year or two of sunspot maximum and diminishes greatly during the few years surrounding sunspot minimum. Nevertheless, intense events can occur virtually any time within the solar cycle except at solar minimum.

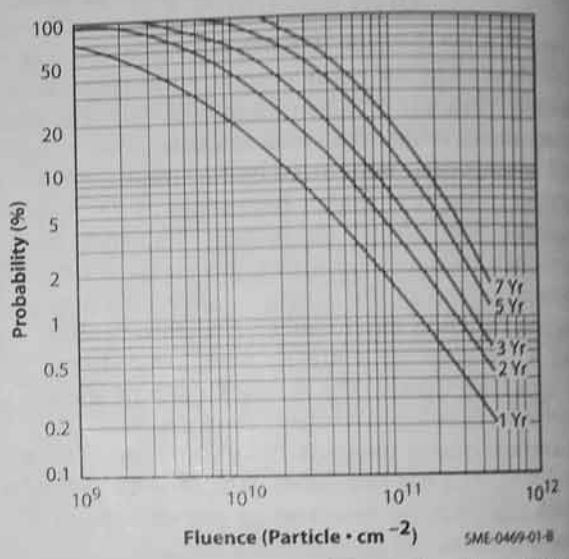


Fig. 7-14. A Plot of Probabilities of Exceeding Given Fluence Levels for Satellite Mission Durations of 1, 2, 3, 5, and 7 Years. The plot represents protons with energies above 10 MeV. These values pertain to satellite missions near the solar activity maxima [Feynman et al., 1988].

The intensities of typical solar proton events closely follow a log-normal distribution [Jursa, 1985; King, 1974; Feynman et al., 1988]; thus, a few individual events can dominate the total proton fluence observed over a complete solar cycle. A typical solar proton event above 10 MeV has a fluence of $10^{8.27}$ cm^{-2}, whereas an extreme (3σ) event would contribute $10^{(8.27 + 3 \times 0.59)}$ cm^{-2} $\approx 1.1 \times 10^{10}$ cm^{-2}. Figure 7-14 shows the probability of the proton fluence (energy >10 MeV) exceeding a given value over time intervals of 1–7 years (typical of the durations of many satellite missions). Solar array outputs typically degrade by a few percent following exposure to fluences above ~10^9 cm^{-2} at energies over ~1 MeV, but actual degradation rates depend on cell type, cover glass thickness, and cell age.

Galactic Cosmic Rays

Galactic cosmic rays (GCR) are particles which reach the vicinity of the Earth from outside the Solar System. The number and type of nuclei in these particles are pro-

Table 7-0, Fig. 7-14, Eq. 7-1◀

portional to those in Solar System material. Cosmic rays pose a serious hazard because a single particle can cause a malfunction in common electronic components such as random access memory, microprocessors, and hexfet power transistors. When a single passing particle causes this malfunction, we call radiation effects *single-event phenomena,* or *SEP.*

A galactic cosmic ray loses energy mainly by ionization. This energy loss depends chiefly on the square of the particle's charge, Z, and can be increased if the particle undergoes nuclear interactions within an electronic part. Thus, the lower-Z (and more abundant) ions deposit as much energy in a device as the less abundant, higher-Z ions. When the galactic cosmic ray generates electron-hole pairs (i.e., excited electrons that move to a higher energy state) in a depletion (i.e., insulating) region of an electronic device, the electric field in that region sweeps away the pairs [Vampola, 1996].

Single-event phenomena include three different effects in electronic components. The first is the so-called *bitflip,* or *single-event upset (SEU),* which neither damages the part nor interferes with its subsequent operation. The second is *single-event latchup (SEL).* In this case, the part hangs up, draws excessive current, and will no longer operate until the power to the device is turned off and then back on. The excessive current drawn in the latched condition can destroy the device if the power supply cannot handle the current. When latchup demands too much current for the power supply, it may drag down the bus voltage, or even damage the supply. The third effect is *single-event burnout (SEB).* This causes the device to fail permanently.

The frequency of single-event phenomena for a given part depends on three things: the external environment; how the incident energy spectrum and particle intensity change as a particle passes through the spacecraft to the sensitive device; and how the electronic device responds to ionizing radiation. These are difficult to evaluate because of the complex interactions between the radiation environment and the device's circuit elements. On-orbit failure rates can be predicted primarily for single-event upsets in memory devices, with well-defined sensitive volumes, in which the galactic cosmic rays produce electron-hole pairs. A useful equation developed by Petersen [1995] expresses the upset rate R as follows:

$$R = 200\,\sigma_L / L^2_{0.25} \qquad (7\text{-}2)$$

where R is the number of upsets or errors per bit day, σ_L is the limiting cross section (sensitive area) of the device in cm^2, and $L_{0.25}$ is the *linear energy transfer* (LET) at 25% of the limiting cross section in units of MeV mg^{-1} cm^{-2}. If experimental cross section data is not available, but device modeling data is, then geometric data can be used in conjunction with the predicted critical charge, and now

$$R = 2 \times 10^{-10}\, abc^2 / Q_c^2 \qquad (7\text{-}3)$$

where R is the number of errors per bit day, a and b are device surface dimensions in mm, c is the device depth in mm, and Q_c is the critical charge in pC. These two equations have been shown to predict upset rates in the geosynchronous orbit for solar minimum conditions with reasonable accuracy. Scale factors for estimating error rates for other orbits and other calculation methods may be found in Petersen [1995].

Single-event upset rates in complex devices such as microprocessors, or single-event latchups or burnouts in any devices, cannot be reliably predicted. One must resort to predictions based on simulated accelerator test observations and flight performance of similar devices.

Galactic cosmic rays can also generate background noise in various satellite subsystems such as star sensors, infrared detectors, and components employing charge-coupled devices. In addition to increased noise signals, these rays create spurious events, which can masquerade as real signals. The spurious signals can affect satellite subsystems depending on the genuine signals' frequency of occurrence, time duration, and repetition, as well as the sophistication of the sensor system. Galactic cosmic rays are a potential source of background noise, which must be taken into account when designing a satellite system.

7.4 Microgravity

James R. Wertz, *Microcosm, Inc./USC*

Microgravity, also called *zero-g, weightlessness,* or *free fall,* is the nearly complete absence of any of the effects of gravity. In the microgravity environment in space objects don't fall, bubbles don't rise, particles don't settle out of solution, and convection currents don't occur. Yet in low-Earth orbit, where all of these things happen, the gravitational force is about 90% of its value at the Earth's surface. Indeed, it is the Earth's gravitational field that keeps the satellite in its orbit.

In space, microgravity comes about because a satellite is in *free fall*—i.e., it is continuously falling through space and all of the parts of a satellite are falling together. In a circular orbit the forward velocity of the satellite (perpendicular to the direction to the Earth) is just enough that the continual falling of the satellite toward the Earth keeps it at the same distance from the Earth's center. Moving in a circular orbit requires a continuous acceleration toward the center, just as does twirling a rock tied to a string.

The term "microgravity" is used in the space environment because, in practice, zero gravity cannot actually be achieved. Two objects traveling very near each other in orbit will not travel in quite the same path due to differences in the gravitational forces or external nongravitational forces acting on them. Two objects held side by side and dropped from a tall building will both accelerate toward the center of the Earth and, as they fall, will converge slightly toward each other. From the point of view

Table 7-0, Fig. 7-14◀, Eq. 7-3

Table 7-1. Equations for Microgravity Level. ω is the orbital angular velocity. $\omega_{LEO} = (\mu/a^3)^{1/2} = 0.061$ deg/sec, assuming spacecraft at 475 km. x, y, and z are the distances from the spacecraft center of mass.

Source	x direction (velocity)	y direction (orbit normal)	z direction (nadir)
Aerodynamic Drag	$\ddot{x} = -0.5(C_D A/m)\rho a^2\omega^2$	$\ddot{y} = 0$	$\ddot{z} = 0$
Gravity Gradient	$\ddot{x} = -x\omega^2$	$\ddot{y} = -y\omega^2$	$\ddot{z} = 2z\omega^2$
Centrifugal (due to spacecraft rotation in inertial frame)	$\ddot{x} = x\omega^2$	$\ddot{y} = 0$	$\ddot{z} = z\omega^2$
Sinusoidal Vibration Along X Axis of Frequency f and Amplitude A	$\ddot{x} = A(2\pi f)^2$	—	—
Coriolis Force from Material Moving in the Spacecraft Frame	$\ddot{x} = 2\dot{z}\omega$	$\ddot{y} = 0$	$\ddot{z} = -2\dot{x}\omega$

of the objects, there is a small component of Earth gravity that pulls them toward each other. An orbiting spacecraft under the influence of atmospheric drag or solar radiation pressure will feel a very small force due to this external pressure. This force can mimic the effect of gravity, causing heavy particles in solution to settle toward the front end of a spacecraft under the influence of atmospheric drag, just as loose cargo in a car appears to be "pulled" toward the front when the car suddenly brakes. Similarly, a rotating spacecraft produces "artificial gravity" due to centrifugal force. Finally, *tidal forces*, sometimes called *gravity-gradient* forces, come about because of very small differences in the force of gravity over an extended object. For a spherical bubble drifting in orbit, the force of gravity on the lower edge of the bubble will be stronger than at the center of mass and weaker at the far edge of the bubble. This very small difference in forces results in "tides" which will distort the shape of the bubble and elongate it toward and away from the direction to the Earth.

For most practical applications, microgravity effects in low-Earth orbit can be reduced to the level of 10^{-6} g (= 1 μg). A level of 10^{-7} g can be achieved over a very small region near the center of mass of the spacecraft. Table 7-1 provides formulas for the most common forces in the microgravity environment. In this table, z is the direction toward nadir and $\ddot{z}$ is the acceleration in the nadir direction. ω is the angular velocity of the satellite in its orbit. Note that for a nadir-pointing spacecraft, the spacecraft rotates in inertial space at a rate of one rotation per orbit and thus adds to the acceleration environment due to the centrifugal force of this rotation. In low-Earth orbit, the lowest microgravity level can be achieved in an inertially-oriented spacecraft rather than a nadir-oriented spacecraft. The last row in Table 7-1 is the *coriolis force* which is an apparent sidewise force that occurs whenever objects move (e.g., fluid flow in a tube) in a rotating reference frame, such as a nadir-oriented spacecraft.

Table 7-2 uses the formulas in the previous table to compute the conditions under which 1 μg is achieved. This gives an idea of the scale over which specific microgravity values can be obtained. Microgravity levels could be substantially lowered by conducting microgravity experiments well away from the vicinity of the Earth or other massive objects.

Microgravity leads to a wide variety of chemical and manufacturing processes that cannot occur on the surface of the Earth. Heavier particles in a solution do not settle out and bubbles do not rise to the surface. This allows uniform, universal mixing and permits chemical reactions to occur that could not occur on Earth because settling would hinder completion of the reaction or hardening of the material. Surface tension and other inter-molecular forces can take over that would otherwise be dominated by settling due to gravity. Similarly, convection does not occur in space and heated materials do not churn or boil. This allows differential heating to take place to provide other chemical reactions. Because settling does not occur, very large crystals can be formed in space which have a variety of industrial applications.

Table 7-2. Microgravity Levels. Each entry gives the conditions under which a microgravity level of 1 μg will be achieved, assuming a gravity-gradient stabilized spacecraft at 700 km. CM = Center of Mass.

Source	x direction	y direction	z direction
Aerodynamic Drag	Altitude of 360 km at solar max, 260 km at solar min for $m/C_D A$ = 65 kg/m²	—	—
Gravity Gradient	7.3 m from CM	7.3 m from CM	3.6 m from CM
Centrifugal	7.3 m from CM	—	7.3 m from CM
Sinusoidal Vibration	$A = 2 \times 10^{-9}$ m at 10 Hz	—	—
Coriolis Force	$\dot{z}$ = 4.6 mm/s	—	$\dot{x}$ = –4.6 mm/s

Table 7-2, Fig. 7-14◀, Eq. 7-3◀

Containerless processing and working in the vacuum of space can lead to extremely pure chemicals for use in both pharmaceuticals and manufacturing. Microgravity has a wide variety of applications that are only beginning to be explored and exploited.

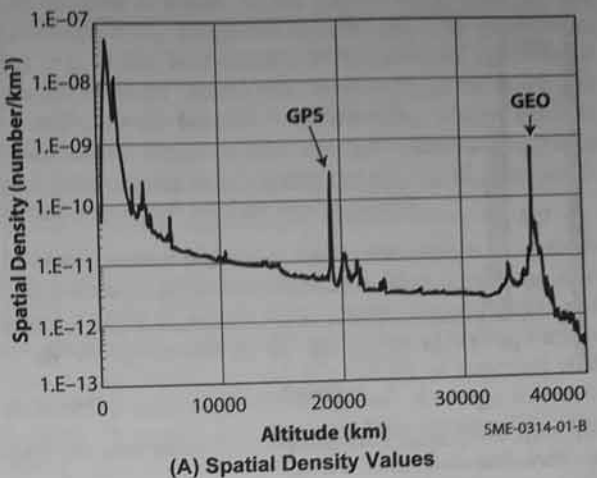
7.5 Orbital Debris—A Space Hazard

David B. Spencer, *Pennsylvania State University*
Ronald A. Madler, *Embry-Riddle University*

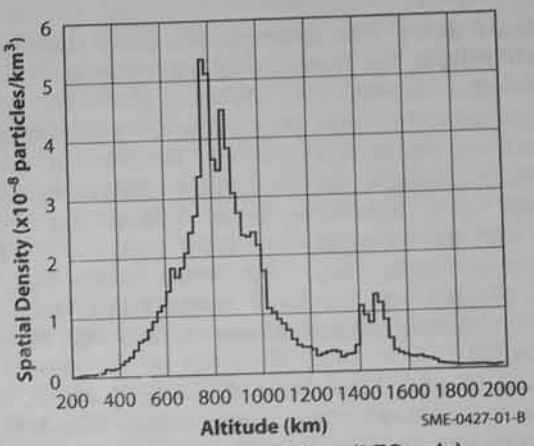
Orbital debris is any non-operational object in orbit. This can be further broken down between *naturally occurring debris* (e.g. micrometeoroids, asteroid particles) and *human-generated debris* from both normal and abnormal spacecraft operations. As an object is launched into orbit, each launch creates payloads, rocket bodies, and mission-related debris. *Mission-related debris* includes: large trackable objects, such as fairings and interstage adapters; small trackable objects, such as Ed White's glove from Gemini 4; and small debris not detectable from the Earth, such as explosive bolt fragments and lens caps [Johnson and McKnight, 1987].

The high expected collision velocity between orbiting objects is one reason to be concerned about space debris. Another is that we cannot detect, much less track and predict an orbit for objects smaller than approximately 10 cm for LEO (about the size of a softball) and 1 meter for GEO. The United States Space Command (USSPACECOM) is tracking over 15,000 objects by late 2010 through its Space Surveillance Network (SSN), a job for which it was not originally intended. This secondary mission of the worldwide network of radar, optical and data processing facilities is to track, identify, and catalog data on the largest orbiting objects. Although many of these objects may not seriously threaten space operations, it is known that a significantly larger number of untrackable objects smaller than 10 cm exist in low-Earth orbit as a result of satellite breakups and operational practices. Because these objects can't be effectively tracked, they can't be avoided by maneuvering. These fragments will threaten spacecraft for tens, hundreds, and even thousands of years. The GEO region is of special concern because it has no natural mechanism that removes debris. Nevertheless, various scientific and space operations organizations (such as NASA, the Air Force, ESA (European Space Agency), and the Russian Space Agency) have been collecting statistical data on untrackable, yet detectable, objects since the early 1990s to aid the modeling of the debris environment and the sources of debris.

All the man-made orbital material in space has come from human activity, so the orbital debris issue is really about how that mass is distributed and managed (i.e., distribution of objects and their orbits). We can quantify how crowded space has become by using the *spatial density*, i.e., the number of objects per volume of space. Figure 7-15a estimates the spatial density of objects in LEO and GEO as a function of altitude for January 2011,



(A) Spatial Density Values



(B) Spatial Density Values (LEO only)

Fig. 7-15. (A) Spatial Density Values of All Catalogued Objects for January 2011, (B) Spatial Density Values of All Catalogued Objects for January 2011 (LEO Only). Densities in GEO are approximately 10 times less than in LEO. The probability of a spacecraft getting hit by something big is small, while the chance of getting hit by something small is big. See text for discussion. [Courtesy of Orbital Debris Program Office, NASA/JSC.]

while Fig. 7-15b zooms into the lower altitude region. The GEO data represents the spatial density within 1 deg of the equatorial plane. The smallest trackable objects may weigh tens to hundreds of grams. We cannot shield against objects of this size. In GEO the average spatial density is one to two orders of magnitude less than LEO. The **average** relative velocity in LEO between orbiting objects is 9 to 10 km/s with maximum values above 14 km/s due to eccentric and retrograde orbits. In comparison, the relative velocity between debris and satellites in GEO ranges between almost nothing (coplanar objects moving in the same direction) to 2.5 km/s (objects with significant different inclination/right ascension combinations). The distribution of objects around the geosynchronous band is grouped by inclination in Fig. 7-16. Note that most objects are in nearly equatorial orbits, although a few objects are inclined by

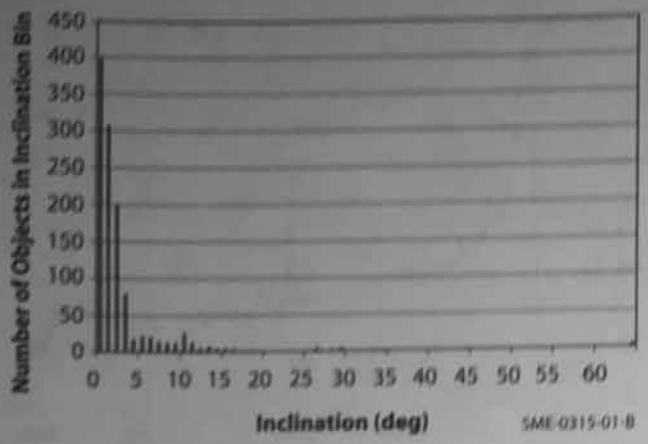


Fig. 7-16. Inclination Distribution of Objects Near the Geosynchronous Belt (as of 1/1/11).

as much as 64°. The difference in relative velocities is due mainly to the lower orbital velocities and smaller inclination distribution of objects in GEO. The key to understanding the damaging effects of a debris impact is understanding the kinetic energy. For example, a 1 kg object impacting something at 10 km/s has the equivalent energy of 25 sticks of dynamite. A Ferrari traveling at 200 mph has the equivalent kinetic energy of a 1.5 kg object moving at 3 km/s. This kinetic energy comparison, coupled with the lower spatial density values in GEO, make the collision hazard much smaller in GEO than LEO.

Note that the LEO density is averaged over all latitudes and that the true density varies with latitude because space objects have various inclinations. Over time, the spatial densities and flux rates will change due to natural decay (see Sec. 9.4.4), new launches, and fragmentation. Presently the only sink for objects in LEO is natural decay, which is dependent on altitude and solar activity as shown in Figs. 9-15 and 9-19, with the exception of controlled reentry of satellites. End-of-life burns are becoming a common removal mechanism for controllable objects in orbit, such as the controlled reentry over the Pacific Ocean of the Russian Mir space station).



For NASA missions, the NASA Safety Standard [NASA, 2007], requires end-of-life maneuvers that result in reentry within 25 years or objects being "reorbited" out of the high traffic orbital regimes, including LEO, GEO, and semisynchronous orbit, but no more than 30 years after launch. Many other spacecraft operators have similar practices. Chapter 30 provides further discussion on the end-of-life process.

7.5.1 Orbital Debris Environment

Meteoroids and orbital debris are now well recognized as a potential hazard to spacecraft. This is largely due to recent fragmentations in low-Earth orbit and the resulting media coverage. Debris awareness during mission design and operations is necessary for mission success and also to lower the risk to present and future spacecraft. An early example of the design implications of orbital debris was the RADARSAT program, where concerns about meteoroids and orbital debris caused the program to add 17 kg of shielding to increase their probability of success from 50% to 87% over the five year mission [NRC, 1995]. In addition to debris impact hazards, end-of-life satellite disposal requirements may be a driving requirement in mission design. Taking into account debris hazard and mitigation requirements early in the design process will allow for a more robust mission design and lessen the risk due to orbital debris. This section will provide a brief overview of the orbital debris environment as well as the implications of orbital debris for mission and spacecraft design. Table 7-3 provides a summary of the process for determining the mission impacts due to debris.

Table 7-3. Mission Design Considerations for Orbital Debris.

Step	Notes
Define Operational Regime	LEO, MEO and GEO will have very different environments, concerns and requirements.
Define Applicable Policies	US Government, commercial, and international vehicles fall under different regulatory regimes.
Determine Applicable Models	Programs may have defined or preferred environment models and procedures depending upon customer and policies.
Review Resources	Extensive resources are available from national and international members of the Interagency Debris Committee. Many references are publicly available.
Determine Hazard and Policy Compliance	
Lessons Learned	Several papers discuss lessons learned by satellite operators.

The hazard due to the natural environment was foreseen as early as the 1940s due to the high expected collision velocity. Meteoroid impact speeds can average about 20 km/s but can be as high as 70 km/s. At these speeds even small particles have high kinetic energies and can thus cause significant damage.† While these particles are not trackable, they can cause a cumulative degradation of surfaces of a spacecraft, which may require additional protection to be designed into the spacecraft.

Over the full spectrum of sizes of orbital debris, collision hazards have varying effects on the spacecraft. This hypervelocity‡ impact damage can range from surface erosion and degradation to damage of mission-critical systems or breakup of the spacecraft. Small impacts on a spacecraft may not even be noticeable unless the space-

† A 1 mm meteoroid at 20 km/s has as much kinetic energy as a person swinging a baseball bat.

‡ A hypervelocity impact is defined as having a higher impact velocity than the speed of the shockwave through the object being hit.

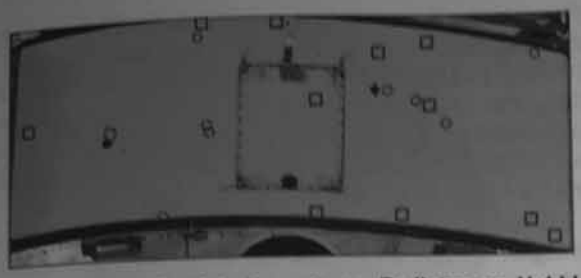


Fig. 7-17. Small Particle Impacts on Radiator on Hubble Space Telescope. Circles indicate impacts identified from the 2002 servicing mission, while squares identify new impacts observed during the post-mission inspections. (Courtesy Orbital Debris Program Office, NASA/JSC.)

craft is recovered. A visual examination of the Hubble Space Telescope, as seen in Fig. 7-17, shows small impacts that occurred during 16 years of exposure. None of these impacts affected the performance of the telescope and it is likely that they would have gone unnoticed had a close visual examination not been conducted. However, it was purely by chance that these impacts occurred in non-critical areas—had an impact occurred at a critical location, performance could have been affected.

The first confirmed on-orbit collision between two tracked objects occurred in 1996 between the Cerise satellite and a piece of debris [Alby et al. 1997]. This impact separated the gravity gradient boom from the spacecraft. A more recent collision between IRIDIUM 33 and COSMOS 2251 destroyed both spacecraft and is an example of a catastrophic impact that has left many large objects and an unknown amount of untrackable debris still in orbit such that the remaining debris will pose a hazard for many years to come [ODQN, 2009]. As of 2010, 203 satellite fragmentations and collisions have generated a large growth of debris and create an additional hazard for space missions [Liou, 2010]. An international awareness has developed of the long-term impact of fragmentations and mission operations on mission hazard, which has led to debris mitigation guidelines that have been accepted by most spacefaring nations.

It is important for space mission designers to assess the impact of their design on the space environment early in the design process. Figure 7-18 outlines the lifecycle of a space mission. There are several phases of a space mission where orbital debris issues need to be addressed. During the launch phase, launch vehicles are delivering payloads to their operational orbits. During this phase, the system is subject to potential collision hazards with existing orbital objects. During satellite operations, assessments of the hazard to the space environment from the spacecraft and hazards to the spacecraft from the space environment should be made. Lastly, at the end-of-life of the mission, considerations as to disposal options should be addressed. Each of these steps should be analyzed in context with any national, international or satellite owner/operator debris mitigation guidelines. Should the designs not meet the desired debris environmental effects, the designer iterates the design until a satisfactory solution is found. At the present time, there are not universal debris mitigation regulations, only guidelines, and it is in the best interest of the satellite operator to operate a "clean" spacecraft, as the spacecraft is subject to self-induced hazards. Additionally, many satellite operators use the same orbit for replenishment, so it is in their best interest to minimize the effects of decommissioned spacecraft on their replacements.

Small natural objects in space are known as *meteoroids* and are in orbit around the Sun. Meteoroids are composed mainly of debris from comets and asteroids and are continually passing through space. Having various sources, meteoroids consist of a variety of materials with densities ranging from 0.2 to 8 g/cm^3. Average values of 2 g/cm^3, 1 g/cm^3, and 0.5 g/cm^3 for masses less than 10^{-6} g, between 10^{-6} and 0.01 g and above 0.01 g, respectively, have been recommended [Anderson, 1994 and NASA, 1991]. The meteoroid environment is usually expressed in terms of an average flux of objects per area per unit time. The actual meteoroid flux consists of sporadic meteoroids and streams of meteoroids. The former have no identifiable parent object or pattern; the latter have a common parent object and orbit. Streams typically occur whenever the Earth encounters debris along the orbital path of a comet. Occasionally, the Earth will pass through an especially dense region of a natural debris stream, causing meteor storms which may threaten spacecraft operations. The risk from storm particles over several hours can match that of the background meteoroid environment for a year. Meteoroid models generally do not take meteor storms into account although some do, but manned missions presently do consider this threat.

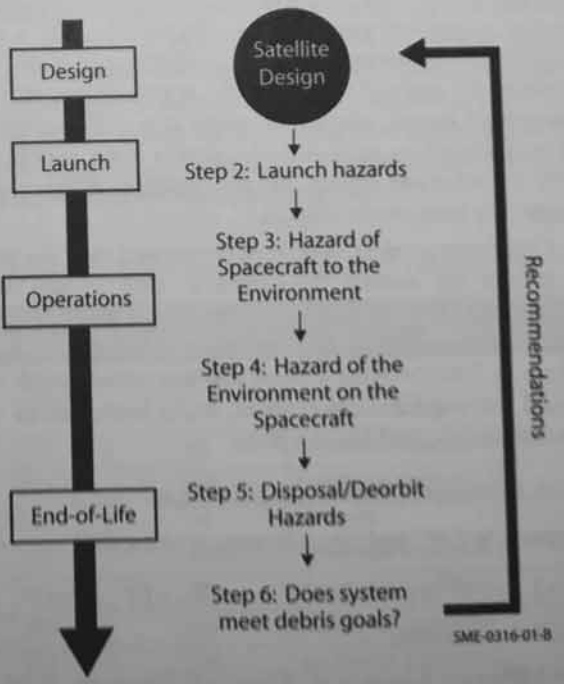


Fig. 7-18. Satellite Design Lifecycle for Orbital Debris Assessment.

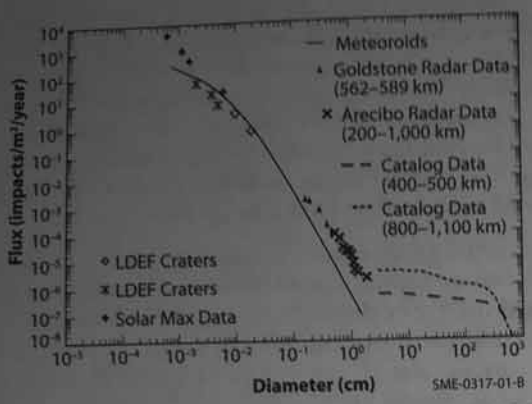


Fig. 7-19. Comparison of the Fluxes of Meteoroids and Orbital Debris for Low Altitude Orbits (adapted from the NRC, 1995). This figure shows that human-generated orbital debris has much higher impact rates than meteoroids for large and small debris diameters.

The flux of meteoroids is less than the flux of orbital debris in LEO for most sizes, but is on the same order as man-made debris between 0.001 and 0.1 cm. Figure 7-19 compares the fluxes of meteoroids and orbital debris. Meteoroid hazard becomes comparable to the hazard due to man-made orbital debris at about 1 mm in size, yet can damage unprotected surfaces due to the high relative velocities. As meteoroids come in close proximity to the Earth (especially in close proximity to where satellites fly), they pose a collision hazard. Estimates put the amount of meteoroids entering the Earth's atmosphere between 20,000 and 60,000 metric tons of material per year [Love and Brownlee, 1993]. Most of these meteoroids are small and light and rarely are recovered. An advantage of meteoroids over man-made orbital debris is that individual objects are not in orbit around the Earth, so that individual particles are a hazard for only a very short time. However, the average velocity of a meteoroid is about 17 km/s [Kessler, 1969], which means that the amount of kinetic energy involved in a collision with a natural object is substantially greater than a collision with a man-made object, but the duration of this risk is longer for man-made objects.

Computer models have been developed over the years to model the natural environment. One of the more widely used models was created by Grün et al [1985]. In this empirical model, they estimate the flux of natural debris at 1 AU. Grün's empirical micrometeoroid flux model for a given size particle, m, is based on in situ measurements, and has the form

$$F_1(m,r_0) = (c_1 m^{\gamma_1} + c_2 m^{\gamma_2} + c_3)^{\gamma_3} + (c_4 m^{\gamma_4} + c_5)^{\gamma_5}$$

where $r_0 = 1AU$ and the constants, $c_1 = 4\times10^{29}$,

$c_2 = 1.5\times10^{44}, c_3 = 1.1\times10^{-2}, c_4 = 2.2\times10^{3}, c_5 = 15$

and the exponents,

$\gamma_1 = 1.85, \gamma_2 = 3.7, \gamma_3 = -0.52, \gamma_4 = 0.306, \gamma_5 = -4.38$ (7-4)

The units on this model are (number/m^{2}s). The terms in the first bracket represent the small particles ($m \leq 10^{-10}$ g) and the second part describes the large particle flux. The exponents control the slopes of the particle populations. This model was originally developed for interplanetary space, and is usable for all regions, including the region around Earth's orbit.

While the Grün model provides a model that allows a designer to estimate the meteoroid flux, it does not take into account sporadic meteoroids. NASA has supported development of a new meteoroid environment model (MEM) [Jones, 2004] that identifies sporadic meteoroids from sources such as comets, provides users with accurate fluxes and directionality for spacecraft from 0.2 AU to 2 AU. The model also incorporates observation data and compares that with theoretical models. The computer models are available through NASA's [2011], Meteoroid Environment Office website.

Table 7-4. Space Object Origination (as of October 6, 2010, Cataloged by the US Space Surveillance Network). (Adopted from ODQN—October 2010, Courtesy of NASA/JSC.)

Country or Organization	Payloads	Rocket Bodies & Debris	Total
China	98	3,395	3,493
CIS	1,406	4,600	6,006
ESA	39	44	83
France	49	426	475
India	41	133	174
Japan	113	76	189
USA	1,124	3,701	4,825
Other	479	115	594
TOTAL	3,349	12,490	15,839

The US satellite catalogue is maintained by tracking data from the SSN. As of October 6, 2010, there were 15,839 catalogued objects in Earth orbit. We know that there exists a much larger population of smaller objects. This has been confirmed by many measurement campaigns with more sensitive radars and evaluation of objects returned from space. Studies have shown that there is approximately an order of magnitude more objects in the 1 cm size than what exists in the catalog [OSTP, 1995]. The 1 cm size is significant because it is the largest size fragment that we can effectively shield against. Thus, there are a significant number of objects that we cannot track, but which can cause substantial spacecraft damage [Spencer et al., 2000].

The man-made debris population has become an important satellite operations issue over the past few decades. The increasing utilization of space has created additional risk to satellites from collision. The orbital debris population has been created by all users of space. The contribution to the orbital population broken down by countries and organizations is shown in Table 7-4. This table shows the origins of rocket bodies and debris as well as payloads.

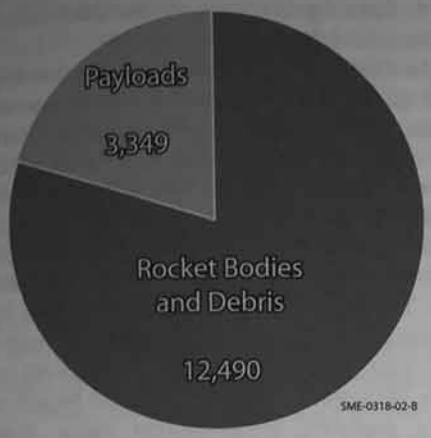


Fig. 7-20. Catalogued Low-Earth Orbit Population.

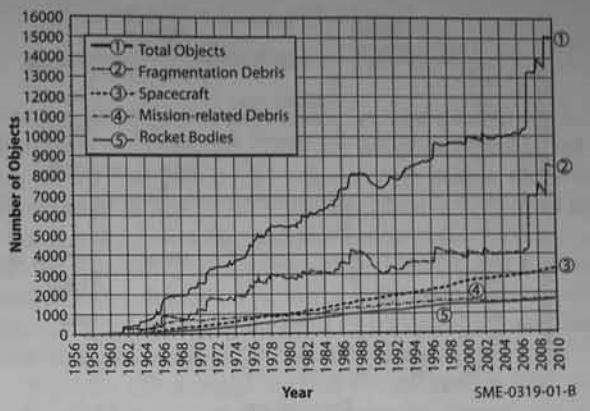


Fig. 7-21. Historical Growth of the Space Population. (Courtesy of Orbital Debris Program Office, NASA/JSC.)

Man-made objects in orbit consists of operational and derelict spacecraft, operational debris, break-up debris and rocket bodies. Derelict spacecraft are intact spacecraft that are no longer functioning. Operational debris includes items that were designed to be released during some phase of operations. Examples of this include lens covers, yo-yo despin devices, and explosive bolt pieces. The break-up debris constitutes the majority of catalogued objects in space, and come from both explosions and collisions. Rocket bodies are the delivery vehicle for spacecraft. As seen in Fig. 7-20, nearly 80% of this population is rocket bodies and debris.

Historically, the orbital population has grown. Figure 7-21 shows this cumulative growth since the beginning of the space age in 1957. Note that the population of active satellites has grown nearly linearly over the past couple of decades. The significant increase in the total population has come from fragmentation debris (break-up events are evident from this figure by identifying step increases in the fragmentation debris curve). At present trends, one theory projects a "runaway" scenario—known as the Kessler syndrome or collisional cascading [Kessler and Cour-Palais, 1978], where the debris from one breakup collides with another satellite and then, debris created, in turn, collides with another satellite, and so on. Concerns as to whether this scenario is plausible is one of the driving factors in managing the space debris environment.

The significant increase in the orbital debris population can be traced back to several fragmentation events. The Orbital Debris Program Office at NASA/JSC has compiled a list of the top 10 fragmentation producing events, and this is shown in Table 7-5. Approximately 1/3 of the orbiting objects in the catalog come from these 10 events.

Spacecraft designers handle the orbital debris environment in various ways. For the very small objects which are untrackable, the spacecraft designers may try to design the spacecraft with sufficient shielding to survive an impact. At the other end of the size spectrum, large objects in space are tracked using two forms of sensors: radar and optical. Geographically spread out radar sites perform the majority of tracking of objects in low Earth orbit. As the altitude increases, optical sensors track more of the objects. In the region in between shielding and tracking there is a vulnerable region where there are on-going efforts to close the gap between what can be shielded against and what can be tracked. Figure 7-22 graphically represents the approximate boundaries between these regions.

Table 7-5. Top 10 Historical Breakups (As of May 2010). (Source: Orbital Debris Quarterly News [July 2010, NASA/JSC])

Common Name	Year of Breakup	Altitude of Breakup	Catalogued Debris	Tracked Debris in Orbit	Cause of Breakup
Fengyun-1C	2007	850 km	2841	2756	Intentional Collision
Cosmos 2251	2009	790 km	1267	1215	Accidental Collision
STEP 2 Rocket Body	1996	625 km	713	63	Accidental Explosion
Iridium 33	2009	790 km	521	498	Accidental Collision
Cosmos 2421	2008	410 km	509	18	Unknown
SPOT 1 Rocket Body	1986	805 km	492	33	Accidental Explosion
OV 2-1/LCS 2 Rocket Body	1965	740 km	473	36	Accidental Explosion
Nimbus 4 Rocket Body	1970	1075 km	374	248	Accidental Explosion
TES Rocket Body	2001	670 km	370	116	Accidental Explosion
CBERS 1 Rocket Body	2000	740 km	343	189	Accidental Explosion
		TOTAL	7903	5172	

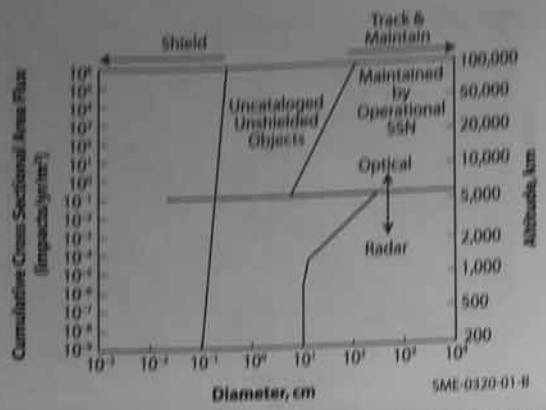


Fig. 7-22. Shielding, Tracking and Cataloguing Thresholds.

To assess the collision risk between a spacecraft and the orbital debris environment, computer models must be employed. There are two types of modeling techniques available: deterministic and stochastic. *Deterministic models* treat orbital debris objects as individual pieces, and examine the close approach possibilities. Many of these deterministic models are limited due to uncertainties in object locations, and, depending on the confidence level in close approach predictions, typically called *conjunctions* or *conjunction events*, spacecraft operators can chose to take a number of actions that range from doing nothing to performing a collision avoidance maneuver. While in the majority of potential collision events a spacecraft operator chooses to do nothing, there are cases where the operators choose to perform a collision avoidance maneuver. Table 7-6 shows collision avoidance maneuvers that were performed in 2009.

Stochastic models, on the other hand, treat the debris environment as a continuum of objects, much like gas particles in an enclosed volume. Here, the concern isn't a distinct object versus another distinct object collision, but a collision between a distinct object and a "cloud" of objects.

Several organizations have developed software that is available for use. The most widely used software packages are those developed by NASA and ESA. As this software is being continually updated, the reader is directed to these agency websites for information on the most up-to-date features.

There are several different categories of software that would be of interest to different technical communities. The spacecraft mission designer would be interested in software to assess the effects of the orbital debris environment on their spacecraft and the effects of their spacecraft on the orbital debris environment. The environment community would be interested in predicting the short-term and long-term behavior of the orbital debris environment. The observation community would be interested in models that convert information about what is observed into information about the current environment. The operations community would be interested in improving methods of predicting close approaches between objects, and reducing the number of false alarms that predict a close approach or a collision.

Many of these modeling efforts have led to international collaboration, and to the development of debris mitigation guidelines. Since the early 1990s, the Inter-agency Space Debris Coordination Committee (IADC) has led the effort to coordinate orbital debris measurements, modeling, protection and mitigation activities underway in various national space agencies. In addition to other organizations, such as the ISO [2010], (International Organization for Standardization), the IADC [2010] has developed debris mitigation guidelines and standards that are being accepted as standard design practices throughout the world [IADC, 2002]. This work has led to the development of United Nations' guidelines on orbital debris mitigation [UN, 2010]. As new technologies become available, these recommended practices and guidelines are continually being updated.

Get Std

7.5.2 Implications of Orbital Debris for Orbit and Spacecraft Design

The space mission that has undertaken the most extensive assessment of the effects of the orbital debris environment is the International Space Station (ISS). The ISS has a unique mission both in duration and in

Table 7-6. Collision Avoidance Maneuvers in 2009 (Source: ODQN, Jan. 2010).

Spacecraft	Maneuver Date	Object Avoided
TDRS 3	27 January	Proton rocket body
ISS	22 March	CZ-4 rocket body debris
Cloudsat	23 April	Cosmos 2251 debris
EO-1	11 May	Zenit rocket body debris
ISS	17 July	Proton rocket body debris
Space Shuttle	10 September	ISS debris
PARASOL (France)*	29 September	Fengyun-1C debris
Aqua	25 November	Fengyun-1C debris
Landsat 7	11 December	Formosat 3D

* Operating in NASA-led Earth observation network

payload. The mission is currently planned to operate until at least 2020 and the payload is the most valuable of cargos—humans. With these two considerations, an extensive assessment of the risk posed to the ISS from the orbital debris environment has been conducted [NRC, 1997]. Orbital debris environment models are used to assess the risk, and determine the collision hazard posed by the orbital debris environment. Small object collision risk can be assessed statistically, while large object collision risk can be assessed deterministically.

Statistical Collision Hazard

The stochastic modeling treats the orbital debris environment as a statistical distribution of particles. We can approximate the probability, P_c, of a piece of debris impacting a space system using the kinetic theory of gases:

$$P_c = 1 - e^{(-SPD \bullet AC \bullet T \bullet VREL)} \tag{7-5}$$

where *SPD* is the spatial density of debris objects (i.e., average number of objects per volume in space), *AC* is the collision cross-sectional area, *T* is the mission duration, and *VREL* is the relative velocity between the satellite and debris population. Detailed analysis should use a more accurate debris environment model (for example, early development of NASA and ESA models can be found in Kessler et al. [1996], and Klinkrad et al. [1997], respectively), as well as a more refined representation of the spacecraft's cross-sectional area. Also space satellite designers must take the meteoroid environment into account. Meteoroids dominate the hazard for sub-millimeter up to millimeter size, and can penetrate thin honeycomb structures and other delicate structures. Debris flux from natural and man-made sources can simply be added together to create a total flux.

NASA's standard orbital debris model uses the yearly averaged flux, *F*, and a reference surface of a randomly rotating plate. Thus, the orbital debris flux, which is a statistical estimate of the debris density, becomes the number of impacted objects per unit surface during a unit of time. The cumulative particle fluence, *N*, on specific spacecraft surfaces can be found by using the "k-factor method" [Anderson, 1994] along with the flux value,

$$N = \int_{t}^{t+T} \sum_{i} k_i F A_i \, dt \tag{7-6}$$

where i represents each surface of the spacecraft, $k_i F$ is the actual flux on the surface i, and A_i is the area of each surface. The Poisson process to calculate the probability of exactly n impacts (P_n) occurring on the spacecraft in the time period is

$$P_n = \frac{N^n}{n!} e^{-N} \tag{7-7}$$

The flux can be derived empirically [Anderson, 1994] for a randomly tumbling surface as:

$$F(d,h,i,t,S) = H(d)\ \varphi(h,S)\ \psi(i) \left[F_1(d) g_1(t) + F_2(d) g_2(t)\right] \tag{7-8}$$

where F is the flux in impacts/m^2/yr, and d is the debris' diameter in centimeters (cm), for which acceptable values are 10^{-4} to 500 cm. The date is t (in years), and h is the altitude in km, valid when less than 2,000 km. The term S is the 13-month smoothed solar-radio flux, $F_{10.7}$, for the previous year (expressed in 10^4 Jy), and i is the inclination in degrees. The factor is the ratio between the flux on a spacecraft in an orbit of inclination i, and the flux on a spacecraft at 50 deg. inclination.

Because the damage caused by an impact with an object is a function of the kinetic energy, the velocity of a particular impact can be found using NASA's standard model for the velocity distribution $f(v)$ as

$$f(v) = \left\{2vv_0 - v^2\right\} \left\{ \begin{array}{l} G \exp\left(-\left(\dfrac{v - Av_0}{Bv_0} \right)^2 \right) \\ +F \exp\left(-\left(\dfrac{v - Dv_0}{Ev_0} \right)^2 \right) \end{array} \right\} + \left\{HC(4vv_0 - v^2)\right\} \tag{7-9}$$

where v is the collision velocity in km/s. The remaining terms are either constants or functions of inclination, and are detailed further in Anderson [1994].

Lastly, the number of impacts on a given spacecraft surface over the mission lifespan can be found by integrating the flux over the surface,

$$N_{impacts}(d) = T_{mission} \int_{Surface} F(d) n_A \, dA \tag{7-10}$$

where F is the total flux at the particle size of interest d, A is the exposed surface area, n_A is the local surface's normal vector corresponding to the area element dA, and $T_{mission}$ is the mission lifetime [Rainey, 2004]. Based on evaluation of the satellite catalog of objects in low Earth orbit, the collision probability for a 10 m^2 object is shown as a function of altitude in Fig. 7-23 [Ailor et al., 2010].

Deterministic Collision Hazard

In addition to shielding, spacecraft operators worry about whether there is an expected *conjunction* (close approach) in the next few days. The Center for Space Standards and Innovation (CSSI) has developed a service known as SOCRATES (Satellite Orbital Conjunction Reports Assessing Threatening Encounters in Space) [CSSI, 2011]. This service provides information on when and where potential conjunctions will occur in the next week. Satellite operators can then use this information to decide whether there is sufficient confidence in a future conjunction to warrant the operator executive collision avoidance maneuver. For example, SOCRATES

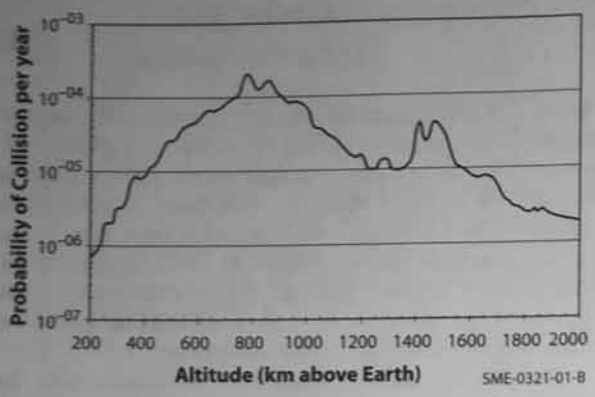


Fig. 7-23. Collision Probability for a 10m² Object in Low Earth Orbit. [Ailor, Womack, Peterson, and Lao 2010.]

identified that there was a close approach between the Iridium 10 satellite and a Delta 1 rocket body on January 15, 2011 at 10:13:04.632 UTC. The two objects were forecast to pass within 116 m of each other, traveling at a relative velocity of 14.729 km/s. The probability of collision was estimated to be less than 0.3%. Based on this low probability, Iridium chose not to perform a collision avoidance maneuver.

Damage Assessment

Impacts with orbital debris can damage spacecraft in various ways. Figure 7-24 shows three levels of damage from hypervelocity impacts. The first picture (a) is a scanning electron microscope image of an area eroded by impacts with very small objects (on the order of microns). This is the most frequent impact on spacecraft. This type of object typically degrades performance, such as decreasing a solar panel's efficiency or interferes with an image taken from a camera. The second picture (b) shows an object penetrating the surface in a hypervelocity impact. Depending on the location of the impact on a spacecraft, the effects can range from small (penetrating an antenna or solar panel) to catastrophic (penetrating a space suit or a fuel tank). The colliding object's size for this level of damage varies—overall damage depends on the impact velocity, the physical characteristics of the object it is hitting, and what the impacting object is made of. The last picture (c) shows a spacecraft-like structure destroyed in a hypervelocity test. Although statistically rare in space, a collision of this intensity would likely be fatal to both colliding objects, as what happened to both the Iridium 33 and COSMOS 2251 on February 10, 2009.

Spacecraft designers generally take into account the possibility of collision with objects in orbit. Risk reduction techniques can range from judicious placement of sensitive components (using other parts of the spacecraft to shield such things as pressurized tanks) to adding shielding to the spacecraft. For example, the International Space Station uses what is known as a stuffed Whipple shield. The Station uses multiple customized shields to protect sensitive structures, especially habitable components and associated support components. An example stuffed Whipple shield is shown in Fig. 7-25 [NRC, 1997]. This shield uses a sandwich of Nextel and Kevlar inserts between the exterior and interior aluminum walls. This design was thoroughly tested on the ground to survive an impact of a 3.5 g aluminum sphere traveling at 7 km/s.

Design Considerations and Debris Mitigation

There are several analysis software packages used to perform orbital debris assessments. NASA has developed a software package, *Debris Assessment Software* (DAS), which was designed to help NASA programs in performing orbital debris assessments [Opiela et al., 2007]. While the software has a NASA focus (including discussion of NASA standards and orbital debris mitigation practices), the software can be used for non-NASA programs, and the results can be used to assess a program's compliance with their own debris policies. Although the overall orbital debris collision hazard analysis partially drives the design of space missions, and is a significant consideration that any mission designer should take into account. In many cases, debris mitigation and debris protection adds to the

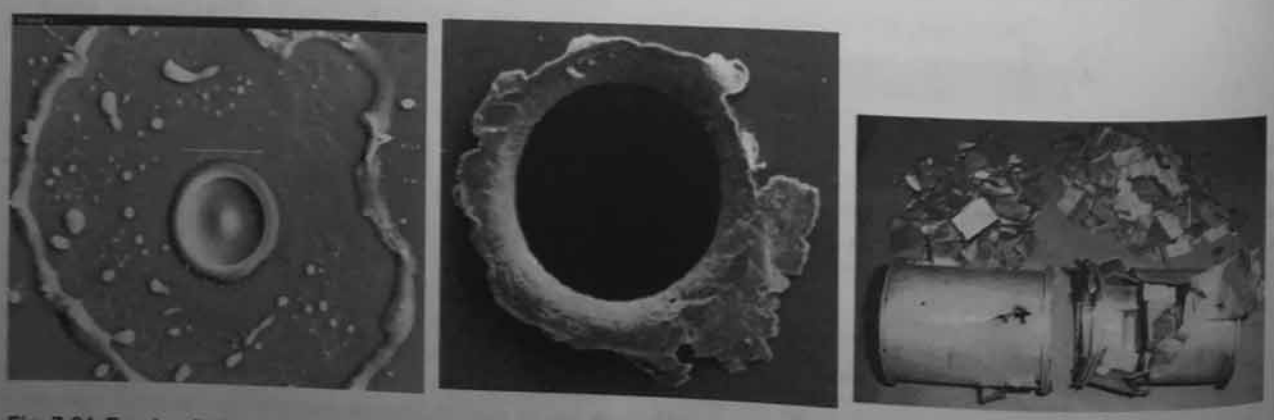


Fig. 7-24. Erosion (Left), Penetration (Middle), and Catastrophic Effects from Impacts with Debris (Right). (Courtesy of NASA)

Table 7-6◀, Fig. 7-24, Eq. 7-10◀

overall cost (both in added mass and added financial considerations). However, the cost of doing nothing can be a premature end to a satellite mission. A balanced approach should be undertaken where the best design practices are performed to minimize the impact of the space mission on the orbital debris environment and to minimize the impact of the environment on the space mission. The overall goal should be to make the spacecraft safe by focusing on well thought out design and operation practices, not to simply meet some arbitrary standard or guideline.

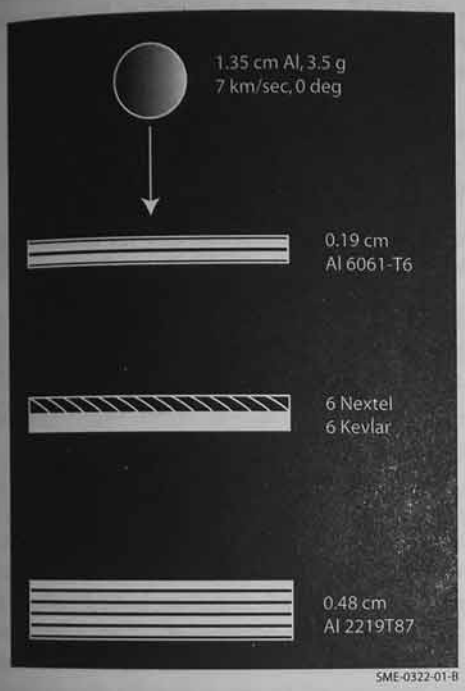


Fig. 7-25. A Stuffed Whipple Shield Used by the International Space Station. This shield design is a multi-layer shield with various distances between the layers. Non-structural insulation can fill the voids between the layers. (Courtesy of NASA)

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Table 7-6◀, Fig. 7-25◀, Eq. 7-10◀

8 Space Mission Geometry

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8.1 Introduction to Space Mission Geometry
Sample Problem: Eclipse Analysis; Coordinate Systems; Basic Spherical Geometry and Unit Vector Formulas

8.2 Applications
Solving Mission Geometry Problems; Example 1: Field of View of a Square Sun Sensor; Example 2: Measurements Made by a Rectangular Array Sensor; Example 3: Calculating the Duration of an Eclipse

8.3 Looking at the Earth from Space
Transforming Between Earth and Spacecraft Coordinates; Earth Features Viewed from Space; Directions, Shapes, and Areas; Projection of Sensor Fields of View onto the Earth

8.4 Computing Parameters for a Single Target or Ground Station Pass
Circular LEO Orbits; Elliptical Orbits

8.5 Satellite Relative Motion

8.6 Mapping and Pointing Budgets
Representative Mapping Budget; Representative Pointing Budget; Minimizing Cost

Most spacecraft observations including both payload sensing and orbit and attitude measurements are angles—i.e., they are measurements made on the sky as seen by the spacecraft, called the *celestial sphere*. This chapter provides an introduction to measurements and observations on the celestial sphere. It begins by explaining why spherical geometry is important to this process and establishes the relationship between spherical geometry and vector analysis with which many analysts are more familiar. It then introduces basic algorithms and provides several sample applications. Wherever possible, we provide summary formulas in both vector and spherical forms that can be readily applied to practical problems.

While this chapter introduces the concepts and mathematics associated with spacecraft sensing and angular measurements, the fundamental purpose of the chapter is to evaluate the space mission geometry, i.e., what the spacecraft sees and how other observers see the spacecraft and its motion, for a particular mission. This mission geometry evaluation process is summarized in Table 8-1 and discussed further in Table 8-2 and Table D-1 in App. D.

There is a substantial literature base on orbit and attitude dynamics and the general equations of motion for spacecraft. There is no comparable literature base for the theory of angular measurements made by spacecraft. Consequently, background information must be assembled from a variety of sources. *Spherical Astronomy* [Green, 1985] provides a good summary of spherical geometry, but only from an astronomical perspective. *Spacecraft Attitude Determination and Control* [Wertz, 1978], provided the first detailed discussion of mission geometry, but only for attitude analysis. (These techniques are broadly applicable, however, and form the basis for much of the current work). An extension and update to this work is *Orbit & Constellation Design & Management* [Wertz, 2009], called OCDM in this chapter. This work extends the description of space mission geometry to all types of directions-only measurements, provides the background for all of the results presented in this chapter, and extends classical spherical geometry to "full-sky spherical geometry" in which any or all of the angles and arc lengths can cover the full range of 0 to

Table 8-1. The Space Mission Geometry Evaluation Process. See Table 8.2 for additional mission geometry topics, summary tables, and where derived and Table D-1 for a list of both geometry background and application areas and where they are discussed. Earth coverage over time by satellites is discussed in Sec. 10.2.

Step	Where Discussed
Evaluate target and ground station coverage and apparent motion as seen from the spacecraft (also sky coverage for systems that are looking away from the central body)	Secs. 8.4, 10.2
Determine sensor field-of-view in space and on the ground	Secs. 8.2.2, 8.2.3, 8.3.4
Determine Sun angle and eclipse conditions	Secs. 8.1, 8.2.4
Evaluate satellite motion as seen from the Earth	Sec. 8.4
Evaluate relative motion of satellites in a constellation or if intersatellite links are needed	Sec. 8.5
Develop mapping and pointing budgets	Sec. 8.6

Table 8-2. Mission Geometry Areas of Analysis. OCDM = *Orbit & Constellation Design & Management* [Wertz, 2009]. See also Table D-1.

Subject	Where Discussed	Summary Tables	Derivation
Fundamentals of mission geometry — what the spacecraft sees	Secs. 8.0, 8.1	App. D, Inside rear cover	OCDM Sec. 6.1*
Appearance of the Earth from space	Sec. 8.3.2	—	OCDM Sec. 9.1.2
Ground track computations	Sec. 8.4	—	OCDM Secs. 8.3.3, 9.3
Earth coverage = Target and ground station appearance and motion	Secs. 8.4, 10.5	Tables 8-11, 8-12, App. C, Inside rear cover	OCDM Secs. 9.1, 9.2, 9.5
Motion of the satellite as seen from Earth	Sec. 8.4	—	OCDM Sec. 9.4
Array Sensor field-of-view (FOV)	Secs. 8.2.2, 8.2.3	Table 8-8	OCDM Sec. 6.3.1
Shapes and angles projected onto the Earth and as seen from space	Sec. 8.3.3	—	OCDM Sec. 9.1.3
Projection of FOV onto the Earth	Sec. 8.3.4	Table 8-10	OCDM Sec. 9.1.4
Sun geometry** (Sun interference, eclipses, power and thermal analysis)	Secs. 8.1.1, 8.2.4	—	OCDM Sec. 6.1
Relative and apparent motion of other satellites	Sec. 8.5	Table 8-13	OCDM Chap. 12
Spacecraft-to-spacecraft viewing	Sec. 8.5	—	OCDM Sec.10.1
Constellation design and analysis	Sec. 10.9, OCDM Chap. 13	Table 10-24	OCDM Chap. 13
Measurement singularities and accuracy analysis	OCDM Chap. 7	—	OCDM Chap. 7
Mapping and pointing budgets	Sec. 8.6	Tables 8-16, 8-17, Figs. 8-53, 8-54	OCDM Sec. 5.5

* *Full Sky Geometry* is spherical geometry in which the angles and arc lengths can go from 0 to 360 deg. thus resolving the quadrant ambiguities that make classical spherical geometry challenging to implement. The basic formulas are derived and applications discussed in OCDM Chap. 6.
** Also used for inertial targets (i.e., space telescopes).

360 deg. This alleviates the problem with classical spherical trig of having to keep track of multiple quadrants in trying to solve spherical triangles. *Space Mission Analysis and Design, 3rd edition* [Wertz and Larson, 1999] gives an overview of the application of spherical geometry to problems of space mission analysis but provides only a top-level introduction. Shuster [1993] gives an excellent and complete assessment of coordinate transformations and their applications. Similarly, Seidelmann [2006] provides a very complete discussion of both coordinate frames and time systems used in essentially all astronomical and astronautical measurement systems. This has been updated in an excellent summary by Kaplan [2005]. From a historical perspective, Nathaniel Bowditch's *American Practical Navigator* [2010] provides one of the most complete discussions of spherical geometry as applied to terrestrial navigation. A large number of authors from the mid-twentieth century [Brink, 1942; Palmer et al., 1950; Small, 1952; and Newcomb, 1906] provide derivations of the general rules of spherical geometry. Note, however, that these rules are less complete and substantially less useful for computer implementation than those which are introduced here and given in detail in App. A of OCDM.

8.1 Introduction to Space Mission Geometry

As listed in Table 8-2, many space mission analysis problems (orbit, attitude, and payload related) are fundamentally geometrical. In analyzing the behavior of spacecraft sensors and the data they produce, we are more interested in what the sensors see than the forces which move either the spacecraft or the objects that we are looking at. For example, in analyzing star camera data, we are interested in where the stars are relative to some object we are observing, but have little interest in their distances or the forces that keep the stars moving about the center of the galaxy.*

Most spacecraft sensors measure only angles. Fundamental to this angles-only geometry is the concept of the *celestial sphere*, illustrated in Fig. 8-1, which is the apparent bowl of the sky, such as that seen by anyone on a dark starry night in a large open area. We know that the distances to the Moon, planets, and stars are dramatically different. But this is not relevant to looking at them with our eye or a camera nor to building clock drives for telescopes or analyzing astronomical photographs. Points on the celestial sphere such as stars, planets, or the outline of the horizon, represent directions in space irrespective of distance. Just as unit vectors, which are also used to represent direction, have a length of one, the celestial

* I apologize to our astronomy customers and colleagues. We really are interested in the motion of the stars. It's just that this motion makes relatively little difference for most practical problems in space mission engineering. Note, however, that the assumption that the stars are essentially fixed begins to break down as we try to achieve more accurate measurements. As high accuracy becomes more and more important, the relative motion of the stars themselves and of the inertial coordinate frame becomes a critical part of spacecraft measurements.

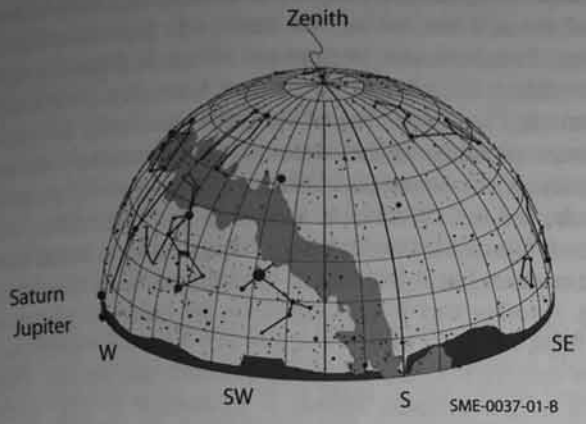


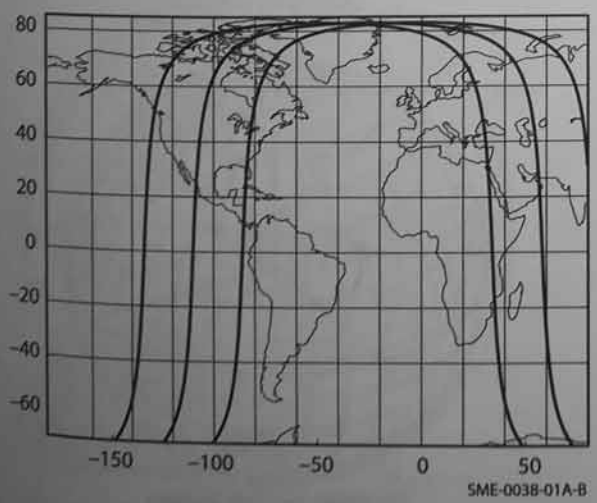
Fig. 8-1. The Celestial Sphere is the Bowl of the Sky Centered on the Observer who is Assumed to be at the Center. Points on the sphere, such as planets or the outline of the Milky Way, represent directions in space, irrespective of distance.

sphere is assumed to have a radius of one so that only the angular coordinates on the sphere are relevant.

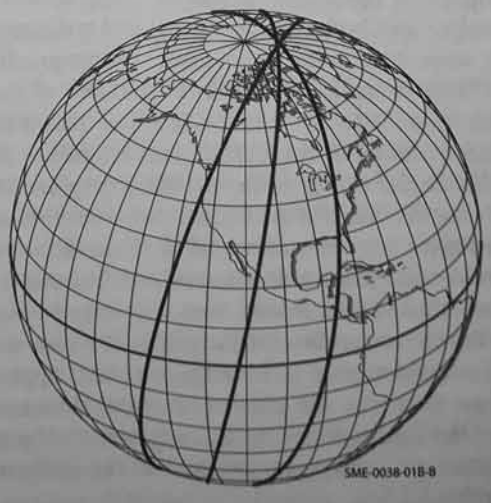
Another way to think of Fig. 8-1 is as the dome of a planetarium as seen from the outside. The planetarium projector and, more importantly, the observer, are at the center of the sphere. We would like to put ourselves at the center of the sphere as well, but that is significantly more difficult to draw. The principal inconvenience in looking at the celestial sphere from the outside is that east and west are reversed in orientation. Thus, in Fig. 8-1, east is to the right of south and, for those who are amateur astronomers, Orion and Canis Major are reversed in their orientation as we would normally see them from "inside" the celestial sphere. If you think of the view from inside the sphere, east will be to the left of south and west will be to the right, as they should be.

How we choose to display the celestial sphere, the Earth, or any spherical surface depends on our objectives. No representation is perfect. Figure 8-2 illustrates the two most common alternatives. On the left is a typical latitude-longitude plot of the Earth's surface common in many ground station displays. On the right is an orthographic projection which we call a *globe plot*. We have drawn the same satellite ground trace on both. The lat-long plot has the advantage of allowing us to see the entire Earth at one time but the disadvantage of distorting shapes and sizes. Is the ground trace on the left a circle? In looking at it, we cannot easily tell because of the distortions introduced by the plot itself. The shapes of figures in a lat-long plot tell us more about the plot projection than about the shapes we're looking at. In the globe plot on the right, we only see half of the surface of the world. But shapes and directions are much more easily distinguished. It is clear in the plot on the right that the orbit is basically circular and we can easily envision the Earth rotating under the orbit in inertial space such that the orbit traces out a slowly moving spiral over the Earth's surface.

Figure 8-3 illustrates comparable alternative representations of the celestial sphere showing a rainbow in the sky and a large, nearby balloon with a gondola attached. As always, the observer is assumed to be at the center of the sphere such that he or she is looking at the balloon and rainbow against the background of the sky. In Fig. 8-3A, it is clear that both the rainbow and the balloon are round. We could easily measure both the diameter and the location of the center. Figure 8-3B contains the identical information plotted in an azimuth-elevation plot such that the horizon is along the bottom, and the *zenith*, or point directly overhead, is at the top. The fundamental problem with the Az-El plot is that it simply doesn't convey as much information to the person who is looking at it. Is the balloon round? Which has the larger diameter, the balloon or the rainbow? We could, of course, analyze Fig. 8-3B to determine at least some of the answers. The arc of the rainbow is 100 deg across; the

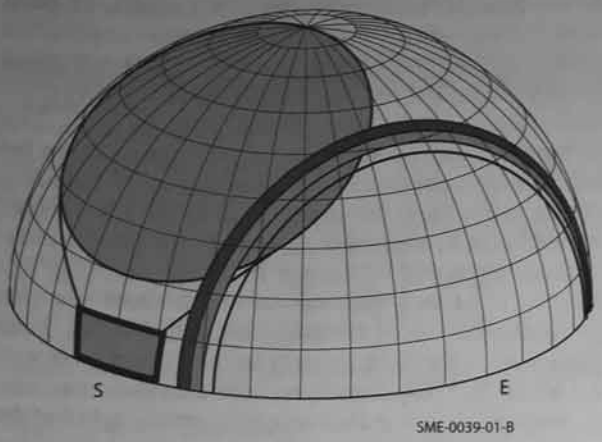


(A) Latitude-Longitude Plot

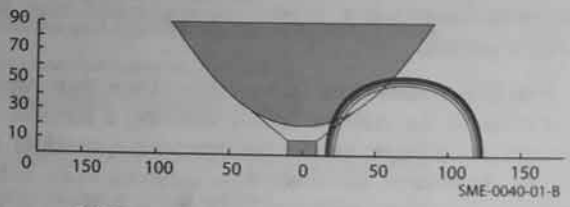


(B) *Orthographic Projection* or *Globe Plot*

Fig. 8-2. Alternative Representations of the Earth's Surface. Our objective is to use the globe plot on the right not as a projection of the surface, but as a "drawing" of a globe. We want to do our arithmetic on the surface of the globe itself and not on any projection.



(A) Globe Plot of a Balloon and a Rainbow



(B) Two-Dimensional Azimuth/Elevation of the Same Scene

Fig. 8-3. The Celestial Sphere Provides Distortion-Free Geometry on the Sky. The key to using the plot in (A) is to think of it as a sphere with the observer at the center and to work on the curved surface of the sphere.

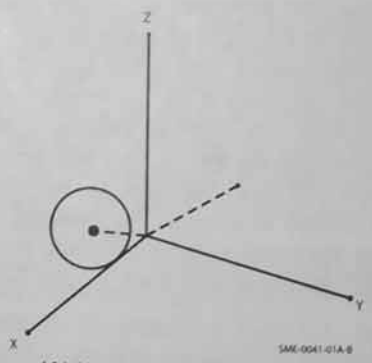
diameter of the balloon, if we believe it to be round goes from an elevation of 20 deg to 90 deg and, therefore, is only 70 deg in diameter. Thus, the rainbow is larger. Nonetheless, this is not immediately apparent from the plot and it takes some work to find what we can see immediately by simply looking at the picture in Fig. 8-3A.

The key issue is that we don't want to use a projection at all. We want to work directly on the curved surface of the Earth or of the celestial sphere. This allows us to completely avoid both the conceptual and arithmetic distortions associated with projections. Consequently, we want to think of the globe plot as a "picture" of a sphere and work directly on the sphere itself. For example, look at the intersections of the latitude and longitude lines in Fig. 8-2B. In the center of the drawing they are at right angles to each other. At the edges, they intersect at very oblique angles. However, if we think of this figure not as a flat projection, but as a drawing of a globe, and see it as round and three dimensional, then it is easy to perceive that on the curved surface of the globe, the intersections are still at right angles at the edge, and will appear that way when we rotate the globe to move them nearer the center of the figure. Working on the globe itself provides truly distortion free geometry, and is the **only** way to achieve this.*

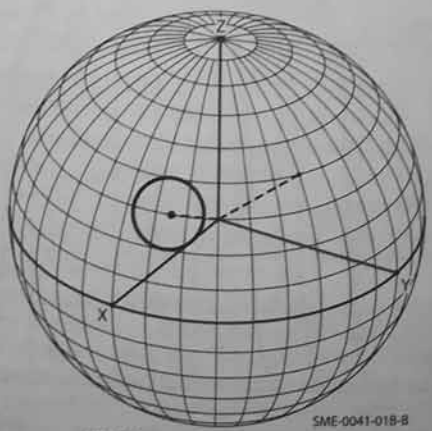
* For a discussion of alternative projections, see, for example, Canters and Decleir [1989].

Most scientists and engineers are familiar with vector analysis and the use of unit vectors to represent directions. Two such unit vectors are shown in a typical representation of a three dimensional Cartesian coordinate frame in Fig. 8-4A. One of the difficulties with this type of representation is that it is difficult to interpret unambiguously. For example: What is the radius of the small circle, and is it centered to the left or the right of the X-Z plane? Given the coordinates of the points, we could find the answer, but it is difficult to interpret from the drawing. Another way to conceptualize the celestial sphere is to think of it as a plot of the tips of all possible unit vectors centered on the origin of a coordinate frame. This idea of the celestial sphere is illustrated in Fig. 8-4B where the coordinate axes and vectors are identical to those in Fig. 8-4A, but we have now added a sphere of unit radius centered on the origin. In the spherical plot, we are interested only in the points on the surface representing the tips of the axes and the unit vectors. In this plot, it is clear that the small circle has a radius of 10 deg, and its center is 15 deg over and 30 deg up from the X axis. While the points plotted in Fig. 8-4A and Fig. 8-4B are identical, we can learn considerably more from the spherical coordinate plot in Fig. 8-4B.

We will use globe plots frequently throughout the chapter and the book. However, in most cases, we will



(A) Unit Vector in 3-D Space



(B) Point on Unit Sphere

Fig. 8-4. Alternate Representations of Unit Vectors. Global geometry plots provide an unambiguous interpretation of directional geometry.

omit the origin, the axes and other lines inside the sphere and draw simply the points on the surface of the sphere itself. Nonetheless, these points represent directions in space just as the unit vectors do. The celestial sphere is simply a good way to think about directional geometry much as vectors are a good way to think about forces and velocity. Doing the mathematics as vectors or in spherical geometry gives the same results. We are interested here in the angular measurements themselves, and the information they contain, not in how they are processed.

Finally, we have discussed thus far principally the celestial sphere as seen by an observer on the ground. However, our principal interest is what is seen from space. Consequently, as illustrated in Fig. 8-5, we need to introduce the concept of the *spacecraft-centered celestial sphere*, showing the direction to objects seen from a spacecraft. The principal difference between this and one for an observer on the ground is that on the ground we are looking only at the sky overhead, and the celestial "sphere" is really just a hemisphere overhead. Spacecraft, on the other hand, see the entire sky, although portions of it may be blocked by the Earth or another planet. While the globe plot approach eliminates the distortion of map projections, it only shows half the sky. Therefore, in most of our plots, we will try and show the half of the sky containing the objects of most interest for the problem at hand. But we need to keep in mind that there is always a second half as well and that for the spacecraft, the celestial sphere is the sky both "above" and "below" the spacecraft.

The differences between the spacecraft-centered celestial sphere and the tradition Az-El plots can be seen in Fig. 8-6. The figures on both the top and bottom show the Earth as seen from a low-Earth orbiting spacecraft. How large is the Earth? Is it round? Where is the center? Both plots contain the same information, but these questions

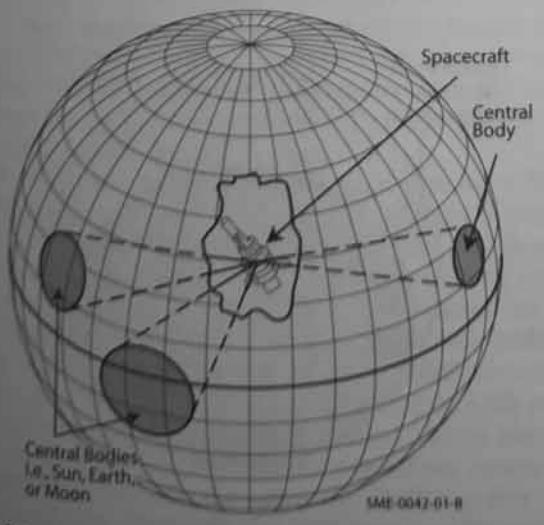
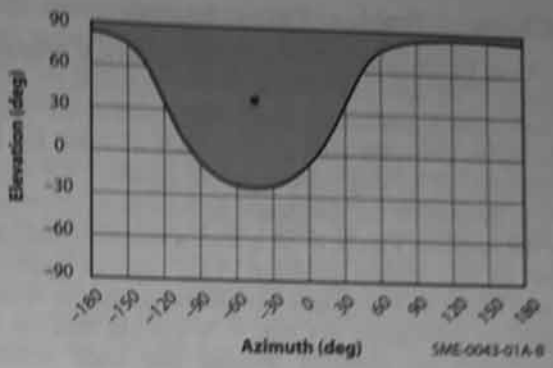
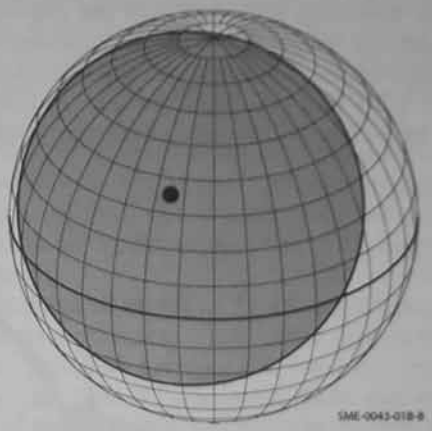


Fig. 8-5. The Spacecraft-Centered Celestial Sphere is the Bowl of the Sky Centered on the Spacecraft. The "observer" at the center may be an instrument, the center of the spacecraft, or any point in space.



(A) Az-El Plot



(B) Globe Plot

Fig. 8-6. The Disk of the Earth as Seen From Low-Earth Orbit. Globe plots of the sky provide distortion-free geometry as seen from the spacecraft.

are not easily answered from an Az-El plot because of the distortion of the plot itself. In effect, we are seeing the consequences of the plot rather than the reality of what is seen. In the globe plot, these problems go away. We have eliminated the distortion by plotting directly on the sphere. Here it is clear that the disk of the Earth is indeed round, has a radius of approximately 60 deg, and is centered about 35 deg above the equator of the plot.

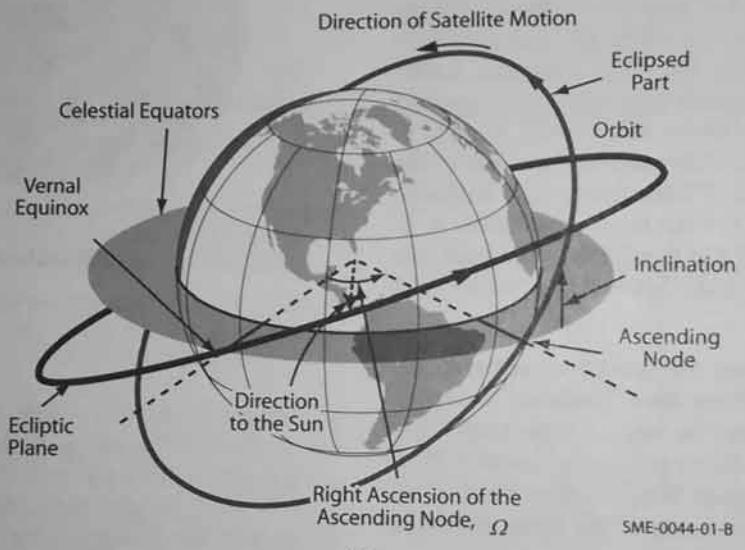
As with unit vectors, the underlying coordinate frame for globe plots can represent a variety of coordinate frames, either fixed or moving, depending on the problem we are interested in analyzing. For the celestial sphere for an observer on the Earth, the two most common coordinate frames are one with the equator along the horizon and the pole at the zenith, or one which is aligned with celestial coordinates with the North Star at the pole and the celestial equator along the globe's equator. Both of these can be used in spacecraft-centered celestial coordinates as well. In addition, common coordinate frames for space mission analysis include spacecraft-fixed coordinates, in which the axes are fixed in the spacecraft frame of reference, and axes which are momentarily fixed relative to the stars, relative to the direction of the Earth, or relative to some target being observed.

8.1.1 Sample Problem: Eclipse Analysis

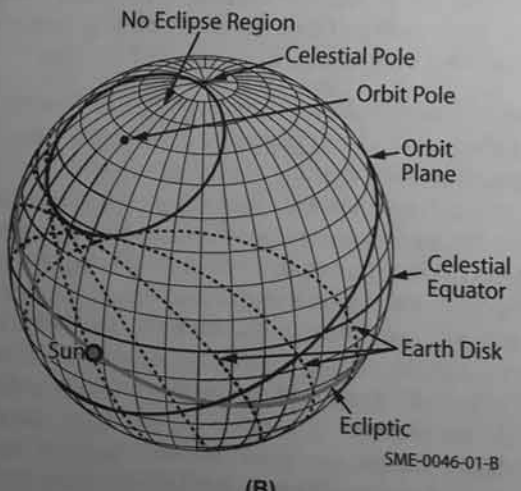
As an example of the utility of the spherical geometry approach, we will analyze the eclipse duration for a satellite in low-Earth orbit. Specifically, assume we have a satellite in a circular orbit at an altitude of 1,000 km and inclination of 32 deg relative to the equator. We would like to determine the frequency and duration of eclipses for the satellite. Figure 8-7A shows a standard isometric view of the eclipse problem. This type of approach would allow us to set up a simulation using the various equations introduced in Chap. 9. We could then run the simulations for different seasons and node crossing times, and determine a statistical distribution. However, we get no numerical estimates from the drawing itself, and no physical insight into what is happening until we have completed the simulation and subsequent analysis.

Figure 8-7B provides better insight by plotting the problem on the spacecraft-centered celestial sphere in inertial coordinates. By inertial coordinates we mean that the sphere is attached to the fixed stars, such that the celestial pole (near the North Star) is at the pole of the coordinate system and the celestial equator is along the coordinate equator. The *ecliptic*, or path of the Sun in the sky, is inclined 23.5 deg to the equator, and is also shown on the plot. The orbit plane is inclined at 32 deg to the equator. Note that the orbit plane is the plane of the spacecraft's orbit for an observer at the center of the Earth. However, for an observer on the spacecraft, it can also be thought of as the orbit of the Earth around the spacecraft.

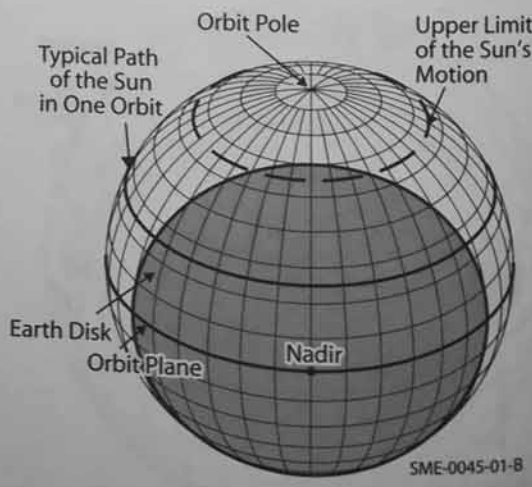
If I am sitting on the spacecraft, I see the Earth going around me relative to the background stars, in the same path as an observer at the center of the Earth would see the spacecraft move against the same stars, only 180 deg out of phase. At any given time the apparent disk of the Earth is a circle on the celestial sphere, centered at a



(A)



(B)



(C)

Fig. 8-7. Eclipse Geometry. (A) Traditional representation of eclipse geometry. (B) Eclipse geometry on *spacecraft-centered celestial sphere* improves results. (C) Eclipse geometry on *Earth-referenced spacecraft-centered celestial sphere provides* the most information and quantitative results. See text for discussion. Note that in (A) the center of the Earth is at the center of the picture, whereas in (B) and (C) the spacecraft is at the center of the sphere.

Table 8-2◀, Fig. 8-7, Eq. 8-0

point somewhere along the orbit plane. As we will determine in Sec. 8.3, the angular radius of the Earth as seen from 1,000 km altitude is approximately 60 deg [see Eq. (8-26)]. The outline of the disk of the Earth at a series of locations along the orbit is shown by the sequence of dotted circles of 60 deg radius centered on the orbit plane. As the Earth moves around the spacecraft (as seen from the spacecraft) it will sweep across the sky and from time to time cover the Sun, thus causing an eclipse to occur from the perspective of the spacecraft. It is the frequency and duration of these eclipses that we would like to determine. Notice on Fig. 8-7B that as the Earth moves along the orbit plane, there will be a region in the vicinity of the orbit pole which is never covered by the disk of the Earth. This "no eclipse region" centered on the orbit pole and a similar one, centered at the opposite pole, will be 30 deg in radius, because the angular diameter of the Earth is 60 deg at this altitude. Whatever is in either of these "polar caps" will not be eclipsed, including the Sun, Moon, and some set of stars. For the sample plotted in Fig. 8-7B, the Sun is not in the "no eclipse region," and therefore, an eclipse will occur at some period during the orbit.

We can learn somewhat more about the eclipses from Fig. 8-7B. The Sun moves along the ecliptic which is inclined 23.5 deg to the celestial equator. The orbit plane is at 32 deg to the equator but may be oriented with the node (i.e., where the orbit plane crosses the equator) at various places along the equator, depending both on how the spacecraft was launched and the motion of the node as described in Chap. 9. Wherever the node is, the orbit plane will still be inclined at 32 deg to the equator. Since the ecliptic is inclined at 23.5 deg to the equator, the most that the ecliptic can be inclined to the orbit plane will be 23.5 + 32 = 55.5 deg. The Earth's angular radius at the assumed altitude is 60 deg. Consequently, at this altitude and inclination, the Sun can never be in the "no eclipse region" surrounding either orbit pole. Thus, at 1,000 km and 32 deg inclination, there will **always** be an eclipse on every orbit, at all seasons, irrespective of the longitude or time at which the satellite is launched. Although it would still be difficult to calculate the duration of a particular eclipse, we have gained substantial insight just by the way we have constructed the figure.

We can further increase our insight into the eclipse problem by making the simple coordinate transformation shown in Fig. 8-7C. Here, we plot the Earth and Sun on the spacecraft centered celestial sphere in *Earth referenced coordinates*, i.e., in coordinates centered on the spacecraft in which the direction of the Earth remains fixed. Many spacecraft fly with one axis fixed toward the direction of the Earth, and one axis toward the orbit pole. This is the coordinate frame we would like to use, although, since we are only interested in eclipse geometry, we don't care whether the spacecraft itself is in this attitude or not. On the figure, the orbit pole is at the pole of the plot, and the orbit plane is along the equator. *Nadir*, or the direction of the center of the Earth, is marked by a dot on the plane. Because our coordinate frame is fixed in the direction of the Earth, the apparent disk of the Earth is fixed and has been shaded in the figure.

In this figure we are at the center of the sphere sitting on the spacecraft, facing the Earth, and watching the Sun and the stars go by behind it. The coordinate frame we have chosen has an axis fixed in inertial space in the direction of the orbit pole. In order to maintain the other axis in the direction of the Earth, the coordinate frame is rotating in inertial space, at one rotation per orbit in order to keep the Earth's disk fixed in the coordinate frame. In this coordinate frame, objects fixed in space will appear to rotate once per orbit as seen in Fig. 8-7C. The stars, the Moon, and the Sun will all move in small circles centered on the orbit pole as the coordinate frame rotates to follow the Earth. The typical path of the Sun in one orbit is shown as the heavy solid line. The Moon and the stars all follow similar paths. Depending on where the Sun is along the ecliptic, and the orientation between the ecliptic and the orbit plane, the Sun will move slowly (over the course of months) up and down on the plot but can never get more than 55.5 deg above or below the orbit plane, as shown by the dashed line at 35 deg from the orbit pole.

As the Sun moves in this coordinate frame, eclipses occur whenever the Sun is behind the disk of the Earth, i.e., within the shaded region. The maximum eclipse occurs when the Sun is in the orbit plane and will cover 120 deg of the orbit or 1/3 of the orbit period. Since the orbit period at this altitude is approximately 105 min (see Sec. 9.1 or the tables on the inside rear cover), the maximum eclipse duration for a satellite in a circular orbit at 1,000 km will be 35 min. The minimum eclipse will occur when the Sun is at the upper limit of the Sun's motion, at the dashed line. Here, the Sun covers 60 deg in azimuth as can be seen by the figure. Consequently, the minimum eclipse duration will be approximately half of the maximum duration, or about 17 min.

Using Fig. 8-7C, we can calculate not only the maximum and minimum duration of the eclipses, but also get substantial physical insight into how they occur. Notice that in the vicinity of the orbit plane, the eclipses remain long and only gets to be shorter as we get very close to the extremes. Therefore, if we were to ask for the average eclipse duration, we would not be far off by saying that it was 35 min. The average eclipse duration is very close to the maximum. We have now gained substantial physical insight into the problem. If we were to increase the orbit inclination by 5 deg, the minimum eclipse would drop to zero, but there would be little impact on the average eclipse. Simply by constructing the plot, we are able to gain considerable insight into the problem, and to provide at least some quantitative estimates. We will use a plot very similar to this to calculate eclipse durations in the example in Sec. 8.2.4, after we've introduced spherical geometry computations.

8.1.2 Coordinate Systems

In principle, any coordinate system will work for space mission geometry problems. However, as we have

Table 8-3. Common Coordinate Systems Used in Space Applications. Figure 8-8 shows the most common ones. See Table 8-6 in Sec. 8.2 for further information on the first two rows.

Coordinate Name	Fixed with Respect to	Center	Z Axis or Pole	X Axis or Ref. Point	Applications
Celestial (Inertial)	Inertial space*	Earth† or spacecraft	Celestial pole	Vernal equinox	Orbit analysis, astronomy, inertial motion
Earth-Fixed	Earth	Earth	Earth pole = celestial pole	Greenwich meridian	Geolocation, apparent satellite motion
Spacecraft-Fixed	Spacecraft	Defined by engineering drawings	Spacecraft axis toward nadir‡	Spacecraft axis in direction of velocity vector‡	Position and orientation of spacecraft instruments
Local Horizontal (RPY)*	Orbit	Spacecraft	Nadir	Perpendicular to nadir and orbit pole (toward velocity vector)	Earth observations, attitude maneuvers††
Ecliptic	Inertial space	Sun	Ecliptic pole	Vernal equinox	Solar System orbits, lunar/solar ephemerides

* Actually rotating slowly with respect to inertial space. See text for discussion.
† Earth-centered inertial coordinates are frequently called *GCI (Geocentric Inertial)*.
‡ For on Earth-oriented mission.
** Also called *LVLH (Local Vertical/Local Horizontal)*, *RPY (Roll, Pitch, Yaw)* or *Local Tangent Coordinates*.
†† An important and environment characteristic of roll pitch, yaw coordinates is that they do not commute (i.e., the result depends on the order in which they are specified). For a complete discussion See OCDM, pg. 154-5.

seen in the previous section, selection of the proper reference coordinates is often critical to developing a good physical understanding, obtaining analytic expressions for key mission parameters, and reducing the probability of error.

To define a *coordinate system* for space applications, we must first specify two characteristics: the location and motion of the origin and what the coordinate system is fixed with respect to (i.e., its attitude and attitude rate). Typically, we choose the Earth's center as the origin for problems in orbit analysis or geometry on the Earth's surface and the spacecraft's position for problems concerning the apparent position and motion of objects as seen from the spacecraft or analysis of payload observations. Occasionally, coordinates are centered on a specific spacecraft instrument when we are interested not only in viewing the outside world but also in obstructions of the field of view by other spacecraft components. Typical ways to fix the rotational orientation of a coordinate system are with respect to inertial space, to the direction of the Earth or some other object being viewed, to the spacecraft, or to an instrument on the spacecraft. Table 8-3 lists the most common coordinate systems used in space mission engineering and their applications. These are illustrated in Fig. 8-8. If you are uncertain of the coordinate system to select, I recommend beginning with the following:

- Earth-centered inertial for orbit problems
- Spacecraft-centered local horizontal for missions viewing the Earth
- Spacecraft-centered inertial for missions viewing anything other than the Earth

For a more detailed discussion of coordinate systems and a complete set of coordinate transformations, see OCDM Sec. 6.1.2 and App. B.

Unfortunately, the inertial coordinate system which everyone uses, called *celestial coordinates*, is not truly fixed with respect to *inertial space*—that is, the mean position of the stars in the vicinity of the Sun. Celestial coordinates are defined by the direction in space of the Earth's pole, called the *celestial pole*, and the direction from the Earth to the Sun on the first day of spring, when the Sun crosses the Earth's equatorial plane going from south to north. This fundamental reference direction in the sky is known as the *vernal equinox* or *First Point of Aries.** Unfortunately for mission engineers, the Earth's axis and, therefore, the vernal equinox precesses around the ecliptic pole (the pole of the Earth's orbit) with a period of 26,000 years. This *precession of the equinoxes* results in a shift of the position of the vernal equinox relative to the fixed stars at a rate of 0.014 deg/year. Because of this slow drift, celestial coordinates require a date to accurately define the position of the vernal equinox. The most commonly used systems are *2000 coordinates* and *true of date*, or *ToD*. The latter coordinates use the same epoch as the orbit parameters and are traditionally used for spacecraft orbit analysis. The small corrections required to maintain ToD coordinates are conveniently done by standard computer subroutines. They are important for precise numerical work, but are not critical for most problems in mission analysis.

Once the underlying coordinate system has been defined most work is done in either rectangular (x, y, z) or

* The position of the vernal equinox in the sky has been known since before the naming of the constellations. When the zodiacal constellations were given their current names several thousand years ago, the vernal equinox was in Aries, the Ram. Consequently the zodiacal symbol for the Ram, ♈, or sometimes a capital T (which has a similar appearance), is used for the vernal equinox. Since that time the vernal equinox has moved through the constellation of Pisces and is now slowly entering Aquarius, ushering in the "Age of Aquarius."

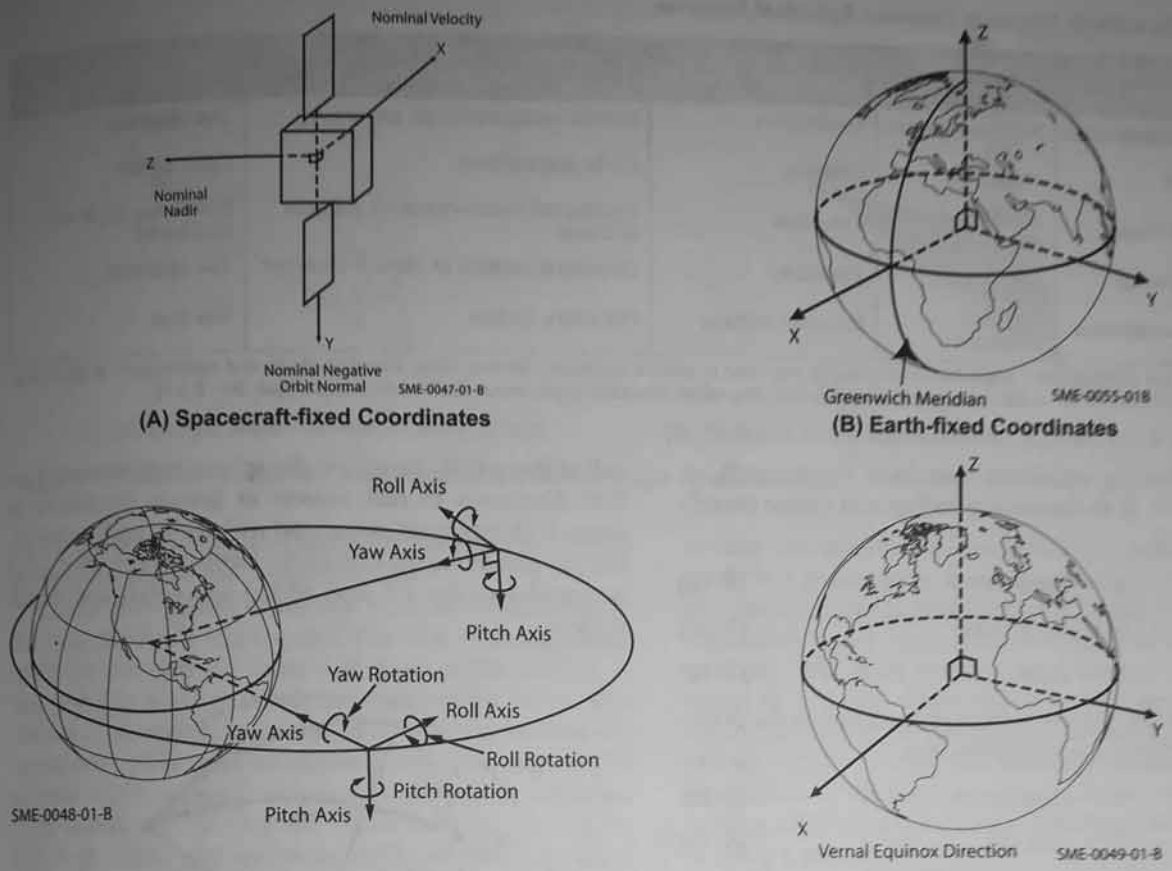


Fig. 8-8. Coordinate Systems in Common Use. See Table 8-4 for characteristics.

spherical (ϕ, θ) coordinates or some combination of the two. A given problem may be easier in one or the other, but so long as the computations are done correctly and we do not introduce new singularities, the accuracy of the solution will be the same in either set of coordinates. Transforming between them is both conceptually and numerically straightforward. As shown in Fig. 8-9, the +X axis is normally toward the reference point on the equator, and the +Z axis is toward the positive or North pole. The great circles through the poles and perpendicular to the equator are called *meridians*. The meridian through any point on the sphere determines the *azimuth coordinate*, ϕ, of that point. *Azimuth* is the equivalent of longitude on the Earth's surface, and is measured along the equator. The azimuth is also equivalent to the rotation angle measured counterclockwise about the pole from the reference point to the point in question. The second coordinate which determines the position of any point on the sphere is the *elevation*, θ, or latitude component. It is the arc-length distance above or below the equator. The *co-latitude* or *co-elevation* is the arc length from the positive pole to the point in question. Small circles at a constant elevation are called *parallels*. Because a parallel of constant elevation is not a great circle (except at the equator), the arc length along a parallel will be larger than the arc-length separation between two points. (For example, 1 deg of arc length on the surface of the Earth

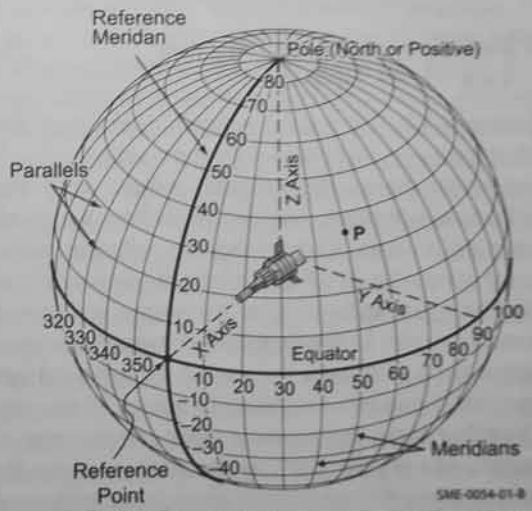


Fig. 8-9. Definition of a Spherical Coordinate System on the Unit Sphere. The point **P** is at an azimuth, ϕ, of 50 deg and elevation, θ, of 35 deg, normally written as (50°, 35°).

is about 111 km. Thus, at the equator 2 points 1 deg apart will be 111 km apart, but two points 1 deg apart in longitude that are the same latitude, but not at the equator, will be less than 111 km apart and will be very close to each other near the pole.) As Table 8-4 shows, several spherical coordinate systems in common use have special names for the azimuth and elevation coordinates.

Table 8-4. Coordinate Names in Common Spherical Systems

Coordinate System	Azimuth Coordinate	Elevation Coordinate (Z Axis)	Applications	Typical Origin
Celestial Coordinates	Right ascension	Declination	*Inertial measurements, astronomy*	*The observer*
Earth-Fixed	Longitude	Latitude	*Earth applications*	*Earth center*
Spacecraft-Fixed	Azimuth or clock angle	Elevation	*Spacecraft measurements, attitude analysis*	*Reference mark on spacecraft*
Local Horizontal	Azimuth	Elevation*	*Directions relative to central observer*	*The observer*
Ecliptic Coordinates	Celestial longitude	Celestial latitude	*Planetary motion*	*The Sun*

* Also used are *zenith angle* = angle from point directly overhead to point in question = 90 deg *minus elevation angle*, and *nadir angle* = *angle at the observer from the center of Earth to point in question* = 90 deg *minus elevation angle minus Earth central angle* (See Sec. 8.3.1)

The following equations transform the azimuth, ϕ, and elevation, θ, to the corresponding unit vector coordinates (x, y, z):

$$x = \cos\phi \cos\theta \qquad (8\text{-}1a)$$

$$y = \sin\phi \cos\theta \qquad (8\text{-}1b)$$

$$z = \sin\theta \qquad (8\text{-}1c)$$

Similarly, to transform from unit vectors to the corresponding spherical coordinates, use:

$$\phi = \text{atan2}(x, y) \qquad (8\text{-}2a)$$

$$\theta = \text{asin}(z) \qquad (8\text{-}2b)$$

where atan2 is the computer function with output defined over 0 to 360 deg and the asin function is evaluated over −90 deg to +90 deg.

8.1.3 Basic Spherical Geometry and Unit Vector Formulas

Section 8.1.1 discussed the broad concept of the celestial sphere and explained why we want to use it. This section reintroduces the celestial sphere as a formal mathematical construct, and describes how we go about working on it. App. D provides the specific rules for spherical geometry. We discuss here the major differences between plane and spherical geometry, and introduce spherical triangles as the principal computational tool.

Figure 8-10 illustrates the use of the celestial sphere to represent directions to objects in space. These objects may be either very close, such as other components of the spacecraft, or very far, such as the surface of the Earth, planets, or stars. As can be seen from the spherical triangle in Fig. 8-10, there are three types of objects on the celestial sphere. Points, lines, and angles on the sphere represent directions, arc lengths and rotation angles as seen from the observer at the center. *Points on the celestial sphere* represent directions in space, such as the direction to the Sun, Moon, or the rotation axis of the spacecraft. The direction opposite a given direction is called the *antipode* or *antipoint* and frequently has a −1 superscript. Thus S^{-1} is the direction opposite the Sun, and is called the *antisolar point*. *Nadir* is the direction to the center of the Earth. The direction opposite nadir is called the *zenith*. Points on the sphere may represent either directions to real objects or simply directions in space with no object associated with them, such as the direction parallel to the axis of the Earth (the *celestial pole*) or parallel to the +Z axis of the spacecraft coordinate system.

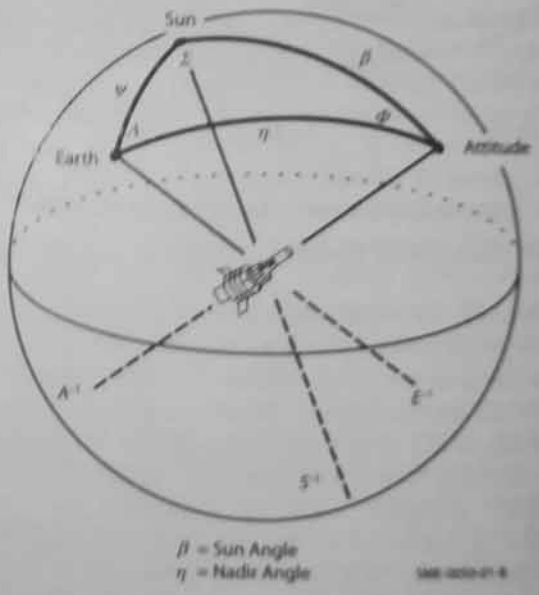
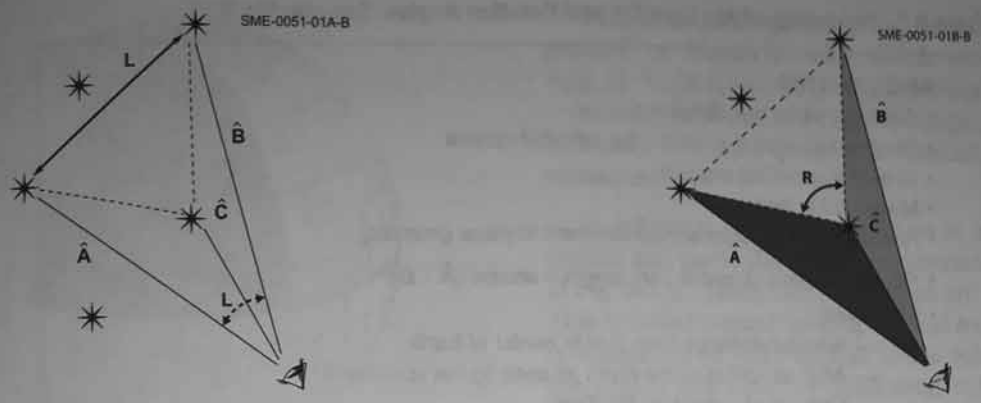


Fig. 8-10. The Spacecraft-Centered Celestial Sphere Specifies Direction in Space. The sides of the triangle are arc lengths. The angles of the triangle are rotation angles.

Specifying a direction in space requires two independent numbers such as the azimuth and elevation of a star in a spherical coordinate system.* "Straight lines" on the celestial sphere are great circle arcs. A *great circle* on the celestial sphere is any circle which divides the sphere into two equal hemispheres. Any part of a great circle is called an *arc* or *arc segment* and is equivalent to a straight line segment in plane geometry. Thus, the shortest path connecting two stars on the celestial sphere is the *great circle arc* connecting the stars. Any two points

* In contrast, orientation or attitude requires three numbers—the direction to the center, and the rotation about the axis.



(A) Arc Length Measurement, L, from, Â to B̂ (B) Rotation Angle Measurement, R, from, Â to B̂ about Ĉ

Fig. 8-11. Distinction between Arc Length and Rotation Angle Measurements.

which are not antipoints of each other determine a unique great circle arc on the celestial sphere.

Given three points in the sky, such as the direction to the Sun, the Earth and the spin axis of a spinning spacecraft, we can connect them with great circle arcs (η, β, and ψ in Fig. 8-10) to construct a *spherical triangle*. **The angles Λ, Σ, and Φ at the vertices of the spherical triangle are called *rotation angles* or *dihedral angles*.** The lengths of arc segments and the size of rotation angles are both measured in degrees. However, as shown in Fig. 8-11, these are distinctly different measurements. The arclength represents the side of the spherical triangle, and is equal to the angular separation between two points as seen on the sky, such as the angle between the Sun and the center of the Earth. The rotation angle, which is always measured about a point on the sphere represents the angle in a spherical triangle and is equal to the dihedral angle between two planes for the observer in the center. For example, assume that we see the Earth, Sun, and Moon on the spacecraft sky. The arc length between the Sun and the Moon is the angular separation between them as seen by the observer at the spacecraft. The rotation angle about the Earth between the Sun and the Moon is equal to the angle between two planes. The observer, Earth, and Sun form the first plane, and the observer, Earth and Moon form the second.

An alternative way to think about the difference, is that a rotation angle is the angle between the hour hand and the minute hand on a clock, or about 90 deg at 12:15. The angular length of the minute hand or the hour hand as I look at the clock are both arc length measurements. Both types of angles are important for spacecraft applications, and we must clearly understand the distinction between them. Table 8-5 lists the properties of these two basic measurement types.

Most work in spherical geometry is done with great circle arcs which are the equivalent of straight lines in plane geometry. However, another important figure on the celestial sphere is the *circle* which is the locus of all points equidistant from a given point. (The center and all of the points on the circle are assumed to be on the celestial sphere.) If the radius of the circle is 90 deg, the circle divides the sphere into two equal hemispheres and the circle is a *great circle*, as discussed above. Otherwise it is a *small circle*. In this context a "small circle" is not necessarily small. It could have a radius of 89.9 deg, for example. We could also construct a small circle with a radius of 110 deg, which would be the same as a small circle centered on the antipode of the center with a radius of 180 – 110 = 70 deg. The lines of constant latitude on the surface of the Earth are small circles, except for the equator which is a great circle, and the poles themselves which are points.

In plane geometry, straight lines which are parallel do not intersect. However, this is not possible with great circles on the sphere. Consider, for example, the equator on a globe and any two lines of constant longitude, which are great circles perpendicular to the equator. Irrespective of the separation between the lines of longitude, they always intersect at the two poles. Any two great circles on the sphere will either lie on top of each other, or intersect at two points 180 deg apart. The orange slice-shaped figure formed by the intersection of two great circles is called a *lune*. The lune has two sides, both of which are 180 deg long. The rotation angle between the sides, at either of the intersection points is equal to the angle between the two planes that form the great circles and pass through the center of the sphere.

Because great circle arcs (which are the spherical equivalent of straight lines) always intersect, triangles in spherical geometry are substantially more common than in plane geometry problems, and take a more important role in computations on the sphere. Nearly all problems in spherical geometry can be reduced to the solution of spherical triangles. Thus, spherical trigonometry becomes the principal computational method for most space mission geometry problems. These computations are described further in OCDM Sec. 8.3, and the rules of spherical trigonometry are given in App. D in this volume. Note that one of the most common errors in spherical trigonometry is to construct a "triangle" in which one of the sides is a small circle arc rather than a great circle arc. Just as all of the sides of a plane triangle must be straight lines, all of the sides of a spherical triangle

Table 8-5. Properties of Arc Lengths and Rotation Angles. See also Fig. 8-11.

ARC LENGTHS

- Angle between two directions
- Segment of a great circle on the celestial sphere
- Side of a spherical triangle
- Measured in degrees
- Analogous to a distance measurement in plane geometry
- For unit vectors $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$, arc length = $\arccos(\hat{\mathbf{A}} \cdot \hat{\mathbf{B}})$
- Examples:
  - — Angular distance from Sun to center of Earth
  - — Angular radius of the Earth as seen by the spacecraft
  - — Latitude of a point on the Earth
- Expressed as "Angle from **A** to **B**" or "Arc length between **A** and **B**"

ROTATION ANGLES

- Angle about a point on the sky from one arc to another
- Dihedral angle between two planes
- Angle in spherical triangle
- Also measured in degrees
- Analogous to an angular measurement in plane geometry
- For unit vectors $\hat{\mathbf{A}}$, $\hat{\mathbf{B}}$, and $\hat{\mathbf{C}}$,

$$\text{rotation angle from } \hat{\mathbf{A}} \text{ to } \hat{\mathbf{B}} \text{ about } \hat{\mathbf{C}} = \arctan\left[\frac{\hat{\mathbf{C}}\cdot\left(\hat{\mathbf{A}}\cdot\hat{\mathbf{B}}\right)}{\hat{\mathbf{A}}\cdot\hat{\mathbf{B}}-\left(\hat{\mathbf{C}}\cdot\hat{\mathbf{A}}\right)\left(\hat{\mathbf{C}}\cdot\hat{\mathbf{B}}\right)}\right]$$

- Examples:
  - — Difference in longitude between two points on the Earth
  - — Angle between the orbit plane and the Earth's equator (i.e., the orbit inclination)
  - — Angular amount that the Earth rotates on its axis in a fixed amount of time
- Expressed as "Rotation angle from **A** to **B** about **C**"

The arc length and rotation angle between two points are equal if and only if the rotation angle axis is 90 deg from both points.

must be great circle arcs. We cannot apply plane trigonometry directly to a piece of pie. Similarly,

The sides of spherical triangles can ***not*** *be small circles.*

Although most of the work in mission geometry is done with spherical triangles and small and great circles, there are other more complex figures as well. The *dual-axis spiral* is described in Sec. 10.4, the *analemma* is introduced in Sec. 8.5, and the *rhumb line* is discussed in OCDM Sec. 6.3.

Another concept in spherical geometry that is different from plane geometry is illustrated in Fig. 8-12, in which we have constructed a spherical triangle using the equator and two lines of longitude. The intersections of the longitude lines with the equator are both right angles, such that the sum of the angles of the triangle exceeds 180 deg by an amount equal to the angle at the pole. The sum of the angles of any spherical triangle is always larger than 180 deg. The amount by which this sum exceeds 180 deg is called the *spherical excess* and is directly proportional to the area of the spherical triangle. Thus, small triangles on the sphere have a spherical excess near zero, and are very similar to plane triangles. Large spherical triangles, however, are very different as in Fig. 8-12 with two right angles.

A *radian* is the angle subtended if I take a string equal in length to the radius of a circle, and stretch it along the circumference. Similarly, if I take an area equal to the square of the radius, and stretch it out on the surface of a sphere (which requires some distortion, since the surface of a sphere cannot fit on a flat sheet), the resulting area is called a *steradian.* Since the area of a sphere is $4\pi r^2$, there are 4π steradians in a full sphere, and 2π steradians in a hemisphere. **These units are convenient for area problems, because in any spherical triangle the spherical excess, expressed in radians, is equal to the area of the triangle expressed in steradians.** In general, the area, A, of any spherical polygon (a figure with sides which are great circle arcs) expressed in steradians is given by:

$$A = \Sigma - (n-2)\,\pi \tag{8-3}$$

where n is the number of sides, and Σ is the sum of the rotation angles expressed in radians. This is used, for ex-

Table 8-5, Fig. 8-11◀, Eq. 8-3

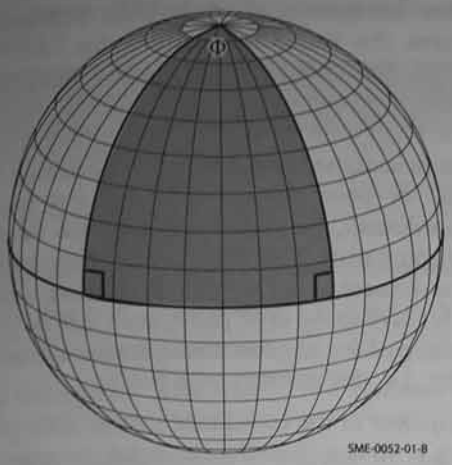


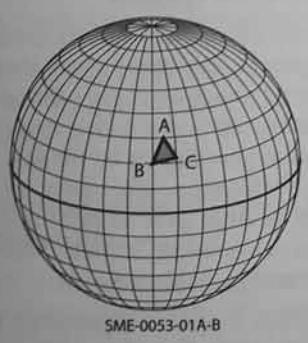
Fig. 8-12. The Sum of the Angles in a Spherical Triangle is Always Greater than 180 deg. The sum of the angles minus π radians is called the *spherical excess* and equals the area of the triangle in steradians.

ample, in measuring the angular area subtended by an array sensor, such as a digital camera on the spacecraft.

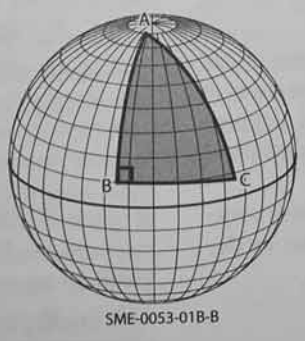
Figure 8-13 shows a variety of spherical triangles. Note that all of the triangle sides are great circle arcs. Figure 8-13A is a nearly plane triangle, for which the sum of the angles is approximately 180 deg and plane geometry is a close approximation. Figure 8-13B is called a *right spherical triangle* because the angle at B is a right angle. Just as right plane triangles have particularly simple relationships among the angles and sides, right spherical triangles also have exceptionally simple relationships between the sides and angles. These are expressed by *Napier's rules* which are written out in App. D. Right spherical triangles are common in mission geometry problems, and provide simple, straightforward solutions for many problems associated with angular measurements.

In spherical trigonometry, there is a second type of special and particularly simple spherical triangle shown in Fig. 8-13C. Here, side A-B has an arc length of 90 deg. This is called a *quadrantal spherical triangle*. An equally simple set of rules applies to the relationship among the angles and sides in quadrantal spherical triangles. These are also summarized in App. D. Between them, right and quadrantal spherical triangles provide solutions to most of the problems encountered in mission analysis.

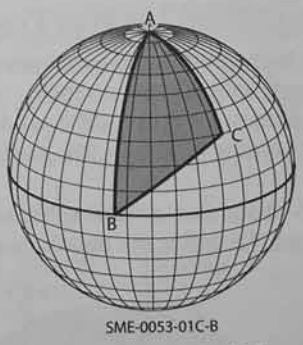
Figure 8-13D shows an *obtuse isosceles triangle* with two equal rotation angles larger than 90 deg. Clearly, this cannot exist in plane geometry. A similar strange triangle for plane geometry is Fig. 8-13E which shows an *equilateral right triangle* in which all three angles and all three sides are 90 deg. This triangle represents 1/8 of the surface of the celestial sphere, and has an area of 0.5 π steradians, which can be seen either by examination or from the spherical excess rule. Finally, Fig. 8-13F shows a very large spherical triangle. Note that this triangle is remarkably similar in appearance to the small triangle in 8-13A. This is because the triangle can be thought of either as a small triangle with three angles of approximately 60 deg, or as a very large one, with three angles of



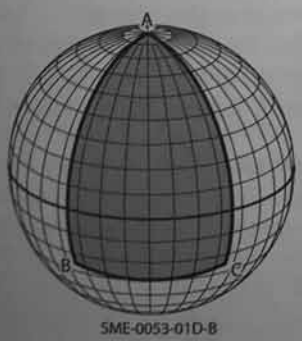
(A) Nearly Plane Triangle



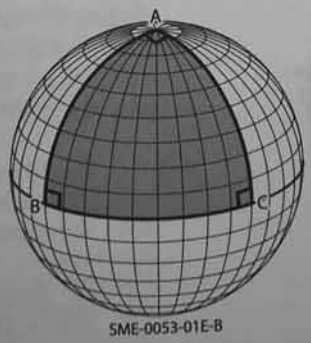
(B) Right Spherical Triangle



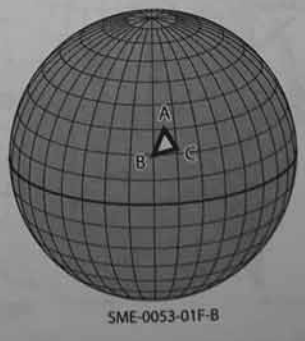
(C) Quadrantal Spherical Triangle



(D) Obtuse Isosceles Triangle



(E) Equilateral Right Triangle



(F) Very Large Spherical Triangle

Fig. 8-13. Types of Spherical Triangles. See text for discussion.

approximately 300 deg. That is, the area between A, B, and C can be thought of either as the inside of a small triangle or as the outside of a large one which covers all of the surface of the sphere except for the small area between A, B, and C. In this large spherical triangle, the rules of spherical geometry still apply and can be used in very convenient fashions as described further in OCDM.

The ideas above lead to defining three major areas of geometry on the surface of the sphere:

- *Spherical trigonometry* is concerned with the solution of triangles on the celestial sphere. It is used for almost all computations
- *Spherical geometry* is concerned with triangles plus great and small circles and their intersection. This expands spherical trigonometry to cover angular area and circles contained in one hemisphere, such as a typical antenna pattern.
- *Global geometry* is concerned with more complex figures on the sphere, and relationships covering the entire sphere. This expands spherical geometry to include problems covering the entire celestial sphere, such as constellation design for satellites or the ground track of a satellite in a circular orbit. This ground track is called a *dual axis spiral* because the satellite is rotating about the orbit pole and the orbit pole is rotating once per day with respect to the surface of the Earth (See Sec. 10.4.)

8.2 Applications

8.2.1 Solving Mission Geometry Problems

The key to most spherical geometry problems is to define the coordinate system correctly. Given the correct set of coordinates, most problems can be solved algebraically very quickly, using the techniques described below.

Figure 8-14 shows a general spherical geometry coordinate system. To define the coordinates, we start by defining the orientation of either pole. For geographic coordinates, this would ordinarily be the Earth's poles. For a coordinate frame attached to the stars, it is often taken to be the *celestial pole* which is the projection of the Earth's pole onto the celestial sphere. For an observer on the surface of the Earth, the pole is frequently taken as the *zenith* or point straight overhead. For spacecraft-fixed coordinates, the pole is typically the +Z axis defined by some mark inscribed on the spacecraft body. It is also convenient to use spacecraft-centered coordinates defined relative to the orbit plane, such that the pole is in the direction of the angular momentum of the spacecraft in its orbit, i.e., the positive orbit pole.

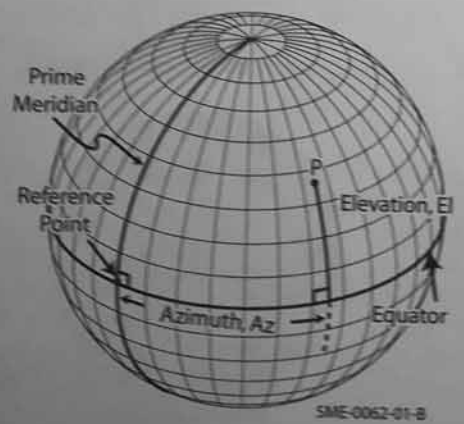


Fig. 8-14. The Spherical Geometry Coordinate System. See Table 8-6 for definitions of principal components. (See also Fig. 8-9.) P is at (65 deg, 35 deg).

The *equator* of the coordinate system is the great circle which is 90 deg from the two poles. In order to fully define the coordinates we need to define where to start our azimuthal coordinate measured along the equator. This is determined by a *reference point* along the equator, or equivalently, by a *prime meridian,* which is one of the lines of constant azimuth going through a well-defined point, not necessarily on the equator. In geographic coordinates, the prime meridian goes through the Royal Greenwich Observatory, near downtown London, and is often called the *Greenwich meridian*. It represents the starting point for longitude on the surface of the Earth. On the celestial sphere, relative to the stars, we use one of the two *equinoxes*, or intersections between the ecliptic (the path of the Sun) and the equator. The one chosen as the zero point on the sky is the *vernal equinox*, which is the location of the Sun on the first day of spring. (The other intersection is called the *autumnal equinox*.) See Table 8-3 for other common systems.

The two coordinates in a general spherical coordinate system are called *azimuth*, measured along the equator, and *elevation*, measured above and below the equator. In geographic coordinates, these are called *longitude* and *latitude*, and in celestial coordinates on the sky, they are *right ascension* and *declination*. *Meridians* are the arcs of great circles running from pole to pole, and latitude lines are small circles at a constant latitude. Naming conventions are summarized in Table 8-6.

Virtually all problems in spherical geometry are done by solving spherical triangles, both because great circles always intersect, and because any three components, including the three rotation angles provide a solution. The specific algebraic rules for solving all types of spherical triangles are summarized in App. D. The rules to be applied to specific types of figures are as follows.

- **Right spherical triangles**: If one of the rotation angles is a right angle, and any two of the remaining components (rotation angles or sides) are known, then the three unknown components can be determined immediately by Napier's rules* which are a very simple set of relationships between the sines and cosines of the various components. These are explicitly defined for all possible combinations of right spherical triangles in App. D.

Table 8-5◀, Fig. 8-14, Eq. 8-3◀

Table 8-6. Names of Principal Components in Common Spherical Coordinate Systems. See Table 8-3 for other coordinate systems.

Component	Coordinate System		
	Generic	Geographic	Celestial
Fixed with Respect to	Various	Earth	"Fixed" stars
Pole	Pole	North and South poles	Celestial poles (projection of Earth's poles onto the sky)
Equator	Equator	Equator	Celestial equator
Reference Meridian	Prime meridian	Greenwich meridian	0 of right ascension
Reference Point	Not named	Not named	Vernal equinox
Azimuthal Coordinate	Azimuth	Longitude	Right ascension
Vertical Coordinate	Elevation	Latitude	Declination

- **Quadrantal spherical triangles:** If one of the sides is equal to 90 deg, and any two of the remaining components are known, then the three unknown components can be determined by rules analogous to Napier's rules for right spherical triangles.
- **Oblique (general) spherical triangles**: Use either the law of cosines or law of sines set out in App. D. The law of cosines is preferred, since the inverse cosine is uniquely defined between 0 and 180 deg.
- **Area of any figure**: Use the spherical excess rule for polygons and the formulas in App. D for the areas of small circles or fractions of small circles.

To illustrate the applications of spherical geometry, we will present two sample problems:

- The field of view of a "square" Sun sensor
- Eclipse duration in a circular orbit (corresponding to the sample problem set out conceptually in Sec. 8.1.1.)

8.2.2 Example 1: Field of View of a "Square" Sun Sensor

Sensors used on spacecraft are often said to have square or rectangular fields of view, i.e., an "8 × 8 deg star sensor" or "128 × 128 Sun sensor." Unfortunately, "square" and "rectangular" are not as well defined as we might like in terms of sensor fields of view on the sky. If the edge of a sensor field of view is defined by a straight line, such as an array of pixels on a focal plane, then this straight line will be projected onto the spacecraft sky as an arc of a great circle. Two parallel edges on the field of view will be projected as two great circle arcs on the sky. We can, for example, think of these two edges as being projected out along two meridians, or lines of longitude in a celestial coordinate system. However, since these are great circle arcs, they do not remain the same distance apart, and would ultimately intersect if extended far enough. Thus, a square or rectangular field of view when projected onto the spacecraft sky does not have edges which remain a fixed distance apart. In this section we define what is meant by a "square" field of view, and determine the parameters of such a field of view projected onto the sky and then extend the computations to rectangular fields of view in Sec. 8.2.3.

To make our example specific, we will look at a "128 deg square" field of view Sun sensor, and ask what is the actual field of view projected onto the sky. Figure 8-15 shows how such a sensor could be constructed. We begin with a flat square array 8.2 cm on a side. We mount the array parallel to an opaque sheet and 2 cm behind it. As shown in the figure, we put a pinhole in the center of the opaque sheet such that it is directly above the center of the array. We use a pinhole because this provides perfect geometrical optics, i.e., there are no dis-

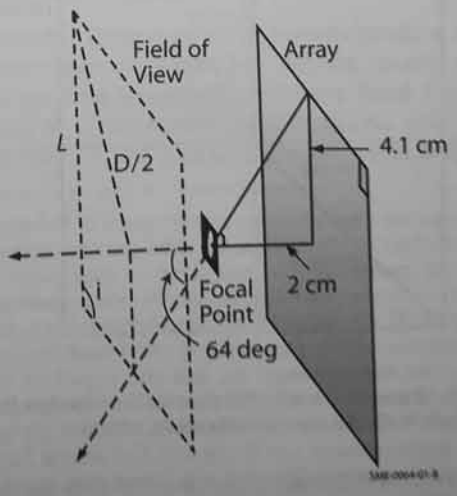


Fig. 8-15. Construction of a "128 × 128 deg" Sun Sensor. The "128 deg" is measured between opposite sides, through the center.

* Not too surprisingly, Napier's rules were introduced by the Scottish mathematician and Baron of Merchiston, John Napier, who also invented logarithms and in 1617 introduced the decimal system as it is used today. During the early part of the 20th century Napier's rules for spherical triangles were taught as a part of high school mathematics and a variety of convenient mechanisms for memorizing them were introduced. Needless to say, relatively few of today's high school students are required to do so.

tortions introduced by the pinhole itself and we are interested in creating a "perfect" sensor, although, perhaps, not a very practical one. As the array looks through the pinhole out into the sky, the angle from the central axis to the center of any one of the edges will be equal to atan (4.1 /2.0) = 64 deg, such that the field of view of the sensor is 128 deg across as measured through the center, perpendicular to any edge. This is what we traditionally mean by "128 deg square" field of view. What we would like to determine for this perfect sensor is:

- The length, L, of each side
- The angle of intersection, i, between two adjacent sides
- The length of the diagonal, D, and
- The projected area, A, of the sensor on the sky.

To show how the problem is done, we will solve it using plane geometry, vectors, and spherical geometry.

The plane geometry solution is shown in Fig. 8-16. Here the arithmetic is particularly simple. In the plane geometry approximation, the distance across the center is 128 deg. Therefore the length of each side, L, is 128 deg. The angle of intersection at the corners is 90 deg, the area, A, is $128^2 = 16{,}384$ deg^2 (equals 39.7% of the celestial sphere) and the diagonal, D, is $128\sqrt{2} = 181.0$ deg. Clearly, there is a very fundamental problem with this solution. Although our sensor may not be a practical one, it is certainly capable of being constructed. And yet, if it is constructed, it is highly unlikely that the field of view on the sky would be more than 180 deg across, since this means that the sensor would be seeing behind itself. The solution using plane geometry is simple to compute, but, unfortunately, wrong.

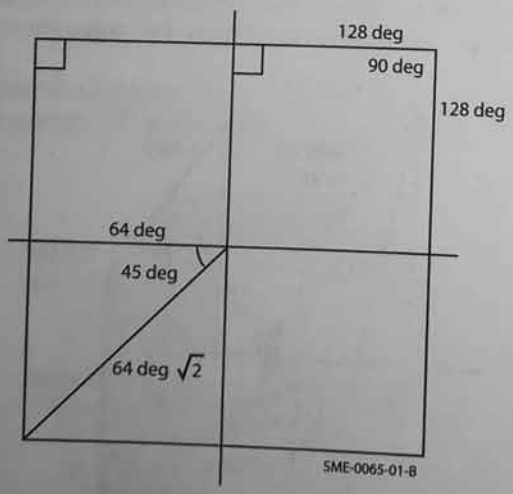


Fig. 8-16. Plane Geometry Solution to Sun Sensor Problem. The answer is simple, but, unfortunately, wrong.

Figure 8-18 illustrates a vector solution to the Sun sensor problem. Here we construct unit vectors to the center of the field of view, to the center of one edge, and to one of the corners. We then use the vector algebra illustrated

Table 8-6◀, Fig. 8-17, Eq. 8-5

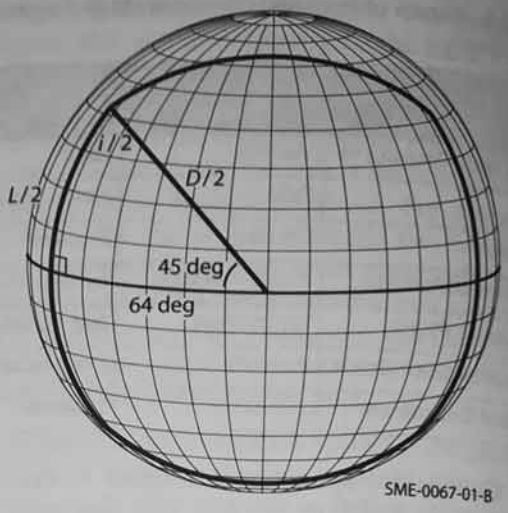


Fig. 8-17. Spherical Geometry Solution to "128 Deg Square Sun Sensor" Problem. Solution is complete, correct, and simple.

in the figure to determine the angles among the various vectors and the rotation angle formed at the corner of the sensor. The solution presented in the figure is computationally correct, but is both awkward and complex. In addition, the area of the field of view is not easily computed using vectors. Finally, the solution provides us with almost no physical insight into the shape of the field of view, or the implications of the shape for either measurements or creating overlapping fields of view with other sensors.

Figure 8-17 shows the spherical geometry solution to the Sun sensor problem. The sphere represents the celestial sphere as seen by the sensor. We construct the edges of the field of view by first defining the center axis as a point on the equator, then going 64 deg along the equator in either direction and drawing the meridians at right angles to the equator. These lines represent the sides of the field of view. The top and bottom are produced by rotating about the center by 90 deg and constructing two similar lines. This is the real shape of the sensor field of view projected onto the sky.

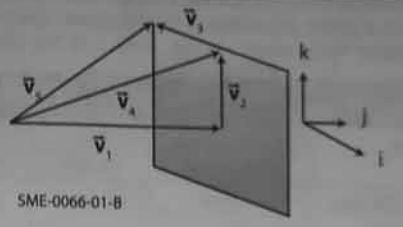
To determine the values of the parameters of interest, we first construct the diagonal going from the center to one of the corners, and the line from the center to the middle of one of the edges. The length of this line is 64 deg by the definition of our problem. The rotation angle at the center between the two edges of the triangle is 45 deg by symmetry, and the line through the center to the middle of the edge intersects the edge at a right angle, again by definition of our problem. Consequently, we have formed a right spherical triangle in which two of the components are known. Using Napier's rules from App. D, we simply write down the formulas for the various components:

$$\tan(L/2) = \sin 64 \text{ deg} \ \tan 45 \text{ deg}$$
$$L = 83.9 \text{ deg} \qquad (8\text{-}4)$$

$$\cos(i/2) = \cos 64 \text{ deg} \ \sin 45 \text{ deg}$$
$$i = 143.9 \text{ deg} \qquad (8\text{-}5)$$

$\frac{L}{2}$ = angle between $(\vec{V}_4, \vec{V}_5)$

$\frac{D}{2}$ = angle between $(\vec{V}_1, \vec{V}_5)$



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$$\cos\left(\frac{L}{2}\right) = \frac{\vec{V}_4 \cdot \vec{V}_5}{|\vec{V}_4||\vec{V}_5|}$$

$$= \frac{(2)\cdot(2)+(4.1)(4.1)}{\sqrt{(2)^2+(4.1)^2}\sqrt{(-4.1)^2+(2)^2+(4.1)^2}}$$

$$= 0.7437$$

$$\frac{L}{2} = 41.95 \text{ deg}, \quad L = 83.9 \text{ deg}$$

$$\vec{V}_1 = 2\hat{j} \quad \vec{V}_2 = 4.1\hat{k} \quad \vec{V}_3 = -4.1\hat{i}$$
$$\vec{V}_4 = \vec{V}_1 + \vec{V}_2 = 2\hat{j} + 4.1\hat{k}$$
$$\vec{V}_5 = \vec{V}_1 + \vec{V}_2 + \vec{V}_3 = -4.1\hat{i} + 2\hat{j} + 4.1\hat{k}$$

$$\frac{D}{2} = 70.97 \text{ deg}, \quad D = 141.9 \text{ deg}$$

$$\cos\left(\frac{D}{2}\right) = \frac{\vec{V}_1 \cdot \vec{V}_5}{|\vec{V}_1||\vec{V}_5|}$$

$$= \frac{4}{(2)\sqrt{(-4.1)^2+(2)^2+(4.1)^2}}$$

$$= 0.3261$$

$$I = \text{vector}\left\{\frac{|\vec{V}_5|\cdot\vec{V}_5(\vec{V}_3\times\vec{V}_2)}{|\vec{V}_5|^2(\vec{V}_3\cdot\vec{V}_2)-(\vec{V}_5\cdot\vec{V}_3)(\vec{V}_5\cdot\vec{V}_2)}\right\}$$

I = 143.9 deg

A = not normally computed with vectors

Fig. 8-18. Vector Solution to Sun Sensor Problem. The answer is correct, but both incomplete (since the area cannot be computed) and complex.

$$\tan(D/2) = \tan 64 \text{ deg} \cos 45 \text{ deg}$$
$$D = 141.9 \text{ deg} \qquad (8\text{-}6)$$

From Eq. (8-3), the Spherical Excess (8-7)

$$= 4 \times 143.9 \text{ deg} - 360 \text{ deg} \qquad (8\text{-}8)$$
$$= 215.6 \text{ deg}$$
$$= 3.763 \text{ rad}$$

$$A = 3.763 \text{ steradians}$$
$$= 29.9\% \text{ of the sphere} \qquad (8\text{-}9)$$

Here the solution is exact, simple, and gives us good physical insight into how the solution will vary with different defining parameters. The sensor is 128 deg across at the middle, but the edges are only 84 deg long. In addition, two edges which are at right angles on the focal plane array intersect at an angle of 144 deg on the celestial sphere. While the rows and columns of the physical array may be orthogonal, the projected fields of view onto the sky are not orthogonal and indeed, are highly correlated.* Rows and columns near the edge of the sensor are no longer independent measurements. The large scale of the Sun sensor exaggerates the size of the effect. Nonetheless, sensors with smaller fields of view typically require much higher accuracy which may require the spherical geometry of the sensor field of view be taken into account. Typical magnitudes of plane geometry errors for various size fields of view are given in Table 8-7. Whether the differences between plane and spherical geometry are important for any given application will depend upon the required accuracies. This table allows us to judge whether a plane geometry approximation is appropriate. In any case, the correct computations in spherical geometry are extremely simple.

8.2.3 Example 2: Measurements Made by a Rectangular Array Sensor

As shown in Fig. 8-19, there are two basic types of rotation angle measurements:

- Type I—rotation about a known axis from a known reference direction to the unknown direction
- Type II—rotation about an unknown axis between two known points

A remarkably large number of sensors produce rotation angle measurements—rotating sensors, nearly all sensors on spinning spacecraft, and many fixed 3-axis sensors such as a two-dimensional array. As discussed in OCDM Sec. 7.1, the type of measurement is determined

* *Independent* or *uncorrelated* measurements are ones that do not depend on each other. In an ordinary rectangular coordinate frame, such as the layout of array sensor, the x and y measurements are orthogonal and uncorrelated — changing the value of x has no impact on the value of y. When this same array coordinate frame is projected onto the celestial sphere, that is no longer true and the measurements are said to be *correlated* or related to each other. As we move along the top row of the array for example, both the x and y values on the celestial sphere will change. If two measurements are *fully correlated*, they are simply two different ways of measuring the same thing, such as specifying a location on the Earth by giving the latitude and the distance from the North Pole. This gives us two measurements of the north-south location and no measurement of the East-West location.

Table 8-7. Representative Errors Using Plane Geometry Approximation for Sensor Fields of View. The subscript "plane" means a plane geometry approximation. The subscript "true" means the actual spherical geometry value. In the "128 deg Square Sun Sensor" in Fig. 8-17, L is the SIDE and *i* is the ANGLE.

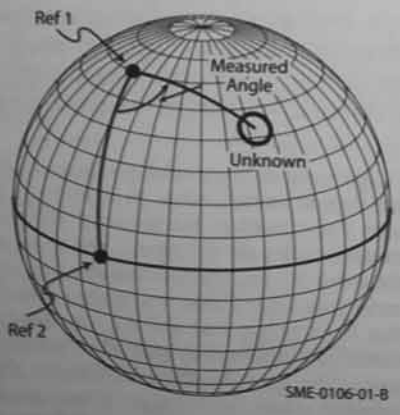
FOV size (deg)	128 × 128	64 × 64	32 × 32	16 × 16	8 × 8	1 × 1	0.1 × 0.1
Application	Sun sensor; Earth at LEO	Sun sensor	General photographs	Earth at GEO	Star sensor	Small telescope	Larger telescope
Equivalent 35 mm Camera Lens	7 mm	25 mm	50 mm	85 mm	200 mm	1.7 m	17 m
$SIDE_{plane}$ (deg)	128.00	64.00	32.00	16.00	8.00	1.00000	0.1000
$SIDE_{true}$ (deg)	83.90	55.84	30.82	15.85	7.98	0.99996	0.9999996
$ANGLE_{plane}$ (deg)	90.00	90.00	90.00	90.00	90.00	90.0000	90.000
$ANGLE_{true}$ (deg)	143.88	106.31	94.36	91.11	90.28	90.0044	90.000044

by the solution loci on the celestial sphere. It is the shape of the loci that determines how the measurements should be processed, the arithmetic associated with analyzing and combining measurements, and the impact of sensor or mission geometry on the accuracy.

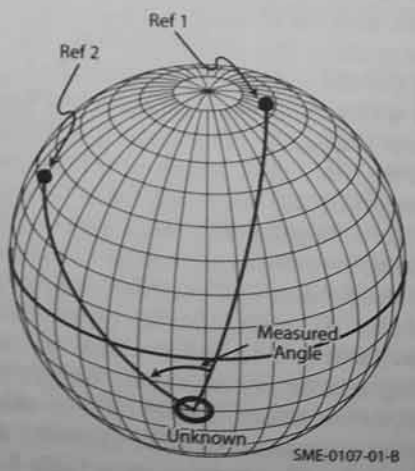
As we will demonstrate shortly, the Type I rotation angle is the basic measurement made by an array processor. In Sec. 8.2.2, we showed the projection of 128 × 128 deg field of view Sun sensor onto the celestial sphere, largely because the size of the sensor dramatically increased the impact of the spherical geometry effects and therefore made the nature of the results more apparent. Nonetheless, the same fundamental geometry and formulas apply to star sensors, payload array sensors, cameras, and any other instrument which uses a generally rectangular array with its field of view projected onto the sky. Because this is such a fundamental type of instrument, we will provide all the relevant formulas for a generic rectangular array.

As illustrated in Fig. 8-20, the basic assumption throughout is that the array is perfectly rectangular, i.e., has straight edges on the focal plane, and that the optical system itself is perfect, i.e., it behaves like a pinhole camera. Of course, a real camera adds distortion and absorbs light differently in various parts of the field of view. The effects that we are concerned with here are purely geometrical, i.e., taking a flat focal plane and projecting it onto the "curved" surface of the celestial sphere. Consequently, these effects apply to any sensor that uses a flat focal plane. This should be the starting point for any systematic calibration because it will minimize the magnitude of the variations. Larger arrays will be used to illustrate the effects more clearly, but numerical values will also be given for smaller fields of view.

Figure 8-21 shows the view projected onto the celestial sphere for the rectangular physical array illustrated in Fig. 8-20. Specifically, we assume that the focal plane array is perfectly rectangular and flat, with dimensions $2x_a$ by $2x_b$; that is, x_a is the distance from the center of the array to the edge of the array along the long axis, and x_b is the distance from the center to the edge along the



(A) Type I



(B) Type II

Fig. 8-19. Two Types of Rotation Angle Measurements. (A) In *Type I* the measurement is made about a known reference axis from a second reference to the unknown direction. (B) In *Type II* the measurement is made about the unknown axis from one known reference to another.

Table 8-7, Fig. 8-19, Eq. 8-9◀

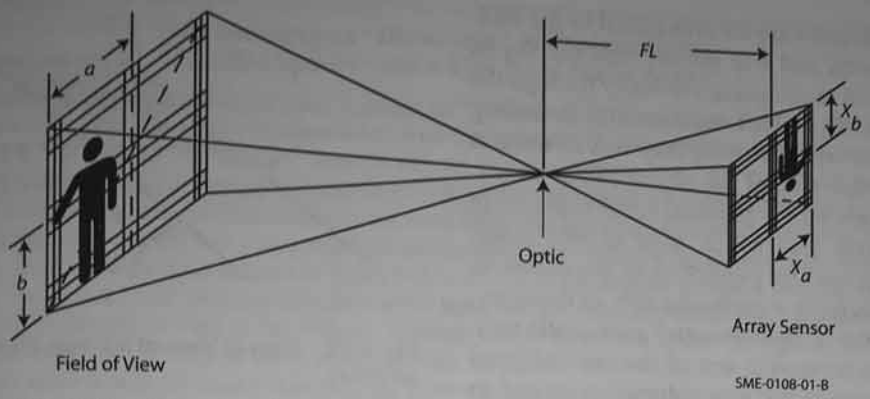


Fig. 8-20. Geometry of an Array Sensor. A rectangular sensor on the right looks through a perfect optical system. Pixels on the right are projected into space on the left and the stick figure in space is projected onto its corresponding location on the array. The projection onto the celestial sphere is shown in Fig. 8-21.

short axis. Since our camera or detector is assumed to be focused at infinity, the array is at a distance from the camera's focal point equal to the focal length, FL. Consequently, we can determine the angular dimensions of the array on the celestial sphere, a and b, by:

$$\tan a = x_a / FL \qquad (8\text{-}10a)$$

$$\tan b = x_b / FL \qquad (8\text{-}10b)$$

where a and b are now the angular dimensions projected onto the sky corresponding to the linear dimensions x_a and x_b. For example, a normal 35 mm camera produces a

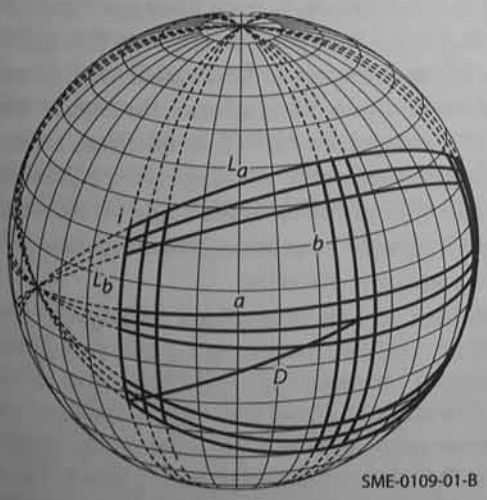


Fig. 8-21. Projection of a Rectangular Array onto the Celestial Sphere. The assumption here is that the projection is a perfect optical system. This is the image seen on the left in Fig. 8-20.

negative on the focal plane with an image size 36 mm × 24 mm. (It is called a 35 mm camera because the film is 35 mm wide, in the 24 mm direction because of the sprocket holes on either side.) This means that x_a = 18 mm and x_b = 12 mm. Projected onto the sky using a normal 50 mm focal length lens, a = atan (18/50) = 20 deg and b = atan (12/50) = 13.5 deg such that the total field of view on the sky is 40 deg wide by 27 deg high, measured through the center perpendicular to the edges.

Given a and b in Fig. 8-21, we want to determine the formulas for the edge lengths L_a and L_b, the angle of intersection of the edges at the corner, i, the distance from the corner to the center (i.e., half the diagonal), D, and the area of the rectangular array. We also wish to calculate the basic parameters for the pixel size and shape at any place within the array. **Note**: Depending upon our application, we may think of a and b either as the dimensions of the entire array or as the coordinates of a given point on the array. Thus, x_a and x_b can be either the size of the physical array on the focal plane or can be the two coordinates of some pixel within that array whose position, size and shape we are trying to calculate.

Section 8.2.2 provided the geometry and formulas for a square array. Here we will describe the general formulas for a rectangular array with the notation shown in Fig. 8-22. The important characteristic here is that a straight line on the focal plane projects onto the sky as a great circle arc. Thus the rows and columns in our perfectly rectangular array project onto the sky as a series of

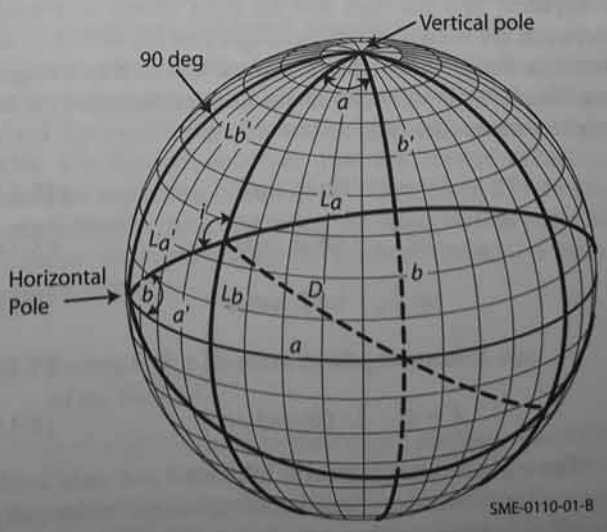


Fig. 8-22. Definition of Parameters for a Rectangular Array.

great circle arcs. The rows are all orthogonal to the middle column in the array and will project onto the sky as orthogonal to the great circle going vertically through the sensor, as shown in Fig. 8-21. Consequently, extending these great circle arcs on either side, they will all meet at a pole 90 deg on either side of the center of the sensor. Similarly, the columns on the physical focal plane array project as great circle arcs which meet at two poles, 90 deg above and below the projection of the center of the camera. By identifying a particular row in the array, we are identifying one of the horizontal great circle arcs. A particular column represents one of the vertical great circle arcs. Consequently, the row-column specification on the focal plane array corresponds to specifying two great circle arcs on the sky. The angle b represents the angular displacement of the point in question above or below the horizontal center line. The angle a represents the angular displacement of the column right or left of the vertical center line. These angles are measured along the two central axes and, therefore, are equal to the rotation angle measured at the pole of those central axes. Consequently, the measurement b implies that the pixel in question lies at a rotation angle b about the horizontal sensor pole upward from the reference axis. Similarly, a specification of a for the point in question means that the point lies along the great circle arc at rotation angle a, measured about the top or bottom pole relative to the reference axis. Thus, a and b are Type I rotation angle measurements.

Once we have recognized the nature of the array geometry, doing the arithmetic to calculate the various values is straightforward. First, note that we have constructed multiple quadrantal spherical triangles since the arc length between the horizontal and vertical poles is 90 deg, as is the arc length between either pole and the center of the field of view. Given the known quantity a, the angle in the quadrantal triangle, a', is just 90 deg – a, and similarly, $b' = 90$ deg – b. Consequently, we know two of the three rotation angles in the quadrantal triangle, and the remaining rotation angle and sides can be easily computed. We can then use the right spherical triangle between the corner of the rectangle and the center to determine the length of the diagonal. Given this triangle, we simply write down by inspection the formulas for the various components as follows:

$$\cos i = \sin a \ \sin b \tag{8-11}$$

$$\tan L_a = \tan a \ \cos b \tag{8-12}$$

$$\tan L_b = \tan b \ \cos a \tag{8-13}$$

$$\cos D = \cos L_a \cos b = \cos L_b \cos a \tag{8-14}$$

$$A = 4i - 2\pi \text{ (steradians)} \tag{8-15}$$

For a sensor array, we are interested not only in the general coordinates of various components in the array, but also in the size, shape, and area of individual pixels projected onto the sky. Here the geometry is shown in

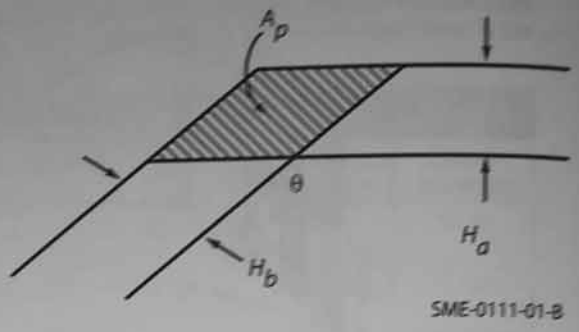


Fig. 8-23. Field of View of a Single Pixel Within a Rectangular Array.

Fig. 8-23. Because the spherical geometry effects are very small near the center, pixels near the center of the array have the shape that one would expect from plane geometry. However, this becomes more distorted as pixels get further from the center. In particular, the angular height of a pixel **along the central axes** is given by:

$$H_{a_\Delta} = \left(\Delta_{x_a}/FL\right)\cos^2 a \tag{8-16}$$

$$H_{b_\Delta} = \left(\Delta_{x_b}/FL\right)\cos^2 b \tag{8-17}$$

where H_{a_Δ} and H_{b_Δ} are the angular dimensions of the pixel in radians along the a and b directions, respectively, Δ_{x_a} and Δ_{x_b} are the linear dimensions of the pixels on the array, and FL is the focal length of the optical system. This is due simply to each pixel covering a smaller and smaller angle as it gets further from the optical axis; i.e., when I am in the center horizontally and move above or below the central horizontal axis, the angular height of each pixel will get shorter. If, in addition, I also move horizontally along a row the height of the pixel will get shorter still (by the cosine of the angle off the central axis) just as longitude lines come closer together as I move from the equator to the pole.

In addition, the angle at which the edges of the pixel intersect gets further from a right angle as it approaches the diagonal corners. In this case, the correlation angle, θ, or angle at which the edges of the pixel intersect is the same angle at which the sides of the image intersect. It is therefore equal to the angle i above. The area of the pixel, A_p, can be computed with a plane geometry approximation. Thus the pixel equations are as follows:

$$H_a = H_{a_\Delta}\cos^2 a \cos L_b \tag{8-18}$$

$$H_b = H_{b_\Delta}\cos L_a \cos^2 b \tag{8-19}$$

$$\cos\theta = \cos i = \sin a \sin b \tag{8-20}$$

$$A_p = H_{a_\Delta} H_{b_\Delta} \cos L_a \cos L_b \cos^2 a \ \cos^2 b \ / \sin\theta \tag{8-21}$$

where H_a is the angular height of the pixel in the a direction, H_{a_Δ} is the angular height of the pixel along the ref-

The Meaning of "Distortion" in Projecting the Celestial Sphere onto a Flat focal Plane Array

It is important to clearly understand what is meant by "distortion" in this context. Assume that we take a picture of a brick wall from some distance away with the optical axis of the camera pointed precisely perpendicular to the wall. In our "perfect" camera with no "optical irregularities," the picture we take of the brick wall will be an exact, though reduced, duplicate of the wall itself. The rows of bricks will line up precisely with the edges of the film. On the print, each brick will have the same area, and so on. However, for a brick that is 45 deg off the central axis both horizontally and vertically the angular area subtended by that brick, from the right hand column in Table 8-8, is 19.2% of the angular area of a brick on the optical axis. This means that I will have less than 20% of the light and less than 20% of the image intensity relative to those bricks directly in front of the camera. Another way to think of it is that the light coming from a brick in the corner of the field-of-view, which appears very small when we look at it, must be spread out to cover the same area on the negative as the light from a brick right in front of us that covers a much larger angular area.

That parallel lines get "closer together" as they get further away is known to anyone that does much drawing or photography. This is usually expressed as "parallel lines converge at infinity" and is most easily seen in railroad tracks or the edges of a highway. Again, this is expressed mathematically in the columns labeled H_a and H_b in Table 8-8.

Finally, the correlation between "orthogonal" directions seen off-axis is much less apparent in everyday life, but can be observed. Try standing very close to a wall and looking at the corner where two walls and the ceiling come together. This intersection represents three orthogonal axes. As your eye gets very close to the wall, the two perpendicular edges of the wall you're next to begin to form nearly a straight line at the corner. A bug walking at the corner will appear to be moving in nearly the same direction if he is going up the seam between the two walls or toward you along the seam between the wall and the ceiling. This is a manifestation of the correlation between orthogonal components expressed in the "i" column in Table 8-8.

Although these effects are apparent only in extreme cases, they are still present in any array image. Whether they are important depends on the precision required. Thus, Table 8-8 should be used as a guide to the size of the effects so that you determine what needs to be taken into account in any given problem.

erence axis through the center of the sensor, and A_p is the area of the pixel.

Table 8-8 gives both the array size and pixel size and shape for a variety of rectangular arrays. This gives a sense of the geometric distortion that comes about, not from any optical irregularities, but simply from the process of projecting the celestial sphere onto a flat image plane.

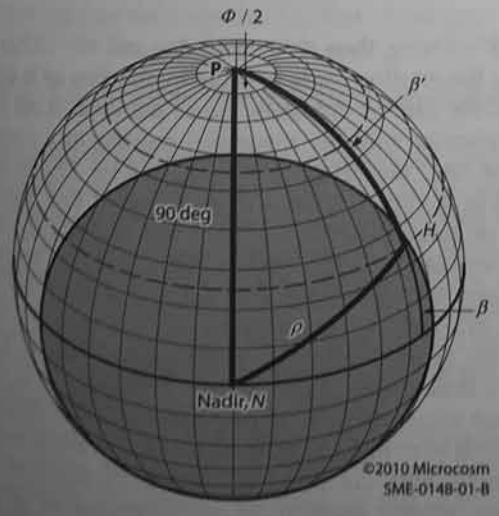


Fig. 8-24. Computation of Eclipse Duration. Compare with Fig. 8-7C in Sec. 8.1.1.

Note that by the time I've gotten to the corner of a 60 deg × 60 deg sensor, the half-length of the sides is only 41 deg and the correlation angle at which the sides intersect is nearly 140 deg, such that a "rectangular" pixel is very distinctly diamond shaped. The distance from the center of the sensor is only 68 deg. Equally important, the pixel that was 1 deg by 1 deg at the center is now only 0.2 deg × 0.2 deg. In addition to its rather elongated shape, the angular area has now been reduced to 5% of the area of one of the pixels at the center. It is this effect that produces the dimmer image at the sides and corners of photographs or electronic images.

Whether or not the spherical geometry calculations are necessary for your particular problem can be determined by examining the size of the effects shown in Table 8-8. Notice, however, that the correct spherical geometric equations in Eqs. (8-11) through (8-21) are extremely simple, such that doing the computation correctly, rather than incorrectly in plane geometry, is very straightforward.

8.2.4 Example 3: Calculating the Duration of an Eclipse

This example returns to the eclipse problem illustrated in Fig. 8-7 in Sec. 8.1.1. The computational geometry is shown in Fig. 8-24. Recall that the altitude of the satellite was 1,000 km, which implies that the angular radius of the Earth $\rho = 59.8$ deg [see Eq. (8-26) in Sec. 8.3.1].

Table 8-7◀, Fig. 8-24, Eq. 8-21◀

Live Calc

Table 8-8. Coordinates and Dimensions on the Celestial Sphere for a 1 deg × 1 deg Pixel (Measured at the Center of the Array). See Fig. 8-22 and Fig.8-23 for the definition of variables. The most important parameters are i for which a value different from 90 deg indicates a correlation between the sensor rows and columns and A_p (the projected area of the pixel) which shows the reduction in coverage by pixels away from the center. Note that this assumes a pixel size that gives a 1 deg × 1 deg field of view on the optical axis of the sensor. The dimensions and area in the 3 right columns can be proportionally adjusted for other pixel sizes.

a	b	L_a	L_b	D	$\theta = i$	H_a	H_b	A_p
0	0	0.00	0.00	0.00	90.00	1.00	1.00	1.000
5	0	5.00	0.00	5.00	90.00	0.99	1.00	0.989
5	5	4.98	4.98	7.05	90.44	0.99	0.99	0.977
10	0	10.00	0.00	10.00	90.00	0.97	0.98	0.955
10	5	9.96	4.92	11.14	90.87	0.97	0.98	0.945
10	10	9.85	9.85	14.00	91.73	0.96	0.96	0.913
15	0	15.00	0.00	15.00	90.00	0.93	0.97	0.901
15	5	14.95	4.83	15.74	91.29	0.93	0.96	0.892
15	10	14.78	9.67	17.78	92.58	0.92	0.94	0.863
15	15	14.51	14.51	20.75	93.84	0.90	0.90	0.818
30	0	30.00	0.00	30.00	90.00	0.75	0.87	0.650
30	5	29.91	4.33	30.28	92.50	0.75	0.86	0.644
30	10	29.62	8.68	31.12	94.98	0.74	0.84	0.627
30	15	29.15	13.06	32.48	97.44	0.73	0.81	0.600
30	30	26.57	26.57	39.23	104.48	0.67	0.67	0.465
45	0	45.00	0.00	45.00	90.00	0.50	0.71	0.354
45	5	44.89	3.54	45.11	93.53	0.50	0.70	0.352
45	10	44.56	7.11	45.44	97.05	0.50	0.69	0.345
45	15	44.01	10.73	45.99	100.55	0.49	0.67	0.335
45	30	40.89	22.21	49.11	110.70	0.46	0.57	0.281
45	45	35.26	35.26	54.74	120.00	0.41	0.41	0.192
60	0	60.00	0.00	60.00	90.00	0.25	0.50	0.125
60	5	59.91	2.50	60.03	94.33	0.25	0.50	0.125
60	10	59.62	5.04	60.13	98.65	0.25	0.49	0.124
60	15	59.13	7.63	60.29	102.95	0.25	0.48	0.122
60	30	56.31	16.10	61.29	115.66	0.24	0.42	0.111
60	45	50.77	26.57	63.43	127.76	0.22	0.32	0.089
60	60	40.89	40.89	67.79	138.59	0.19	0.19	0.054

For specificity we will calculate the duration of the eclipse when the Sun is 25 deg out of the orbit plane, i.e., $\beta = 25$ deg in Fig. 8-24.

We use the pole, **P**, of the coordinate system, which is also the orbit pole, as one of the corners of a spherical triangle. This is one of the most common spherical geometry computation techniques, and the pole of the coordinate system is often chosen such that it can be a corner of a spherical triangle. The second corner is the point, **H**, at which the Sun crosses the horizon of the Earth's disk, i.e., at the beginning or end of the eclipse, and the third point is the direction to nadir, **N**, or the center of the Earth. The angle from the orbit pole to nadir is 90 deg by definition, so we have constructed a quadrantal spherical triangle. The angular radius of the Earth, $\rho = 59.8$ deg, is known. Similarly, the side $\beta' = 90\,\text{deg} - \beta = 65$ deg is also known. Consequently, we can use Napier's rules for quadrantal triangles to determine any of the three rotation angles. In particular, we would like to determine $\Phi/2$ at the orbit pole, which is given by:

$$\cos(\Phi/2) = \cos\rho / \sin\beta' \qquad (8\text{-}22)$$

If $\beta' = 65$ deg, then $\Phi/2 = 56.3$ deg and $\Phi = 112.6$ deg. This is the rotation angle covered by the Sun as it passes behind the disk of the Earth. Since the orbit is assumed to be circular with the spacecraft moving at a constant angular rate, the eclipse duration will be $112.6/360 = 31.3\%$ of the orbit. With an orbit period of 105 min, this gives an eclipse duration of 32.9 min for this example. The formulation is extremely simple, and can be easily evaluated for other angles of the Sun out of the orbit plane. In addition, we could take into account the 0.5 deg diameter of the Sun itself, or the angular height of the Earth's atmosphere, and calculate, for example, the time at which eclipse begins, when it reaches full darkness, or the length of twilight at either end. None of these computations requires complex simulations or detailed orbit analysis. Simple geometrical calculations give us good insight into both the duration and character of eclipses seen from space.

Table 8-8, Fig. 8-24◀, Eq. 8-22

For eccentric orbits the situation is more complex, because the eclipse fraction varies depending upon where the eclipse occurs relative to apogee and perigee. Nonetheless, we can relatively easily determine the eclipse extremes that occur at apogee and perigee. First, note that even though the orbit is noncircular, the projection of the orbit plane onto the celestial sphere, and therefore, the apparent path of the Earth will still be a great circle. The differences from a circular orbit are twofold. The Earth moves at a nonuniform rate along this great circle and the apparent size of the Earth's disk changes with changing altitude. As shown in Fig. 8-25 the no-eclipse region about either pole will change from circular to an approximately oval shape with a much larger no-eclipse region in the direction of apogee. The maximum eclipse at both apogee and perigee will, of course, occur when the Sun is in the orbit plane, and is given by:

$$E_{A/P} \approx \left(\frac{\rho_{A/P}}{180}\right)\left(\frac{R_E + H_{A/P}}{a}\right) \qquad (8\text{-}23)$$

$$a = R_E + (H_A + H_P)/2 \qquad (8\text{-}24)$$

where $E_{A/P}$ is the fraction of the orbit period spent in eclipse at apogee and perigee when the Sun is in the orbit plane, $\rho_{A/P}$ is the angular radius of the Earth at apogee or perigee in degrees, R_E is the radius of the Earth, $H_{A/P}$ is the height of apogee and perigee and a is the semimajor axis. The angular radius of the Earth can be evaluated using Eq. (8-25) in Sec. 8.3.1. The effect of the Sun not being in the orbit plane can be determined using Eq. (8-22), just as for circular orbits. Note that the excursion range is not as large as one might anticipate, since at apogee the Earth's disk is much smaller, but the spacecraft is moving more slowly, thus providing some compensation. For example, in an extremely elliptical orbit with perigee at 500 km and apogee at geosynchronous altitude of 35,768 km, the maximum eclipse at perigee is 10.6% of the orbit period, vs. 8.3% at apogee. However, as the Sun moves out of the orbit plane, it will enter the no-eclipse region at apogee far more quickly than at perigee.

8.3 Looking at the Earth from Space

This section is intended as a cookbook for Earth coverage. Our goal is to provide a practical set of formulas and algorithms for analyzing Earth coverage from space. We will use the preceding sections for both motivation and derivations as appropriate. Formulas for a single ground station or target pass and what the spacecraft sees are covered here. Formulas for the ground track over the Earth and coverage over time are discussed in the Earth Coverage section in 10.6.

Throughout the section, we give vector formulations wherever they are significantly different from the spherical geometry formulation. To make the text easier to follow, the vector formulations have been put in inline notes, separated from the remainder of the text by a rule at the top and bottom, and a substantial philosophical gulf. Those who have reviewed the preceding two sections may safely ignore the inline notes. On the other hand, those who remain staunchly unconverted (direct descendants of those who refused to look through Galileo's telescope, no doubt) will find the necessary formulas in the inline notes and can ignore the rest of the text.

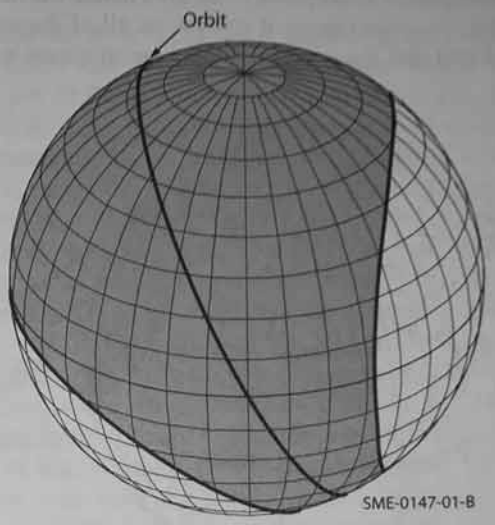


Fig. 8-25. Motion of the Disk of the Earth in an Eccentric Orbit. Shaded area is the portion of the celestial sphere blocked by the disk of the Earth. Compare with Fig. 8-7B in Sec. 8.1.1.

A spherical Earth model is used throughout the chapter. For problems which represent pure coordinate transformations, the spherical approximation to the Earth's surface is exact so long as geocentric coordinates are used for the Earth. For problems involving the physical surface of the Earth, Earth's oblateness may need to be taken into account, depending upon the accuracy required. In low-Earth orbit, Earth oblateness can shift the position of the horizon by as much as a few tenths of a degree on an angular radius of 60 to 70 deg. Consequently, oblateness may be important for data analysis, but is rarely of concern for mission analysis or planning. Alternative methods for taking oblateness into account are discussed in OCDM Sec. 8.1.5.

8.3.1 Transforming Between Earth and Spacecraft Coordinates

The most common problem in space mission geometry is to transform back and forth between Earth coordinates and the view as seen from the spacecraft. This section provides all of the fundamental formulas in both spherical geometry and vector forms for making this transformation. An example of problems of this type is to use the given coordinates of a target on the Earth to determine its coordinates in the spacecraft field of view. Another is to determine the intercept point on the surface of the Earth corresponding to a given direction in space-

craft coordinates or a given location in the field of view of a spacecraft camera.

Figure 8-26 shows the relationship between geometry on the Earth's surface and as seen from the spacecraft. A vector from the spacecraft will be tangent to the surface of the Earth at the *true* or *geometric horizon*. (The *effective horizon* introduced in Sec. 10.6 will be nearer to the spacecraft subsatellite point due to both foreshortening and atmospheric absorption.) The area inside the horizon is called the *access area*; it represents all of the surface of the Earth that the spacecraft can communicate with or

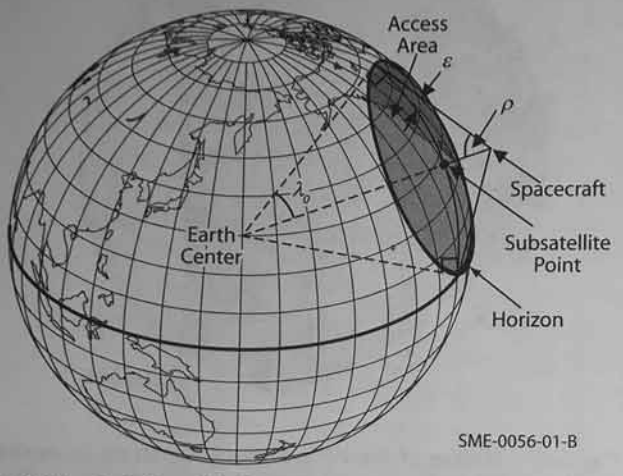


Fig. 8-26. Relationship Between Geometry as Viewed from the Spacecraft and from the Center of the Earth.

observe at this time. For a spherical Earth, the true horizon will be a small circle. The angular radius of this circle is called the *maximum Earth Central Angle*, λ_0, when measured from the center of the Earth, and the *angular radius of the Earth*, ρ, when measured from the spacecraft. The axis of the two cones formed by the horizon and either the Earth's center or the spacecraft is the *spacecraft position vector* when it goes from the center of the Earth to the spacecraft, and the *nadir vector* when it goes from the spacecraft to the center of the Earth. This line intersects the surface of the Earth at the *subsatellite point, SSP*. Finally, the direction toward the center of the Earth is called *nadir*, and the opposite direction is called *zenith*. Thus, for an observer standing at the subsatellite point, the center of the Earth is in the nadir direction and the spacecraft is at the zenith. (Nadir and zenith are well defined for an observer on the spacecraft or on the Earth's surface, but not for an observer at the center of the Earth).

For both conceptual and computational purposes, it is convenient to define an access area coordinate frame on the Earth's surface corresponding to the nadir-centered coordinate frame on the spacecraft. As shown in Fig. 8-27, the elevation component is the Earth Central Angle, λ, measured from the SSP and the azimuthal angle, Φ_E, is the rotation angle measured about SSP from a reference direction to the point in question. Depending upon the particular problem, the reference direction may be either North or the projection of the velocity vector down onto the Earth's surface. The transformation between latitude and longitude and access area coordinates is straightforward and is given in the boxed example at the end of this section.

Our fundamental geometry problem has now been reduced to transforming back and forth between access area coordinates on the surface of the Earth and spacecraft coordinates centered on nadir. This is particularly easy to do for the azimuthal component, since it is the same azimuth angle measured in either frame of reference, i.e., the azimuth is the angle between two planes—one containing the spacecraft, the center of the Earth, and the pole, and the other containing the spacecraft, the center of the Earth, and the target. In order to keep the rotation right handed, as seen from the spacecraft and on the Earth's surface, we introduce a minus sign so that:

$$\Phi_{spc} = -\Phi_E \tag{8-25}$$

In Lat/Long Coordinates:
Target at: Lat'_T, $Long_T$
SSP at: Lat'_{SSP}, $Long_{SSP}$
$\Delta L = Long_{SSP} - Long_T$

In Access Area Coordinates:
Target at: λ, Φ_E
SSP at: 0, 0

In Spacecraft Coordinate:
Target at: η, $\Phi_{spc} = \eta, -\Phi_E$
SSP at: 0,0

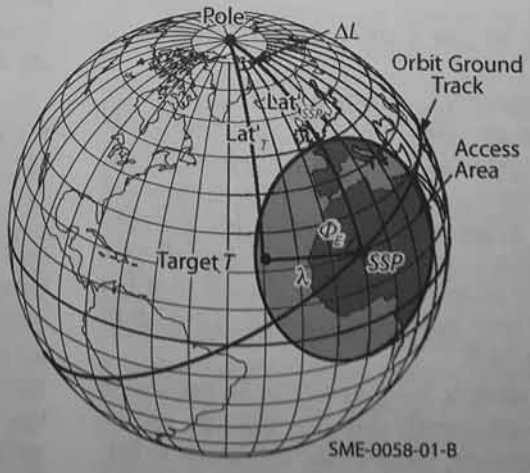


Fig. 8-27. Definition of the Access Area Coordinate Frame. See the box at the end of Sec. 8.3.1 for transformation equations.

Table 8-8◀, Fig. 8-27, Eq. 8-25

where Φ_{spc} is the azimuthal angle about nadir from the reference direction to the target point, and Φ_E is the same angle measured on the surface of the Earth about the subsatellite point.

The remaining problem is to transform the radial component. Depending on the problem at hand, it is convenient to measure this radial component either in terms of angles at the satellite, at the Earth's center, or at the target point on the surface of the Earth. As shown in Fig. 8-28, the *nadir angle*, η, is measured at the spacecraft from the subsatellite point (= nadir direction) to the target. The Earth central angle, λ, is measured at the center of the Earth from the subsatellite point to the target, and the *grazing angle* or *spacecraft elevation angle*, ε, is the angle measured at the target between the spacecraft and the local horizontal. First, we find the angular radius of the Earth, ρ, and the maximum Earth Central Angle, λ_0, from:

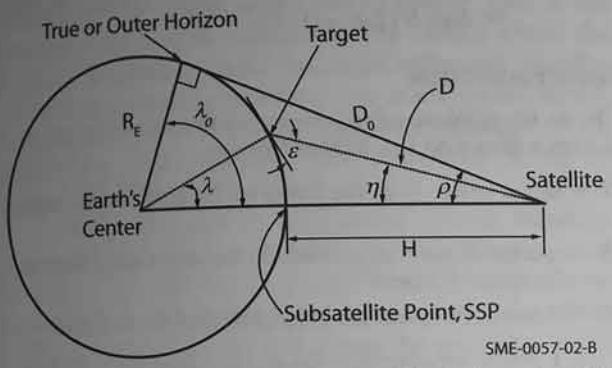


Fig. 8-28. Definition of Angular Relationships Between the Satellite, Target, and Earth's Center.

$$\sin\rho = \cos\lambda_0 = \frac{R_E}{R_E + H} \qquad (8\text{-}26)$$

where R_E is the radius of the Earth = 6,378.136 6 km and H is the altitude of the spacecraft. Next, if λ is known, we find η from:

$$\tan\eta = \frac{\sin\rho\,\sin\lambda}{1 - \sin\rho\,\cos\lambda} \qquad (8\text{-}27)$$

Or, if η is known, we find ε from:

$$\cos\varepsilon = \frac{\sin\eta}{\sin\rho} \qquad (8\text{-}28)$$

Or, if ε is known, we find η from:

$$\sin\eta = \cos\varepsilon\,\sin\rho \qquad (8\text{-}29)$$

Finally, the remaining angle is determined from:

$$\eta + \lambda + \varepsilon = 90 \text{ deg} \qquad (8\text{-}30)$$

The distance D, to the target, is:

$$D = R_E\,(\sin\lambda / \sin\eta) \qquad (8\text{-}31)$$

and the distance to the true horizon D_0 is given by:

$$D_0 = R_E / \tan\rho \qquad (8\text{-}32)$$

Vector Formulation

For the corresponding vector equations, see the boxed example at the end of this subsection.

As an example, consider a satellite at an altitude of 1,000 km. From Eq. (8-26), the angular radius of the Earth, ρ = 59.8 deg. From Eqs. (8-26) and (8-32), the horizon is 30.2 deg in Earth Central Angle from the subsatellite point and is at a line-of-sight distance of 3,709 km from the satellite. We will assume a ground station at Hawaii (D_r=22deg, L_r = 200 deg) and a subsatellite point at δ_s = 10 deg, L_s = 185 deg. From Eqs. (8-33) and (8-34), the ground station is a distance λ = 18.7 deg from the subsatellite point, and has an azimuth relative to North = 48.3 deg. Using Eqs. (8-27) and (8-31) to transform into spacecraft coordinates, we find that at the spacecraft from the subsatellite point to the target is 56.8 deg up from nadir (η) at a line of sight distance, D, of 2,444 km. From Eq. (8-30), the elevation of the spacecraft as seen from the ground station is 14.5 deg. The substantial foreshortening at the horizon can be seen in that at ε = 14.5 deg we are nearly half way from the horizon to the subsatellite point (λ = 18.7 deg vs. 30.2 deg at the horizon).

The spherical Earth approximation is adequate for most mission geometry applications. However, for precise work, we must apply a correction for oblateness as described in OCDM Sec. 8.15. The Earth's oblateness has two distinct effects on the shape of the Earth as seen from space. First, the Earth appears slightly oblate rather than round, and second, the center of the visible oblate Earth is displaced from the true geometric center of the Earth. Except when we are specifically dealing with oblateness, we will use spherical coordinates throughout this chapter, both on the Earth and in the spacecraft frame. Computationally, we can treat both oblateness and surface irregularities as simply the target's altitude above or below a purely spherical Earth or, equivalently, as a small variation in the radius of the Earth. That the Earth's real surface is both irregular and oblate is immaterial to the computation and, therefore, the results are exact.

8.3.2 Earth Features Viewed from Space

What does the surface of the Earth look like from space? This is a key question, not only for understanding what an astronaut sees, but for developing an understanding of how to design space missions and analyze data from them. We are so used to looking at projections onto the Earth that it is easy to forget that it is the geometry as seen from the spacecraft that has a major impact on the capacity of the spacecraft to fulfill its objectives.

Transformations Between Latitude/Longitude, Access Area Coordinates, and Spacecraft Nadir Coordinates

To transform between the various coordinate sets, we first define coordinates for the subsatellite point and for the *target* (i.e., the point on the surface of the Earth we wish to determine or point at). As shown in Fig. 8-27 in the text, the geocentric coordinates (latitude and longitude) for these points are:

Target Point, **T**, at $(Lat_T, Long_T)$
Subsatellite Point (*SSP*), **S**, at $(Lat_{SSP}, Long_{SSP})$
$\Delta L \equiv Long_{SSP} - Long_T$

In Area Access coordinates:

Target Point, **T**, at (λ, Φ_E)
Subsatellite Point, **S**, at $(0, 0)$

and, in Spacecraft Nadir coordinates:

Target Point, **T**, at $(\eta, \Phi_{spc}) = (\eta, -\Phi_E)$
Subsatellite Point, **S**, at $(0, 0)$

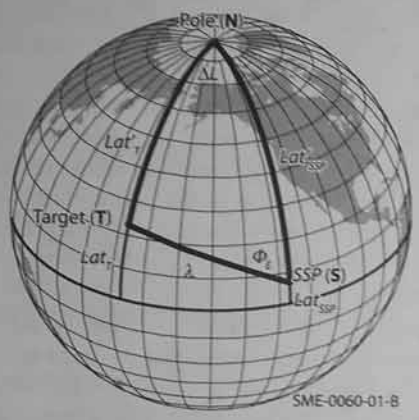


Fig. 8-29. Relationship Between the Target and the Subsatellite Point on the Earth's Surface.

Using spherical coordinates on both the Earth's surface and the spacecraft, and given the latitude and longitude for the target and *SSP*, we have:

$$\lambda = \text{acos}\left[\cos Lat'_T \cos Lat'_{SSP} + \sin Lat'_T \sin Lat'_{SSP} \cos \Delta L\right] \quad (8\text{-}33)$$

$$\Phi_E = \text{acos2}\left[\frac{\cos Lat'_T - \cos Lat'_{SSP}\cos\lambda}{\left(\sin Lat'_{SSP}\sin\lambda\right)}, H\left(\Delta L\right)\right] \quad (8\text{-}34)$$

where $Lat' \equiv (90 \text{ deg} - Lat)$ is the angle measured from the pole and is frequently more convenient to use. The acos function is evaluated over range 0 to 180 deg, and the acos2 and *hemisphere function, H,* are defined in App. D.4.* Given the area access coordinates (λ, Φ_E), we have:

$$\eta = \text{atan}\left[\sin\rho \sin\lambda / (1 - \sin\rho \cos\lambda)\right]$$
point is on visible side if $\lambda < (90 \text{ deg} - \rho)$ (8-35)

$$\varepsilon = \text{acos}\left(\sin\eta / \sin\rho\right) \quad (8\text{-}36)$$

$$\lambda = 90 \text{ deg} - \eta - \varepsilon \quad (8\text{-}37)$$

$$Lat'_T = \text{acos}\left[\cos\lambda \cos Lat'_{SSP} + \sin\lambda \sin Lat'_{SSP} \cos\Phi_E\right] \quad (8\text{-}38)$$

$$\Delta L = \text{acos}\left[\frac{\cos\lambda - \cos Lat'_T \cos Lat'_{SSP}}{H\left(Lat'_{SSP}\right)\sin Lat'_{SSP}\sin Lat'_T}\right] - 90 \text{ deg}\left[H\left(Lat'_{SSP}\right) - 1\right] \quad (8\text{-}39)$$

Vector Formulation

To do the problem using vectors, we first define the vector quantities shown on Fig. 8-29 as follows:

T = target vector from the center of the Earth to the target point

S = spacecraft position vector from the center of the Earth to the spacecraft, and

O = observation vector from the spacecraft to the target.

Given **T** and **S**, we determine the observation vector, **O**, from the spacecraft by:

$$\mathbf{O} = \mathbf{T} - \mathbf{S} \quad (8\text{-}40)$$

Given the observation unit vector, $\hat{\mathbf{O}}$, from the spacecraft, first determine the distance to Earth's surface, *D*:

$$D = -\left(\mathbf{S}\cdot\hat{\mathbf{O}}\right) \pm \sqrt{\left(\mathbf{S}\cdot\hat{\mathbf{O}}\right)^2 - |\mathbf{S}|^2 + R_E{}^2} \quad (8\text{-}41)$$

where R_E is the Earth radius (6,378.136 km), and the smaller value of *D* corresponds to the intersection on the visible portion of the Earth. Then,

$$\mathbf{O} = \left(D\cdot\hat{\mathbf{O}}\right) \quad (8\text{-}42)$$

$$\mathbf{T} = \mathbf{S} + \mathbf{O} \quad (8\text{-}43)$$

To compute the observability parameters, ρ, η, ε, λ, and Φ_E.

$$\eta = \text{acos}\left(-\hat{\mathbf{S}}\cdot\hat{\mathbf{O}}\right) \quad (8\text{-}44)$$

$$\rho = \text{asin}\left(R_E / |\mathbf{S}|\right) \quad (8\text{-}45)$$

$$\varepsilon = \text{acos}\left(\hat{\mathbf{T}}\cdot\hat{\mathbf{O}}\right) - 90 \text{ deg} \quad (8\text{-}46)$$

$$\lambda = \text{acos}\left(\hat{\mathbf{S}}\cdot\hat{\mathbf{T}}\right) \quad (8\text{-}47)$$

The point is visible if $\lambda < (90 \text{ deg} - \rho)$. Finally,

$$\Phi_E = \text{acos}\left[\left(\hat{\mathbf{S}}\times\hat{\mathbf{N}}\right)\cdot\left(\hat{\mathbf{S}}\cdot\hat{\mathbf{T}}\right) / \left|\hat{\mathbf{S}}\times\hat{\mathbf{N}}\right|\cdot\left|\hat{\mathbf{S}}\cdot\hat{\mathbf{T}}\right|\right] \quad (8\text{-}48)$$

where $\hat{\mathbf{N}}$ is the unit vector from the center of the Earth to the celestial pole.

* For complete treatment of full-sky spherical geometry see Wertz [2009], Chap. 8 and App. A.

Figure 8-30 is a photo of the Earth taken by the Apollo 17 astronauts on their way to the Moon. Northern Africa, Arabia, the Red Sea, and the Mediterranean are all clearly visible at the top of the photo. The Antarctic polar ice cap is visible at the bottom although it blends in with the clouds somewhat. The brightest feature by far is the cloud tops such that weather patterns become much more visible than features on the surface. There is also good contrast between the desert in northern Africa and the vegetation in central Africa. Although this view is nearly a "full Earth," the terminator, or sunrise/sunset line, can be seen by the slightly fuzzier, right-hand edge of the Earth's disk. Also, notice the bright spot in the clouds just above and to the left of center. This is the specular reflection of the Sun off the clouds where the angle of incidence equals the angle reflection. Finally, notice that around the perimeter you don't see any detail, even where it is cloud-free. This is an example of the geometrical foreshortening that dominates much of spacecraft viewing of the Earth.

To clearly illustrate the geometrical effects, Fig. 8-31 shows the geometry on the surface of the Earth and as seen from spacecraft for a satellite at 1,000 km over Mexico's Yucatan Peninsula at the southern edge of the Gulf of Mexico. The left side shows the geometry on the surface of the Earth. The right side shows the geometry as seen by the spacecraft projected onto the spacecraft-centered celestial sphere, i.e. what the spacecraft really sees. As computed in Sec. 8.3.1, the maximum Earth central angle will be approximately 30 deg from this altitude, such that the spacecraft can see from northwestern South America to Maine on the East Coast of the U.S. and Los Angeles on the West Coast. The angular radius of the Earth as seen from the spacecraft will be $90 - 30 = 60$ deg as shown in Fig. 8-31B. Because the spacecraft is over 20 North latitude, the direction to nadir in spacecraft-centered celestial coordinates will be 20 deg south of the celestial equator. (The direction from the spacecraft to the Earth's center is exactly opposite the direction from the Earth's center to the spacecraft.)

Even after staring at it a bit, the view from the spacecraft in Fig. 8-31B looks strange. First, recall that we are looking at the spacecraft-centered celestial sphere **from the outside**. The spacecraft is at the center of the sphere. Therefore, the view for us is reversed right-to-left relative to the view as seen by the spacecraft. Consequently, the Atlantic is on the left and the Pacific on the right. Nonetheless, there still appear to be distortions in the view. Mexico has an odd shape and South America has almost disappeared. All of this is due to the very strong foreshortening at the edge of the Earth's disk. Notice, for example, that on the Earth, Jacksonville, FL, is about halfway from the subsatellite point to the horizon. This means that only 1/4th of the area seen by the spacecraft is closer to the subsatellite point than Jacksonville. Nonetheless, as seen from the perspective of the spacecraft (Fig. 8-31B and B′), Jacksonville is 54 deg from nadir, i.e., 90% of the way to the horizon with 3/4ths of the visible area beyond it.



Fig. 8-30. Photo of the Earth from Apollo 17 on the Way to the Moon. See text for discussion. (Photo courtesy of NASA)

The rectangle in the upper left of Fig. 8-31B is the field of view of a 35 mm camera with a 50 mm focal length lens (a normal lens that is neither wide angle nor telephoto). The camera person on our spacecraft has photographed Florida and the eastern seaboard of the U.S. to approximately Maine. The region photographed on the Earth is shown in Fig. 8-31A′ and 8-31B′ and an enlargement of a portion of the photo from Georgia to Maine is shown in Fig. 8-31B″. Note the dramatic foreshortening as Long Island and Cape Cod become little more than horizontal lines (Fig 8-30B″), even though they are some distance from the horizon. This distortion does not come from the plotting style, but is what the spacecraft sees. We see the same effect standing on a hilltop or a mountain. (In a sense, the spacecraft is simply a very tall mountain, even when the spacecraft is Apollo 17 on the way to the Moon.) Most of our angular field of view is taken up by the mountain top we are standing on. For our satellite, most of what is seen is the Yucatan and Gulf of Mexico directly below. There is plenty of real estate at the horizon, but it appears very compressed.

As shown quantitatively in Fig. 8-51 in Sec. 8.6.2 we can point an antenna approximately equally well at Cancun on the Yucatan below us or at Boston, on the horizon. However, as shown in Fig. 8-50, mapping can be done extremely well directly below us, but is nearly impossible near the horizon. From the perspective of the satellite, we can draw an excellent map of the Yucatan or the Gulf of Mexico. We can point our antenna at Boston. But we can not realistically map Boston or Massachusetts from the current location of the satellite.

8.3.3 Directions, Shapes, and Areas

Figure 8-31 provides a good conceptual view of the geometry as seen from space. However, for analysis and computation, it is most convenient to work in terms of

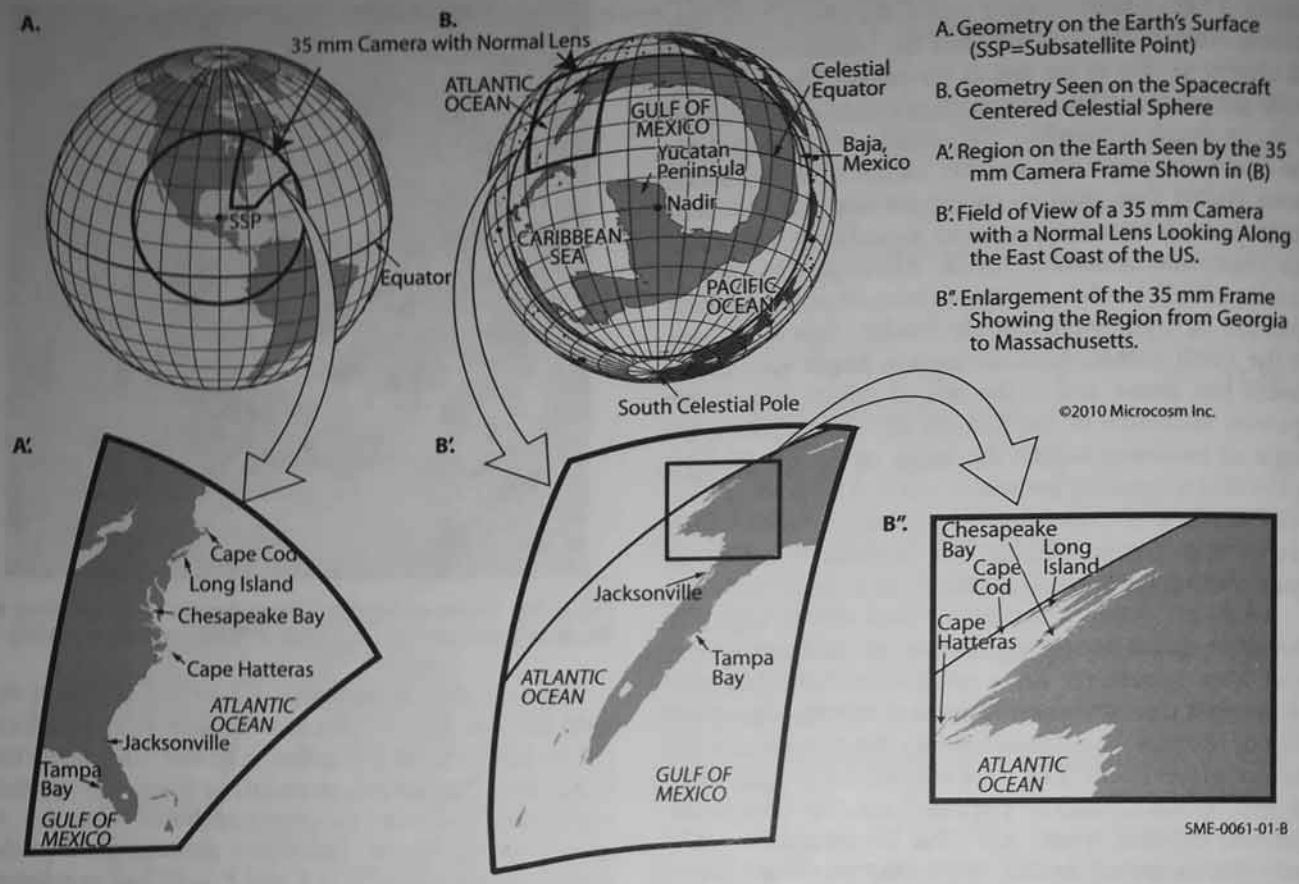


Fig. 8-31. Viewing Geometry for a Satellite at 1,000 km over the Yucatan Peninsula at 90 deg West Longitude and 20 deg N Latitude. (A) Geometry projected onto the Earth's surface. (B) Geometry of (A) as seen on the spacecraft-centered celestial sphere. See text for discussion.

points, directions, shapes and areas on the Earth's surface, and to ask how these are viewed from the spacecraft. The formulation for transforming points was given in Sec. 8.3.1. Directions, shapes and areas are described below. For viewing geometry computations, it is very convenient to introduce horizontal and vertical to refer to directions when looking at the Earth. Specifically, *horizontal* refers to the direction parallel to the Earth's horizon, either in the spacecraft coordinate frame, or on the surface of the Earth. Similarly, *vertical* refers to the direction perpendicular to the horizontal, i.e., toward or away from nadir or radial with respect to the subsatellite point. With these definitions, horizontal and vertical mean the same for the spacecraft, the access area coordinate system, and an observer standing at the subsatellite point. (See Fig. 8-32.)

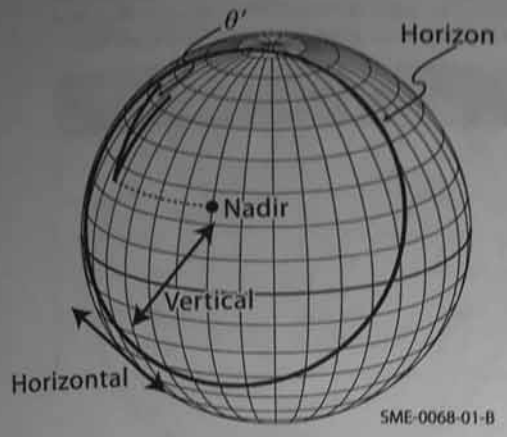
Although horizontal and vertical remain the same as seen from the spacecraft and in the access area coordinate system, angular directions relative to horizontal and vertical do not. As shown in Fig. 8-32, directions are most conveniently described relative to the horizontal. If we define θ as the angle relative to the horizontal on the surface of the Earth and θ' as the angle relative to the horizontal as seen from the spacecraft, then these are related by:

$$\tan \theta' = \tan \theta \sin \varepsilon \qquad (8\text{-}49)$$

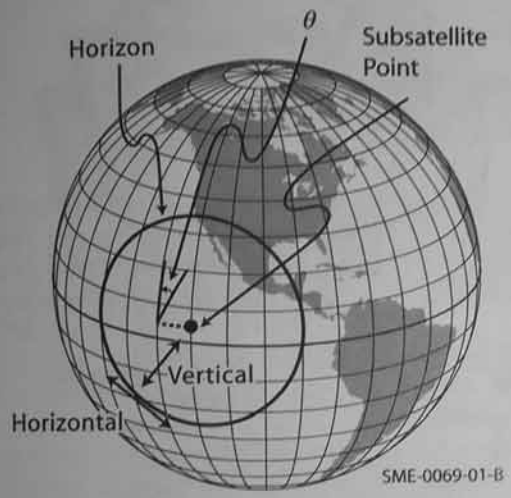
where ε is the elevation angle at the point in question. We can see from this equation another effect of foreshortening. As ε goes to 0 near the horizon, any line that isn't purely horizontal becomes nearly vertical. Thus, essentially all curves on the spacecraft centered celestial sphere, when projected onto the Earth, will intersect the horizon nearly perpendicular to it. Similarly, essentially all curves on the surface of the Earth near the horizon will become nearly horizontal when seen from the spacecraft. This can be seen clearly by looking at the shape of the coastline near the horizon in Fig. 8-31B".* For a discussion of the projection of lines from the spacecraft onto the Earth or lines on the Earth as see by the spacecraft, see OCDM Sec. 8.1.

Finally, the projection of area is also of interest. Given a number of deg^2 on the Earth's surface, what angular area in deg^2 does this represent as seen from the spacecraft? In general, the only solution is to integrate the pro-

* The effect is easily seen by taking a square piece of paper and drawing a diagonal between 2 opposite corners. Hold the paper so that you're looking at a very shallow angle and the angle between the line and the side nearest you will appear to be much less than 45 deg.



(A) On the Spacecraft-Centered Celestial Sphere



(B) On the Earth

Fig. 8-32. Directions are Most Conveniently Described Relative to the Horizontal, or Equivalently, with Respect to the Subsatellite Point or Nadir.

jected area. However, there are several convenient approximations that make estimating the angular area reasonably easy. For shapes which have a high degree of symmetry about nadir, exact formulas are readily available. These formulas are given in Table 8-9 with the definition of variables shown in Fig. 8-32.

Finally, there is a convenient approximation that can be made as accurate as desired for odd shaped areas. Angular segments, i, on the Earth centered on nadir, project into annular segments as seen from the spacecraft and the exact area, A_i, of each is easily computed.

$$A_i = \Phi_i(\cos\lambda_i - \cos\lambda_{i+1}) \tag{8-50a}$$

where Φ_i defines the width of the segment, λ_i is the inner radius, and λ_{i+1} the outer radius. The total area, A, is then simply

$$A = \Sigma A_i \tag{8-50b}$$

Since Φ_i is the same on the Earth and as seen from the spacecraft, the comparable formulas for the angular area on the spacecraft centered celestial sphere, A_i', are:

$$A_i' = \Phi_i(\cos\eta_i - \cos\eta_{i+1}) \tag{8-51a}$$

$$A' = \Sigma A_i' \tag{8-51b}$$

where η_i is related to λ_i by the previously derived equations:

$$\tan\eta_i = \frac{\sin\rho\sin\lambda_i}{1-\sin\rho\cos\lambda_i} \tag{8-52}$$

If η_i is known, we find ε_i from:

$$\cos\varepsilon_i = \frac{\sin\eta_i}{\sin\rho} \tag{8-53a}$$

If ε_i is known, we find η_i from:

$$\sin\eta_i = \cos\varepsilon_i\sin\rho \tag{8-53b}$$

Finally, the remaining angle is obtained from:

$$\eta_i + \lambda_i + \varepsilon_i = 90 \text{ deg} \tag{8-54}$$

Consequently, as shown in Fig. 8-33, any arbitrary area can be conveniently transformed by simply breaking it down into appropriately sized annular segments.

Vector Formulation

In the unit vector formulation, there is no convenient method for working with directions shapes, or areas on the celestial sphere except as collections of points. Consequently, all of the computational work in a vector approach would be done using the point transformations given in Sec 8.3.1. The only additional computational element required is to determine when a point on the Earth is beyond the horizon. This occurs whenever:

$$\hat{\mathbf{x}}\cdot\hat{\mathbf{s}} < \sin\lambda \tag{8-55}$$

where $\hat{\mathbf{x}}$ is the unit vector from the Earth's center to the point in question, $\hat{\mathbf{s}}$ is the unit vector from the Earth's center to the subsatellite point, and λ is the angular radius of either the true horizon or the effective horizon, as appropriate.

8.3.4 Projection of Sensor Fields of View onto the Earth

We now wish to work the Earth geometry problem the other way from Sec. 8.3.2, and project shapes as seen from the spacecraft (i.e., on the spacecraft-centered celestial sphere) down onto the Earth. Once again, points transform as given in Sec. 8.3.1, and directions transform as given in Sec. 8.3.3. For convenience, the transformation equations used for projecting onto the surface of the Earth are as follows:

$$\Phi_{spc} = \Phi_{Earth} = \Phi \tag{8-56}$$

$$\cos\varepsilon = \sin\eta / \cos\rho \tag{8-57}$$

$$\lambda = 90 \text{ deg} - \eta - \varepsilon \tag{8-58}$$

$$\tan\theta = \tan\theta' / \sin\varepsilon \tag{8-59}$$

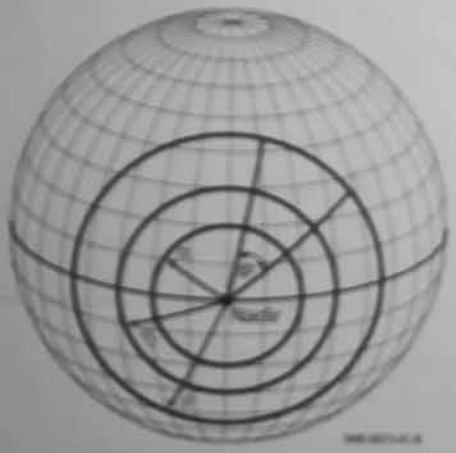
Table 8-8◀, Fig. 8-32, Eq. 8-59

Table 8-8. Formulas for the Areas of Shapes Symmetric About Nadir. See Fig. 8-32 for the definition of variables. See Eqs. (8-27) to (8-30) for transformation between η and λ. For area computations, $\Phi_{Earth} = \Phi_{SC} = \Phi$. **Note that for area formulas, all angles should be expressed in radians and angular areas in steradians.**

Shape	Area on the Surface of the Earth	Area on the Spacecraft-Centered Celestial Sphere
Entire Access Area	$2\pi (1 - \cos \lambda_0)$ $= 2\pi (1 - \sin \rho)$	$2\pi (1 - \cos \rho)$ $= 2\pi (1 - \sin \lambda_0)$
Area within Effective Horizon	$2\pi (1 - \cos \lambda)$	$2\pi (1 - \cos \eta)$
Pie Shaped Piece	$\Phi (1 - \cos \lambda)$	$\Phi (1 - \cos \eta)$
Annulus Centered on Nadir	$2\pi (\cos \lambda_1 - \cos \lambda_2)$	$2\pi (\cos \eta_1 - \cos \eta_2)$
Annular Segment Centered on Nadir	$\Phi (\cos \lambda_1 - \cos \lambda_2)$	$\Phi (\cos \eta_1 - \cos \eta_2)$

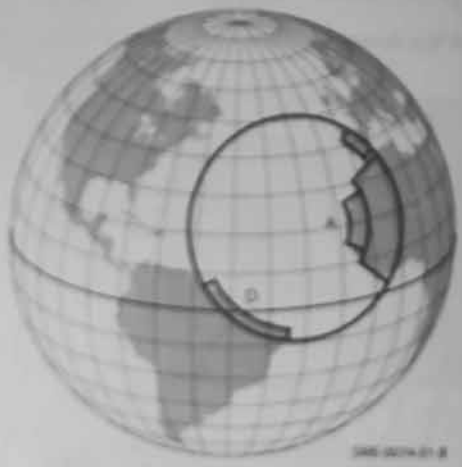


(A) On the Surface of the Earth

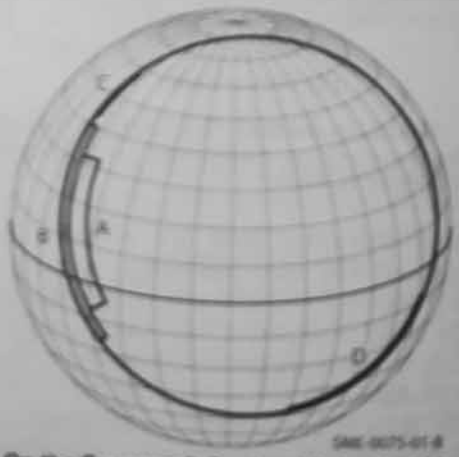


(B) On the Spacecraft-Centered Celestial Sphere

Fig. 8-33. Definition of Variable for Annular Shapes Centered on Nadir. See Table 8-9 for area formulas.



(A) On the Earth



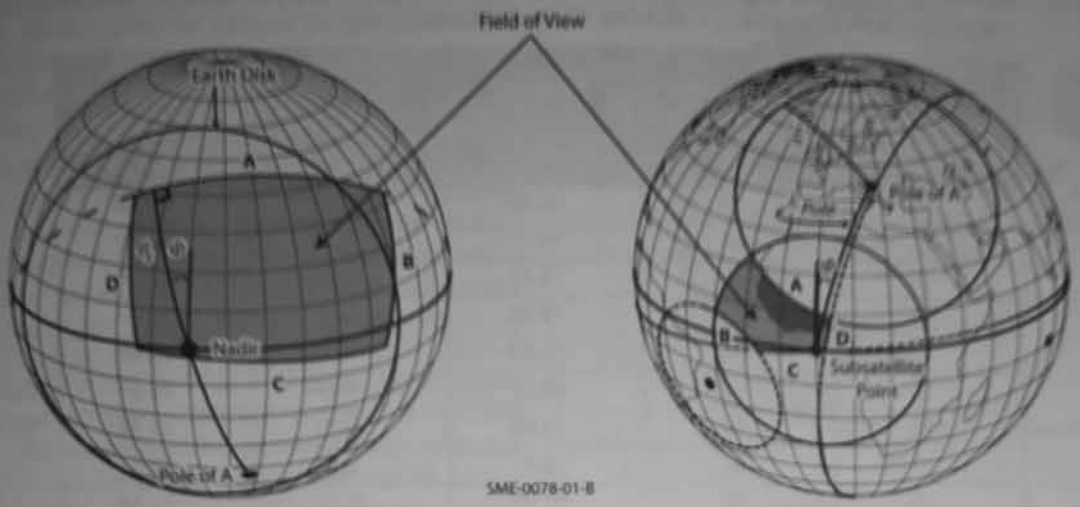
(B) On the Spacecraft-Centered Celestial Sphere

Fig. 8-34. Areas are Conveniently Transformed From Earth to Spacecraft Coordinates by Using Annular Segments for Which the Area can be Transformed Exactly.

where Φ is the azimuthal orientation of the point in question and is the same both from the spacecraft and on the surface of the Earth, ε is the elevation angle at the target point from the horizontal up to the spacecraft, η is the nadir angle measured at the spacecraft from the subsatellite point to the target, ρ is the angular radius of the Earth determined from Eq. (8-26), λ is the Earth Central Angle measured at the center of the Earth from the subsatellite point to the target, θ is the direction relative to the horizontal measured on the Earth's surface, and θ' is the direction relative to the horizontal measured on the spacecraft-centered celestial sphere. The equations in vector form are given in the box at the end of Sec. 8.3.1; there is no convenient vector form for determining the direction relative to the horizontal.

Most practical shapes (i.e., instrument fields of view and antenna patterns) are lines, circles, or rectangles, as

Table 8-9, Fig. 8-34, Eq. 8-59 ◀



(A) On the Spacecraft-Centered Celestial Sphere **(B) On the Earth**

Fig. 8-35. Projection of a Rectangular Field of View as seen from the Spacecraft onto the Earth. The field of view is equivalent to that of a 50 mm lens on a 35 mm camera. (A) as seen from the spacecraft, (B) as projected onto the Earth's surface. See Table 8-10 for Live Calc.

seen from the spacecraft. Examples include the field of view of a conical scanner, a circular antenna pattern, or an array sensor pointed at the Earth. The rectangular array is discussed here and the circular FOV is discussed in OCDM Sec. 9.1.4.

The edge of any rectangular or linear field of view from a spacecraft instrument projects onto the sky as a great circle arc and projects into space as a plane surface. The pole of the great circle on the celestial sphere is in the direction of the vector normal to the plane in space. This great circle is characterized by two numbers: the azimuthal orientation, Φ, of the pole relative to some reference and either θ_p, the angle from nadir to the pole, or θ_E, the angle from nadir to the arc of the great circle (= the edge of the field of view). Note that θ_E is measured along the great circle arc connecting nadir and the pole and is perpendicular to the great circle representing the edge. If the coordinates of the pole of the great circle representing the edge in any coordinate frame are (α_p, δ_p) and the coordinates of nadir in the same coordinate frame are (α_N, δ_N) then:

$$\sin \theta_E = \cos \theta_p = \sin \delta_p \sin \delta_N + \cos \delta_p \cos \delta_N \cos (\alpha_p - \alpha_N)$$

$$\theta_E = |\, 90 \text{ deg} - \theta_p| \qquad (8\text{-}60)$$

where θ_E is the angle from nadir to the nearest point on the edge.

The plane representing the edge of the field of view will miss the Earth altogether if:

$$\theta_E > \rho \quad \text{(great circle misses Earth)} \qquad (8\text{-}61)$$

where ρ is the angular radius of the Earth. If $\theta_E = 0$, then the plane passes through nadir and the projection of the plane onto the surface of the Earth will be a great circle through the subsatellite point. If $0 < \theta_E < \rho$, then the plane intersects the spherical surface of the Earth in a small circle. Therefore, in general, **the edges of a rectangular field of view from the spacecraft will intersect the surface of the Earth in a set of four intersecting small circles**. A small circle on the surface of the Earth is fully defined by the center and radius. As always, the azimuthal orientation of the center of the small circle is the same as seen from the spacecraft and as projected onto the surface of the Earth. (Note that the azimuth changes sign if we wish to maintain a right-handed coordinate system in both cases.) The distance λ_{pole} from the pole of the small circle to nadir is given by:

$$\lambda_{pole} = 90 \text{ deg} - \theta_E = \theta_p \qquad (8\text{-}62)$$

and finally, the radius of the small circle, ξ, is given by:

$$\cos \xi = \frac{(R_E + H) \sin \theta_E}{R_E} \qquad (8\text{-}63)$$

where R_E is the radius of the Earth and H is the altitude of the spacecraft.

An alternative approach to projecting the field of view is to project the corners of the field of view onto the Earth using the point transformation equations at the end of Sec. 8.3.1 and the radius of the small circle defined by Eq. (8-63) to construct the four sides. We could also use any three points along the small circle such as two of the corners and either the nearest point to nadir or the point at which the small circle intersects the horizon.

The equations above were used to construct Fig. 8-35, which shows the projection of a rectangular field of view down onto the surface of the Earth. Note that the azimuthal orientation of all of the points remains the same on both the spacecraft-centered celestial sphere and on the surface of the Earth (to within the sign change for opposite directions), but that the radial distortion caused by

Table 8-10. Planes or Edges Projected onto the Surface of the Earth. Altitude = 1,000 km. Therefore, $\rho = 59.8$ deg and $\lambda_0 = 30.2$ deg. All values in deg.

Live Calc

On Spc Dist. from Nadir to Edge (θ_E)	On Spc. Dist. from Nadir to Pole (θ_P)	On Earth Radius (φ)	On Earth Dist. from Edge to SSP (λ_{Pole})	On Earth Dist. from Edge to Horizon ($\lambda_0 - \lambda_{Pole}$)
0	90	90.00	0.00	30.18
5	85	84.21	0.79	29.39
10	80	78.41	1.59	28.59
15	75	72.58	2.42	27.76
20	70	66.69	3.31	26.87
25	65	60.73	4.27	25.91
30	60	54.66	5.34	24.84
35	55	48.43	6.57	23.61
40	50	41.96	8.04	22.14
45	45	35.12	9.88	20.30
50	40	27.61	12.39	17.79
55	35	18.63	16.37	13.81
56	34	16.46	17.54	12.64
57	33	14.03	18.97	11.21
58	32	11.18	20.82	9.36
59	31	7.45	23.55	6.63
59.5	30.5	4.64	25.86	4.32

projection onto the surface of the Earth transforms the great circle arcs on the spacecraft-centered celestial sphere into small circle arcs on the surface of the Earth.

Most sensors will have fields of view which are more symmetric with respect to nadir than that in Fig. 8-35, and the angular parameters with respect to nadir can frequently be determined by inspection. The approximately rectangular shape is largely retained when the field of view is oriented toward nadir. The more strongly curved edges occur when the edge of the sensor lies near or crosses the edge of the disk of the Earth. Thus, in Fig. 8-35, sides C and D pass relatively close to nadir and, therefore, are nearly great circles when projected onto the Earth. Side A is further from nadir and has a distinctly curved edge on the Earth. Side B is nearly parallel to the horizon and, as seen from the spacecraft, crosses the horizon at a very shallow angle. However, when projected onto the Earth it still crosses the horizon at essentially a right angle and is strongly curved. Since a corner of the FOV is off the Earth, the projection onto the Earth has a fifth side consisting of the horizon itself. Of course each of the rows and columns within the FOV could also be represented by corresponding small circles projected onto the Earth. Table 8-10 shows the angular radius for various small circles and the distance from the edge of the small circle to the subsatellite point for straight lines projected onto the surface of the Earth from an altitude of 1,000 km. As we approach the edge of the disk of the Earth, the curvature of the projection becomes very strong.

Table 8-10, Fig. 8-35◀, Eq. 8-63◀

8.4 Computing Parameters for a Single Target or Ground Station Pass

We define a *target* as an object on or near the Earth's surface that we wish to observe or track. A *ground station* is an antenna fixed on the Earth with which we wish to communicate. However, for purposes of the relative geometry and motion there is essentially no distinction between them and we will use the two terms interchangeably throughout this section. Overall, it makes very little difference whether the target is stationary or moving over the surface of the Earth, such as airplanes flying in and out of a busy airport. The satellite velocity is far greater than any Earth targets, such that nearly all targets are essentially stationary in terms of coverage analysis as seen from spacecraft.* (We can pick out moving targets using Doppler radar but they do not move very far in the time that the satellite is observing them.) The only realistic exception is rocket launches, which, of course, approach orbital velocities.

8.4.1 Circular LEO Orbits

This section looks at the characteristics of a ground station pass for a satellite in low-Earth orbit. We begin by looking at the qualitative characteristics of this pass and then compute the relevant parameters for the pass based

* For the motion of a satellite in geosynchronous orbit that is nearly fixed with respect to the Earth's surface, see Sec. 9.5.1.

on the assumption that the ground trace is a great circle arc. Finally, we compute the ground station parameters using the substantially more accurate approximation of the ground trace as a small circle centered on the instantaneous rotation axis.

Figure 8-36A shows the geometry of a ground station pass as seen on the surface of the Earth. Three representative orbit passes are drawn, one which passes directly overhead the ground station at Hawaii, one which passes within 5 deg of the ground station, and one which passes within 15 deg of the ground station. The appearance of these three orbits on the ground station-centered celestial sphere is shown in Fig. 8-36B. The general parameters for the shape of this orbit are defined in OCDM Sec. 6.3.4. The satellite which passes directly over the station will travel along a great circle arc which passes through the zenith. The others will have the appearance of an orbit viewed from nearby. Although the satellite moves uniformly along the orbit path in Fig. 8-36A, it does not move uniformly across the sky as seen from the ground station (Fig. 8-36B). Generally, the spacecraft will appear to travel slowly when it is near the horizon due to both the greater distance and the substantial foreshortening. It will move most rapidly in apparent angular velocity as it passes the point nearest the ground station. It then repeats the process in a symmetric fashion and leaves slowly over the "forward horizon" (i.e., in the direction the satellite is moving).

Because the apparent satellite path is not a simple geometrical figure, it is best computed using a simulation program. A variety of commercial tools are available to do this. These programs also work with elliptical orbits so they are convenient, along with the appropriate formulas from this chapter, for evaluating specific orbit geometry. Unfortunately, a simulation does not provide the desired physical insight into the apparent motion of satellites, nor does it provide a rapid method for evaluating geometry in the general case, as is most appropriate when first designing a mission or trying to determine the most common parameters for a ground station pass. For these problems, we are interested in bounding or approximating the apparent motion rather than in computing the detailed path precisely.

Throughout this section, we use the notation shown in Fig. 8-37, which shows the geometry of a ground station pass where we have approximated the ground track by a great circle centered on the orbit pole. (Later in this section, we will correct that approximation by shifting to a ground track centered on the instantaneous rotation axis.) The small circle centered on the ground station represents the subsatellite points at which the spacecraft elevation, ε, seen by the ground station is greater than some *minimum working elevation angle* ε_{min}. The nature of the communication or observation will determine the value of ε_{min}. For ground station communications, the satellite typically must be more than 5 deg above the horizon, so $\varepsilon_{min} = 5$ deg. The size of this circle of accessibility strongly depends on the value of ε_{min}, as emphasized in

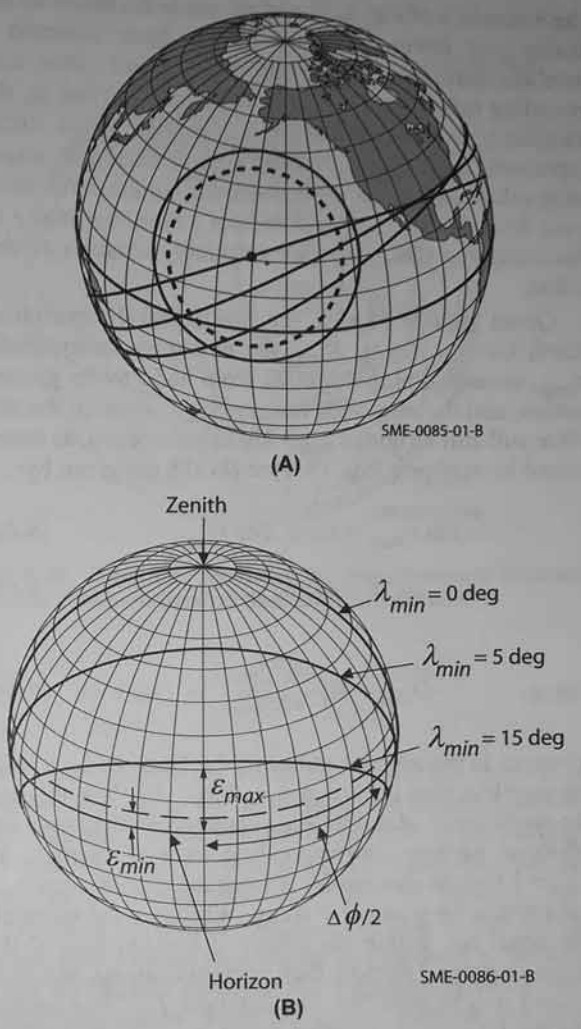


Fig. 8-36. Apparent Motion as Seen From the Earth of a Satellite in 1,000 km Circular Orbit. (A) Geometry on the Earth's surface. (B) Geometry on the ground station-centered-celestial sphere.

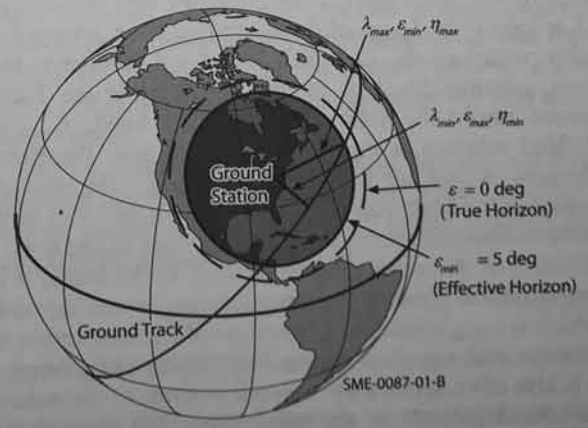


Fig. 8-37. Geometry of Satellite Ground Track Relative to an Observer on the Earth.

the discussion of Fig. 8-31 and shown in the tables on the inside rear cover. In Fig. 8-37 we have assumed a satellite altitude of 1,000 km. The dashed circle surrounding the ground station is at $\varepsilon = 0$ deg (that is, the satellite's true outer horizon), and the solid circle represents $\varepsilon_{min} = 5$ deg. In practice we often select a specific value of ε_{min} and use that number for the entire analysis. However, you should remain aware that many of the computed parameters are extremely sensitive to this value.

Given a value of ε_{min}, we can define the *maximum Earth Central Angle*, λ_{max}, the *maximum nadir angle*, η_{max}, measured at the satellite from nadir to the ground station, and the *maximum range*, D_{max}, at which the satellite will still be above ε_{min}. These parameters, as determined by applying Eqs. (8-4) to (8-31), are given by:

$$\sin \eta_{max} = \sin \rho \cos \varepsilon_{min} \tag{8-64}$$

$$\lambda_{max} = 90 \text{ deg} - \varepsilon_{min} - \eta_{max} \tag{8-65}$$

$$D_{max} = R_E \frac{\sin \lambda_{max}}{\sin \eta_{max}} \tag{8-66}$$

where ρ is the angular radius of the Earth as seen from the satellite; that is, $\sin \rho = R_E/(R_E + H)$. We will call the small circle of radius λ_{max} centered on the target the *effective horizon*, corresponding in our example to $\varepsilon_{min} = 5$ deg, to distinguish it from the *true* or *geometrical horizon* for which $\varepsilon = 0$ deg. Whenever the subsatellite point lies within the effective horizon around the target or ground station, then communications or observations will be possible. The duration, T, of this contact and the maximum elevation angle, ε_{max}, of the satellite will depend on how close the ground station is to the satellite's ground track on any given pass.

As described in Chap. 9, the plane of a spacecraft's orbit and, therefore, the ground track, is normally defined by the inclination, i, and either the right ascension, Ω, or longitude, L_{node}, of the ascending node. Except for orbit perturbations, Ω, which is defined relative to the stars, remains fixed in inertial space while the Earth rotates under the orbit. On the other hand, L_{node} is defined relative to the Earth's surface and, therefore, changes by 360 deg in 1,436 min, which is the rotation period of the Earth relative to the stars. (Recall that orbit perturbations affect the exact rotation rate of the node.) Because of this orbit rotation relative to the Earth, it is convenient to speak of the *instantaneous ascending node* which is L_{node} evaluated at the time of an observation or passage over a ground station. For purposes of geometry it is also often appropriate to work in terms of the *instantaneous orbit pole*, or the pole of the orbit plane at the time of the observation. The coordinates of this pole are:

$$lat_{pole} = 90 \text{ deg} - i \tag{8-67}$$

$$long_{pole} = L_{node} - 90 \text{ deg} \tag{8-68}$$

A satellite will pass directly over a target or ground station (represented by the subscript gs) on the Earth's surface if and only if:

$$\sin (long_{gs} - L_{node}) = \tan lat_{gs} / \tan i \tag{8-69}$$

There are two valid solutions to the above equation, corresponding to the satellite passing over the ground station on the northbound leg of the orbit or on the southbound leg. To determine when after crossing the equator the satellite passes over the ground station for a circular orbit, we can determine μ, the arc length along the instantaneous ground track from the ascending node to the ground station, from:

$$\sin \mu = \sin lat_{gs} / \sin i \tag{8-70}$$

Again, the two valid solutions correspond to the northbound and southbound passes.

Figure 8-37 defines the parameters of the satellite's ground station pass in terms of λ_{min}, the minimum Earth central angle between the satellite's ground track and the ground station. This will be 90 deg minus the angular distance measured at the center of the Earth from the ground station to the instantaneous orbit pole at the time of contact. If we know the latitude and longitude of the orbit pole and ground station (gs), then the value of λ_{min} will be given by:

$$\sin \lambda_{min} = \sin lat_{pole} \sin lat_{gs} + \cos lat_{pole} \cos lat_{gs} \cos (\Delta long) \tag{8-71}$$

where $\Delta long$ is the longitude difference between gs and the orbit pole. At the point of closest approach, we can compute the minimum nadir angle, η_{min}, maximum elevation angle, ε_{max}, and minimum range, D_{min} as:

$$\tan \eta_{min} = \frac{\sin \rho \sin \lambda_{min}}{1 - \sin \rho \cos \lambda_{min}} \tag{8-72}$$

$$\varepsilon_{max} = 90 \text{ deg} - \lambda_{min} - \eta_{min} \tag{8-73}$$

$$D_{min} = R_E \frac{\sin \lambda_{min}}{\sin \eta_{min}} \tag{8-74}$$

At the point of closest approach, the satellite will be moving perpendicular to the line of sight to the ground station. Thus, the *maximum angular rate* of the satellite as seen from the ground station, $\dot{\theta}_{max}$ will be:

$$\dot{\theta}_{max} = \frac{V_{sat}}{D_{min}} = \frac{2\pi (R_E + H)}{P D_{min}} \tag{8-75}$$

where V_{sat} is the orbital velocity of the satellite, and P is the orbit period from Chap. 9 or the inside rear cover.

Finally, it is convenient to compute the *total azimuth range*, $\Delta\phi$, which the satellite covers as seen by the ground station, the *total time in view*, T, and the azimuth, ϕ_{center}, at the center of the viewing arc where the elevation angle is a maximum:

Table 8-10◀, Fig. 8-37◀, Eq. 8-75

$$\cos\frac{\Delta\phi}{2} = \frac{\tan\lambda_{min}}{\tan\lambda_{max}} \tag{8-76}$$

$$T = \left(\frac{P}{180\ \text{deg}}\right) \text{acos}\left(\frac{\cos\lambda_{max}}{\cos\lambda_{min}}\right) \tag{8-77}$$

where the arc cos is in degrees. ϕ_{center} is related to ϕ_{pole}, the azimuth to the direction of the projection of the orbit pole onto the ground, by

$$\phi_{center} = 180\ \text{deg} - \phi_{pole} \tag{8-78}$$

$$\cos\phi_{pole} = (\sin lat_{pole} - \sin\lambda_{min}\sin lat_{gs}) / (\cos\lambda_{min}\cos lat_{gs}) \tag{8-79}$$

where $\phi_{pole} < 180$ deg if the orbit pole is east of the ground station and $\phi_{pole} > 180$ deg if the orbit pole is west of the ground station.

Table 8-11 summarizes the computations for ground station coverage and provides a worked example. As indicated above, T is particularly sensitive to ε_{min}. If we assume a mountain-top ground station with $\varepsilon_{min} = 2$ deg, then the time in view for the example increases by 15% to 14.27 min.

Thus far, we have ignored the impact of the rotation of the Earth on ground station coverage. While this is a reasonable simplification for some cases, the assumption becomes less valid as the altitude of the satellite becomes higher. In addition, the rotating Earth under the satellite changes not only the velocity of the satellite relative to the Earth's surface but also its direction of motion. The dual-axis spiral formulation developed in OCDM Chap. 8 and summarized here, provides the mechanism to readily take this rotation into account. The geometry of doing so is shown in Fig. 8-38. Fundamentally, we want to replace the great circle centered on the orbit pole with a small circle of appropriate radius centered on the instantaneous rotation axis.* This will correctly model the direction, the velocity, and the curvature of the orbit as it passes the ground station, all of which are slightly different than the motion over a non-rotating Earth.

We begin by noting that the *instantaneous rotation axis* or *Euler axis*, **E**, lies on the great circle containing the orbit pole and the Earth's pole, with a latitude given by

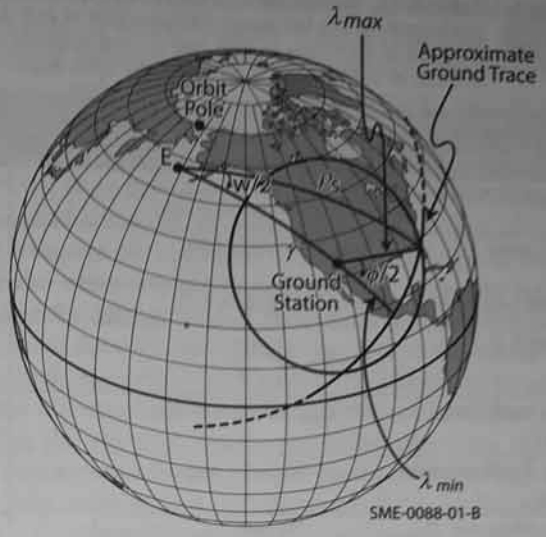


Fig. 8-38. Geometry for Using the Instantaneous Rotation Axis to Compute Ground Station Parameters.

$$\delta_E = \text{lat}_E = \text{atan}\left(\frac{\omega_{Earth} + n\cos i}{n\sin i}\right) \tag{8-80}$$

where $n = 6 / P$ is the mean motion of the satellite in deg/sec, P is the period in minutes, i is the orbit inclination, and ω_{Earth} is the inertial rotation rate of the Earth on its axis = 0.004 178 075 deg/sec. [To take the node regression rate into account see OCDM Eq. (9-66).]

There is a potential complication here in that the instantaneous rotation axis should be evaluated at the time when the satellite passes at the minimum distance from the ground station. Since we cannot compute this time precisely, we would in principle be left to an iterative process to determine the right time to evaluate both the orbit pole for the previous calculation and the instantaneous rotation axis for this calculation. However, the orbit pole and the instantaneous rotation axis are rotating about the Earth's pole at a rate of approximately 0.004 deg/sec. Consequently, providing the correct time to evaluate the position of the rotation axes to within even a few minutes is sufficiently accurate for most analysis purposes.

Having determined the position of the pole, we now determine the angular rotation rate, ω_E, about the instantaneous rotation axis, **E**, and the arc length distance, ρ_S, from **E** to the subsatellite point by:

$$\omega_E = n(\sin i / \cos\delta_E) \tag{8-81}$$

$$\rho_S = \text{acos}(\sin\delta\ \sin\delta_E + \cos\delta\ \cos\delta_E\ cos\ \Delta\alpha) \tag{8-82}$$

We then use the standard formula for arc lengths on the surface of the Earth to determine γ, the distance from the instantaneous rotation axis to the ground station:

$$\cos\gamma = \sin\delta_E\sin\delta_{gs} + \cos\delta_E\cos\delta_{gs}\cos(\alpha_E - \alpha_{gs}) \tag{8-83}$$

Table 8-10◀, Fig. 8-38, Eq. 8-83

* We can think of the satellite as being fixed in a rigid coordinate frame rotating in inertial space. Similarly, the surface of the Earth is also a rigid coordinate frame rotating about a different axis in inertial space. The most general motion of two rigid, rotating coordinate frames is a rotation about an *instantaneous rotation axis* or *Euler axis*, **E**, that is fixed instantaneously, but moving over time in both the orbit frame and the Earth frame. (Similarly, the wheel on a car is rotating instantaneously about the point of contact with the ground, but that point is continuously moving with respect to both the wheel and the ground.) See OCDM Chap. 8 for a detailed discussion and development of the relevant equations.